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超对称规范理论中的指标计算

Index Computations in Supersymmetric Gauge Theories

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中文摘要

量子场论的一个核心问题是如何理解强耦合区域中的动力学，尤其是微扰展开无法捕捉的非微扰效应。瞬子、孤子、磁单极、拓扑扇区、真空隧穿和强耦合对偶等现象决定了低能有效作用量、真空结构、质量谱、相变和受保护可观测量。超对称规范理论为研究这些问题提供了特别有效的框架：全纯性、短表示和受保护指标约束量子修正，而超对称局域化、Seiberg–Witten 理论、D 膜工程和 ADHM 构造把无限维路径积分化为有限维留数积分、模空间积分或组合对象的求和。本文围绕超对称指标计算，研究两个方向：由四维理论在 S^2 上紧致化得到的二维对偶理论，以及由 ADHM 瞬子计数、规范折纸和高维杨图引出的壳公式。

第一类问题是如何用受保护指标检验或发现不同量子场论之间的对偶关系。对偶性是强耦合物理中的基本结构：两个拉氏量、自由度和作用量看似不同的理论，在红外极限或受保护扇区中可能描述同一个物理系统。本文研究由四维超对称理论紧致化到二维后得到的 $\mathcal{N} = (2, 2)$ 和 $\mathcal{N} = (0, 2)$ 理论，并用椭圆亏格作为检验对偶的精确量。相应的计算链为

受保护指标 $\longrightarrow$ Jeffrey–Kirwan 留数 $\longrightarrow$ 反常、 R 荷与超势一致性 $\longrightarrow$ 对偶理论。

第二类问题是如何把 ADHM 瞬子量子力学中的极点组织为统一的固定点组合语言。五维瞬子计数、含定向反演平面或非基本表示物质的 ADHM 模型、D0–D6/D0–D8 系统以及 Donaldson–Thomas 型计数都表明，普通二维杨图只是更一般高维组合结构的一个截面。对于这些系统，本文使用壳、壳荷和 $\mathcal{J}$ -因子来描述固定点贡献。相应的计算链为

超对称局域化 $\longrightarrow$ 受保护指标 $\longrightarrow$ Jeffrey–Kirwan 留数 $\longrightarrow$ 杨图固定点 $\longrightarrow$ 壳公式。

第二章回顾全文使用的受保护指标和留数技术。在一维超对称量子力学中，Witten 指标计数受保护基态；在二维理论中，椭圆亏格是环面上的 Ramond–Ramond 迹，局域化后成为椭圆函数的留数积分。本文采用的基本记号为

$$\mathcal{J}_{1d} = \text{Tr}_{\mathcal{H}}(-1)^F e^{-\beta H} \prod_a y_a^{F_a},$$
$$\mathcal{J}_{2d}^{\mathcal{N}=(2,2)}(q, y) = \text{Tr}_{\text{RR}}(-1)^F q^{L_0 - c/24} \bar{q}^{\bar{L}_0 - \bar{c}/24} \prod_a y_a^{F_a}.$$

局域化后, 规范理论指标具有如下 Jeffrey–Kirwan 留数形式:

$$\mathcal{Z} = \frac{1}{|\mathcal{W}|} \text{JK-Res}_\eta \left[\mathcal{J}_{1\text{-loop}}(\mathbf{u}) d\mathbf{u}_1 \wedge \cdots \wedge d\mathbf{u}_r \right]$$

其中 $\mathcal{J}_{1\text{-loop}}$ 是一圈行列式。在一维量子力学中, 基本一圈因子由双曲函数 $\text{sh}(x) = e^{x/2} - e^{-x/2}$ 组成; 在二维环面指标中, 相应因子由 ϑ 函数和 Dedekind η 函数组成, 并且必须满足规范逸度的双周期性。第二章还统一介绍箭图记号: 圆形节点表示规范群, 方形节点表示味对称, 实线和虚线分别表示手征多重态与费米多重态或相应的一维自由度; 不同箭头、方向和表示决定被积函数中的双曲函数或 ϑ 函数贡献。Jeffrey–Kirwan 留数给出规范积分的取极点处方, 并与 Fayet–Iliopoulos 参数、 η -室、退化极点正则化以及非紧模空间中的库仑连续谱有关。

第三章回顾四维与五维规范理论背景, 并为第五章的瞬子与规范折纸计算作准备。本文首先讨论四维 $\mathcal{N} = 2$ 理论的低能有效作用量、库仑分支、Seiberg–Witten 几何和 BPS 谱, 再说明这些结构在五维 $\mathcal{N} = 1$ 理论中的推广。五维理论中瞬子数成为粒子荷, 规范耦合、Chern–Simons 项和质量参数共同决定 BPS 粒子配分函数。引入 Ω -背景后, ADHM 量子力学的 Witten 指标化为有限维 Jeffrey–Kirwan 留数积分, 并进一步局域化到由杨图标记的离散固定点。典型瞬子配分函数写为

$$\mathcal{Z}_{\text{inst}}(\mathbf{a}, \mathbf{m}, \epsilon_i, \mathbf{q}) = \sum_{k \geq 0} \mathbf{q}^k \oint_{\text{JK}} \prod_{I=1}^k \frac{d\phi_I}{2\pi i} \mathcal{J}_k(\phi, \mathbf{a}, \mathbf{m}, \epsilon_i) = \sum_{\vec{\lambda}} \mathbf{q}^{|\vec{\lambda}|} \mathcal{Z}_{\vec{\lambda}}(\mathbf{a}, \mathbf{m}, \epsilon_i).$$

其中 $\vec{\lambda}$ 是固定点处的杨图数据。膜网、定向反演平面和希格斯化构造在本文中主要作为物理动机: 它们解释为什么普通二维杨图需要推广到三维、四维以及带边界条件的高维杨图, 也解释为什么固定点贡献需要一种比臂长和腿长更统一的组织方式。

第四章研究由四维理论得到的二维超对称对偶。本章首先给出 S^2 上的拓扑扭曲和零模约化字典: 选择无反常的 R 对称与味对称混合, 打开规范或背景味通量, 并用 Riemann–Roch 定理计算四维手征多重态在二维中的手征或费米零模。四维矢量多重态约化为二维矢量多重态, 四维 $\mathcal{N} = 2$ 矢量多重态和超多重态在适当扭曲下分别给出二维 $\mathcal{N} = (2, 2)$ 矢量多重态和手征多重态。随后本文研究 $\mathcal{S}$ 类理论的 $\mathcal{N} = (2, 2)$ 约化。六维 $(2, 0)$ 理论在带标记点黎曼面上的紧致化产生四维 $\mathcal{N} = 2$ 理论, 不同裤分解对应不同四维对偶框架; 在进一步作 S^2 紧致化后, 这些框架变成二维箭图描述。

在 $\mathcal{N} = (2, 2)$ 部分, 本文研究由 $\mathcal{S}$ 类基本构件 $U_N^{(2,2)}$ 胶合得到的二维箭图

理论。该基本构件的椭圆亏格贡献为

$$\mathcal{F}_{\mathrm{U}_N}^{(2,2)}(\mathbf{a}, \mathbf{b}, \mathbf{c}) = \prod_{i,j=1}^N \frac{\vartheta_1(y a_i b_j c)}{\vartheta_1(a_i b_j c)}.$$

对 $\mathrm{SU}(N)$ 味对称作规范化时，需要加入矢量多重态贡献：

$$\mathcal{F}_{\mathrm{SU}(N)}^{(2,2),\mathrm{V}}(\zeta) = \frac{1}{N!} \left(\frac{\eta(q)^3}{\vartheta_1(y)} \right)^{N-1} \prod_{\substack{i,j=1 \\ i \neq j}}^N \frac{\vartheta_1(\zeta_i/\zeta_j)}{\vartheta_1(y \zeta_i/\zeta_j)}$$

并通过 Jeffrey–Kirwan 留数完成规范化。对于亏格一旦带 n 个极小标记点的理论，椭圆亏格化为简单乘积

$$\mathcal{F}_{g=1,n}^{(2,2),N}(c_1, \dots, c_n) = \prod_{s=1}^n \frac{\vartheta_1(y^N c_s^N) \vartheta_1(y)}{\vartheta_1(c_s^N) \vartheta_1(y^N)}.$$

对于亏格二，本文比较了不同箭图胶合方式给出的留数积分，并得到相同结果，例如

$$\mathcal{F}_{g=2}^{(2,2),N} = N \frac{\vartheta_1(y) \vartheta_1(y^N d^{2N})}{\vartheta_1(y^N) \vartheta_1(d^{2N})}.$$

这些公式说明，相关二维 $(2,2)$ 箭图虽然具有不同拉氏量描述，但在椭圆亏格层面表现出对偶框架无关性和类似拓扑量子场论的结构。

在 $\mathcal{N} = (0,2)$ 部分，本文以二维 SU 型 Seiberg 对偶作为基本输入和检验标准。一边是

$$\mathrm{SU}(n_3), \quad n_3 = \frac{N_1 + N_2 - N_3}{2},$$

含 N_1 个基本表示手征多重态、 N_2 个反基本表示手征多重态、 N_3 个反基本表示费米多重态以及单态介子；另一边是

$$\mathrm{SU}(n_2), \quad n_2 = \frac{N_1 + N_3 - N_2}{2},$$

含相应的对偶手征多重态、费米多重态和介子，并带有相容的 $(0,2)$ 超势。基于这一基本对偶，本文进一步提出并检验若干新的三重对偶和张量物质对偶。反对称张量方向得到

$$\mathrm{Sp}(2N) + (2N+2)\square \iff \mathrm{SU}(2N) + 1\mathbf{AS} + (2N+2)\square + \Psi \iff \text{朗道-金兹堡模型}(A, \Psi),$$

其中超势分别包含 $\mathcal{W} = \Psi \mathrm{Pf} X$ 或 $\mathcal{W} = \Psi \mathrm{Pf} A$ 。对称张量方向得到

$$\mathrm{SO}(N) + (N-2)\square + X + \Psi \iff \mathrm{SU}(N) + 1\mathrm{Sym} + (N-2)\square + \Psi \iff \text{朗道-金兹堡模型}(A, \Psi),$$

其中超势包含 $\mathcal{W} = \Psi(X^2 + \det A)$ 或 $\mathcal{W} = \Psi \det A$ 。

利用这些基本对偶，本文还构造了更一般的 SU 张量物质对偶。对于带一个反对称张量手征多重态的理论，本文得到

$$\mathrm{SU}(N_A) + 1\mathbf{AS} + n_1\overline{\square} + N\square + m_1\overline{\square}_F \iff \mathrm{SU}(N_B) + 1\mathbf{AS} + n_2\overline{\square} + N\square + m_2\overline{\square}_F,$$

其中

$$N_A = N + n_1 - m_1 - 2, \quad N_B = N + n_2 - m_2 - 2, \quad m_i = 0, 1.$$

对于带一个对称张量手征多重态的理论，对应的秩变为

$$N_A = N + n_1 - m_1 + 2, \quad N_B = N + n_2 - m_2 + 2.$$

此外，本文还讨论了 $\mathrm{Sp}(2N)$ 、 $\mathrm{SO}(2N)$ 、 $\mathrm{SO}(2N+1)$ 带伴随手征多重态的对偶，例如

$$\mathrm{Sp}(2N) + 1\mathbf{Adj} \iff N \text{ 个自由手征多重态} + N \text{ 个自由费米多重态}.$$

这些候选对偶均通过椭圆亏格、Jeffrey–Kirwan 留数、规范反常、味反常、引力反常、 R 荷分配以及 E, J -项一致性进行检验。对于可以解析化简的情形，本文给出椭圆亏格恒等式；对于更复杂的秩和逸度依赖，本文只将有限秩留数计算和有限 q -展开作为强证据，而不将其误写成一般解析证明。

第五章研究瞬子配分函数、规范折纸与壳公式。普通五维 ADHM 瞬子计数是这一部分的入口。对于五维 $\mathcal{N} = 1$ $\mathrm{U}(N)$ 规范理论，Nekrasov 瞬子配分函数写为

$$\mathcal{Z}_{\mathrm{inst}}^{\mathrm{U}(N)}(v_1, \dots, v_N) = \sum_{k=0}^{\infty} \mathbf{q}^k \mathcal{Z}_{N,k}^{\mathrm{U}}(v_1, \dots, v_N),$$

其中

$$\begin{aligned} \mathcal{Z}_{N,k}^{\mathrm{U}}(v_1, \dots, v_N) &= \oint_{\mathrm{JK}} \prod_{i=1}^k \frac{d\phi_i}{2\pi i} \mathcal{J}_{N,k}^{\mathrm{U}}, \\ \mathcal{J}_{N,k}^{\mathrm{U}} &= \frac{1}{k!} \prod_{i \neq j}^k \mathrm{sh}(\phi_i - \phi_j) \prod_{i,j=1}^k \frac{\mathrm{sh}(\phi_i - \phi_j - \epsilon_{12})}{\mathrm{sh}(\phi_i - \phi_j - \epsilon_{1,2})} \prod_{i=1}^k \prod_{\alpha=1}^N \frac{1}{\mathrm{sh}(\phi_i - v_\alpha) \mathrm{sh}(v_\alpha - \epsilon_{12} - \phi_i)}. \end{aligned}$$

这里 $\mathrm{sh}(x) = e^{x/2} - e^{-x/2}$ 。应用 Jeffrey–Kirwan 留数后，极点由 N 个二维杨图

$$\vec{\lambda} = (\lambda_1, \dots, \lambda_N)$$

标记。对于 λ_α 中的盒子 $\mathbf{x} = (i, j)$ ，对应极点为

$$\mathcal{X}_\alpha(\mathbf{x}) = v_\alpha + (\mathbf{x} - \mathbf{1}) \cdot \boldsymbol{\epsilon}_{12} = v_\alpha + i\epsilon_1 + j\epsilon_2 - \epsilon_{12}.$$

论文中的 Nekrasov 因子记作

$$N_{\alpha,\beta}^{\vec{\lambda}}(\mathbf{x}) \equiv v_\alpha - v_\beta + L_{\lambda_\alpha}(\mathbf{x})\epsilon_1 - A_{\lambda_\beta}(\mathbf{x})\epsilon_2 - \epsilon_2,$$

从而

$$\mathcal{Z}_{N,k}^U(v_1, \dots, v_N) = \sum_{||\vec{\lambda}||=k} \prod_{\alpha,\beta=1}^N \prod_{\mathbf{x} \in \lambda_\alpha} \frac{1}{\text{sh}(-N_{\alpha,\beta}^{\vec{\lambda}}(\mathbf{x})) \text{sh}(N_{\alpha,\beta}^{\vec{\lambda}}(\mathbf{x}) + \epsilon_{12})}.$$

这个表达展示了普通五维瞬子计数中的二维杨图、臂长和腿长结构，也说明了为什么在更高维固定点问题中需要更统一的语言。

当引入定向反演平面、非基本表示物质、D0–D6、D0–D8 或更一般的规范折纸系统时，普通二维杨图的臂长/腿长语言不再足够。固定点会变成三维、四维或带无穷腿和渐近边界的高维杨图，并伴随额外的极点选择、符号、离散 θ 角、取向符号和重数问题。本文使用壳公式统一这些固定点贡献。对于一个带标签

$$\mathcal{A} = (A, \alpha) = (a_1 a_2 \dots a_d, \alpha)$$

的 d -维杨图 $\mathbf{Y}_{\mathcal{A}}$ ，其坐标函数定义为

$$\mathcal{X}_{\mathcal{A}}(\mathbf{x}) \equiv v_{\mathcal{A}} + (\mathbf{x} - \mathbf{1}) \cdot \epsilon_A = v_{\mathcal{A}} + \sum_{i=1}^d (x_i - 1)\epsilon_{a_i}.$$

壳和壳荷分别为

$$\mathcal{S}(\mathbf{Y}) \equiv (\mathbf{Y} + \mathbf{B}_d) \setminus \mathbf{Y}, \quad Q_{\mathbf{Y}}(\mathbf{x}) = \sum_{\substack{\mathbf{b} \in \mathbf{B}_d \\ \mathbf{x} - \mathbf{b} \in \mathbf{Y}}} (-1)^{|\mathbf{b}|},$$

其中 $\mathbf{B}_d = \{\mathbf{b} = (b_1, \dots, b_d) \mid b_i \in \{0, 1\}\}$ 。核心的 $\mathcal{J}$ -因子定义为

$$\mathcal{J}(x | \mathbf{Y}_{\mathcal{A}}) \equiv \prod_{\mathbf{y} \in \mathcal{S}(\mathbf{Y}_{\mathcal{A}})} \text{sh}(x - \mathcal{X}_{\mathcal{A}}(\mathbf{y}))^{Q_{\mathbf{Y}_{\mathcal{A}}}(\mathbf{y})}, \quad \mathcal{J}(x | \emptyset_{\mathcal{A}}) \equiv \frac{1}{\text{sh}(x - \mathcal{X}_{\mathcal{A}}(\mathbf{1}))}.$$

在普通 $U(N)$ 五维纯超杨–Mills 理论中，这一记号把瞬子配分函数改写为

$$\mathcal{Z}_{N,k}^U(v_1, \dots, v_N) = \sum_{||\vec{\lambda}||=k} \prod_{\alpha,\beta=1}^N \prod_{\mathbf{x} \in \lambda_\alpha} \frac{\mathcal{J}(\mathcal{X}_\alpha(\mathbf{x}) | \lambda_\beta)}{\text{sh}(-\mathcal{X}_\alpha(\mathbf{x}) + \mathcal{X}_\beta(\mathbf{0}))}.$$

因此，本部分的主要结论是：由杨图标记分类的固定点贡献可以用壳公式清晰组织。五维普通瞬子对应二维杨图；D0–D6 与 Donaldson–Thomas 三维计数对应三维杨图；D0–D8、所谓 “Magnificent Four” 系统与 Donaldson–Thomas 四维计数对应四维杨图；Spiked 瞬子和规范折纸则要求带有多方向渐近数据的高维组合结构。壳公式把复杂的一圈行列式、相互作用项和边界贡献压缩为壳荷与

$\mathcal{J}$ -因子的组合。对于 D0–D8 系统，本文还引入 $\mathcal{J}_{\geq}$ 处理四维杨图中需要比较最后一个坐标的情形；对于带渐近边界的 Donaldson–Thomas 型问题，本文采用截断处方：先截断无穷方向，计算有限杨图的壳和壳荷，再在 $m \rightarrow \infty$ 极限中去掉只由人工边界带来的因子。

总之，本文从二维对偶和瞬子配分函数两个方向研究超对称规范理论中的非微扰扇区。前者说明椭圆亏格是检验二维红外对偶的精确工具，后者说明壳公式是组织瞬子、规范折纸和 Donaldson–Thomas 型计数的有效语言。二者共同展示了受保护指标、超对称局域化和 Jeffrey–Kirwan 留数在强耦合物理与组合计数之间的桥梁作用。未来值得继续研究的问题包括：一般秩下椭圆亏格恒等式的解析证明、更系统的 $(0, 2)$ 对偶分类、精细化 Donaldson–Thomas 四维理论中的取向和重数、规范折纸的轨形/晶体推广、涡旋折纸、三维中半指标以及相关量子环面代数结构。

关键字：超对称指标；Jeffrey–Kirwan 留数；二维对偶；规范折纸；壳公式；杨图
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Abstract

This thesis studies non-perturbative effects in supersymmetric gauge theories through two parallel uses of protected indices. One uses the elliptic genus to test two-dimensional dualities. The other uses the Witten index of ADHM-type quantum mechanics to study instanton partition functions and gauge origami, where localization produces Young-diagram fixed points and the shell formula reorganizes them.

The first part tests two-dimensional dualities obtained from compactifying four-dimensional supersymmetric theories on S^2 . The reduction depends on the topological twist, background fluxes, and R -symmetry assignments, which determine the chiral and Fermi zero modes. For the $\mathcal{N} = (2, 2)$ class- $\mathcal{S}$ examples, the elliptic genus exhibits gluing independence for quiver descriptions with the same topological data. For the $\mathcal{N} = (0, 2)$ examples, the thesis treats the SU Seiberg-like duality, Sp/SO triality, SU/Sp quiver dualities, tensor-matter dualities, and adjoint examples. Anomaly cancellation, flavor and gravitational anomalies, R -charges, E, J -terms, fugacity maps, and elliptic-genus equalities are treated as necessary but distinct checks. Some identities are proved analytically; others are supported by finite-rank JK residues or finite q -expansions, with the verified range stated explicitly.

The second part studies the fixed-point organization of ADHM instanton counting and gauge origami. Matter, brane-web Higgsing, and orientifold projections motivate a language beyond ordinary two-dimensional Young diagrams. D0–D4 systems give two-dimensional Young diagrams, while D0–D6, D0–D8, spiked-instanton, and DT-type systems involve plane partitions, solid partitions, and asymptotic boundary conditions. The shell formula replaces arm-leg data by the shell of a Young diagram, an integer charge on each shell box, and the associated $\mathcal{J}$ -factor. It gives a uniform notation for ordinary five-dimensional instantons, tetrahedron instantons, the magnificent four, spiked instantons, and DT3/DT4-type counting.

Keywords: supersymmetric index; Jeffrey–Kirwan residue; two-dimensional duality; gauge origami; shell formula; Young diagram

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Chapter 1

Introduction

Quantum field theory is most transparent when a weakly coupled expansion is available. Perturbation theory organizes amplitudes around a chosen classical vacuum, and the resulting Feynman-diagram expansion gives a powerful description of short-distance physics. Many of the most important phenomena, however, are intrinsically non-perturbative. Vacuum degeneracy, tunneling, confinement, dynamical symmetry breaking, BPS solitons, wall crossing, and infrared dualities cannot in general be seen at any finite order in a coupling constant. Supersymmetry provides a rare setting in which such effects can be studied exactly: holomorphy, BPS shortening, and protected indices restrict the answer strongly enough that exact non-perturbative information can often be extracted^[1–4].

This thesis studies non-perturbative effects in supersymmetric gauge theories from two complementary directions. The first direction is the study of duality. If two ultraviolet theories flow to the same infrared fixed point, their protected observables must agree, even when the elementary fields and the Lagrangians look completely different. In two-dimensional supersymmetric quantum field theories, elliptic genera are especially efficient probes of such equivalences: they are invariant under renormalization-group flow and can be computed by localization as Jeffrey–Kirwan residue integrals^[5–7]. In this direction, the thesis uses compactification of four-dimensional supersymmetric theories on S^2 to produce two-dimensional $\mathcal{N} = (2, 2)$ and $\mathcal{N} = (0, 2)$ dualities, and then tests them by elliptic genera.

The second direction is the fixed-point organization of instanton and gauge-origami indices. In four- and five-dimensional gauge theories with eight supercharges, instantons give non-perturbative corrections to the low-energy effective action. Nekrasov’s Ω -background turns the equivariant integral over instanton moduli space into a sum over fixed points labelled by Young diagrams^[4,8]. ADHM quantum mechanics with matter, fivebrane/O-plane constructions, and Higgsing identities motivate this direction, but in this thesis they serve primarily as the path toward a uniform shell-formula language rather than as a separate main line of results.

The shell-formula viewpoint continues the second direction but abstracts its combinatorics. Ordinary Nekrasov factors are naturally written in terms of arm and leg lengths of two-dimensional Young diagrams. Gauge origami, however, involves plane partitions, solid partitions, and Young diagrams with asymptotic boundaries. The shell formula replaces arm-leg data by the shell of a Young diagram, a charge assigned to each shell box, and the corresponding $\mathcal{J}$ -factor. This gives a uniform description of ordinary instantons, the magnificent four, tetrahedron instantons, spiked instantons, and DT3/DT4 counting^[9–13].

For two-dimensional dualities the protected index is the elliptic genus. For five-dimensional instantons and gauge origami it is the Witten index of the ADHM quantum

mechanics. In both cases, the non-perturbative information is encoded in exact contour integrals whose poles are selected by the same JK-residue logic.

This point of view also explains the practical value of the computations. On the physical side, exact indices test infrared dualities, determine BPS degeneracies, and constrain non-perturbative corrections to effective couplings. On the mathematical side, the same formulas appear as equivariant volumes of instanton moduli spaces, refined topological-string amplitudes, Donaldson–Thomas type invariants, and characters of algebras acting on fixed-point bases. The thesis is therefore organized not only as a collection of exact evaluations, but as an attempt to make the common mechanism behind these applications explicit.

Duality as a Probe of Non-Perturbative Physics Two-dimensional supersymmetric theories serve as simplified but highly non-trivial laboratories for quantum field theory. Many of them admit exact observables, phase structures, and infrared descriptions that would be hard to see directly from the ultraviolet Lagrangian. The foundational work of Witten on gauged linear sigma models showed how $\mathcal{N} = (2, 2)$ theories interpolate between different geometric phases and Landau–Ginzburg descriptions^[5]. Later developments in non-Abelian two-dimensional gauge dynamics led to dualities for $\mathcal{N} = (2, 2)$ theories^[14–15], and the discovery of $\mathcal{N} = (0, 2)$ triality^[16] opened a parallel direction for theories with only right-moving supersymmetry.

The two-dimensional part of the thesis has two related but distinct roles. First, Lagrangian class $\mathcal{S}$ theories of type A , obtained from the six-dimensional $(2, 0)$ theory on a punctured Riemann surface^[17], provide a controlled source of $\mathcal{N} = (2, 2)$ examples. After compactification on S^2 with the twist that mixes the S^2 holonomy with the appropriate R-symmetry and flavor symmetry, the resulting two-dimensional $\mathcal{N} = (2, 2)$ quiver theories retain the frame-independence of the parent class- $\mathcal{S}$ construction. This gives elliptic-genus identities for different gluing descriptions of the same surface, and these identities can be evaluated by localization and JK residues.

The other is the $\mathcal{N} = (0, 2)$ duality sector. In this case the relevant dualities are not consequences of class $\mathcal{S}$ frame-independence. Rather, they arise from twisted S^2 compactifications of four-dimensional dual pairs and related four-dimensional SCFT data, producing genuinely chiral two-dimensional theories with Fermi multiplets, singlet mesons, and non-trivial E, J -term constraints^[18–19]. The thesis therefore treats the SU Seiberg-like duality as a basic input and then studies the resulting $(0, 2)$ Sp/SO trialities, SU/Sp quiver dualities, tensor-matter dualities with symmetric or antisymmetric chirals, and adjoint examples. For these candidate dualities, equality of elliptic genera is only one part of the check: it is supplemented by gauge, flavor, and gravitational anomaly matching, R-charge assignments and fugacity maps.

Instantons and Young Diagrams The other route to non-perturbative physics is to compute instanton effects directly. In four-dimensional $\mathcal{N} = 2$ gauge theory, the Seiberg–Witten solution determines the low-energy prepotential from a family of algebraic curves and periods^[2–3]. Nekrasov’s partition function gives a microscopic derivation by summing equivariant contributions over instanton moduli spaces^[4]. In five dimensions, instantons become BPS particles, and the instanton counting parameter is the fugacity for their charge. The ADHM construction packages the k -instanton moduli space into the Higgs branch of a supersymmetric quantum mechanics, so the instanton partition function is a Witten index.

For the purposes of this thesis, the instanton material is used as a bridge to fixed-point combinatorics. The first description is the ADHM quantum mechanics, whose contour integral is evaluated by JK residues^[20–23]. The second is the fivebrane-web and orientifold description, where Higgsing and representation-theoretic isomorphisms predict identities among partition functions^[24–27]. Together they explain why two-dimensional Young diagrams are insufficient for all examples and why higher-dimensional Young diagrams and shell charges enter naturally.

Gauge Origami and Shell Formula The appearance of Young diagrams in instanton counting is not accidental. Fixed points of torus actions on instanton moduli spaces are combinatorial objects, and the local contribution of a fixed point is a product over weights of tangent and matter complexes. Gauge origami enlarges this perspective by allowing intersecting branes in several complex planes. D0–D4 systems lead to ordinary instantons, D0–D6 systems to tetrahedron instantons, D0–D8 systems to the magnificent four, and configurations with asymptotic legs to DT3/DT4 counting and spiked instantons^[10–12,28–29].

The shell formula gives a common expression for these systems. A Young diagram of arbitrary dimension has a shell, and each shell box carries an inclusion-exclusion charge. The $\mathcal{J}$ -factor is a product of sh-functions over shell boxes raised to these charges. It reproduces the ordinary Nekrasov factor in two dimensions and continues to make sense for three- and four-dimensional Young diagrams. The shell formula is therefore the final organizing principle of this thesis: it expresses the same non-perturbative index data in a dimension-independent language.

Original Contributions of the Thesis Several ingredients used in this thesis come from standard tools in supersymmetric field theory, localization, and brane engineering. The original contributions of the thesis lie in applying these tools to new two-dimensional duality checks and in reorganizing Young-diagram fixed-point formulas for BPS counting problems.

First, the thesis computes a series of elliptic genera for two-dimensional $\mathcal{N} = (2, 2)$ theories obtained from compactifying Lagrangian class $\mathcal{S}$ theories on S^2 . In this part, class $\mathcal{S}$ is used as a source of $(2, 2)$ quiver theories with different gluing descriptions. The thesis evaluates the elliptic genera of the corresponding building blocks and glued quivers, verifies their frame-independence in the examples considered, and records the associated consistency conditions such as anomaly cancellation, R-charge assignments, and the dependence on flavor fugacities. These computations give explicit two-dimensional tests of the duality-frame independence inherited from the four-dimensional parent theories.

Second, the thesis finds and checks several families of two-dimensional $\mathcal{N} = (0, 2)$ dualities. Starting from the SU Seiberg-like duality as a basic input, it constructs new duality chains and trialities involving classical gauge groups and tensor matter. These include the Sp/SU/LG trialities associated with antisymmetric matter, the SO/SU/LG trialities associated with symmetric matter, SU/Sp quiver dualities obtained through duality moves and Higgsing, dualities for SU gauge theories with one antisymmetric or one symmetric chiral multiplet, adjoint-matter examples for Sp and SO gauge groups, and recursive dualities for SU(N) theories with both symmetric and antisymmetric chiral multiplets. For these proposed $(0, 2)$ dualities, the thesis compares elliptic genera using Jeffrey–Kirwan residues and finite q -expansions where necessary, while also checking

gauge, flavor, and gravitational anomalies, R -charge assignments, fugacity maps, and E, J -term consistency. Whenever a general analytic proof is not available, the finite-rank or finite-expansion scope of the evidence is stated explicitly.

Third, the thesis defines and develops the shell formula as a uniform way to express BPS counting problems whose fixed points are labeled by Young diagrams and their higher-dimensional analogues. The shell formula replaces the usual arm-leg data by the shell of a Young diagram, an integer charge assigned to each shell box, and the corresponding $\mathcal{J}$ -factor. This formalism is used to rewrite the fixed-point contributions of ordinary five-dimensional instantons for classical gauge groups, including the effects of O-plane projections and discrete choices, and it is further applied to gauge-origami systems such as tetrahedron instantons, the D0–D8 magnificent four, spiked instantons, and DT3/DT4-type counting. The thesis also clarifies the cutoff prescriptions, boundary conditions, recursion relations, sign choices, and orientation or multiplicity issues needed to compare the original residue formulas with the corresponding Young-diagram and shell-formula expressions.

Organization of the Thesis Chapter 2 reviews the protected indices, one-loop factors, quiver rules, and JK-residue prescription used later. Chapter 3 reviews the four- and five-dimensional gauge-theory background and explains how ADHM localization leads to Young-diagram fixed points. Chapter 4 studies two-dimensional dualities from four dimensions, beginning with the S^2 -compactification dictionary that produces the relevant quivers. Chapter 5 combines ADHM fixed-point technology with the shell formula. Chapter 6 summarizes the common non-perturbative theme and discusses future directions. The appendices collect notation, theta and sh-function conventions, basic shell computations, additional two-dimensional duality checks, and detailed shell-formula examples.

Chapter 2

Protected Indices and Jeffrey–Kirwan Residues

This chapter fixes the protected-index and residue technology used throughout the thesis. Only two kinds of indices are needed later: the elliptic genus of two-dimensional quiver theories and the Witten index of ADHM-type quantum mechanics. Both are computed by the same schematic localization formula

$$\mathcal{J} = \frac{1}{|W|} \text{JK-Res}_\eta [Z_{1\text{-loop}}(u) du_1 \wedge \cdots \wedge du_r]. \quad (2.1)$$

In a 2d elliptic genus, u is a gauge flat connection on T^2 , the one-loop form is built from elliptic theta and eta functions, and the output is a duality invariant. In a 1d ADHM index, u is a complexified holonomy or equivariant scalar, the one-loop form is built from sh-factors, and the output is a fixed-point sum over Young diagrams. The chapter is therefore a toolkit rather than a general review of low-dimensional supersymmetry.

2.1 1d SUSY and Witten Index

2.1.1 Algebra and protected trace

The minimal complex supersymmetric quantum mechanics has a Hamiltonian H , a fermion number F , and supercharges $Q, Q^\dagger$ satisfying

$$Q^2 = (Q^\dagger)^2 = 0, \quad \{Q, Q^\dagger\} = H, \quad [H, Q] = [H, Q^\dagger] = 0. \quad (2.2)$$

Equivalently one may use two real supercharges Q_1, Q_2 with

$$\{Q_\alpha, Q_\beta\} = 2\delta_{\alpha\beta}H, \quad \alpha, \beta = 1, 2. \quad (2.3)$$

Every state with $E > 0$ is paired with another state of the same energy and opposite fermion number, while supersymmetric ground states are annihilated by both supercharges. The Witten index ^[1,30]

$$\mathcal{J}_W(\beta) = \text{Tr}_{\mathcal{H}}(-1)^F e^{-\beta H} \quad (2.4)$$

is therefore independent of β when no states escape to infinity. It is the Euler characteristic of the Q -cohomology,

$$\mathcal{J}_W = n_B^{(0)} - n_F^{(0)}. \quad (2.5)$$

In gauge quantum mechanics the trace is refined by flavor fugacities u_a ,

$$\mathcal{J} = \text{Tr}_{\mathcal{H}}(-1)^F e^{-\beta H - \sum_a u_a F_a}, \quad (2.6)$$

where F_a are Cartan generators of global symmetries.

2.1.2 Superfields and Lagrangian terms

A convenient superspace uses coordinates $(t, \theta, \bar{\theta})$ and supercovariant derivatives

$$D = \frac{\partial}{\partial \theta} - i\bar{\theta}D_t, \quad \bar{D} = -\frac{\partial}{\partial \bar{\theta}} + i\theta D_t, \quad \{D, \bar{D}\} = 2iD_t. \quad (2.7)$$

Here D_t is the gauge-covariant derivative along the time direction. On a field X in a representation R ,

$$D_t X = \partial_t X - iA_t^a T_R^a X,$$

with background flavor connections added in the same way. It reduces to ∂_t on a neutral field.

A 1d $\mathcal{N} = 2$ chiral multiplet obeys $\bar{D}\Phi = 0$ and contains a complex boson ϕ , a complex fermion ψ , and, off shell, an auxiliary field F . In the on-shell convention used here,

$$\Phi = \phi + \sqrt{2}\theta\psi + \theta\bar{\theta}iD_t\phi. \quad (2.8)$$

The non-propagating field F_i has already been integrated out by its algebraic equation of motion. For a canonical kinetic term, $F_i = -\bar{\partial}_i \bar{W}$, and substituting it back gives the potential $V_W = \sum_i |\partial_i W|^2$; hence no independent F_i appears in (2.8). Its kinetic term and a holomorphic superpotential are

$$L_{\text{kin}} = \int d\theta d\bar{\theta} \bar{\Phi} e^V \Phi, \quad (2.9)$$

$$L_W = \int d\theta W(\Phi) + \int d\bar{\theta} \bar{W}(\bar{\Phi}). \quad (2.10)$$

In addition to chiral multiplets, one can introduce 1d $\mathcal{N} = 2$ Fermi multiplets. A Fermi multiplet Λ_a is a fermionic superfield whose constraint takes the form

$$\bar{D}\Lambda_a = \sqrt{2}E_a(\Phi), \quad (2.11)$$

where $E_a(\Phi)$ is a holomorphic function of the chiral multiplets, transforming in the same gauge representation as Λ_a . When $E_a = 0$, its component expansion may be written as

$$\Lambda_a = \lambda_a + \sqrt{2}\theta G_a + \theta\bar{\theta}iD_t\lambda_a, \quad (2.12)$$

where λ_a is a complex fermion and G_a is an auxiliary boson. Thus a Fermi multiplet contains no propagating bosonic scalar; it contributes fermionic degrees of freedom and auxiliary fields.

The kinetic term of a Fermi multiplet is

$$L_\Lambda = \frac{1}{2} \int d\theta d\bar{\theta} \bar{\Lambda}_a \Lambda_a. \quad (2.13)$$

One may also introduce holomorphic J -type couplings,

$$L_J = \int d\theta \Lambda_a J^a(\Phi) + \int d\bar{\theta} \bar{\Lambda}_a \bar{J}^a(\bar{\Phi}). \quad (2.14)$$

Supersymmetry requires the E - and J -terms to satisfy

$$\sum_a E_a(\Phi) J^a(\Phi) = 0. \quad (2.15)$$

After eliminating the auxiliary fields, the Fermi multiplets contribute bosonic potential terms of the schematic form

$$V_{E,J} = \sum_a (|E_a(\phi)|^2 + |J^a(\phi)|^2), \quad (2.16)$$

together with Yukawa couplings between the Fermi fermions λ_a and the chiral fermions ψ_i , controlled by $\partial_i E_a$ and $\partial_i J^a$.

2.1.3 One-loop determinants and hyperbolic functions

For a gauge group G , the vector multiplet contains the gauge field A_t , a real scalar σ , gauginos, and an auxiliary field D . The chiral and Fermi kinetic operators depend on the complexified gauge holonomy

$$\phi \equiv \beta(\sigma + iA_t), \quad (2.17)$$

as well as on the complexified flavor holonomies

$$u \equiv \beta(m + iA_t^f), \quad (2.18)$$

where m is a real flavor mass and A_t^f is the background gauge field for the flavor symmetry. More generally, for a field of gauge weight q and flavor charge f , the combination entering the kinetic operator is

$$x = q\phi + fu. \quad (2.19)$$

For non-abelian gauge and flavor symmetries, this should be read as the corresponding weight pairing, $x = \rho(\phi) + \varrho(u)$.

The origin of the sh-function in the 1d index can be seen from the Fourier modes on the Euclidean time circle. A charged field $\varphi(t)$ has the mode expansion

$$\varphi(t) = \sum_{n \in \mathbb{Z}} \varphi_n \exp\left(\frac{2\pi i n t}{\beta}\right). \quad (2.20)$$

In the presence of the background gauge and flavor holonomies, the covariant derivative acting on a mode is shifted as

$$\beta D_t \longrightarrow 2\pi i n + \mathbf{q} \cdot \boldsymbol{\phi} + \mathbf{f} \cdot \mathbf{u}. \quad (2.21)$$

Thus the quantity $\mathbf{q} \cdot \boldsymbol{\phi} + \mathbf{f} \cdot \mathbf{u}$ is simply the total complexified chemical potential seen by that field: $\mathbf{q} \cdot \boldsymbol{\phi}$ comes from its gauge charge and $\mathbf{f} \cdot \mathbf{u}$ from its flavor charge.

A complex boson of total charge x contributes the inverse product over Matsubara modes,

$$Z_{\text{bos}}(x) \propto \prod_{n \in \mathbb{Z}} \frac{1}{2\pi i n + x}, \quad (2.22)$$

whereas a complex fermion of the same total charge contributes

$$Z_{\text{ferm}}(x) \propto \prod_{n \in \mathbb{Z}} (2\pi i n + x). \quad (2.23)$$

Using the product formula

$$\text{sh}(x) \equiv 2 \sinh \frac{x}{2} = x \prod_{n=1}^{\infty} \left(1 + \frac{x^2}{4\pi^2 n^2} \right), \quad (2.24)$$

one obtains the elementary 1d building blocks

$$\mathcal{J}_{\text{chiral}}^{1d}(x) = \frac{1}{\text{sh}(x)}, \quad \mathcal{J}_{\text{Fermi}}^{1d}(x) = \text{sh}(x), \quad (2.25)$$

Here the chiral multiplet contributes a bosonic zero-mode denominator, while the Fermi multiplet contributes the corresponding fermionic numerator. These are the factors appearing in ADHM quantum mechanics and in the shell formula.

For a $U(k)$ vector multiplet, the off-diagonal modes in the adjoint representation have weights $\phi_I - \phi_J$, and therefore give

$$\mathcal{J}_{\text{vec}}^{1d} = \prod_{I=1}^k \frac{d\phi_I}{2\pi i} \prod_{I \neq J} \text{sh}(\phi_I - \phi_J). \quad (2.26)$$

For quantum mechanics with a compact space of supersymmetric vacua this index is an ordinary graded trace. ADHM quantum mechanics is subtler because its classical Higgs branch is non-compact. In the applications below the index is understood equivariantly: the Ω -background, flavor masses, and FI chamber regularize the flat directions and turn the computation into a well-defined contour integral. Possible continuum contributions are excluded by working in this equivariant chamber; when such contributions are present, the resulting index should be interpreted as the localized, chamber-dependent ADHM contribution.

2.2 2d SUSY and Elliptic Genera

2.2.1 Algebras and multiplets

Two-dimensional supersymmetry is naturally expressed in light-cone coordinates $x^\pm = x^0 \pm x^1$. A theory with $\mathcal{N} = (p, q)$ supersymmetry has p left-moving and q right-moving supercharges. For $\mathcal{N} = (2, 2)$ the supercharges are $Q_\pm, \bar{Q}_\pm$, with

$$\{Q_+, \bar{Q}_+\} = 2(H + P), \quad \{Q_-, \bar{Q}_-\} = 2(H - P), \quad (2.27)$$

and possible central terms. The theory often has vector and axial R-symmetry $U(1)_V \times U(1)_A$. A $(2, 2)$ chiral multiplet obeys

$$\bar{D}_+ \Phi = \bar{D}_- \Phi = 0, \quad (2.28)$$

and its Lagrangian is the sum of a D-term kinetic piece and an F-term superpotential,

$$L_{\text{chiral}}^{(2,2)} = \int d^4\theta \, \bar{\Phi} e^V \Phi + \left(\int d\theta^+ d\theta^- \, W(\Phi) + \text{c.c.} \right). \quad (2.29)$$

For a vector multiplet the twisted chiral field strength is denoted Σ , and the Fayet–Iliopoulos and theta angle combine into $t = r - i\vartheta$:

$$L_{\text{FI}}^{(2,2)} = \frac{1}{2} \int d\theta^+ d\bar{\theta}^- t \Sigma + \text{c.c.} \quad (2.30)$$

These are the standard GLSM ingredients of Witten and of Hori–Tong^[5,14–15].

For $\mathcal{N} = (0, 2)$ one keeps the right-moving supercharges Q_+ , $\bar{Q}_+$. The basic multiplets are a chiral multiplet Φ with $\bar{D}_+ \Phi = 0$, a Fermi multiplet Ψ with

$$\bar{D}_+ \Psi = E(\Phi), \quad (2.31)$$

and a vector multiplet. Superpotential interactions are written as

$$L_J = -\frac{1}{\sqrt{2}} \int d\theta^+ \Psi_a J^a(\Phi) + \text{c.c.}, \quad (2.32)$$

and supersymmetry requires

$$\sum_a E_a(\Phi) J^a(\Phi) = 0. \quad (2.33)$$

A $(2, 2)$ chiral multiplet decomposes into a $(0, 2)$ chiral multiplet and a $(0, 2)$ Fermi multiplet; a $(2, 2)$ vector multiplet decomposes into a $(0, 2)$ vector and a $(0, 2)$ chiral multiplet in the adjoint representation.

2.2.2 Protected trace on the torus

The elliptic genus is a protected Ramond–Ramond trace, originally developed as a supersymmetric refinement of the index on loop space^[31],

$$\mathcal{J}(q, y, \mathbf{z}) = \text{Tr}_{\text{RR}}(-1)^F q^{L_0 - c/24} \bar{q}^{\bar{L}_0 - \bar{c}/24} y^{J_R} \prod_a z_a^{F_a}. \quad (2.34)$$

Only states annihilated by the chosen right-moving supercharge contribute, so the result is invariant under RG flow. This makes elliptic genera sharp tests of two-dimensional dualities^[6–7].

More concretely, choosing $\bar{Q}_+$ gives

$$\{\bar{Q}_+, Q_+\} = 2(H - P), \quad (2.35)$$

so all states with $H \neq P$ pair up in the trace and cancel. The remaining right-moving ground states are counted with their left-moving oscillator excitations and global charges. This is why the elliptic genus is holomorphic in q after the cancellation of the $\bar{q}$ -dependence, up to the familiar anomalies that determine its modular and elliptic transformation laws. In gauge theories, those same anomaly constraints are precisely what makes the localized integrand a globally defined meromorphic form on the gauge torus.

The theta function arises because a two-dimensional field has two independent towers of modes on the torus rather than a single tower on a circle. Let $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$. For a field of gauge weight ρ , flavor charges f_a , and R-charge R , the additive chemical potential entering the kinetic operator is

$$u_{\text{tot}} = \rho(u) + \sum_a f_a z_a + \frac{R}{2} z_R, \quad X = e^{2\pi i u_{\text{tot}}}, \quad q = e^{2\pi i \tau}. \quad (2.36)$$

Here u , z_a , and z_R are additive holonomies on the gauge, flavor, and R-symmetry tori. Equivalently, the field obeys twisted boundary conditions

$$\Phi(w + 1) = e^{2\pi i \alpha} \Phi(w), \quad \Phi(w + \tau) = e^{2\pi i \beta} X \Phi(w), \quad (2.37)$$

where α, β encode the spin structure. Expanding on the two cycles of the torus gives modes labelled by $(m, n) \in \mathbb{Z}^2$, and the covariant Cauchy–Riemann operator is shifted schematically as

$$\bar{\partial} \longrightarrow m + \alpha + (n + \beta)\tau + u_{\text{tot}}. \quad (2.38)$$

Thus the one-loop determinant is a regularized product over the lattice $\mathbb{Z} + \tau\mathbb{Z}$. A complex boson contributes the inverse product

$$Z_{\text{bos}}^{2d}(X) \propto \prod_{m, n \in \mathbb{Z}} \frac{1}{m + n\tau + u_{\text{tot}}}, \quad (2.39)$$

whereas a complex fermion with the same total fugacity contributes the corresponding product in the numerator.

The appearance of the theta function can be seen by performing one of the two products first. For fixed n , set $z_n = u_{\text{tot}} + n\tau$. The one-dimensional product formula gives, up to a regularization-dependent constant,

$$\prod_{m \in \mathbb{Z}} (m + z_n) \propto 2i \sin(\pi z_n) = -e^{-\pi i z_n} (1 - X q^n). \quad (2.40)$$

The remaining product over $n \in \mathbb{Z}$ therefore has the zero structure

$$\prod_{n \in \mathbb{Z}} (1 - X q^n) \sim (1 - X) \prod_{n=1}^{\infty} (1 - X q^n) (1 - X^{-1} q^n), \quad (2.41)$$

where the divergent monomial in X and q is fixed by the standard zeta-function regularization. This is precisely the X -dependent part of the odd theta function. With the normalization conventions collected in Appendix A, the regularized charged determinant is

$$\prod_{m, n \in \mathbb{Z}} (m + n\tau + u_{\text{tot}})_{\text{reg}} \propto \frac{\vartheta_1(X)}{\eta(q)}. \quad (2.42)$$

Thus the 1d factor $\text{sh}(x)$ is replaced on the torus by the elliptic factor $\vartheta_1(X)/\eta(q)$. The elementary $(0, 2)$ one-loop factors are therefore

$$Z_{\Phi}^{(0,2)}(X) = \frac{i\eta(q)}{\vartheta_1(X)}, \quad Z_{\Psi}^{(0,2)}(X) = \frac{\vartheta_1(X)}{i\eta(q)}. \quad (2.43)$$

The ratio of $\eta(q)$ functions records the oscillator modes that do not depend on the gauge fugacity. The theta function records the charged zero-mode lattice and hence determines the pole structure of the elliptic genus.

2.2.3 R-charges, anomalies, and elementary factors

If a field has R-charge R and flavor weight ρ , then the argument is shifted by the R-symmetry fugacity. It is useful to distinguish the additive chemical potentials from the multiplicative fugacities used by the theta functions. In the conventions used for the elliptic genera in Chapter 4, set

$$x_\rho = e^{2\pi i \rho(u)}, \quad z_a = e^{2\pi i z_a}, \quad y = e^{2\pi i z_R}. \quad (2.44)$$

A multiplet with gauge weight ρ , flavor charges q_a , and R-charge R then carries the total multiplicative fugacity

$$X = x_\rho \prod_a z_a^{q_a} y^{R/2} = \exp \left[2\pi i \left(\rho(u) + \sum_a q_a z_a + \frac{R}{2} z_R \right) \right]. \quad (2.45)$$

Thus the R-charge determines how the same gauge and flavor weight is embedded into the torus of fugacities. A change of R-charge is not a harmless normalization. The elliptic genus is a meromorphic section on the gauge torus, and its building blocks are theta functions rather than ordinary rational functions. Therefore shifting the R-charge multiplies X by a power of y , and this affects the quasi-periodic transformation of the full integrand under large gauge transformations.

In multiplicative notation, the relevant quasi-periodicity of the odd theta function is

$$\vartheta_1(qX) = -q^{-1/2} X^{-1} \vartheta_1(X), \quad (2.46)$$

$$\vartheta_1(q^n X) = (-1)^n q^{-n^2/2} X^{-n} \vartheta_1(X), \quad n \in \mathbb{Z}. \quad (2.47)$$

Hence a large gauge transformation

$$u \longrightarrow u + \tau \alpha, \quad X \longrightarrow q^{\rho(\alpha)} X,$$

produces factors depending on $\rho(\alpha)$, on the flavor fugacities, and on the R-charge contribution $y^{R/2}$. These factors do not disappear by a redefinition of notation. Thus an R-charge assignment is part of the physical data of the elliptic genus, not just a convention for writing the same answer.

For a $(2, 2)$ chiral multiplet, let X denote the gauge/flavor fugacity before the relative vector R-symmetry shift. The product of its $(0, 2)$ chiral and Fermi components gives the familiar ratio

$$Z_{\Phi}^{(2,2)}(X; y) = \frac{\vartheta_1(yX)}{\vartheta_1(X)}, \quad (2.48)$$

up to the convention for the vector R-charge fugacity y . The vector multiplet gives the product over roots; for example,

$$Z_{\text{SU}(N)}^{(2,2),\text{vec}} = \frac{1}{N!} \left(\frac{\eta(q)^3}{\vartheta_1(y)} \right)^{N-1} \prod_{i \neq j} \frac{\vartheta_1(x_i/x_j)}{\vartheta_1(yx_i/x_j)}, \quad x_i = e^{2\pi i u_i}. \quad (2.49)$$

2.3 Quiver Diagrams

We use quiver diagrams as a dictionary from fields to one-loop factors. The same drawing can be read in three closely related ways: as a 1d $\mathcal{N} = 2$ quantum-mechanics index, as a 2d $\mathcal{N} = (0, 2)$ elliptic genus, or, when the $(0, 2)$ chiral and Fermi fields are paired, as a 2d $\mathcal{N} = (2, 2)$ elliptic genus. A circular node is a gauge group, a square node is a flavor symmetry, a solid arrow is a chiral multiplet, and a dashed arrow is a Fermi multiplet. If an arrow points from a node with additive holonomies b_β to a node with additive holonomies a_α , write

$$A_\alpha = e^{2\pi i a_\alpha}, \quad B_\beta = e^{2\pi i b_\beta}, \quad M_{\chi, \alpha\beta} = e^{2\pi i m_{\chi, \alpha\beta}}, \quad \chi = \Phi, \Psi.$$

Here $m_{\chi, \alpha\beta}$ denotes the additive mass, equivariant shift, or flavor/R-symmetry shift carried by the arrow. The corresponding 1d additive weight and 2d multiplicative fugacity are

$$\mu_{\chi, \alpha\beta} = a_\alpha - b_\beta + m_{\chi, \alpha\beta}, \quad X_{\chi, \alpha\beta} = A_\alpha B_\beta^{-1} M_{\chi, \alpha\beta}.$$

Gauge-node variables are integrated, whereas flavor-node variables are kept fixed. In the arrow factors below, $\prod_{\alpha, \beta}$ means $\prod_{\alpha=1}^{N_a} \prod_{\beta=1}^{N_b}$.

- A circular node represents a gauge group G . If G has Weyl group W , rank r , and root system Δ , write $X^\alpha = e^{2\pi i \alpha \cdot u}$ for the 2d root fugacity. The vector multiplet contributes

$$\begin{aligned} \text{Diagram: } \bigcirc \text{ with } G \text{ inside and } u \text{ below it} &\implies \\ \mathcal{J}_G^{1d} &= \frac{1}{|W|} \prod_{\alpha \in \Delta} \text{sh}(\alpha \cdot u), \\ \mathcal{J}_G^{(0,2)} &= \frac{(-i\eta(q)^2)^r}{|W|} \prod_{\alpha \in \Delta} \frac{\vartheta_1(X^\alpha)}{i\eta(q)}, \\ \mathcal{J}_G^{(2,2)} &= \frac{1}{|W|} \left(\frac{\eta(q)^3}{\vartheta_1(y)} \right)^r \prod_{\alpha \in \Delta} \frac{\vartheta_1(X^\alpha)}{\vartheta_1(yX^\alpha)}. \end{aligned}$$

The $(2, 2)$ expression is the $(0, 2)$ vector contribution multiplied by the adjoint chiral contained in the $(2, 2)$ vector multiplet shifted by the $(2, 2)$ R-symmetry fugacity y .

- A square node represents a flavor group. It supplies fixed background fugacities but no contour integral. A solid arrow from N_b to N_a denotes a chiral multiplet Φ_α^β , fundamental under the node at the arrow tip and anti-fundamental under the node at the tail. Its one-loop factor is

$$\begin{aligned} \text{Diagram: } \boxed{N_a} \xleftarrow{m_{\Phi, \alpha\beta}} \boxed{N_b} &\implies \\ \mathcal{J}_\Phi^{1d} &= \prod_{\alpha, \beta} \frac{1}{\text{sh}(\mu_{\Phi, \alpha\beta})}, \\ \mathcal{J}_\Phi^{(0,2)} &= \prod_{\alpha, \beta} \frac{i\eta(q)}{\vartheta_1(X_{\Phi, \alpha\beta})}, \\ \mathcal{J}_\Phi^{(2,2)} &= \prod_{\alpha, \beta} \frac{\vartheta_1(yX_{\Phi, \alpha\beta})}{\vartheta_1(X_{\Phi, \alpha\beta})}. \end{aligned}$$

The last line is the product of a $(0, 2)$ chiral and its $(0, 2)$ Fermi partner, with the Fermi argument shifted by the $(2, 2)$ R-symmetry fugacity y .

- A dashed arrow with the same orientation convention denotes a Fermi multiplet Ψ_α^β . It contributes the numerator counterpart of the chiral factor:

$$\boxed{N_a} \text{ --- } \overset{m_{\Psi, \alpha\beta}}{\leftarrow} \text{ --- } \boxed{N_b} \quad \Rightarrow \quad \begin{aligned} \mathcal{J}_\Psi^{1d} &= \prod_{\alpha, \beta} \text{sh}(\mu_{\Psi, \alpha\beta}), \\ \mathcal{J}_\Psi^{(0,2)} &= \prod_{\alpha, \beta} \frac{\vartheta_1(X_{\Psi, \alpha\beta})}{i\eta(q)}. \end{aligned}$$

Multiplying all node and arrow factors gives the localized integrand. In 1d this integrand is built from sh-functions and is evaluated by a JK residue. In 2d it is built from theta functions and must be a well-defined meromorphic form on the gauge torus. This is why the $(0, 2)$ anomaly cancellation and the E, J -term condition (2.33) are checked before comparing elliptic genera in Chapter 4.

2.4 Jeffrey–Kirwan Residue

The computation of the Witten index requires explicit evaluation of contour integrals. The JK-residue prescription^[6–7,19,32–33] provides the correct method for performing these integrals. It is applied to the integrands of Sec. 5.2–5.3; the resulting poles are classified by the Young diagrams introduced in Sec. 5.1. Here we review the JK-residue procedure, following the shell-formula derivation.

We consider a gauge theory with a rank- k gauge group. The index is expressed as an integral of a meromorphic k -form over a specific cycle:

$$\mathcal{Z} = \oint_{\text{JK}} \prod_{I=1}^k \frac{du_I}{2\pi i} \mathcal{J}(\mathbf{u}) = \oint_{\text{specific cycles}} \mathcal{J}(\mathbf{u}) du_1 \wedge \cdots \wedge du_k, \quad (2.50)$$

where $\mathbf{u} = (u_1, \dots, u_k)$ denotes the complexified gauge variables. The integrand is periodic in each u_I :

$$\mathcal{J}(\dots, u_I, \dots) = \mathcal{J}(\dots, u_I + T, \dots). \quad (2.51)$$

where, T denotes a period of the integrand. For example, in the Witten index, the period of the sh function is $T = 2\pi i$, whereas in the EG, the two periods are $T = 1$ or τ , respectively.

The poles of the integrand arise from the denominator, which takes the schematic form

$$\begin{aligned} \mathcal{J}(\mathbf{u}) &\propto \prod_a \frac{1}{\text{sh}\left(\sum_{I=1}^k Q_a^I u_I + m_a\right)^{N_a}} && \text{in 1d} \\ \mathcal{J}(\mathbf{u}) &\propto \prod_a \frac{1}{\vartheta_1\left(e^{2\pi i(\sum_{I=1}^k Q_a^I u_I + m_a)}\right)^{N_a}} && \text{in 2d} \end{aligned} \quad (2.52)$$

where $Q_a^I \in \mathbb{Q}$ are charge vectors and m_a are effective masses or equivariant parameters. The poles of $\mathcal{F}$ are located at solutions of

$$\begin{aligned} \sum_{I=1}^k Q_a^I u_I + m_a &= 2\pi i n_a, \quad n_a \in \mathbb{Z}, \quad \text{in 1d} \\ \sum_{I=1}^k Q_a^I u_I + m_a &= n_a + n'_a \tau, \quad n_a, n'_a \in \mathbb{Z}, \quad \text{in 2d} \end{aligned} \quad (2.53)$$

The periodicity leads to multiple copies of poles shifted by T , and the allowed values of n_a and n'_a can be parametrized by an invertible matrix Q_{ai} as

$$\begin{pmatrix} n_1 \\ \vdots \\ n_k \end{pmatrix}, \begin{pmatrix} n'_1 \\ \vdots \\ n'_k \end{pmatrix} = \frac{Q}{|\det Q|} \begin{pmatrix} l_1 \\ \vdots \\ l_k \end{pmatrix}, \quad l_I \in \{0, 1, \dots, |\det Q| - 1\}. \quad (2.54)$$

Given a pole u_* satisfying (2.53), one identifies the associated set of charge vectors $Q_* = \{Q_1, \dots, Q_r\}$ with $r \geq k$. A flag F is constructed from any k -sequence of linearly independent charge vectors $\{Q_{a_1}, \dots, Q_{a_k}\} \subset Q_*$:

$$\{0\} \subset F_1 \subset \dots \subset F_k = \mathbb{R}^k, \quad F_\ell = \text{span}\{Q_{a_1}, \dots, Q_{a_\ell}\}. \quad (2.55)$$

The sequence $\{Q_{a_1}, \dots, Q_{a_k}\}$ is called a basis $\mathcal{B}(F, Q_*)$ of F in Q_* . From each flag and its basis one forms

$$\kappa(F, Q_*) = (\kappa_1, \dots, \kappa_k), \quad \kappa_\ell = \sum_{\substack{Q \in Q_* \\ Q \in F_\ell}} Q. \quad (2.56)$$

Choose a reference vector $\eta = (\eta_1, \dots, \eta_k) \in (\mathbb{R}^k)^*$. A flag contributes only when

$$\kappa(F, Q_*)^T \lambda = \eta, \quad \lambda = (\lambda_1, \dots, \lambda_k) \in \mathbb{R}_+^k. \quad (2.57)$$

Equivalently,

$$\delta(F, \eta) = \begin{cases} 1, & \kappa(F, Q_*) \text{ satisfies (2.57),} \\ 0, & \text{otherwise.} \end{cases} \quad (2.58)$$

The JK residue of the pole is

$$\text{JK-Res}(\eta)_{\mathcal{F}} = \sum_{F} \delta(F, \eta) \frac{\text{sgn det } \kappa(F, Q_*)}{\det \mathcal{B}(F, Q_*)} \text{Res}_{\varepsilon_k=0} \cdots \text{Res}_{\varepsilon_1=0} \mathcal{F} \Big|_{\substack{Q_{a_1} u_* + m_{a_1} + \varepsilon_1 = T n_{a_1} \\ \vdots \\ Q_{a_k} u_* + m_{a_k} + \varepsilon_k = T n_{a_k}}} \quad (2.59)$$

Finally,

$$\oint_{\text{JK}} \prod_{I=1}^k \frac{du_I}{2\pi i} \mathcal{F}(u) = \sum_{u=u_*} \text{JK-Res}(\eta)_{\mathcal{F}}(u). \quad (2.60)$$

The standard ADHM chamber is represented by $\eta = (1, \dots, 1)$, corresponding to a positive FI parameter.

For special unitary gauge groups one removes the overall diagonal gauge fugacity $\sum_i u_i = 0$. Practically, one solves the constraint before applying the JK prescription or, equivalently, works on the rank- $(N - 1)$ Cartan torus with roots and weights projected to the traceless subspace. This convention is used for the SU building blocks in Chapter 4.

When more than k denominator factors vanish at the same point, the pole is degenerate. The flag formula above is the regularized definition used in this situation. In computations one keeps generic masses and equivariant parameters, evaluates the JK residue, and only then takes special limits such as the unrefined limit $\epsilon_2 = -\epsilon_1$ or the Calabi–Yau conditions. If a limit creates higher-order poles, the multiplicity or sign factors must be tracked explicitly; this is precisely the issue addressed in the shell and DT4 examples of Chapter 5 and Appendix C.

Chapter 3

Prepotential, ADHM Construction and Brane

This chapter supplies the bridge from the residue technology of Chapter 2 to the instanton and gauge-origami part of the thesis. We first recall the four- and five-dimensional gauge-theory quantities whose non-perturbative sectors are counted by instantons. We then explain how ADHM localization turns these sectors into residue integrals, how the Ω -background regularizes the computation, and how the selected poles lead to Young-diagram fixed points. This prepares the shell-formula language of Chapter 5. The S^2 compactification dictionary used for the two-dimensional dualities is placed at the beginning of Chapter 4, where it is used directly.

3.1 Prepotential of 4d and 5d theories

3.1.1 4d $\mathcal{N} = 2$ gauge theory and prepotentials

Four-dimensional $\mathcal{N} = 2$ gauge theories have eight supercharges and R-symmetry $SU(2)_R \times U(1)_r$. In a relativistic normalization,

$$\{Q_\alpha^I, \bar{Q}_{J\dot{\alpha}}\} = 2\delta^I_J \sigma_{\alpha\dot{\alpha}}^\mu P_\mu, \quad \{Q_\alpha^I, Q_\beta^J\} = 2\epsilon_{\alpha\beta} \epsilon^{IJ} Z, \quad (3.1)$$

where $I, J = 1, 2$ and Z is the central charge. In $\mathcal{N} = 1$ language, the basic multiplets are

$$\text{vector: } (V, \Phi) \longleftrightarrow (A_\mu, \lambda_\alpha^I, \varphi), \quad \varphi \in \mathfrak{g}_\mathbb{C}, \quad (3.2)$$

$$\text{hyper: } (Q, \tilde{Q}) \longleftrightarrow (q, \tilde{q}^\dagger, \psi_Q, \psi_{\tilde{Q}}), \quad (3.3)$$

where V is the $\mathcal{N} = 1$ vector multiplet and Φ is an adjoint chiral multiplet. The microscopic pure Yang–Mills action contains

$$S_{\text{bos}}^{4d} = \frac{1}{g_4^2} \int_{\mathbb{R}^4} \text{Tr} \left(\frac{1}{2} F \wedge *F + D\varphi \wedge *D\bar{\varphi} + \frac{1}{2} [\varphi, \bar{\varphi}]^2 \text{vol}_4 \right) + \frac{i\theta}{8\pi^2} \int_{\mathbb{R}^4} \text{Tr} F \wedge F. \quad (3.4)$$

On the Coulomb branch,

$$\varphi = \text{diag}(a_1, \dots, a_N), \quad U(N) \longrightarrow U(1)^N.$$

The two-derivative abelian effective action is fixed by a holomorphic prepotential $\mathcal{F}_{4d}$:

$$S_{\text{eff}}^{4d} = \frac{1}{4\pi} \text{Im} \left[\int d^4x d^4\theta \bar{A}_i \frac{\partial \mathcal{F}_{4d}}{\partial A_i} + \frac{1}{2} \int d^4x d^2\theta \tau_{ij}(A) W_i^\alpha W_{j\alpha} \right]. \quad (3.5)$$

where

$$\tau_{ij} = \frac{\partial^2 \mathcal{F}_{4d}}{\partial a_i \partial a_j} \quad (3.6)$$

The fact that a single prepotential controls the abelian low-energy theory is a strong consequence of four-dimensional $\mathcal{N} = 2$ supersymmetry. The vector multiplet contains the gauge field and the Coulomb-branch scalar in the same short multiplet, so the scalar kinetic terms, photon kinetic terms, theta angles, and electric-magnetic couplings cannot be chosen independently. Holomorphy then restricts these couplings to come from one holomorphic function of the chiral special coordinates. Gauge invariance and electric-magnetic duality further imply that different duality frames are related by changing the special-coordinate basis, not by introducing new independent functions. In geometric language the Coulomb branch is special Kahler; in physical language the same function determines the effective gauge couplings, the dual magnetic variables, and the central charges of BPS states. The Seiberg–Witten curve is a method for computing this prepotential through periods of a meromorphic differential; no detailed curve technology is needed below ^[2–3,34–35].

For pure $U(N)$, with

$$\tau_{UV} = \frac{\theta}{2\pi} + \frac{4\pi i}{g_4^2}, \quad \mathbf{q} = e^{2\pi i \tau_{UV}},$$

the prepotential may be organized as

$$\mathcal{F}_{4d}^{U(N)}(\mathbf{a}) = \frac{\tau_{UV}}{2} \sum_{\alpha=1}^N a_\alpha^2 + \frac{i}{8\pi} \sum_{\alpha \neq \beta} (a_\alpha - a_\beta)^2 \log \frac{(a_\alpha - a_\beta)^2}{\Lambda^2} + \sum_{k \geq 1} \mathbf{q}^k \mathcal{F}_k^{U(N)}(\mathbf{a}). \quad (3.7)$$

This is the standard weak-coupling form: the first term is classical, the logarithmic term is one-loop exact in the perturbative expansion, and the last sum contains instanton corrections ^[2–4,36]. For matter hypermultiplets one subtracts the corresponding weight sum

$$\Delta \mathcal{F}_{\text{hyper}} = -\frac{i}{8\pi} \sum_{\rho \in R} (\rho(\mathbf{a}) + m)^2 \log \frac{(\rho(\mathbf{a}) + m)^2}{\Lambda^2}.$$

The instanton terms $\mathcal{F}_k$ will reappear from the regularized instanton partition function in Subsection 3.2.2.

3.1.2 5d $\mathcal{N} = 1$ gauge theory and prepotentials

Five-dimensional $\mathcal{N} = 1$ gauge theories also have eight supercharges and $SU(2)_R$ symmetry. The basic multiplets are

$$\text{vector: } (A_M, \phi, \lambda^I), \quad M = 0, \dots, 4, \quad \phi \in \mathfrak{g}, \quad (3.8)$$

$$\text{hyper: } (q^I, \psi), \quad q^I \in R, \quad I = 1, 2. \quad (3.9)$$

The bosonic part of the microscopic pure gauge action has Yang–Mills and Chern–Simons pieces:

$$S_{\text{bos}}^{5d} = \frac{1}{g_0^2} \int_{\mathbb{R}^5} \text{Tr} \left(\frac{1}{2} F \wedge *F + D\phi \wedge *D\phi \right) + \frac{\kappa}{24\pi^2} \int_{\mathbb{R}^5} \text{Tr} \left(A \wedge F \wedge F - \frac{i}{2} A^3 \wedge F - \frac{1}{10} A^5 \right) + \dots \quad (3.10)$$

Here g_0^2 has dimension of length. The Chern–Simons level κ is quantized and exists for gauge groups with a cubic invariant, most importantly $SU(N)$ with $N \geq 3$. If hypermultiplets are present, their kinetic terms, real-mass couplings, and Yukawa couplings are included in the ellipsis.

At a generic point on the Coulomb branch,

$$\phi = \sum_i \phi_i H_i, \quad G \longrightarrow T_G,$$

the light fields are abelian vector multiplets. Five-dimensional supersymmetry then organizes their two-derivative couplings and abelian Chern–Simons couplings into one real prepotential $\mathcal{F}_{5d}(\phi)$. The reason this is possible is parallel to the 4d case but less holomorphic: the real scalar ϕ_i , the photon A_i , and the gaugino sit in the same short multiplet, so their kinetic terms and Chern–Simons interactions are tied together. Gauge invariance and dimensional analysis further imply that the exact prepotential is piecewise cubic. The quadratic Yang–Mills term is multiplied by g_0^{-2} , which has mass dimension one; the classical Chern–Simons term is cubic; and a massive BPS particle of mass $M(\phi)$ contributes at one loop as $|M(\phi)|^3$. Hence the effective metric is continuous and piecewise linear in ϕ , with possible jumps of second derivatives across walls where W-bosons or matter fields become massless^[37–38].

For hypermultiplets of masses m_f in representations R_f , the general prepotential is

$$\mathcal{F}_{5d}(\phi) = \frac{1}{2g_0^2} h_{ij} \phi_i \phi_j + \frac{\kappa}{6} d_{ijk} \phi_i \phi_j \phi_k + \frac{1}{12} \left(\sum_{\alpha \in \Delta} |\alpha(\phi)|^3 - \sum_f \sum_{\rho \in R_f} |\rho(\phi) + m_f|^3 \right). \quad (3.11)$$

Here $h_{ij} = \text{Tr}(H_i H_j)$ is the invariant quadratic form on the Cartan, $d_{ijk} = \frac{1}{2} \text{Tr}(H_i \{H_j, H_k\})$ is the invariant cubic tensor when it exists, Δ is the root system, and ρ runs over matter weights. The three terms are respectively the classical Yang–Mills term, the classical Chern–Simons term, and the one-loop contribution of W-bosons and hypermultiplets.

For pure SYM with classical gauge group, set the hypermultiplet term to zero. In a Weyl chamber, write $\mathbf{a} = (a_1, \dots, a_r)$. For type A_r , namely $G = SU(r+1)$ with $\sum_{\alpha=1}^{r+1} a_\alpha = 0$,

$$\mathcal{F}_{5d}^{A_r} = \frac{1}{2g_0^2} \sum_{\alpha=1}^{r+1} a_\alpha^2 + \frac{\kappa}{6} \sum_{\alpha=1}^{r+1} a_\alpha^3 + \frac{1}{12} \sum_{\alpha \neq \beta} |a_\alpha - a_\beta|^3. \quad (3.12)$$

For type B_r , $G = SO(2r+1)$,

$$\mathcal{F}_{5d}^{B_r} = \frac{1}{2g_0^2} \sum_{i=1}^r a_i^2 + \frac{1}{6} \sum_{1 \leq i < j \leq r} (|a_i - a_j|^3 + |a_i + a_j|^3) + \frac{1}{6} \sum_{i=1}^r |a_i|^3. \quad (3.13)$$

For type C_r , $G = Sp(2r)$, with the convention where the fundamental representation has dimension $2r$,

$$\mathcal{F}_{5d}^{C_r} = \frac{1}{2g_0^2} \sum_{i=1}^r a_i^2 + \frac{1}{6} \sum_{1 \leq i < j \leq r} (|a_i - a_j|^3 + |a_i + a_j|^3) + \frac{4}{3} \sum_{i=1}^r |a_i|^3. \quad (3.14)$$

For type D_r , $G = \text{SO}(2r)$,

$$\mathcal{F}_{5d}^{D_r} = \frac{1}{2g_0^2} \sum_{i=1}^r a_i^2 + \frac{1}{6} \sum_{1 \leq i < j \leq r} (|a_i - a_j|^3 + |a_i + a_j|^3). \quad (3.15)$$

In these formulas, the quadratic term is the classical Yang–Mills contribution. The κ -dependent cubic term appears only in the A_r case above. The absolute-value cubic terms are the W-boson one-loop contributions from the roots: $\pm(a_i \pm a_j)$ for B, C, D , plus the short roots $\pm a_i$ for B_r and the long roots $\pm 2a_i$ for C_r . The positivity of the Hessian of $\mathcal{F}_{5d}$ inside the chosen chamber is the basic low-energy consistency condition^[39–40].

The same prepotential builds the abelian low-energy action. With

$$G_{ij}(\phi) = \partial_i \partial_j \mathcal{F}_{5d}, \quad c_{ijk}(\phi) = \partial_i \partial_j \partial_k \mathcal{F}_{5d}, \quad T_i = \partial_i \mathcal{F}_{5d},$$

the bosonic terms take the schematic form

$$S_{\text{eff}}^{5d} = \int_{\mathbb{R}^5} \left[\frac{1}{2} G_{ij}(\phi) d\phi_i \wedge *d\phi_j + \frac{1}{2} G_{ij}(\phi) F_i \wedge *F_j + \frac{1}{24\pi^2} c_{ijk}(\phi) A_i \wedge F_j \wedge F_k + \dots \right]. \quad (3.16)$$

The same first derivative T_i gives the tension of the monopole string associated with the simple coroot H_i .

The relation with four dimensions is close but not identical. Both theories have eight supercharges, both admit a Coulomb branch where the gauge group is broken to its maximal torus, and in both cases the low-energy abelian couplings are encoded by derivatives of a prepotential. In four dimensions the prepotential is holomorphic and receives an infinite instanton expansion controlled by the dimensionless coupling $\mathfrak{q}$. In five dimensions the Yang–Mills coupling has dimension of length and the prepotential is a real piecewise-cubic function on the Coulomb branch. The instanton is also reinterpreted: a four-dimensional instanton event becomes a five-dimensional BPS particle with conserved current

$$J_{\text{inst}} = \frac{1}{8\pi^2} \star \text{Tr } F \wedge F, \quad Q_{\text{inst}} = \frac{1}{8\pi^2} \int_{\mathbb{R}^4} \text{Tr } F \wedge F. \quad (3.17)$$

The fugacity for these particles is the same instanton-counting parameter that enters the ADHM index in Section 3.2.

3.2 ADHM Data and Ω background

3.2.1 Localization and ADHM moduli

Supersymmetric localization starts from a Q -exact deformation^[4,36]:

$$Z(t) = \int D\Phi \exp\{-S[\Phi] - t QV[\Phi]\}, \quad \frac{dZ(t)}{dt} = 0. \quad (3.18)$$

Taking $t \rightarrow \infty$ restricts the path integral to the zero locus of the positive bosonic part of QV :

$$\mathcal{L}_Q = \{\Phi \mid (QV)_{\text{bos}} = 0\} / \text{gauge} = \{\Phi \mid Q\psi = 0\} / \text{gauge}. \quad (3.19)$$

For the standard localizing fermion $V = \sum_{\psi} \langle \psi, Q\psi \rangle$, and on the usual positive integration contour, its bosonic variation is a sum of squares,

$$(QV)_{\text{bos}} = \sum_{\psi} \|Q\psi\|^2 + \text{positive auxiliary and gauge-fixing squares.}$$

Consequently it vanishes precisely when every fermionic supersymmetry variation $Q\psi$ vanishes. Geometrically, both descriptions in (3.19) are therefore the fixed-point locus of the vector field generated by Q on field space; the quotient identifies points on the same gauge orbit. For a four-dimensional pure $\mathcal{N} = 2$ gauge theory on $\mathbb{R}^4$, with the fields approaching a fixed Coulomb vacuum at infinity, this locus decomposes into instanton sectors. The gauge field obeys the anti-self-dual equation

$$F^+ = 0, \quad k = \frac{1}{8\pi^2} \int_{\mathbb{R}^4} \text{Tr } F \wedge F \in \mathbb{Z}_{\geq 0}. \quad (3.20)$$

The scalar fields are covariantly constant on the instanton background, and in the pure theory the remaining bosonic matter fields are absent. Therefore the localized four-dimensional path integral is an integral over the instanton moduli space $\mathfrak{M}_{G,k}$ in each topological sector. The Yang–Mills action decomposes as

$$S_{\text{YM}} = \frac{1}{2g_4^2} \int_{\mathbb{R}^4} \text{Tr}(F^+ \wedge *F^+) + \frac{8\pi^2 k}{g_4^2} - i\theta k. \quad (3.21)$$

Thus the path integral has the schematic instanton expansion

$$Z_{\text{gauge}} = Z_{\text{pert}} \sum_{k \geq 0} q^k \int_{\mathfrak{M}_{G,k}} \omega_k, \quad (3.22)$$

where $\mathfrak{M}_{G,k}$ is the k -instanton moduli space and ω_k denotes the induced measure and operator insertions.

For $G = \text{U}(N)$, the ADHM construction gives a systematic construction of all framed instanton solutions. It replaces the PDE (3.20) by finite-dimensional matrix equations: from matrices satisfying the ADHM moment maps and stability condition one constructs a gauge bundle and a connection with $F^+ = 0$, and conversely every framed smooth instanton gives such matrix data^[20–21,41–42]. Let

$$K \simeq \mathbb{C}^k, \quad N \simeq \mathbb{C}^N.$$

The ADHM variables are

$$B_1, B_2 \in \text{End}(K), \quad I \in \text{Hom}(N, K), \quad J \in \text{Hom}(K, N), \quad (3.23)$$

with moment maps

$$\mu_{\mathbb{C}} = [B_1, B_2] + IJ = 0, \quad \mu_{\mathbb{R}} = [B_1, B_1^\dagger] + [B_2, B_2^\dagger] + II^\dagger - J^\dagger J = \zeta \mathbf{1}_K. \quad (3.24)$$

Here ζ is the non-commutative/FI resolution parameter. The framed instanton moduli space is the hyperkahler quotient

$$\mathfrak{M}_{N,k} = \{(B_1, B_2, I, J) \mid \mu_{\mathbb{C}} = 0, \mu_{\mathbb{R}} = \zeta \mathbf{1}_K\} / \text{U}(k), \quad \dim_{\mathbb{R}} \mathfrak{M}_{N,k} = 4Nk. \quad (3.25)$$

Equivalently, one imposes stability: no proper subspace $K' \subset K$ with $I(N) \subset K'$ is preserved by B_1, B_2 .

The ADHM monad gives the gauge field explicitly. For $z = (z_1, z_2) \in \mathbb{C}^2$, define

$$\sigma_z = \begin{pmatrix} -B_2 + z_2 \\ B_1 - z_1 \\ J \end{pmatrix}, \quad \tau_z = \begin{pmatrix} B_1 - z_1 & B_2 - z_2 & I \end{pmatrix}. \quad (3.26)$$

The complex ADHM equation gives

$$\tau_z \sigma_z = [B_1, B_2] + IJ = 0.$$

The fiber of the framed gauge bundle is

$$E_z = \ker \tau_z / \text{im } \sigma_z, \quad \text{rank } E_z = N. \quad (3.27)$$

Equivalently, one packages σ_z and τ_z into the standard ADHM Dirac operator $\Delta^\dagger(z)$. Its kernel is the fiber E_z of the instanton bundle. The matrix $U(z)$ is chosen to have N orthonormal columns spanning this kernel, namely

$$\Delta^\dagger(z)U(z) = 0, \quad U^\dagger U = \mathbf{1}_N.$$

The connection and curvature are

$$A = U^\dagger dU, \quad F = dU^\dagger(1 - UU^\dagger)dU = U^\dagger d\Delta(\Delta^\dagger\Delta)^{-1} \wedge d\Delta^\dagger U. \quad (3.28)$$

The real moment-map equation implies

$$\Delta^\dagger\Delta = f^{-1}(z) \otimes \mathbf{1}_2,$$

so the two-form in (3.28) is anti-self-dual:

$$F^+ = 0, \quad c_2(E) = k. \quad (3.29)$$

Conversely, every framed smooth $U(N)$ solution of (3.20) with charge k determines ADHM data modulo $U(k)$. This is the ADHM theorem^[20–21,41–42].

The same data form the one-dimensional ADHM quiver quantum mechanics shown in Fig. 3-1:

$$B_1, B_2, I, J \text{ chiral}, \quad A_1 \text{ Fermi.}$$

The Fermi multiplet imposes the holomorphic moment map through

$$W_{\text{ADHM}} = \text{Tr}_K A_1 \mu_{\mathbb{C}} = \text{Tr}_K A_1 ([B_1, B_2] + IJ), \quad (3.30)$$

and the D-term is $\mu_{\mathbb{R}} = \zeta \mathbf{1}_K$. Thus ADHM quantum mechanics is not an auxiliary analogy: it is the finite-dimensional supersymmetric system whose Higgs branch is the instanton moduli space of 4d pure $U(N)$ theory.

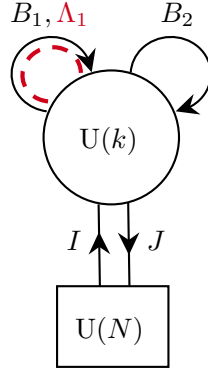


Figure 3-1 ADHM quiver diagram of the D0-D4 system in the infinite adjoint-mass limit.

3.2.2 Omega regularization and prepotentials

The moduli spaces in (3.22) are non-compact. Nekrasov's Ω -background regularizes the integral by equivariant rotation of the two complex planes:

$$\mathbb{C}^2 = \mathbb{C}_{\epsilon_1} \oplus \mathbb{C}_{\epsilon_2}, \quad (z_1, z_2) \mapsto (e^{i\epsilon_1} z_1, e^{i\epsilon_2} z_2).$$

After an accompanying R-symmetry rotation, the preserved supercharge satisfies

$$Q_\Omega^2 = \mathcal{L}_{V_\Omega} + \delta_{\text{gauge}}(\mathbf{a}) + \delta_{\text{flavor}}(\mathbf{m}) + \delta_R(\epsilon_{12}/2). \quad (3.31)$$

Here δ_{gauge} , δ_{flavor} , and δ_R denote infinitesimal gauge, flavor, and R-symmetry transformations with parameters shown in parentheses. The four-dimensional instanton integral becomes

$$Z_{\text{inst}}^{4d}(\mathbf{a}, \mathbf{m}, \epsilon_i, \mathbf{q}) = \sum_{k \geq 0} \mathbf{q}^k \int_{\mathfrak{M}_{N,k}}^T 1 = \sum_{k \geq 0} \mathbf{q}^k \sum_{p \in \mathfrak{M}_{N,k}^T} \frac{1}{e_T(T_p \mathfrak{M}_{N,k})}, \quad (3.32)$$

where

$$T = \text{U}(1)_{\epsilon_1} \times \text{U}(1)_{\epsilon_2} \times \text{U}(1)_{\mathbf{a}}^N \times T_{\text{flavor}}.$$

Here e_T is the T -equivariant Euler class. If the tangent representation at an isolated fixed point decomposes into one-dimensional weight spaces,

$$T_p \mathfrak{M}_{N,k} = \bigoplus_{\ell} \mathbb{C}_{w_\ell}, \quad e_T(T_p \mathfrak{M}_{N,k}) = \prod_{\ell} w_\ell,$$

where each w_ℓ is a linear combination of $\epsilon_{1,2}$, a_α , and flavor masses. Thus $1/e_T$ is the equivariant localization Jacobian, equivalently the one-loop determinant of fluctuations normal to the fixed point. For pure $\text{U}(N)$, the fixed points are N -tuples of Young diagrams $\vec{\lambda} = (\lambda_1, \dots, \lambda_N)$, with box pole

$$\phi_{\alpha, \mathbf{x}} = a_\alpha + (i-1)\epsilon_1 + (j-1)\epsilon_2, \quad \mathbf{x} = (i, j) \in \lambda_\alpha. \quad (3.33)$$

Equivalently, the Ω -background turns the integral over $\mathfrak{M}_{N,k}$ into the Witten index of the ADHM quantum mechanics:

$$Z_k = \frac{1}{k!} \oint_{\text{JK}(\eta)} \prod_{I=1}^k \frac{d\phi_I}{2\pi i} Z_{B_1, B_2, \Lambda_1}^{1d}(\boldsymbol{\phi}, \epsilon_i) Z_{I, J}^{1d}(\boldsymbol{\phi}, \mathbf{a}, \epsilon_i). \quad (3.34)$$

The prepotential is recovered by removing the equivariant volume regulator:

$$\mathcal{F}_{4d}^{\text{inst}}(\mathbf{a}, \mathbf{m}) = - \lim_{\epsilon_1, \epsilon_2 \rightarrow 0} \epsilon_1 \epsilon_2 \log Z_{\text{inst}}^{4d}(\mathbf{a}, \mathbf{m}, \epsilon_i, \mathbf{q}). \quad (3.35)$$

The factor $\epsilon_1 \epsilon_2$ has a simple origin. In the Ω -background, the equivariant volume of $\mathbb{R}^4$ is regularized as

$$\text{Vol}_{\Omega}(\mathbb{R}^4) = \frac{1}{\epsilon_1 \epsilon_2}.$$

The prepotential is the local density of the low-energy effective action on the Coulomb branch. Therefore the leading small- ϵ_i behavior of the instanton partition function is schematically

$$\log Z_{\text{inst}}^{4d} = - \frac{\mathcal{F}_{4d}^{\text{inst}}}{\epsilon_1 \epsilon_2} + \text{subleading terms},$$

which gives (3.35).

For five-dimensional theories on $\mathbb{R}_{\epsilon_1, \epsilon_2}^4 \times S_{\beta}^1$, the same construction is K-theoretic:

$$Z_{\text{inst}}^{5d}(\mathbf{a}, \mathbf{m}, \epsilon_i, \mathbf{q}) = \sum_{k \geq 0} \mathbf{q}^k Z_k^{5d}(\mathbf{a}, \mathbf{m}, \epsilon_i), \quad \mathbf{q} = \exp\left(-\frac{8\pi^2 \beta}{g_0^2} + i\theta\right). \quad (3.36)$$

The elementary one-loop weight is lifted by

$$4d: \quad x, \quad 5d: \quad \text{sh}(\beta x). \quad (3.37)$$

With dimensionful Coulomb parameters, the instanton part of the 5d prepotential is obtained, up to the conventional normalization of β , from

$$\mathcal{F}_{5d}^{\text{inst}} = - \lim_{\epsilon_1, \epsilon_2 \rightarrow 0} \frac{\epsilon_1 \epsilon_2}{\beta} \log Z_{\text{inst}}^{5d}. \quad (3.38)$$

The extra factor of β^{-1} reflects the additional circle: the leading effective-action contribution is integrated over $\mathbb{R}_{\Omega}^4 \times S_{\beta}^1$, so

$$\log Z_{\text{inst}}^{5d} = - \frac{\beta \mathcal{F}_{5d}^{\text{inst}}}{\epsilon_1 \epsilon_2} + \text{subleading terms}.$$

Thus multiplying by $\epsilon_1 \epsilon_2 / \beta$ extracts the five-dimensional prepotential density. The replacement $x \mapsto \text{sh}(\beta x)$ is the K-theoretic uplift of a cohomological weight: it records the tower of modes around S_{β}^1 , and reduces to the 4d weight as $\beta \rightarrow 0$. The small-circle limit reproduces the four-dimensional formula:

$$\text{sh}(\beta x) \sim \beta x, \quad Z_{\text{inst}}^{5d} \xrightarrow{\beta \rightarrow 0} Z_{\text{inst}}^{4d}.$$

Chapter 5 starts from these ADHM indices and rewrites their fixed-point contributions in shell-formula language^[4,8,21,36,43].

3.3 Brane Construction

The brane picture is used here as a dictionary for the quivers, Coulomb parameters, and fixed-point labels appearing later. The starting point is the open-string sigma model with a boundary gauge field,

$$S_{\text{op}} = -\frac{1}{4\pi\alpha'} \int_{\Sigma} d^2\sigma \left(\sqrt{h} h^{ab} g_{MN} + \epsilon^{ab} B_{MN} \right) \partial_a X^M \partial_b X^N + \int_{\partial\Sigma} A_M(X) \dot{X}^M d\tau + \dots \quad (3.39)$$

Here g and B are the NS–NS metric and two-form backgrounds, A is the Chan–Paton gauge field at the string endpoint, F is its field strength, and the dots denote the world-sheet fermions. Although the boundary coupling is written in terms of A , its variation depends on F and gives

$$\delta X^M \left[g_{MN} \partial_{\sigma} X^N + (B + 2\pi\alpha' F)_{MN} \partial_{\tau} X^N \right]_{\partial\Sigma} = 0. \quad (3.40)$$

A Dp -brane is the choice of $p + 1$ Neumann directions and $9 - p$ Dirichlet directions,

$$\mu = 0, \dots, p : \text{Neumann}, \quad I = p + 1, \dots, 9 : \delta X^I|_{\partial\Sigma} = 0.$$

For N coincident D-branes, endpoint labels $i, j = 1, \dots, N$ give Chan–Paton matrices. The massless open strings give a $U(N)$ vector field A_{μ} and adjoint transverse scalars

$$X^I = 2\pi\alpha' \Phi^I, \quad m_{\alpha\beta} = \frac{|X_{\alpha}^I - X_{\beta}^I|}{2\pi\alpha'} + \dots, \quad (3.41)$$

so diagonal brane positions are Coulomb or mass parameters in the gauge theory. On an index circle, Wilson lines combine with positions into complex fugacities such as v_{α} .

The low-energy action on a Dp -brane is the Dirac–Born–Infeld plus Chern–Simons action^[44–45],

$$S_{Dp} = -T_p \int_{Dp} e^{-\Phi} \text{Tr} \sqrt{-\det(P[g + B] + 2\pi\alpha' F)} + \mu_p \int_{Dp} \text{Tr} \sum_q P[C_q] \wedge e^{B + 2\pi\alpha' F}. \quad (3.42)$$

Here T_p is the brane tension, μ_p is the RR charge, $P[\dots]$ denotes pullback to the world-volume, $e^{-\Phi}$ records the string coupling, $F = dA + A^2$ is the Chan–Paton field strength, and C_q are the RR potentials. The determinant is over worldvolume directions; the trace is over Chan–Paton indices. Expanding the DBI term gives the Yang–Mills kinetic term, while the CS exponential records induced lower-dimensional D-brane charges. An anti- Dp -brane has the same DBI tension but opposite RR charge, $\mu_p \rightarrow -\mu_p$, and the opposite RR supersymmetry projector.

Five-dimensional gauge-theory branes

The five-dimensional web description uses type-IIB fivebranes. The D5 action is the $p = 5$ case of (3.42); finite D5 segments carry the gauge fields, while NS5-branes impose boundary conditions on those segments. In a standard web frame,

$$D5 : 012345, \quad NS5 : 012346, \quad O5 : 012345, \quad (3.43)$$

so the common 01234 directions support a 5d $\mathcal{N} = 1$ theory. The fivebrane projectors are schematically

$$\text{D5} : \epsilon_L = F_{012345}\epsilon_R, \quad \text{NS5} : \epsilon_{L,R} = F_{012346}\epsilon_{L,R}, \quad (3.44)$$

leaving eight common supercharges for compatible web angles. A general (p, q) five-brane is the $\text{SL}(2, \mathbb{Z})$ image of D5/NS5 data, with slope in the (x^5, x^6) -plane fixed by the charge vector (p, q) . Internal face areas encode $1/g_5^2$, finite D5 segment positions encode Coulomb parameters, and external leg positions encode masses.

Quantizing the open strings subject to the boundary condition (3.40) gives the 5d spectrum:

$$\begin{aligned} \text{D5--D5} : & \text{ vector multiplet on a D5 segment,} \\ \text{stretched strings} : & \text{ W-bosons or matter,} \\ M_{\alpha\beta} \sim & |a_\alpha - a_\beta|. \end{aligned}$$

NS5-branes do not add D5 Chan–Paton factors; instead they end D5 segments, project the open-string zero modes, and produce the finite 5d gauge node. Moving fivebranes realizes Higgsing and Hanany–Witten transitions^[45–48].

In this D5-brane description, a D1-brane extending x^0, x^5 can be viewed as a 5d instanton particle. The D5 Chern–Simons coupling contains

$$\mu_5 \int_{\text{D5}} C_2 \wedge \text{Tr } F \wedge F, \quad k = \frac{1}{8\pi^2} \int_{\mathbb{R}_{1234}^4} \text{Tr } F \wedge F, \quad (3.45)$$

so D1 charge dissolved in the D5 worldvolume is the gauge instanton number k . A D1 stretched along x^0 and the finite web direction is therefore a particle in the common 5d spacetime. Its D1–D1 and D1–D5 open strings give the one-dimensional ADHM quantum mechanics, T-dual to the D0–D4 description of Section 3.2.1^[42,44].

Orientifold planes implement $\Omega_{\text{ws}} R_\perp$, a worldsheet parity action combined with a spacetime reflection. An O5-plane parallel to the D5 stack preserves the same 5d supercharges as the D5 segment, but the open string must be invariant under the orientifold action. Hence Chan–Paton matrices obey

$$\lambda = \pm \gamma_\Omega \lambda^T \gamma_\Omega^{-1}. \quad (3.46)$$

The sign distinguishes O^- and O^+ : D5-branes on top of an O^- -plane give SO groups, while O^+ -planes give Sp groups. In instanton quantum mechanics the same open-string projection changes the ADHM gauge group:

$$G_{5d} = \text{SO}(N) : \hat{G} = \text{Sp}(2k), \quad G_{5d} = \text{Sp}(2N) : \hat{G} = \text{O}(k), \quad (3.47)$$

with the components of $\text{O}(k)$ corresponding to discrete-theta sectors^[21–23,49].

Gauge-origami branes

For gauge origami, it is convenient to write

$$z_a = x^{2a-1} + ix^{2a}, \quad a = 1, \dots, 4,$$

and to use x^0 as the time or index circle. The branes used later are:

Brane	Worldvolume	Main role
D0	x^0	instanton probe; ADHM gauge node $U(k)$
$D2_a, \overline{D2}_a$	$x^0, \mathbb{C}_a$	boundary/vortex sectors and brane-antibrane complexes
$D4_{ab}, \overline{D4}_{ab}$	$x^0, \mathbb{C}_a, \mathbb{C}_b$	ordinary and spiked instantons; 2d Young diagrams
$D6_{abc}, \overline{D6}_{abc}$	$x^0, \mathbb{C}_a, \mathbb{C}_b, \mathbb{C}_c$	tetrahedron/DT3-type systems; plane partitions
$D8, \overline{D8}$	$x^0, \mathbb{C}_1, \mathbb{C}_2, \mathbb{C}_3, \mathbb{C}_4$	magnificent-four/DT4-type systems; solid partitions

Table 3-1 Gauge-origami branes. Anti-branes have the same worldvolume and opposite RR charge; they are used through tachyon condensation or brane-antibrane complexes.

The open-string mass formula has the schematic form

$$\alpha' M^2 = \frac{L^2}{4\pi^2\alpha'} + N_{\text{osc}} + E_0(\nu, \varphi_a), \quad (3.48)$$

where L is the brane separation, ν is the number of Neumann–Dirichlet directions, and φ_a are angles or B -field rotations. The massless modes become chiral or Fermi multiplets in the one-dimensional quiver. For the ordinary D0–D4 system,

$$\text{D0–D0} : B_1, B_2, A_1, \quad \text{D0–D4} : I, J, \quad J_{A_1} = [B_1, B_2] + IJ. \quad (3.49)$$

Adding D6 or D8 branes adds more adjoint coordinates and Fermi constraints:

$$B_a \in \mathbf{Adj}_{U(k)}, \quad J_{ab} = [B_a, B_b] + \dots, \quad \sum_{a < b} \text{Tr } E_{ab} J_{ab} = 0. \quad (3.50)$$

These equations are the brane origin of the higher-dimensional Young diagrams and $\mathcal{F}$ -factors in Chapter 5.

From the worldvolume viewpoint, the D0-brane is the generalized instanton probe of gauge origami. For a $D2d$ -brane, the Chern–Simons coupling contains

$$\mu_{2d} \int_{D(2d)} C_1 \wedge \text{ch}_d(F), \quad (3.51)$$

so D0 charge is the top Chern character of the gauge bundle on the higher brane. For D4 this is the ordinary instanton number; for D6 and D8 it becomes the generalized instanton/DT-type charge. The D0– $D2d$ open strings then produce the matrix variables whose fixed points are 2d, 3d, or 4d Young diagrams.

A Dp -brane wrapping a complex coordinate plane preserves spinors satisfying

$$\epsilon_L = \Gamma_{0\dots p} \mathcal{M}_p \epsilon_R, \quad (3.52)$$

while an anti-brane reverses its sign. A coincident brane and anti-brane of the same orientation would therefore require both $\epsilon_L = \Gamma_{0\dots p} \mathcal{M}_p \epsilon_R$ and $\epsilon_L = -\Gamma_{0\dots p} \mathcal{M}_p \epsilon_R$, so they preserve no common nonzero spinor. The reversed GSO projection in the brane–anti-brane open-string sector also leaves a tachyonic NS ground state. In gauge origami

such pairs are used as complexes: a holomorphic tachyon map $\mathcal{T} : E_{\overline{D}p} \rightarrow E_{Dp}$ condenses and removes brane–anti-brane pairs away from the zero locus of $\mathcal{T}$, where the remaining lower-dimensional D-brane charge is supported.

A constant B -field changes the boundary condition (3.40). Under T-duality it is equivalent to a constant relative rotation by angles φ_a ,

$$B_{2a-1,2a} = \frac{\tan \varphi_a}{2\pi\alpha'} \implies \mathcal{R}(\varphi) = \exp\left(\frac{1}{2} \sum_a \varphi_a F_{2a-1,2a}\right).$$

The fluxed projector is conjugated by $\mathcal{R}$, and supersymmetry is preserved only when the relative rotation fixes a common spinor. On a spinor with plane chiralities $s_a = \pm 1$, this becomes the angle-sum condition

$$\sum_a s_a \varphi_a = 0 \pmod{2\pi},$$

with the orientation sign modified for an anti-brane^[50–52]. The same angles shift the open-string oscillator moding and the zero-point energy $E_0(\nu, \varphi_a)$ in (3.48). In the quiver, the constant B -field is encoded by a stability/FI parameter, while the holomorphic tachyon data determine the E, J -constraints.

Chapter 4

2d Dualities from 4d

This chapter develops the duality direction of the thesis. Its starting point is the twisted compactification of four-dimensional supersymmetric theories on S^2 , which produces the two-dimensional quiver theories studied below. The protected observable is the elliptic genus, evaluated by the JK prescription reviewed in Chapter 2. The main results are based on^[53], with the surrounding background reorganized for the thesis.

When a proposed duality is not proved by an analytic elliptic-genus identity, the text below states the finite-rank JK check, finite q -expansion, or anomaly/ E , J -term evidence used for it. Thus the chapter keeps the evidence level local to each family instead of collecting it in a separate table.

The chapter is organized as follows. Section 4.1 fixes the S^2 -reduction dictionary: the twist, background fluxes, Riemann–Roch zero-mode counting, and the map from four-dimensional multiplets to two-dimensional $(0, 2)$ or $(2, 2)$ multiplets. Section 4.2 applies this dictionary to Lagrangian class- $\mathcal{S}$ examples and checks gluing independence of their elliptic genera. Section 4.3 then turns to genuinely chiral $(0, 2)$ reductions, where elliptic-genus identities must be supplemented by anomaly matching, R-charge assignments, fugacity maps, and E , J -term consistency.

4.1 Twisted compactification on S^2

Before discussing particular duality families, we recall the compactification dictionary used throughout this chapter. The two-dimensional quivers below are zero-mode reductions of four-dimensional theories on $T^2 \times S^2$. For a 4d $\mathcal{N} = 1$ theory,

$$\mathrm{Spin}(4) \supset \mathrm{U}(1)_{T^2} \times \mathrm{U}(1)_{S^2}.$$

Choose an anomaly-free symmetry

$$\mathfrak{R} = R_0 + \sum_a t_a F_a, \quad \mathfrak{r}_i = R_i + \sum_a t_a q_i^a, \quad \sum_{\Phi_i \in W} \mathfrak{r}_i = 2. \quad (4.1)$$

Here F_a denotes the Cartan generator of an anomaly-free flavor symmetry, q_i^a is the charge of the 4d chiral field Φ_i , and the last condition states that each 4d superpotential term has $\mathfrak{R}$ -charge two. Charges under symmetries commuting with $\mathfrak{R}$ are inherited as two-dimensional flavor or R -charges. Here $F_\perp$ denotes a generator in the unbroken global-symmetry algebra that commutes with the twist generator $\mathfrak{R}$. The subscript $\perp$ means that this symmetry is not used as the background connection implementing the twist; it is a flavor generator, not a field strength. Its charge is inherited according to

$$F_\perp^{2d}(\text{zero mode}) = F_\perp^{4d}(\text{parent field}), \quad [F_\perp, \mathfrak{R}] = 0.$$

The topological twist is

$$D_{S^2}^{\text{tw}} = D_{S^2} - i \frac{\mathfrak{R}}{2} \omega_{S^2}, \quad J'_{S^2} = J_{S^2} + \frac{\mathfrak{R}}{2}, \quad J'_{S^2}(Q) = 0. \quad (4.2)$$

Here D_{S^2} is the ordinary covariant derivative on the sphere, including the spin connection ω_{S^2} , and $D_{S^2}^{\text{tw}}$ is the covariant derivative after adding the background $U(1)_{\mathfrak{R}}$ connection. Thus the preserved supercharges are scalar on S^2 . The $\mathfrak{R}$ -charges must be integral; otherwise the twisted fields are not sections of honest line bundles on $\mathbb{P}^1 \simeq S^2$ [18, 54].

One may also turn on gauge and flavor fluxes

$$\frac{1}{2\pi} \int_{S^2} F_G = \mathfrak{m} \in \text{cochar}(G), \quad \frac{1}{2\pi} \int_{S^2} F_a = \mathfrak{n}_a \in \mathbb{Z}. \quad (4.3)$$

In this line F_G and F_a are field strengths, i.e. curvatures of the gauge and background flavor bundles; this should not be confused with the flavor generators F_a in (4.1). The fluxes $\mathfrak{m}, \mathfrak{n}_a$ determine the degrees of these bundles. For a 4d chiral multiplet Φ of gauge weight ρ , flavor charges q_a , and $\mathfrak{R}$ -charge $\mathfrak{r}_\Phi$, the scalar zero modes are counted by

$$\mathcal{L}_{\Phi, \rho} = K_{\mathbb{P}^1}^{\mathfrak{r}_\Phi/2} \otimes L_G^\rho \otimes \prod_a L_a^{q_a}, \quad (4.4)$$

$$n_{\Phi, \rho} = \deg \mathcal{L}_{\Phi, \rho} = \rho(\mathfrak{m}) + \sum_a q_a \mathfrak{n}_a - \mathfrak{r}_\Phi. \quad (4.5)$$

Here $K_{\mathbb{P}^1} \simeq \mathcal{O}(-2)$ is the canonical bundle, and $K_{\mathbb{P}^1}^{1/2} \simeq \mathcal{O}(-1)$ fixes the spin-bundle convention. The scalar BPS equation on S^2 is the holomorphicity condition $\bar{\partial}_{D^{\text{tw}}} \phi = 0$, so scalar zero modes are sections of $\mathcal{L}_{\Phi, \rho}$, while the conjugate fermionic zero modes are counted by the first cohomology of the same complex. Riemann–Roch on $\mathbb{P}^1$ gives

$$h^0(\mathbb{P}^1, \mathcal{O}(n)) = \max(n+1, 0), \quad h^1(\mathbb{P}^1, \mathcal{O}(n)) = \max(-n-1, 0), \quad (4.6)$$

and hence the 4d chiral reduces as

$$\Phi_{4d, \rho} \longrightarrow \begin{cases} (n_{\Phi, \rho} + 1) \Phi_\rho^{(0,2)}, & n_{\Phi, \rho} \geq 0, \\ 0, & n_{\Phi, \rho} = -1, \\ (-n_{\Phi, \rho} - 1) \Psi_\rho^{(0,2)}, & n_{\Phi, \rho} \leq -2. \end{cases} \quad (4.7)$$

In the vanishing gauge-flux sector used most often below, $\mathfrak{m} = 0$, and for no extra flavor flux this becomes

$$\Phi_{4d}(\mathfrak{r}) \longrightarrow \begin{cases} (1 - \mathfrak{r}) \text{ 2d } (0, 2) \text{ chirals,} & \mathfrak{r} < 1, \\ 0, & \mathfrak{r} = 1, \\ (\mathfrak{r} - 1) \text{ 2d } (0, 2) \text{ Fermi multiplets,} & \mathfrak{r} > 1. \end{cases} \quad (4.8)$$

For vectors there is no harmonic one-form on S^2 , hence

$$V_G^{4d, \mathcal{N}=1} \longrightarrow V_G^{2d, (0,2)}, \quad (4.9)$$

$$V_G^{4d, \mathcal{N}=2} = V_G^{4d, \mathcal{N}=1} \oplus \Phi_{\text{Adj}} \quad (\mathfrak{r}_{\Phi_{\text{Adj}}} = 0) \longrightarrow V_G^{2d, (0,2)} \oplus \Phi_{\text{Adj}}^{(0,2)} = V_G^{2d, (2,2)}. \quad (4.10)$$

For the flavored reductions of 4d $\mathcal{N} = 2$ theories,

$$\mathfrak{R}_{(0,2)} = R + \frac{1}{2}(r - f), \quad \mathfrak{R}_{(2,2)} = 2R - f,$$

where $R \subset \text{SU}(2)_R$, r is the 4d $\text{U}(1)_r$, and f separates the two half-hypermultiplets. In the $(2, 2)$ twist,

$$\mathcal{Q}_-^1, \mathcal{Q}_+^2, \tilde{\mathcal{Q}}_+^1, \tilde{\mathcal{Q}}_-^2 : \quad J'_{S^2} = 0, \quad (4.11)$$

$$\text{U}(1)_V = \text{U}(1)_{-R+f/2}, \quad \text{U}(1)_A = \text{U}(1)_r. \quad (4.12)$$

The basic field maps are then

$$4\text{d } \mathcal{N} = 2 \text{ vector} \longrightarrow 2\text{d } \mathcal{N} = (2, 2) \text{ vector}, \quad (4.13)$$

$$(q, \tilde{q}) \text{ hypermultiplet } (\mathfrak{r}_q, \mathfrak{r}_{\tilde{q}}) = (0, 2) \longrightarrow q^{(0,2)} \oplus \tilde{q}_{\text{Fermi}}^{(0,2)} = 2\text{d } \mathcal{N} = (2, 2) \text{ chiral}. \quad (4.14)$$

The index relation used below is schematically

$$\mathcal{I}_{T^2 \times S^2}^{4d}(\mathfrak{R}, \mathfrak{m}, \mathfrak{n}) = \mathcal{I}_{\text{ell}}(\mathcal{T}_{2d}[\mathfrak{R}, \mathfrak{m}, \mathfrak{n}]), \quad (4.15)$$

where $\mathcal{T}_{2d}$ is the zero-mode theory defined by the rules above. The same twist and flux data must be used on both sides of a 4d duality. The following sections test the reduced two-dimensional theories: first the $(2, 2)$ class- $\mathcal{S}$ examples, and then the chiral $(0, 2)$ dualities where anomaly, E, J -term, fugacity-map, and JK-residue checks all enter.

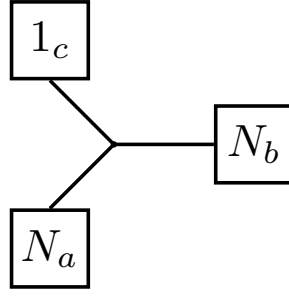
4.2 Class $\mathcal{S}$ and $\mathcal{N} = (2, 2)$

Class $\mathcal{S}$ theories are four-dimensional $\mathcal{N} = 2$ theories obtained by compactifying the six-dimensional $(2, 0)$ theory on a punctured Riemann surface C ^[17]. Pants decompositions of C encode different four-dimensional duality frames; after the S^2 compactification reviewed in Section 4.1, these frames become two-dimensional quiver descriptions whose elliptic genera can be compared.

For the $(2, 2)$ examples we only need the flavored reduction of^[18]. In a four-dimensional $\mathcal{N} = 2$ theory with $\text{SU}(2)_R \times \text{U}(1)_r$ symmetry, the twist uses $\mathfrak{R} = 2R - f$, where $\text{U}(1)_f$ distinguishes the two half-hypermultiplets. The detailed charge table is not repeated here; the only reduction rule used below is

$$\begin{aligned} 4\text{d } \mathcal{N} = 2 \text{ vector multiplet} &\rightsquigarrow 2\text{d } \mathcal{N} = (2, 2) \text{ vector multiplet} , \\ 4\text{d } \mathcal{N} = 2 \text{ hypermultiplet} &\rightsquigarrow 2\text{d } \mathcal{N} = (2, 2) \text{ chiral multiplet} . \end{aligned}$$

In class $\mathcal{S}$ theories of type A_{N-1} , punctures are labeled by partitions of N , and these theories generally lack a Lagrangian description. To perform the $(2, 2)$ reduction, we focus on class $\mathcal{S}$ theories with Lagrangian descriptions. The fundamental building block, in this case, is a sphere with two maximal punctures and one minimal puncture, corresponding to N^2 hypermultiplets with a flavor symmetry group $\text{U}(N^2)$. This group contains a subgroup $\text{SU}(N)_a \times \text{SU}(N)_b \times \text{U}(1)_c$. In the $(2, 2)$ reduction, $\text{U}(1)_c$ plays the role of the flavor symmetry used in the twist. The block reduces to N^2 $\mathcal{N} = (2, 2)$ chiral multiplets. This reduced $(2, 2)$ theory, denoted $U_N^{(2,2)}$, serves as the basic building block for the $\mathcal{N} = (2, 2)$ theories we consider, and its quiver is depicted in Fig. 4-1.

Figure 4-1 Basic building block $U_N^{(2,2)}$ for $SU(N)$ theory

Using elliptic genera, we study $\mathcal{N} = (2, 2)$ quiver gauge theories constructed from the building block $U_N^{(2,2)}$. The necessary review of elliptic genera and quiver building blocks is given in Chapter 2. The contribution to the elliptic genus from the basic building block $U_N^{(2,2)}$, which consists of N^2 chiral multiplets, is

$$\mathcal{J}_{U_N}^{(2,2)}(\mathbf{a}, \mathbf{b}, c) = \prod_{i,j=1}^N \frac{\vartheta_1(y a_i b_j c)}{\vartheta_1(a_i b_j c)}, \quad (4.16)$$

where we impose the condition $\prod_{i=1}^N a_i = 1 = \prod_{i=1}^N b_i$ for the $SU(N)$ fugacities.

We construct $\mathcal{N} = (2, 2)$ quiver gauge theories by gauging the flavor symmetries of the building blocks $U_N^{(2,2)}$. To gauge the $SU(N)$ flavor symmetries, we include the contribution from the $SU(N)$ vector multiplet in the Jeffrey-Kirwan integral of the elliptic genus

$$\mathcal{J}_{SU(N)}^{(2,2),V}(\boldsymbol{\zeta}) = \frac{1}{N!} \left(\frac{\eta(q)^3}{\vartheta_1(y)} \right)^{N-1} \prod_{\substack{i,j=1 \\ i \neq j}}^N \frac{\vartheta_1(\zeta_i / \zeta_j)}{\vartheta_1(y \zeta_i / \zeta_j)}, \quad (4.17)$$

where the condition $\prod_{i=1}^N \zeta_i = 1$ is imposed on the gauge fugacities. We do not introduce a superpotential in a quiver theory.

As in the case of $\mathcal{N} = (0, 4)$ theories^[19], the $\mathcal{N} = (2, 2)$ elliptic genera for quiver types of genus greater than zero also turn out to be remarkably simple. First, consider the theory of genus one with one minimal puncture, which consists of the free chiral plus the $SU(N)$ gauge theory with adjoint chiral. The $SU(N)$ gauge theory with adjoint chiral is indeed the $\mathcal{N} = (4, 4)$ vector multiplet, where the potential for the scalar fields is

$$V = \frac{1}{2e^2} \text{Tr}[\sigma, \sigma^\dagger]^2 + \frac{e^2}{2} \text{Tr}[\phi, \phi^\dagger]^2 + \frac{1}{2} \text{Tr}[\phi^\dagger, \sigma^\dagger][\sigma, \phi] + \frac{1}{2} \text{Tr}[\phi^\dagger, \sigma][\sigma^\dagger, \phi]. \quad (4.18)$$

Here, ϕ is the lowest component of the $\mathcal{N} = (2, 2)$ adjoint chiral while σ is the scalar in the $\mathcal{N} = (2, 2)$ vector multiplet. Therefore, the moduli space is spanned by the mutually commuting eigenvalues of ϕ and σ up to the action of the Weyl group, which leaves the maximal torus $U(1)^{N-1}$ of the gauge group unbroken. Consequently, the vacuum moduli space of the theory is

$$\mathbb{C} \times \frac{\mathfrak{t}_{\mathbb{C}} \times \mathfrak{t}_{\mathbb{C}}}{S_N} \quad (4.19)$$

where $\mathfrak{t}_{\mathbb{C}}$ is the Cartan subalgebra of $\mathrm{SL}(N, \mathbb{C})$. The IR CFT is the orbifold CFT with this target, and therefore the central charge is

$$c_L = c_R = 3(N^2 - (N^2 - 1)) + 6(N - 1) = 3 + 6(N - 1). \quad (4.20)$$

The first term accounts for the contributions from N^2 chiral multiplets and $N^2 - 1$ gauginos, while the second term arises from the unbroken gauge group $\mathrm{U}(1)^{N-1}$. The scope of verification for the elliptic-genus identities in this chapter is summarized in Section 4.3.1. In particular, when a general proof is not available, the text states the finite-rank residue or finite q -expansion evidence explicitly. With this convention, the elliptic genus of the theory is expressed as

$$\begin{aligned} \mathcal{J}_{g=1, n=1}^{(2,2), N} &= \oint_{\mathrm{JK}} \frac{d\mathbf{a}}{2\pi i \mathbf{a}} \mathcal{J}_{U_N}^{(2,2)}(\mathbf{a}, \mathbf{a}^{-1}, c) \mathcal{J}_{\mathrm{SU}(N)}^{(2,2), \mathrm{V}}(\mathbf{a}) \\ &= \frac{\vartheta_1(y^N c^N) \vartheta_1(y)}{\vartheta_1(c^N) \vartheta_1(y^N)} = \frac{\vartheta_1(y c)}{\vartheta_1(c)} \cdot \frac{\vartheta_1(y^N c^N) \vartheta_1(c) \vartheta_1(y)}{\vartheta_1(y c) \vartheta_1(c^N) \vartheta_1(y^N)}. \end{aligned} \quad (4.21)$$

Similar to [19]Eqn. (3.37), the first term represents the contribution from a free hypermultiplet, while the second term corresponds to the elliptic genus of an $\mathcal{N} = (4, 4)$ vector multiplet. The chiral ring of the symmetric product $(\mathfrak{t}_{\mathbb{C}} \times \mathfrak{t}_{\mathbb{C}})/S_N$ is generated by the operators $\mathrm{Tr}(\phi^i \sigma^j)$ for $i + j > 0$. The compact expression is consistent with an interpretation in terms of these chiral operators, together with Fermi fields imposing their relations. The displayed cancellation is an observed feature of the protected index rather than a complete derivation of the chiral-ring relations. For the $\mathrm{SU}(2)$ gauge group, such a chiral-ring relation is given in [55]§2.2.3.

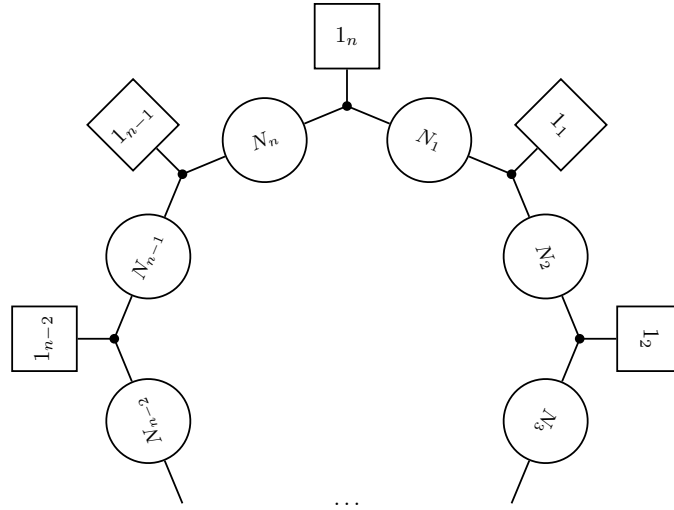


Figure 4-2 Quiver theory of genus one with n punctures.

Let us consider the quiver theory of genus one with n punctures as illustrated in Fig. 4-2. The theory can be understood as the $(2,2)$ reduction of the corresponding Lagrangian class $\mathcal{S}$ theory. At a generic point on the moduli space of chiral multiplets, the diagonal Cartan subgroup $\mathrm{U}(1)^{N-1}$ is unbroken at the infrared. Therefore, the corresponding twisted chiral field can take the vacuum expectation value. Therefore, the complex dimension of the vacuum moduli space is $2(N - 1) + n$, and the central charge of the IR CFT is

$$c_L = c_R = 3n(N^2 - (N^2 - 1)) + 6(N - 1) = 3n + 6(N - 1), \quad (4.22)$$

where the first term accounts for the contributions from n sets of N^2 chiral multiplets and n sets of $N^2 - 1$ gauginos, while the second term arises from the unbroken gauge group $U(1)^{N-1}$. The JK residue integral gives the product form built from (4.21):

$$\mathcal{J}_{g=1,n}^{(2,2),N}(c_1, \dots, c_n) = \prod_{s=1}^n \frac{\vartheta_1(y^N c_s^N) \vartheta_1(y)}{\vartheta_1(c_s^N) \vartheta_1(y^N)} \quad (4.23)$$

where c_i are the $U(1)$ flavor symmetries associated to the n puncture. Thus, this indicates that the elliptic genera receive local contributions from minimal punctures for this class of $\mathcal{N} = (2, 2)$ theories.

Unlike class $\mathcal{S}$ theories, we consider $U(1)$ gauging in 2d $\mathcal{N} = (2, 2)$ theories when gluing the building blocks U_N . However, gauging a $U(1)$ node results in a theory that is no longer a $(2,2)$ reduction of a class $\mathcal{S}$ theory. Nonetheless, as in Fig. 4-3, we can gauge the anti-diagonal part of $U(1)$ while preserving the diagonal $U(1)$ as a global symmetry.

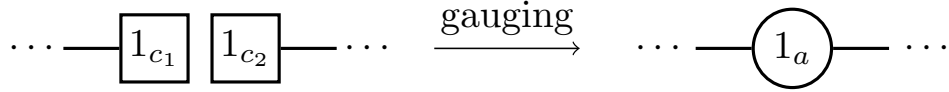


Figure 4-3 U(1) gauging

At the level of the elliptic genus, the $U(1)$ gauging procedure is given by

$$\mathcal{J}_{\mathcal{G}_1}(c_1, \dots) \mathcal{J}_{\mathcal{G}_2}(c_2, \dots) \rightarrow \frac{\eta(q)^3}{\vartheta_1(y)} \oint_{\text{JK}} \frac{da}{2\pi i a} \mathcal{J}_{\mathcal{G}_1}(ad, \dots) \mathcal{J}_{\mathcal{G}_2}(a^{-1}d, \dots) \quad (4.24)$$

where d is the diagonal $U(1)$ flavor fugacity.

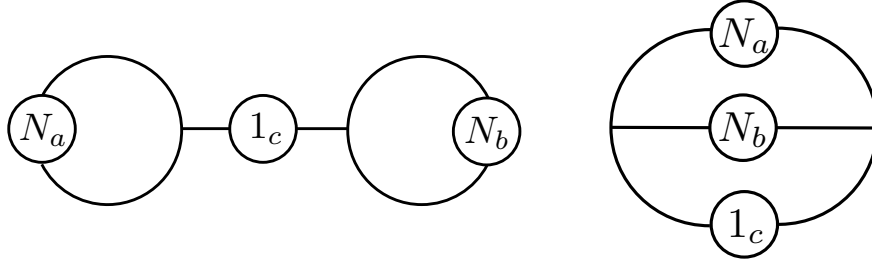


Figure 4-4 Genus two theories.

Let us consider the quiver theories of genus two, which involves the $U(1)$ gauging. There are two types of quivers, as shown in Fig. 4-4. In the theory of genus one with one puncture, the Cartan subgroup $U(1)^{N-1}$ is unbroken. For the genus-two theory, the Cartan subgroup $U(1)^{2(N-1)}$ of the gauge group $SU(N) \times SU(N)$ is unbroken. Consequently, the corresponding twisted chiral fields can take the expectation value, and the central charge of the genus-two theory is

$$c_L = c_R = 3(2N^2 - 2(N^2 - 1) - 1) + 12(N - 1) = 3 + 12(N - 1). \quad (4.25)$$

Equivalently, the central charges follow from the 2d 't Hooft anomalies of the exact superconformal R-current^[56–57]:

$$k_{RR} = \text{Tr } \gamma^3 R^2, \quad k_{\text{grav}} = \text{Tr } \gamma^3, \quad c_R = 3k_{RR}, \quad c_L = c_R - k_{\text{grav}}.$$

These are also the anomaly relations used in the consistency checks of Section 4.3.1. The trace is over all two-dimensional Weyl fermions, with $\gamma^3 = +1$ for right-movers and -1 for left-movers. For the $\mathcal{N} = (2, 2)$ theories considered here, $k_{\text{grav}} = 0$, hence $c_L = c_R$; counting the massless chiral and twisted-chiral coordinates on the infrared branch gives the expression above. Here, the first term accounts for the contributions from two sets of N^2 chiral multiplets and two sets of $N^2 - 1$ gauginos, along with an additional single $U(1)$ gaugino. The second term originates from the unbroken gauge group $U(1)^{2(N-1)}$. Since the two quiver descriptions have different Lagrangians, they have different expressions by JK residue integrals for the elliptic genera

$$\begin{aligned} \mathcal{F}_{\bigcirc-\bigcirc}^{(2,2),N} &= \frac{\eta(q)^3}{\vartheta_1(y)} \int_{\text{JK}} \frac{da}{2\pi i a} \frac{db}{2\pi i b} \frac{dc}{2\pi i c} \mathcal{F}_{U_N}^{(2,2)}(a, a^{-1}; dc) \mathcal{F}_{U_N}^{(2,2)}(b, b^{-1}; dc^{-1}) \mathcal{F}_{\text{SU}(N)}^{(2,2),V}(a) \mathcal{F}_{\text{SU}(N)}^{(2,2),V}(b), \\ \mathcal{F}_{\bigcirc}^{(2,2),N} &= \frac{\eta(q)^3}{\vartheta_1(y)} \int_{\text{JK}} \frac{da}{2\pi i a} \frac{db}{2\pi i b} \frac{dc}{2\pi i c} \mathcal{F}_{U_N}^{(2,2)}(a, b; dc) \mathcal{F}_{U_N}^{(2,2)}(a^{-1}, b^{-1}; dc^{-1}) \mathcal{F}_{\text{SU}(N)}^{(2,2),V}(a) \mathcal{F}_{\text{SU}(N)}^{(2,2),V}(b). \end{aligned} \quad (4.26)$$

However, the explicit evaluation of the JK residue integrals verifies their agreement:

$$\begin{aligned} \mathcal{F}_{\bigcirc-\bigcirc}^{(2,2),N} &= \mathcal{F}_{\bigcirc}^{(2,2),N} = \frac{\eta(q)^3 \vartheta_1(y)}{\vartheta_1(y^N)^2} \oint_{\text{JK}} \frac{da}{2\pi i a} \frac{\vartheta_1(y^N a^N d^N)}{\vartheta_1(a^N d^N)} \frac{\vartheta_1(y^N a^{-N} d^N)}{\vartheta_1(a^{-N} d^N)} \\ &= N \frac{\vartheta_1(y) \vartheta_1(y^N d^{2N})}{\vartheta_1(y^N) \vartheta_1(d^{2N})}, \end{aligned} \quad (4.27)$$

indicating that they are dual to each other. This suggests that the IR theory is independent of specific descriptions of genus-two theories, implying an underlying TQFT structure in this class of theories.

For a $(2, 2)$ quiver theory of genus $g > 0$ constructed from the $U_2^{(2,2)}$ theory, the minimum number of $U(1)$ gauge groups required is $g - 1$. Hence, extending the previous results, we can consider a $(2, 2)$ quiver theory of genus $g > 0$ with n punctures, where the numbers of $SU(N)$ and $U(1)$ gauge groups are $2(g - 1) + n$ and $g - 1$, respectively. In the branch used for the residue computation, the unbroken gauge group is taken to be $U(1)^{g(N-1)}$. Consequently, the central charge of the theory is:

$$c_L = 6g(N - 1) + 3(n + g - 1) = c_R. \quad (4.28)$$

For this class of theories, the elliptic genus depends only on (g, n) , and is independent of the quiver descriptions (or frames), which is expressed as

$$\mathcal{F}_{g>0,n}^{(2,2),N}(c_1, \dots, c_n) = \mathcal{F}_{g=1,n}^{(2,2),N}(c_1, \dots, c_n) \prod_{i=1}^{g-1} \frac{N \vartheta_1(y) \vartheta_1(y^N d_i^{2N})}{\vartheta_1(y^N) \vartheta_1(d_i^{2N})}, \quad (4.29)$$

where $\mathcal{F}_{g=1,n}^{(2,2),N}$ is given by (4.23). Thus, theories with different quiver descriptions are all dual to each other. Moreover, the integral formula (4.27) guarantees that the above form of the elliptic genera is consistent with the TQFT structure as

$$\begin{aligned} \mathcal{F}_{g=g_1+g_2, n_1+n_2-2}^{(2,2),N} &= \frac{\eta(q)^3}{\vartheta_1(y)} \int_{\text{JK}} \frac{da}{2\pi i a} \mathcal{F}_{g_1, n_1}^{(2,2),N}(\dots, d_{g_1-1} a) \mathcal{F}_{g_2, n_2}^{(2,2),N}(d_{g_1-1} a^{-1}, \dots), \\ \mathcal{F}_{g+1, n-2}^{(2,2),N} &= \frac{\eta(q)^3}{\vartheta_1(y)} \int_{\text{JK}} \frac{da}{2\pi i a} \mathcal{F}_{g,n}^{(2,2),N}(\dots, d_g a, d_g a^{-1}). \end{aligned} \quad (4.30)$$

As a result of JK residue integrals, these elliptic genera are all expressed as simple products of theta functions, suggesting that they are dual to $\mathcal{N} = (2, 2)$ Landau-Ginzburg models. The challenge lies in identifying the correct superpotential. The method for this is given by using S^2 partition functions^[58] based on the techniques of^[59–61]. We leave this problem for future investigation.

Furthermore, one can introduce FI parameters for the $U(1)$ gauge groups and discrete theta angles for $SU(N)$ gauge groups. It is also desirable to study the phases of the $\mathcal{N} = (2, 2)$ quiver theories considered in this thesis with respect to these parameters, following the approach of^[5].

The $(2, 2)$ class $\mathcal{S}$ examples above illustrate how four-dimensional duality-frame independence becomes gluing independence of elliptic genera after the flavored S^2 reduction. The rest of the chapter turns to the genuinely chiral $(0, 2)$ reductions. In that setting the same JK-residue technology remains useful, but it must be supplemented by anomaly matching, R-charge assignments, and E, J -term consistency.

4.3 $\mathcal{N} = (0, 2)$ dualities

The general $(0, 2)$ multiplet conventions, anomaly constraints, and elliptic genus factors were fixed in Chapter 2. This section uses those conventions for the main constructions of the paper *2d dualities from 4d*: starting from known four-dimensional dualities compactified on S^2 , we obtain two-dimensional Seiberg-like dualities, extend them to Sp/SO trialities, and then use these elementary moves to build tensor-matter and adjoint dualities. The original unitary triality of^[16] and the compactification results of^[18–19, 55, 62–63] are used as inputs; the new content below is the systematic extension to the families treated in the following subsections.

4.3.1 Consistency checks for proposed $(0, 2)$ dualities

For each proposed $(0, 2)$ duality below, the elliptic genus is only one part of the consistency check. We also require the standard anomaly and superpotential data to match. In the sign convention used in this chapter, right-moving fermions in chiral multiplets and left-moving fermions in Fermi or vector multiplets contribute with opposite signs. The basic checks are summarized in Tab. 4-1.

We use evidence labels A–D in the remaining subsections. A Level A statement is a proof at the level of elliptic genera, while Levels B and C record finite residue or series checks. Level D records the structural checks in Tab. 4-1 when no elliptic-genus equality is evaluated. Whenever the verification range is known explicitly, it is stated near the corresponding formula rather than being promoted to an all-rank proof.

4.3.2 Comments on 2d $(0, 2)$ Seiberg-like dualities with SU gauge groups

Let us first review the duality of 2d $\mathcal{N} = (0, 2)$ SU gauge theories obtained by the twisted compactification of the original 4d $\mathcal{N} = 1$ Seiberg duality^[64] on S^2 . This duality was first obtained in^[18]; here it is used as an input, and the subsection focuses on subtleties of the charge assignments, anomaly checks, and elliptic-genus comparison. The duality is illustrated in Fig. 4-5. Its status in this thesis is Level B/C/D: the parent duality is

Check	Requirement
Gauge anomaly	$\mathcal{A}_G = \sum_{\Phi} T_G(R_{\Phi}) - \sum_{\Psi} T_G(R_{\Psi}) - T_G(\mathbf{Adj}) = 0$ for every gauge node.
Flavor anomaly	The quadratic 't Hooft anomalies $k_{AB} = \sum_{\Phi} q_A q_B - \sum_{\Psi} q_A q_B$ agree across the dual pair, including singlet mesons.
Gravitational anomaly	$c_R - c_L = \text{Tr } \gamma^3$ agrees after including gauge, chiral, and Fermi multiplet fermions.
R-charge assignment	The assigned R-charges make all superpotential terms compatible with the $(0, 2)$ supersymmetry convention and keep the elliptic genus doubly periodic.
E - and J -terms	Every Fermi multiplet satisfies $\bar{D}_+ \Psi = E(\Phi)$ and the full set obeys $\sum_a E_a J^a = 0$.

Table 4-1 Uniform consistency checks used for the $(0, 2)$ dualities discussed in this chapter.

imported from the literature, while the text records finite JK-residue or q -expansion checks together with anomaly, charge, and E , J -term consistency.

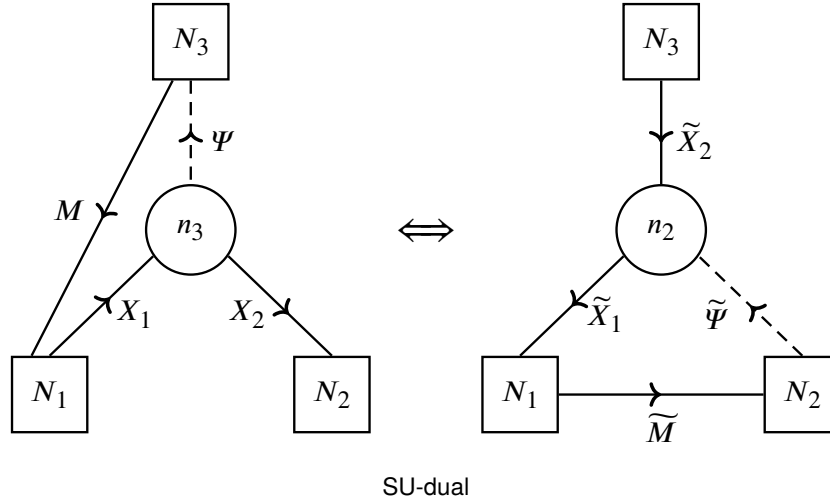


Figure 4-5 2d $(0, 2)$ duality for SQCDs with SU gauge groups, where $n_3 = \frac{N_1 + N_2 - N_3}{2}$ and $n_2 = \frac{N_1 + N_3 - N_2}{2}$ ($N_1 \geq N_2 + N_3$). The solid line represents a chiral multiplet while the dashed line represents a Fermi multiplet.

The two dual theories are described as follows:

- The first theory is an $SU(n_3)$ gauge theory with:
 - N_1 fundamental chiral multiplets X_1 ,
 - N_2 anti-fundamental chiral multiplets X_2 ,
 - N_3 anti-fundamental Fermi multiplets Ψ ,

where $\mathbb{Z} \ni n_3 = \frac{N_1 + N_2 - N_3}{2}$. Additionally, the theory contains a gauge-singlet meson M , which couples via the superpotential:

$$\mathcal{W} = \text{Tr}(\Psi M X_1). \quad (4.31)$$

- The second theory is an $SU(n_2)$ gauge theory with:

- N_1 anti-fundamental chiral multiplets $\tilde{X}_1$,
- N_2 fundamental Fermi multiplets $\tilde{\Psi}$,
- N_3 fundamental chiral multiplets $\tilde{X}_2$,

where $\mathbb{Z} \ni n_2 = \frac{N_1 + N_3 - N_2}{2}$. This theory also contains a gauge-singlet meson $\tilde{M}$, which couples via the superpotential:

$$\mathcal{W} = \text{Tr}(\tilde{\Psi} \tilde{X}_1 \tilde{M}). \quad (4.32)$$

For the duality to hold, we impose the condition:

$$N_1 \geq N_2 + N_3. \quad (4.33)$$

The charges of each multiplet under the global symmetry are summarized in the following table:

	X_1	X_2	Ψ	M	$\tilde{X}_1$	$\tilde{X}_2$	$\tilde{\Psi}$	$\tilde{M}$
$SU(N_1)$	$\overline{\square}$	1	1	$\square$	$\square$	1	1	$\overline{\square}$
$SU(N_2)$	1	$\square$	1	1	1	1	$\overline{\square}$	1
$SU(N_3)$	1	1	$\square$	$\overline{\square}$	1	$\overline{\square}$	1	$\square$
$U(1)_1$	-1	0	0	1	$1 - \frac{N_1}{n_2}$	$\frac{N_1}{n_2}$	$\frac{N_1}{n_2}$	-1
$U(1)_2$	0	1	0	0	0	0	-1	0
$U(1)_3$	0	0	1	-1	0	-1	0	1

Why the triality fails with SU gauge groups: However, unlike in the case of the unitary gauge group, this duality does not extend to a triality. The failure of triality when transitioning from $U(N_c)$ to $SU(N_c)$ can be attributed to the absence of a Fayet-Iliopoulos (FI) term, as explained below.

The 2d $\mathcal{N} = (0, 2)$ triality^[16] is an equivalence of three unitary gauge theories in the infrared. This triality can be understood through the relationship among the target geometry of the non-linear sigma models in the infrared^[65–66]. The target geometry involves the tautological bundle S of rank k and the quotient bundle Q of rank $(n - k)$ on a Grassmannian $\text{Gr}(k, n)$, related by the short exact sequence:

$$0 \longrightarrow S \longrightarrow \mathcal{O}^n \longrightarrow Q \longrightarrow 0.$$

For the 2d $\mathcal{N} = (0, 2)$ triality^[16], an essential bundle isomorphism plays a key role:

$$\begin{array}{ccc} S & & Q^* \\ \downarrow & \cong & \downarrow \\ \text{Gr}(k, n) & & \text{Gr}(n - k, n). \end{array} \quad (4.34)$$

This isomorphism reflects the interchange between the tautological bundle S and the dual of the quotient bundle Q^* under the duality transformation $\text{Gr}(k, n) \longleftrightarrow \text{Gr}(n - k, n)$.

For unitary gauge groups, one can introduce the Fayet-Iliopoulos (FI) term ζ . For a positive FI term ($\zeta > 0$), one of the triality theories flows to the non-linear sigma model with the target space

$$\begin{array}{ccc} S^{\oplus N_3} \oplus Q^{\oplus N_2} & & S^{*\oplus N_2} \oplus Q^{*\oplus N_3} \\ \downarrow & \cong & \downarrow \\ \text{Gr}(n_3, N_1) & & \text{Gr}(n_2, N_1) \end{array} \quad (4.35)$$

where $n_i = \frac{-N_i + N_{i+1} + N_{i+2}}{2}$ (indices are considered mod 3). Conversely, for a negative FI term ($\zeta < 0$), the target space configuration is:

$$\begin{array}{ccc} S^{*\oplus N_3} \oplus Q^{*\oplus N_1} & & S^{\oplus N_1} \oplus Q^{\oplus N_3} \\ \downarrow & \cong & \downarrow \\ \text{Gr}(n_3, N_2) & & \text{Gr}(n_1, N_2) \end{array} \quad (4.36)$$

Permuting the indices (N_1, N_2, N_3) , we see that $(0, 2)$ triality manifests as equivalence among these target spaces. We refer to ^[66]Fig. 5 for a detailed illustration.

However, the triality does *not* hold, namely, the three theories are *inequivalent* in the infrared if we use a special unitary gauge group $\text{SU}(N_c)$ instead of $\text{U}(N_c)$, removing Fermi fields in the determinant representation of the unitary gauge group. In fact, elliptic genera distinguish the three theories if the gauge groups are special unitary groups $\text{SU}(N_c)$.

One key obstruction is that the condition (4.33) for the duality in Fig. 4-5 is incompatible with the triality condition $N_i + N_{i+1} \geq N_{i+2}$ ^[16]. Furthermore, the FI term cannot be introduced for an $\text{SU}(N_c)$ gauge group. Hence, the two geometric configurations described by (4.35) and (4.36) cannot be connected through an $\text{SU}(N_c)$ gauge theory. In fact, this underpins why the triality fails when naively changing from $\text{U}(N_c)$ to $\text{SU}(N_c)$.

Non-linear sigma model: With the SU gauge groups, the dual pair in Fig. 4-5 flows to the non-linear sigma model with the same target manifold

$$\begin{array}{ccc} \det(S) \oplus S^{\oplus N_2} \oplus Q^{\oplus N_3} & & \det(Q^*) \oplus Q^{*\oplus N_2} \oplus S^{*\oplus N_3} \\ \downarrow & \cong & \downarrow \\ \text{Gr}(n_3, N_1) & & \text{Gr}(n_2, N_1) \end{array} \quad (4.37)$$

In fact, the condition (4.33) ensures that the base Grassmannian is defined as the moduli space of subspaces in $\mathbb{C}^{N_1}$. Note that, due to the SU gauge group, the N_1 (anti-)fundamental chirals parametrize

$$\begin{array}{ccc} \det(S) & & \det(Q^*) \\ \downarrow & \cong & \downarrow \\ \text{Gr}(n_3, N_1) & & \text{Gr}(n_2, N_1) . \end{array} \quad (4.38)$$

where $\det(S)$ and $\det(Q^*)$ denote the determinant line bundles of S and Q^* , respectively. Thus, the proposed duality for SU gauge groups can be understood as the equivalence of the target geometries in (4.37).

Central charges: Since the target space is non-compact, one of the fundamental assumptions of c -extremization in [56–57] is violated. Specifically, the issue arises because a non-holomorphic current may exist for the flavor symmetry associated with the non-compact direction. This current cannot mix with the R -symmetry current, which invalidates the naive application of c -extremization [57,63].

To address this issue, it is necessary to identify the chiral operators that parametrize these non-compact directions and assign them an R -charge of 0. In our example, while the Grassmannian itself is compact, the total space (4.38) of the determinant line bundle, parametrized by N_1 (anti-)fundamental chirals, is non-compact. Since these N_1 (anti-)fundamental chirals transform with the same charge under the $U(1)_1$ flavor symmetry, their R -charges must all be set to zero.

Consequently, in both dual theories, the R -charges of all chiral fields must be uniformly set to 0. Additionally, the R -charge of the Fermi field is fixed at 1 to ensure that the superpotential retains an R -charge of 1 [57,63].

With this assignment, the central charges agree for the dual theories:

$$c_L = 2(n_3^2 + N_1 N_3 + 1), \quad c_R = 3(n_3^2 + N_1 N_3 + 1). \quad (4.39)$$

Note that the right-moving central charge is three times the dimension of the target space (4.37).

Let us compare this with the case of $\mathcal{N} = (0, 2)$ triality studied in [16]. For unitary gauge groups, the (anti-)fundamental chiral multiplets in the theory simply parameterize a compact Grassmannian manifold (without the determinant line bundle in this case). In addition, the presence of a superpotential imposes further constraints on the R -charges of the supermultiplets. Consequently, the c -extremization must be performed to determine the R -symmetry. Hence, although the vacuum moduli space is non-compact as seen in (4.35) and (4.36), the application of the c -extremization is still required.

Other checks for the duality: The 't Hooft anomaly comparison follows from the charges listed above and is one of the checks summarized in Table 4-1. In addition, the elliptic genera can be computed using the JK residue integral:

$$\begin{aligned} \mathcal{J}_A &= \frac{1}{n_3!} \oint_{\text{JK}} \frac{d\mathbf{a}}{2\pi i \mathbf{a}} \prod_{i=1}^{n_3} \frac{\prod_{j \neq i} \vartheta_1(a_i a_j^{-1}) \cdot \prod_{j=1}^{N_3} \vartheta_1(a_i^{-1} d_j x_3)}{\prod_{j=1}^{N_1} \vartheta_1(a_i b_j^{-1} x_1^{-1}) \cdot \prod_{j=1}^{N_2} \vartheta_1(a_i^{-1} c_j x_2)} \frac{\eta(q)^{n_3^2 + 3n_3 + N_1 N_3 - 2}}{\prod_{i=1}^{N_1} \prod_{j=1}^{N_3} \vartheta_1(x_1 x_3^{-1} b_i d_j^{-1})}, \\ \mathcal{J}_B &= \frac{1}{n_2!} \oint_{\text{JK}} \frac{d\mathbf{a}}{2\pi i \mathbf{a}} \prod_{i=1}^{n_2} \frac{\prod_{j \neq i} \vartheta_1(a_i a_j^{-1}) \cdot \prod_{j=1}^{N_2} \vartheta_1(a_i c_j^{-1} x_2^{-1} x_1^{\frac{N_1}{n_2}})}{\prod_{j=1}^{N_1} \vartheta_1(a_i^{-1} b_j x_1^{-\frac{n_3}{n_2}}) \cdot \prod_{j=1}^{N_3} \vartheta_1(a_i d_j^{-1} x_3^{-1} x_1^{\frac{N_1}{n_2}})} \frac{\eta(q)^{n_2^2 + 3n_2 + N_1 N_2 - 2}}{\prod_{i=1}^{N_1} \prod_{j=1}^{N_2} \vartheta_1(x_2 x_1^{-1} c_j b_i^{-1})}. \end{aligned} \quad (4.40)$$

The identity $\mathcal{J}_A = \mathcal{J}_B$ has been checked up to the rank-five integral. When $N_1 = N_2 + N_3$, an additional condition is required:

$$x_1^{N_1} = x_2^{N_2} x_3^{N_3}. \quad (4.41)$$

4.3.3 From 2d (0,2) Seiberg-like dualities to new trialities

The triality statements in this subsection have Level B/C/D evidence. The rank maps are derived from the SU Seiberg-like move, and the proposed trialities are checked by anomalies and by the finite elliptic-genus comparisons stated below.

In addition to SU gauge groups, the twisted compactification of 4d $\mathcal{N} = 1$ dualities with Sp gauge groups^[67] on S^2 was first explored in^[18], and later refined in^[63]. This leads to the 2d (0,2) duality between Sp SQCD and a Landau-Ginzburg theory. Following their approach, this subsection presents a new 2d (0,2) duality from the twisted compactification of the Intriligator-Seiberg duality^[68] for the SO gauge group on S^2 and extends these 2d (0,2) Seiberg-like dualities to new trialities.

Sp gauge group: The twisted compactification on S^2 is applied to 4d $\mathcal{N} = 1$ dualities with $\text{Sp}(2N)$ gauge groups^[67], leading to the identification of a 2d $\mathcal{N} = (0, 2)$ duality between $\text{Sp}(2N)$ SQCD (AS-1) and an LG model (AS-3)^[18,63]. We further generalize this result and propose a new triality below.

Before proceeding, let us remark on an important distinction between the 2d $\mathcal{N} = (0, 2)$ Seiberg-like dualities involving the SU and Sp gauge groups. A key difference is that Sp gauge theories only have chiral multiplets in the fundamental representation, whereas SU gauge theories involve both fundamental and anti-fundamental chirals in their duality. Consequently, upon the compactification on S^2 , we obtain a duality between a 2d $\mathcal{N} = (0, 2)$ $\text{Sp}(2N)$ gauge theory and a Landau-Ginzburg (LG) model. However, to our knowledge, a duality between two 2d $\mathcal{N} = (0, 2)$ $\text{Sp}(2N)$ gauge theories has not yet been discovered.

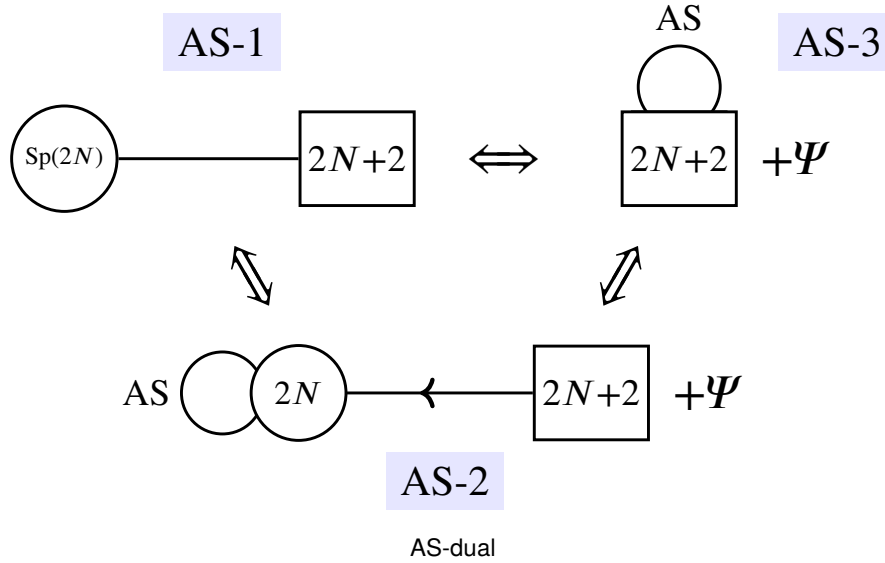


Figure 4-6 A triality among three distinct theories: (AS-1) an $\text{Sp}(2N)$ gauge theory with $2N + 2$ fundamental chirals, (AS-2) an $\text{SU}(2N)$ gauge theory with one antisymmetric and $2N + 2$ fundamental chirals, coupled to a neutral Fermi multiplet Ψ , and (AS-3) a Landau-Ginzburg model consisting of $(N + 1)(2N + 1)$ chiral multiplets coupled to a Fermi multiplet Ψ .

AS-1. $\text{Sp}(2N)$ gauge theory with $2N + 2$ fundamental chirals Z and no superpotential.

AS-2. $\text{SU}(2N)$ gauge theory with one antisymmetric chiral X and $2N + 2$ fundamental chirals Y . Additionally, there is a neutral Fermi multiplet Ψ , forming a superpotential

$$\mathcal{W} = \Psi \text{Pf } X . \quad (4.42)$$

AS-3. LG model of one Fermi Ψ and $(N+1)(2N+1)$ chirals, forming an antisymmetric $(2N+2) \times (2N+2)$ matrix A with a superpotential

$$\mathcal{W} = \Psi \text{Pf } A. \quad (4.43)$$

The $U(1)_x$ charges of the fields are given as follows:

	Ψ	A	Z	X	Y
$U(1)_x$	$-(2N+2)$	2	1	$\frac{2N+2}{N}$	$\frac{1}{N}$

(4.44)

As a simple check, their central charges agree as

$$c_L = 2N(2N+3), \quad c_R = 3N(2N+3). \quad (4.45)$$

The elliptic genera also agree

$$\begin{aligned} \mathcal{F} &= -\frac{\eta(q)^{2N(N+3)}}{2^N N!} \oint_{\text{JK}} \frac{d\mathbf{a}}{2\pi i \mathbf{a}} \prod_{i=1}^N \frac{\vartheta_1(a_i^{\pm 2}) \prod_{j>i} \vartheta_1(a_j^{\pm} a_i^{\pm})}{\prod_{j=1}^{2N+2} \vartheta_1(x a_i^{\pm} b_j)} \\ &= \frac{\eta(q)^{(2N^2+9N-3)} \vartheta_1(x^{-(2N+2)})}{(2N)!} \oint_{\text{JK}} \frac{d\mathbf{a}}{2\pi i \mathbf{a}} \prod_{i=1}^{2N} \frac{\prod_{i \neq j} \vartheta_1(a_i/a_j)}{\prod_{i<j} \vartheta_1(x^{\frac{2N+2}{N}} a_i a_j) \prod_{j=1}^{2N+2} \vartheta_1(x^{\frac{1}{N}} a_i b_j)} \\ &= \frac{\eta(q)^{N(2N+3)} \vartheta_1(x^{-(2N+2)})}{\prod_{i<j} \vartheta_1(x^2 b_i b_j)}. \end{aligned} \quad (4.46)$$

We find this triality by considering the $(0,2)$ reduction of 4d $\mathcal{N} = 2$ SCFTs^[18–19] on S^2 . For a 4d $\mathcal{N} = 2$ theory with gauge group G , the β -function is given by

$$\beta \propto 2T(\mathbf{Adj}) - \sum_i T(R_i), \quad (4.47)$$

where $T(R)$ denotes the Dynkin indices of a representation R for a half-hypermultiplet. On the other hand, the gauge anomaly of the corresponding $(0,2)$ theory is expressed as

$$\text{Tr}(\gamma_3 G^2) = \sum_i T(R_i) - T(\mathbf{Adj}). \quad (4.48)$$

Since only half of the 4d half-hypermultiplets become $(0,2)$ chiral multiplets in the flavored reduction, the condition for the vanishing β -function in 4d $\mathcal{N} = 2$ theory is equivalent to the condition for the vanishing 2d gauge anomaly. Therefore, the $(0,2)$ theory obtained from the Lagrangian 4d $\mathcal{N} = 2$ SCFT is well-defined. In^[19], the $(0,2)$ reduction of 4d $\mathcal{N} = 2$ Lagrangian SCFTs of type A class $\mathcal{S}$ is considered. The elliptic genera of these theories can be expressed in a surprisingly simple form, involving products of theta functions and eta functions. In some cases, the LG dual theories are identified.

Here, we consider the 4d $\mathcal{N} = 2$ SCFTs: $\text{Sp}(2N)$ gauge theory with $2N+2$ fundamental half-hypermultiplets, and $\text{SU}(2N)$ gauge theory with one antisymmetric and $2N+2$ fundamental hypermultiplets. These theories arise from D4-NS5-O6⁻ brane system (See^[69]§3.1). Their $(0,2)$ reductions yield the AS-1 and AS-2 theories shown in Fig. 4-6, and they turn out to be dual to a $(0,2)$ LG model (AS-3). Notably, the integrand of the elliptic genus (4.46) coincides with that of the corresponding 4d $\mathcal{N} = 2$

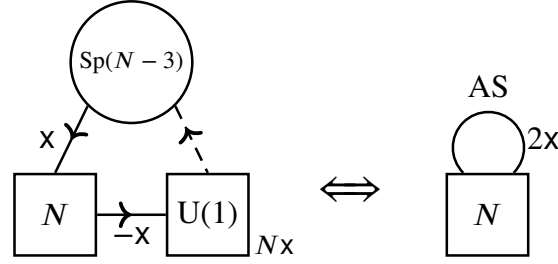


Figure 4-7 A duality between a free theory with $N(N-1)/2$ chirals and $\text{Sp}(N-3)$ gauge theory with N fundamental chirals, one fundamental Fermi multiplet and N neutral mesons.

Schur indices (up to the modification of the $U(1)_x$ -charges due to the superpotential (4.42))^[18–19].

Additionally, we find a similar triality with the flavor group $\text{SU}(N)$ with odd N , which is given in Appendix B.1.2. As another generalization, we also propose a duality between the following theories (Fig. 4-7).

- $\text{Sp}(N-3)$ gauge theory with N fundamental chirals X , one fundamental Fermi multiplet Ψ and N neutral mesons M with a superpotential

$$\mathcal{W} = \omega^{\alpha\beta} \Psi_{\alpha} X_{\beta,i} M^i \quad (4.49)$$

where $\omega^{\alpha\beta}$ is the symplectic form that projects the tensor product $\square \otimes \square$ to the trivial representation for representations of the gauge group $\text{Sp}(N-3)$.

- a free theory with $N(N-1)/2$ chirals

This duality is used frequently in what follows. The elliptic genera are given by

$$\begin{aligned} \mathcal{J} &= \frac{\eta(q)^{(N^2+2N-9)/2}}{2^{(N-3)/2} \frac{N-3}{2}!} \oint_{\text{JK}} \frac{d\mathbf{a}}{2\pi i \mathbf{a}} \prod_{i=1}^{\frac{N-3}{2}} \frac{\vartheta_1(a_i^{\pm} x^{-N}) \vartheta_1(a_i^{\pm 2}) \prod_{j<i} \vartheta_1(a_i^{\pm} a_j^{\pm})}{\prod_{j=1}^N \vartheta_1(x a_i^{\pm} b_j)} \cdot \frac{1}{\prod_{i=1}^N \vartheta_1(x^{N-1} b_i^{-1})} \\ &= \prod_{i<j} \frac{\eta(q)}{\vartheta_1(x^2 b_i b_j)}. \end{aligned} \quad (4.50)$$

This duality is used to derive other dualities in what follows.

SO gauge group: We also consider the twisted compactification of the 4d $\mathcal{N} = 1$ Intriligator-Seiberg duality^[68] on S^2 , which results in a duality between a 2d $\mathcal{N} = (0, 2)$ SO SQCD (Sym-1) and an LG model (Sym-3). This procedure is detailed in Appendix B.1.1. Like in the case of the Sp gauge group, the compactification of the 4d $\mathcal{N} = 1$ duality on S^2 does not produce a duality between two SO gauge theories, as it involves only fundamental chiral multiplets. Moreover, we extend this duality to a triality, as illustrated in Fig. 4-8.

Sym-1. $\text{SO}(N)$ gauge theory with $N-2$ fundamental chirals Y . Additionally, there is a gauge neutral chiral X and Fermi Ψ , forming a superpotential

$$\mathcal{W} = \Psi(X^2 + \det A), \quad (4.51)$$

where A is defined by the projector $\eta^{\alpha\beta}$ of the tensor product $\square \otimes \square$ to the trivial representation for $\text{SO}(N)$ gauge group as

$$A_{ij} := \eta^{\alpha\beta} Y_{\alpha,i} Y_{\beta,j}. \quad (4.52)$$

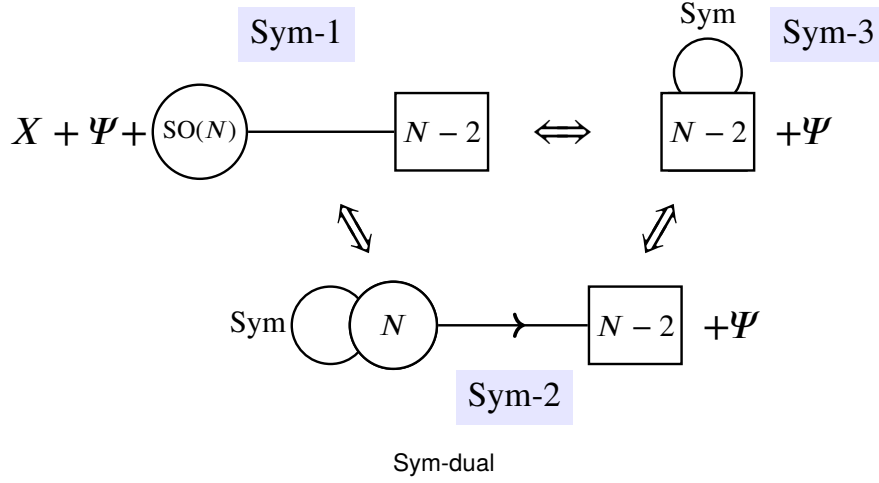


Figure 4-8 A triality among three distinct theories: (Sym-1) an $SO(N)$ gauge theory of SQCD type with $N - 2$ fundamental chirals coupled to a neutral chiral X and Fermi Ψ , (Sym-2) an $SU(N)$ gauge theory with one symmetric and $N - 2$ fundamental chirals, coupled to a neutral Fermi multiplet Ψ , and (Sym-3) a Landau-Ginzburg model consisting of $(N - 2)(N - 1)/2$ chiral multiplets coupled to a Fermi multiplet Ψ .

Sym-2. $SU(N)$ gauge theory with one symmetric chiral Z and $N - 2$ fundamental chirals W . Furthermore, there is a neutral Fermi multiplet Ψ , which forms a superpotential

$$\mathcal{W} = \Psi \det A, \quad (4.53)$$

where

$$A_{ij} = Z_{\alpha\beta} W_i^\alpha W_j^\beta. \quad (4.54)$$

Sym-3. LG model of one Fermi Ψ and $(N - 2)(N - 1)/2$ chirals, forming a symmetric $(N - 2) \times (N - 2)$ matrix A with a superpotential

$$\mathcal{W} = \Psi \det A. \quad (4.55)$$

The $U(1)_x$ charges of the fields are given as follows:

	Ψ	A	X	Y	Z	W	
$U(1)_x$	$-2(N - 2)$	2	$N - 2$	1	$\frac{2(N-2)}{N}$	$\frac{2}{N}$	(4.56)

These theories have the same central charges

$$c_L = N(N - 3), \quad c_R = \frac{3N(N - 3)}{2} \quad (4.57)$$

Furthermore, the elliptic genera agree

$$\begin{aligned}
\mathcal{G} &= \frac{2\eta(q)^{(N^2-3N)/2} \vartheta_1(x^{-2(N-2)})}{2^{[N/2]} [N/2]! \vartheta_1(x^{N-2})} \oint_{JK} \frac{d\mathbf{a}}{2\pi i \mathbf{a}} \frac{\prod_{i < j}^{[N/2]} \vartheta_1(a_i^\pm a_j^\pm)}{\prod_{i=1}^{[N/2]} \prod_{j=1}^{N-2} \vartheta_1(x a_i^\pm b_j)} \left(\frac{\prod_{i=1}^{[N/2]} \vartheta_1(a_i^\pm)}{\prod_{j=1}^{N-2} \vartheta_1(x b_j)} \right)^x \\
&= - \frac{\eta(q)^{\frac{1}{2}(N^2+3N-6)} \vartheta_1(x^{-2(N-2)})}{N!} \oint_{JK} \frac{d\mathbf{a}}{2\pi i \mathbf{a}} \prod_{i=1}^N \frac{\prod_{i \neq j} \vartheta_1(a_i/a_j)}{\prod_{i \leq j} \vartheta_1(x^{\frac{2(N-2)}{N}} a_i a_j) \prod_{j=1}^{N-2} \vartheta_1(x^{\frac{2}{N}} b_j/a_i)} \\
&= \frac{\eta(q)^{N(N-3)/2} \vartheta_1(x^{-2(N-2)})}{\prod_{i \leq j} \vartheta_1(x^2 b_i b_j)} \quad (4.58)
\end{aligned}$$

where $\chi \equiv N \pmod{2}$.

In fact, we arrive at this triality by considering the $(0, 2)$ reduction of the 4d $\mathcal{N} = 2$ SCFTs: $\text{SO}(2N)$ gauge theory with $2N - 2$ fundamental half-hypermultiplets, and $\text{SU}(2N)$ gauge theory with one symmetric and $2N - 2$ fundamental hypermultiplets. These theories arise from D4-NS5-O6⁺ brane system (See^[69]§3.1). Their $(0, 2)$ reductions yield the Sym-1 and Sym-2 theories shown in Fig. 4-8, and they turn out to be dual to a $(0, 2)$ LG model (Sym-3).

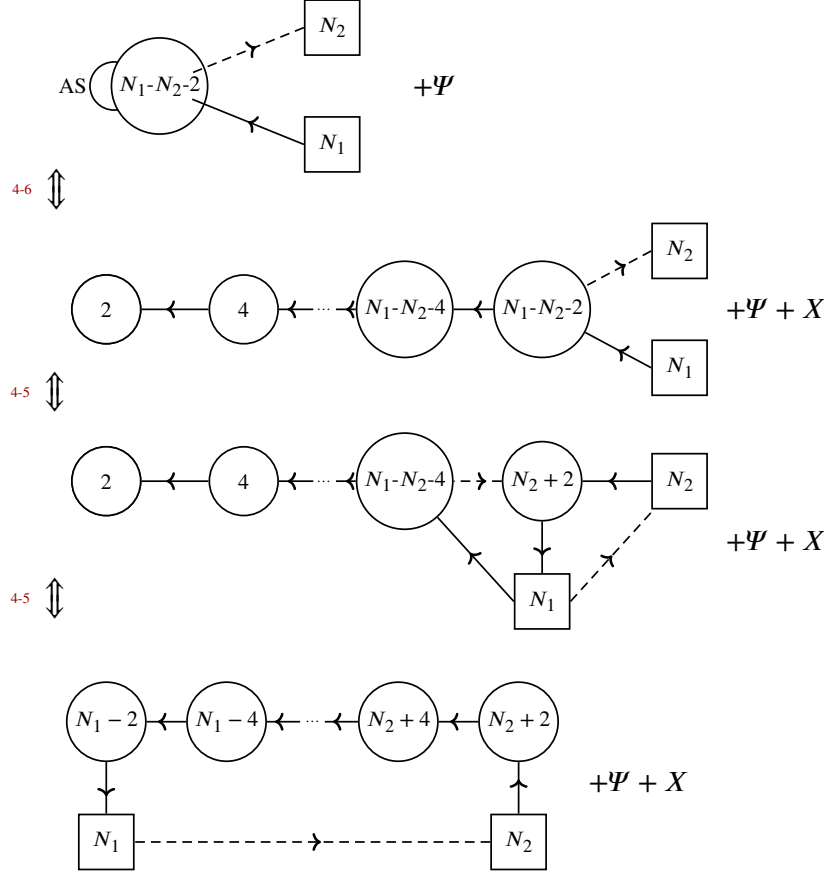


Figure 4-9 Duality chain used to relate the antisymmetric SU theory to a linear quiver.

4.3.4 Duality between SU and Sp gauge theory

The SU/Sp duality proposed in this subsection has Level B/D evidence, upgraded to Level C in the cases where the q -expansion check is displayed. The derivation uses a finite chain of known duality moves followed by Higgsing.

It is natural to ask whether the duality between the Sp and SU gauge theories (i.e. AS-1 and AS-2) in Fig. 4-6 can be generalized to include the fundamental Fermi multiplets. To explore this, adding anti-fundamental Fermi multiplets to the AS-2 theory, we consider the following theory:

- $\text{SU}(N_1 - N_2 - 2)$ gauge theory with one antisymmetric chiral X ($\text{U}(1)_x$ -charge 2), N_1 fundamental chirals Y and N_2 anti-fundamental Fermi's F where $N_1 - N_2$ is even. In addition, there is one free Fermi multiplet Ψ ($\text{U}(1)_x$ -charge $N = N_1 - N_2 - 2$), forming a superpotential

$$\mathcal{W} = \Psi \text{ Pf } X . \quad (4.59)$$

The elliptic genus of the theory is

$$\mathcal{J}_{\text{SU}} = \frac{\vartheta_1(x^N)\eta(q)^{(N^2+9N-6)/2}}{N!} \oint_{\text{JK}} \frac{d\mathbf{a}}{2\pi i \mathbf{a}} \prod_{i=1}^N \frac{\prod_{j \neq i} \vartheta_1(a_i/a_j) \cdot \prod_{j=1}^{N_2} \vartheta_1(a_i^{-1} c_j y_2)}{\prod_{j < i} \vartheta_1(x^2 a_i a_j) \prod_{j=1}^{N_1} \vartheta_1(a_i b_j y_1^{-1})}. \quad (4.60)$$

We aim to establish a duality with an Sp gauge theory by adding anti-fundamental Fermi multiplets to the AS-1 theory. (See the right panel of Fig. 4-10.) However, as we shall see, the resulting Sp gauge theory lacks the $U(1)_x$ symmetry. Therefore, we perform a sequence of duality transformations starting from the theory described above (illustrated in Fig. 4-9) and then apply a Higgsing procedure to eliminate the $U(1)_x$ symmetry. Let us explain this process step by step.

We begin by recursively applying the duality between the AS-2 and AS-3 theories in Fig. 4-6. Specifically, we apply this duality to the SU gauge node that contains a single antisymmetric chiral multiplet, as demonstrated in [63,70]. This transformation yields a linear quiver theory with SU gauge groups, as shown in the second line of Fig. 4-9. It is important to note that for SU(2), the antisymmetric representation is trivial. Consequently, the dual description includes an additional chiral multiplet X and a superpotential term $\mathcal{W} = \Psi X$, where Ψ is a Fermi multiplet. In this intermediate theory, the bifundamental chiral multiplets between gauge groups, as well as Ψ and X , are charged under the $U(1)_x$ symmetry.

Next, we recursively apply the SU Seiberg-like duality from Fig. 4-5 to each gauge node, proceeding from right to left. This sequence of dualities ultimately produces the theory shown at the bottom of Fig. 4-9, where all the theories in Fig. 4-9 are dual to each other.

The final theory is characterized by the superpotential

$$\mathcal{W} = \Psi X + \text{Tr} \left[\Gamma Y^{(1)} \dots Y^{(\frac{N_1-N_2}{2})} \right], \quad (4.61)$$

where Γ denotes the Fermi meson, and each $Y^{(i)}$ is a bifundamental chiral multiplet transforming under the $SU(N_2 + 2i) \times SU(N_2 + 2i - 2)$ gauge or flavor groups for $i = 1, \dots, \frac{N_1-N_2}{2}$. These fields are charged under a linear combination of the $U(1)_{y_1}$ and $U(1)_{y_2}$ global symmetries. At this point, the $U(1)_x$ symmetry can be removed by giving the vacuum expectation value $\langle X \rangle$, effectively decoupling the chiral multiplet X and the Fermi multiplet Ψ . This procedure completes the elimination of the $U(1)_x$ symmetry and ensures that the global symmetry structure of the resulting theory matches that of the target Sp gauge theory.

As a result, we propose a duality between the Sp gauge theory and the quiver gauge theory with SU gauge nodes obtained through this sequence of dualities and Higgsing, as illustrated in Fig. 4-10. The specifics of these dual theories are detailed below:

- The quiver theory with SU gauge groups features a superpotential:

$$\mathcal{W} = \text{Tr} \Gamma Y^{(1)} \dots Y^{(\frac{N_1-N_2}{2})}, \quad (4.62)$$

where Γ is the Fermi meson, and $Y^{(i)}$ represents the bifundamental chiral fields between the $SU(N_2+2i)$ and $SU(N_2+2i-2)$ gauge/flavor groups ($i = 1, \dots, \frac{N_1-N_2}{2}$).

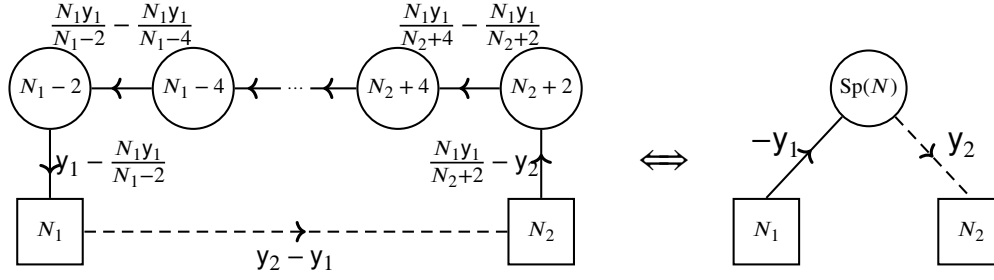


Figure 4-10 A duality between the quiver gauge theory with SU gauge groups and $\text{Sp}(N_1 - N_2 - 2)$ gauge theory. The labels with a linear combination of y_1 and y_2 represent U(1) flavor chemical potentials for the corresponding bifundamental fields.

The elliptic genus of the theory is

$$\mathcal{J}_{\text{quiver}} = \prod_{k=1}^{\frac{N}{2}} \frac{1}{(N_1 - 2k)!} \int_{\text{JK}} \frac{da}{2\pi i a} \prod_{k=1}^{\frac{N}{2}} \frac{\prod_{i \neq j}^{N_1 - 2k} \vartheta_1(a_i^{(k)} / a_j^{(k)})}{\prod_{i=1}^{N_1 - 2k + 2} \prod_{j=1}^{N_1 - 2k} \vartheta_1\left(y_1^{\frac{-2N_1}{(N_1 - 2k + 2)(N_1 - 2k)}} a_i^{(k-1)} / a_j^{(k)}\right)} \frac{\eta(q)^{\frac{1}{4}(3N_1 - N_2)N} \prod_{i=1}^{N_1} \prod_{j=1}^{N_2} \vartheta_1(y_2 y_1^{-1} b_i^{-1} c_j)}{\prod_{i=1}^{N_2 + 2} \prod_{j=1}^{N_2} \vartheta_1\left(a_i^{\frac{N}{2}} c_j^{-1} y_2^{-1} y_1^{\frac{N_1}{N_2 + 2}}\right)} \quad (4.63)$$

where the fugacity $a^{(0)} = b$.

- $\text{Sp}(N_1 - N_2 - 2)$ gauge theory with N_1 fundamental chirals and N_2 fundamental Fermi's. The elliptic genus of the theory is

$$\mathcal{J}_{\text{Sp}} = \frac{\eta(q)^{\frac{N_2}{2} + 3N}}{2^{\frac{N}{2}} \frac{N}{2}!} \oint_{\text{JK}} \frac{da}{2\pi i a} \prod_{i=1}^{\frac{N}{2}} \frac{\vartheta_1(a_i^{\pm 2}) \prod_{j < i} \vartheta_1(a_i^{\pm} a_j^{\pm}) \cdot \prod_{j=1}^{N_2} \vartheta_1(a_i^{\pm} c_j y_2)}{\prod_{j=1}^{N_1} \vartheta_1(a_i^{\pm} b_j^{-1} y_1^{-1})}. \quad (4.64)$$

Note that, throughout this subsection, we assume that $N \equiv N_1 - N_2 - 2$ is even.

In fact, the central charges of both theories match:

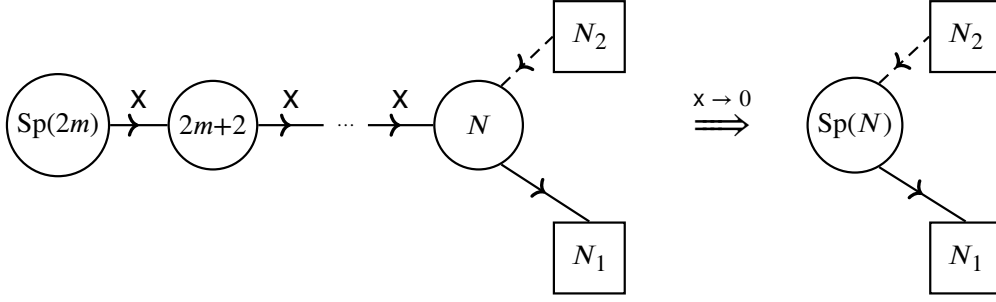
$$c_L = (N_1 - N_2 - 2)(N_1 - N_2 + 1), \quad c_R = \frac{3}{2}(N_1 - N_2 - 2)(N_1 - N_2 + 1). \quad (4.65)$$

Furthermore, we have verified that the elliptic genera agree, $\mathcal{J}_{\text{quiver}} = \mathcal{J}_{\text{Sp}}$, up to the rank-six JK integrals ($N_1 = 7$ and $N_2 = 1$ case).

At the level of elliptic genera, the Higgsing procedure can be understood in the following way. The elliptic genus (4.60) reduces precisely to (4.64) when the $\text{U}(1)_x$ flavor fugacity is turned off:

$$\lim_{x \rightarrow 1} \mathcal{J}_{\text{SU}} = \mathcal{J}_{\text{Sp}}. \quad (4.66)$$

Since all theories shown in Fig. 4-9 are dual to each other, their elliptic genera must coincide with (4.60). Consequently, the identity (4.66) holds equally between the second theory in Fig. 4-9 and the Sp gauge theory. Furthermore, reversing the direction of arrows in the quiver diagrams on both sides establishes a similar Higgsing relation

Figure 4-11 Duality between the linear SU quiver and the $\text{Sp}(N)$ gauge theory after Higgsing.

between the two theories depicted in Fig. 4-11. Specifically, once the $\text{U}(1)_x$ fugacity associated to the bifundamental chials is turned off, the elliptic genus of the quiver theory on the left-hand side in Fig. 4-11 coincides with that of the Sp gauge theory on the right-hand side. This relation will play a role in deriving additional dualities in subsequent subsections. (See Fig. 4-14 and Fig. 4-16.)

4.3.5 SU gauge theories with antisymmetric chiral

The antisymmetric-tensor dualities in this subsection have Level B/C/D evidence. They are obtained by replacing the antisymmetric tensor with an auxiliary Sp node, applying the elementary dualities, and then comparing finite-rank elliptic genera or q -expansions in the ranges stated below.

Using the dualities above, we can derive another duality. This subsection presents a duality between SU gauge theories with one antisymmetric chiral, (anti-)fundamental chirals, and potentially fundamental Fermi multiplets.

We consider an $\text{SU}(N_A)$ gauge theory with the following matter content:

- One antisymmetric chirals Y
- n_1 anti-fundamental chirals X_1
- N fundamental chirals X_2
- m_1 anti-fundamental Fermi Ψ where m_1 is either zero or one
- A Fermi meson Γ , transforming as $(\overline{\square}, \square)$ under the flavor symmetries $\text{SU}(n_1) \times \text{SU}(N)$
- A chiral meson M_1 , transforming as $(-1, \square)$ under the flavor symmetries $\text{U}(m_1) \times \text{SU}(N)$
- A chiral meson M_2 , transforming as $(1, \square)$ under the flavor symmetries $\text{U}(m_1) \times \text{SU}(n_1)$
- A Fermi meson Λ , transforming as $\overline{\square} = \wedge^{n_1-2} \square$ under the flavor symmetry $\text{SU}(n_1)$
- $1 - m_1$ neutral chiral P under the flavor symmetries $\text{U}(m_1)$

The gauge group is determined by anomaly-free condition, which is given by

$$N_A = N + n_1 - m_1 - 2. \quad (4.67)$$

The superpotential takes the form

$$\mathcal{W} = \Gamma X_2 X_1 + \Psi M_1 X_2 + \Lambda^{ij} X_{1,i}^\alpha X_{1,j}^\beta Y_{\alpha\beta} . \quad (4.68)$$

The dual theory has a similar structure: it consists of an $SU(N_B)$ gauge theory, where the matter content is modified by replacing n_1 and m_1 in the original theory with n_2 and m_2 , respectively, where $m_2 \in \{0, 1\}$. The gauge anomaly-free condition in the dual theory is given by

$$N_B = N + n_2 - m_2 - 2 . \quad (4.69)$$

Note that the duality holds only under the condition

$$N \geq m_1 + m_2 + 2 . \quad (4.70)$$

	X_1	X_2	Ψ	M_1	M_2	Γ	Y	Λ	P
$SU(N_A)$	$\overline{\square}$	$\square$	$\overline{\square}$	1	1	1	$\square$	1	1
$SU(N)$	1	$\overline{\square}$	1	$\square$	1	$\square$	1	1	1
$SU(n_1)$	$\square$	1	1	1	$\square$	$\overline{\square}$	1	$\overline{\square}$	1
$U(1)_0$	-1	1	$-N_A - 1$	N_A	$-N_A$	0	2	0	$-N_A$
$U(1)_1$	0	-1	N	$1 - N$	N	1	0	0	N
$U(1)_2$	1	0	$-n_1$	n_1	$1 - n_1$	-1	0	-2	$-n_1$
	$\tilde{X}_1$	$\tilde{X}_2$	$\tilde{\Psi}$	$\tilde{M}_1$	$\tilde{M}_2$	$\tilde{\Gamma}$	$\tilde{Y}$	$\tilde{\Lambda}$	$\tilde{P}$
$SU(N_B)$	$\overline{\square}$	$\square$	$\overline{\square}$	1	1	1	$\square$	1	1
$SU(N)$	1	$\overline{\square}$	1	$\square$	1	$\square$	1	1	1
$SU(n_2)$	$\square$	1	1	1	$\square$	$\overline{\square}$	1	$\overline{\square}$	1
$U(1)_0$	$\frac{N_A}{N_B}$	$-\frac{N_A}{N_B}$	$N_A + \frac{N_A}{N_B}$	$-N_A$	N_A	0	$-2\frac{N_A}{N_B}$	0	N_A
$U(1)_1$	$-\frac{N}{N_B}$	$\frac{N}{N_B} - 1$	$-\frac{N}{N_B}$	1	0	1	$2\frac{N}{N_B}$	0	0
$U(1)_3$	1	0	$-n_2$	n_2	$1 - n_2$	-1	0	-2	$-n_2$

Table 4-2 Summary of symmetries for Theory A (upper one) and Theory B (lower one) in Fig. 4-12.

As illustrated in Fig. 4-12, this duality can be derived through a sequence of duality transformations. In the derivation, an SU antisymmetric chiral multiplet is replaced by an Sp gauge group with a bifundamental chiral multiplet via the duality in either Fig. 4-6 or Fig. 4-7. The process begins by applying one of these dualities, with the choice determined by whether N_A is even or odd. The SU Seiberg-like duality in Fig. 4-5 is then applied, followed by a second Sp replacement chosen according to the parity of the intermediate gauge-group rank. This sequence gives the dual theory in Fig. 4-12.

The symmetries of the theory are summarized in Tab. 4-2. The 't Hooft anomalies of each flavor symmetry match under the duality; this is the first consistency check before comparing elliptic genera.

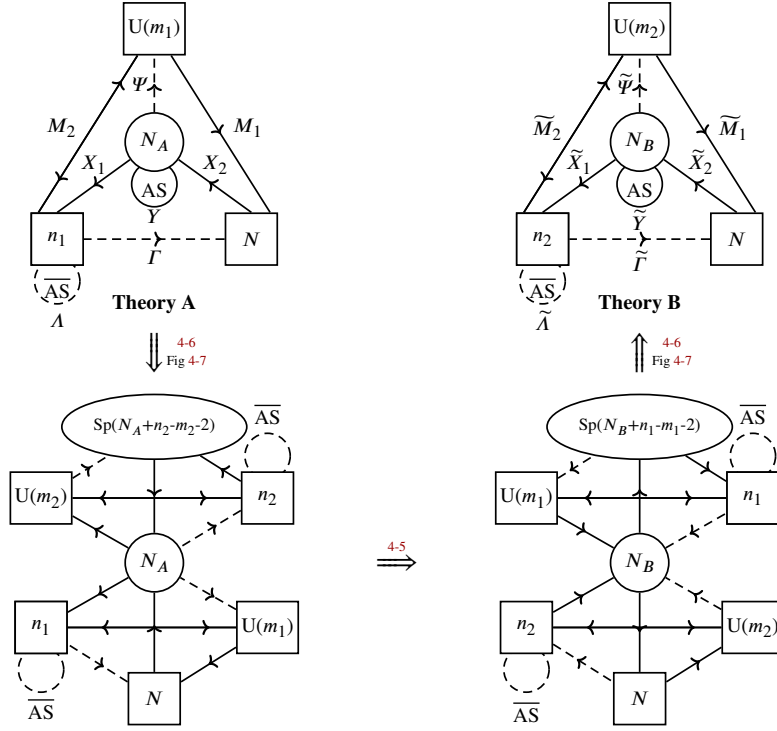


Figure 4-12 Duality between $SU(N_A)$ and $SU(N_B)$ gauge theories with 1AS where $N_A = N + n_1 - m_1 - 2$ and $N_B = N + n_2 - m_2 - 2$. Note that m_1 and m_2 are either zero or one.

Moreover, the elliptic genera for these theories are expressed as

$$\begin{aligned}
 \mathcal{F}_A &= \frac{\eta(q)^{\frac{N_A^2+9N_A-4}{2}}}{N_A!} \oint_{\text{JK}} \frac{d\mathbf{a}}{2\pi i \mathbf{a}} \prod_{i=1}^{N_A} \frac{\prod_{j \neq i} \vartheta_1(a_i a_j^{-1}) \cdot \prod_{j=1}^{m_1} \vartheta_1(a_i^{-1} e_j x_0^{-1})}{\prod_{j < i} \vartheta_1(x_0^2 a_i a_j) \cdot \prod_{j=1}^N \vartheta_1(a_i b_j^{-1} x_0 x_1^{-1}) \cdot \prod_{j=1}^{n_1} \vartheta_1(a_i^{-1} c_j x_0^{-1} x_2)} \\
 &\quad \times \frac{\prod_{i=1}^{n_1} \prod_{j < i} \vartheta_1(x_2^{-2} c_i^{-1} c_j^{-1})}{\eta(q)^{\frac{n_1(n_1-1)}{2} - N(m_1-n_1) - m_1 n_1 - 1 + m_1}} \prod_{i=1}^N \frac{\prod_{j=1}^{n_1} \vartheta_1(b_i c_j^{-1} x_1 x_2^{-1})}{\prod_{j=1}^{m_1} \vartheta_1(b_i e_j^{-1} x_1)} \cdot \frac{\vartheta_1(e_1)^{m_1-1}}{\prod_{i=1}^{m_1} \prod_{j=1}^{n_1} \vartheta_1(e_i c_j x_2)} \\
 \mathcal{F}_B &= \frac{\eta(q)^{\frac{N_B^2+9N_B-4}{2}}}{N_B!} \oint_{\text{JK}} \frac{d\mathbf{a}}{2\pi i \mathbf{a}} \prod_{i=1}^{N_B} \frac{\prod_{j \neq i} \vartheta_1(a_i a_j^{-1}) \cdot \prod_{j=1}^{m_2} \vartheta_1(a_i^{-1} f_j x_0^{-1} y^{-1})}{\prod_{j < i} \vartheta_1(x_0^2 y^2 a_i a_j) \cdot \prod_{j=1}^N \vartheta_1(a_i b_j^{-1} x_0 y x_1^{-1}) \cdot \prod_{j=1}^{n_2} \vartheta_1(a_i^{-1} d_j x_0^{-1} y^{-1} x_3)} \\
 &\quad \times \frac{\prod_{i=1}^{n_2} \prod_{j < i} \vartheta_1(x_3^{-2} d_i^{-1} d_j^{-1})}{\eta(q)^{\frac{n_2(n_2-1)}{2} - N(m_2-n_2) - m_2 n_2 - 1 + m_2}} \prod_{i=1}^N \frac{\prod_{j=1}^{n_2} \vartheta_1(b_i d_j^{-1} x_1 x_3^{-1})}{\prod_{j=1}^{m_2} \vartheta_1(b_i f_j^{-1} x_1)} \cdot \frac{\vartheta_1(f_1)^{m_2-1}}{\prod_{i=1}^{m_2} \prod_{j=1}^{n_2} \vartheta_1(f_i c_j x_3)}
 \end{aligned} \tag{4.71}$$

where $n_1 + n_2 + m_1 + m_2$ is even. The fugacities f_1 , e_1 and y are defined as

$$f_1 = x_0^{N_A} x_3^{-n_2}, \quad e_1 = x_0^{-N_A} x_1^N x_2^{-n_1}, \quad y^{N_B} = x_0^{-N_A - N_B} x_1^N. \tag{4.72}$$

For the special case where $N = m_1 + m_2 + 2$, an additional condition must be satisfied:

$$x_1^{(1-m_1)N} x_2^{-(1-m_1)n_1} x_3^{-(1-m_2)n_2} = x_0^{(m_2-m_1)(n_1+m_2)}. \tag{4.73}$$

This condition originates from (4.41), which is based on the relation $N_1 = N_2 + N_3$ in Fig. 4-5.

The elliptic genus $\mathcal{F}_A$ is independent of the $SU(n_1)$ fugacity and the $U(1)_2$ fugacity x_2 . Likewise, $\mathcal{F}_B$ is independent of the $SU(n_2)$ fugacity and the $U(1)_3$ fugacity x_3 . Furthermore, the agreement between $\mathcal{F}_A$ and $\mathcal{F}_B$ has been verified up to the rank-five JK integrals through an expansion in q .

When $m_1 = 0$ and $N = 2, 3, 4$ in Theory A, Theory B becomes an LG model, as shown in [70]. To obtain the LG dual found in [70], we first transfer the Fermi mesons I and Λ from Theory A to Theory B, where they are reinterpreted as chiral mesons. By appropriately choosing n_2 and m_2 , as described below, the LG duality can be established.

- **$N = 2$ and even n_1 :** Set $n_2 = 2$ and $m_2 = 0$ in Theory B. This results in a gauge group rank $N_B = N + n_2 - m_2 - 2 = 2$. Using the duality in Fig. 4-6, where $SU(2)$ is identified with $Sp(2)$, the corresponding LG dual theory [70]§3.1 is obtained.
- **$N = 2$ and odd n_1 :** Assign $n_2 = 1$ and $m_2 = 0$, reducing the gauge group rank to $N_B = 1$. This corresponds to an LG theory [70]§3.2.
- **$N = 3$ and even n_1 :** Choose $n_2 = m_2 = 0$, yielding $N_B = 1$, which directly describes an LG theory [70]§3.3.
- **$N = 3$ and odd n_1 :** Set $n_2 = 1$ and $m_2 = 0$, giving $N_B = 2$. Applying the duality in Fig. 4-6, we obtain the LG dual [70]§3.4.
- **$N = 4$ and even n_1 :** Select $n_2 = m_2 = 0$, resulting in $N_B = 2$. The LG dual [70]§3.5 follows from the duality in Fig. 4-6.
- **$N = 4$ and odd n_1 :** Assign $n_2 = 0$ and $m_2 = 1$, leading to $N_B = 1$. This implies Theory B is an LG theory [70]§3.6.

The duality holds under the condition in (4.70), which explicitly excludes the cases where the number of fundamental chirals is $N = 1$ or $N = 0$. Alternative dualities for these excluded cases are discussed in Appendices B.1.3 and B.1.4.

4.3.6 SU gauge theories with symmetric chiral

The symmetric-tensor duality has the same Level B/C/D status as the antisymmetric case, with the auxiliary Sp node replaced by an SO node.

In a similar manner, we establish a duality between SU gauge theories containing one symmetric chiral multiplet, as depicted in Fig. 4-13. The gauge groups of the dual theories are given by

$$N_A = N + n_1 - m_1 + 2, \quad N_B = N + n_2 - m_2 + 2. \quad (4.74)$$

The symmetric case is parallel to the antisymmetric construction: the Sp auxiliary node used for an antisymmetric tensor is replaced by an SO auxiliary node for a symmetric tensor. We therefore keep the same fugacity map and display only the changes in the one-loop factors, rather than repeating the full matter content and symmetry discussion.

Essentially, through the duality depicted in Fig. 4-8, an SU symmetric chiral can be replaced by an SO gauge group with a bifundamental chiral. Consequently, as in Fig. 4-12, this duality is derived by sequentially applying the known dualities illustrated in Fig. 4-8 and 4-5.

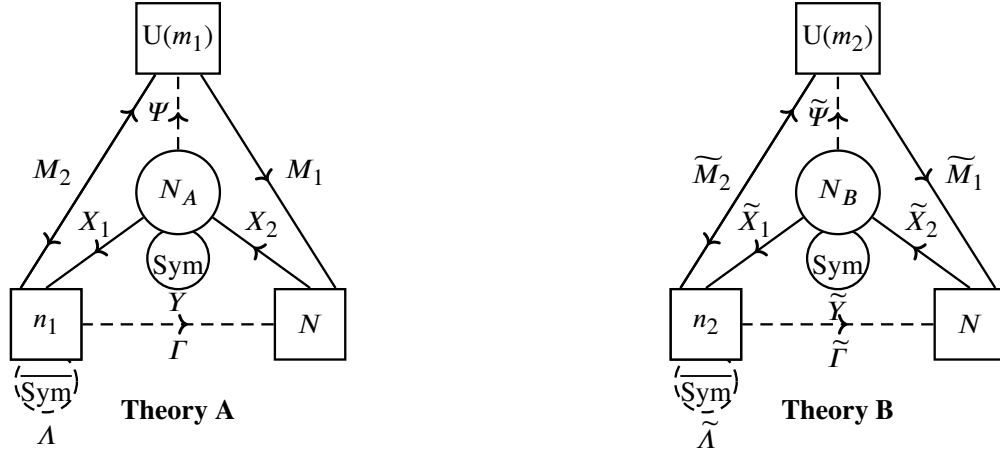


Figure 4-13 Duality for $SU(N_A)$ and $SU(N_B)$ gauge theories with 1Sym where $N_A = N + n_1 - m_1 + 2$ and $N_B = N + n_2 - m_2 + 2$. Note that m_1 and m_2 are either zero or one.

The expressions for the elliptic genera of these theories are provided below:

$$\begin{aligned}
 \mathcal{J}_A &= \frac{\eta(q)^{\frac{N_A^2+3N_A-4}{2}}}{N_A!} \oint_{\text{JK}} \frac{da}{2\pi i a} \prod_{i=1}^{N_A} \frac{\prod_{j \neq i} \vartheta_1(a_i a_j^{-1}) \cdot \prod_{j=1}^{m_1} \vartheta_1(a_i^{-1} e_j x_0^{-1})}{\prod_{j \leq i} \vartheta_1(x_0^2 a_i a_j) \cdot \prod_{j=1}^N \vartheta_1(a_i b_j^{-1} x_0 x_1^{-1}) \cdot \prod_{j=1}^{n_1} \vartheta_1(a_i^{-1} c_j x_0^{-1} x_2)} \\
 &\quad \times \frac{\prod_{i=1}^{n_1} \prod_{j \leq i} \vartheta_1(x_2^2 c_i c_j)}{\eta(q)^{\frac{n_1(n_1+1)}{2} - N(m_1 - n_1) - m_1 n_1 - 1}} \cdot \prod_{i=1}^N \frac{\prod_{j=1}^{n_1} \vartheta_1(b_i c_j^{-1} x_1 x_2^{-1})}{\prod_{j=1}^{m_1} \vartheta_1(b_i e_j^{-1} x_1)} \cdot \frac{\vartheta_1(e_1^2)^{-m_1} \vartheta_1(e_1)^{m_1-1}}{\prod_{i=1}^{m_1} \prod_{j=1}^{n_1} \vartheta_1(e_i c_j x_2)} \\
 \mathcal{J}_B &= \frac{\eta(q)^{\frac{N_B^2+3N_B-4}{2}}}{N_B!} \oint_{\text{JK}} \frac{da}{2\pi i a} \prod_{i=1}^{N_B} \frac{\prod_{j \neq i} \vartheta_1(a_i a_j^{-1}) \cdot \prod_{j=1}^{m_2} \vartheta_1(a_i^{-1} f_j x_0^{-1} y^{-1})}{\prod_{j \leq i} \vartheta_1(x_0^2 y^2 a_i a_j) \cdot \prod_{j=1}^N \vartheta_1(a_i b_j^{-1} x_0 y x_1^{-1}) \cdot \prod_{j=1}^{n_2} \vartheta_1(a_i^{-1} d_j x_0^{-1} y^{-1} x_3)} \\
 &\quad \times \frac{\prod_{i=1}^{n_2} \prod_{j \leq i} \vartheta_1(x_3^2 d_i d_j)}{\eta(q)^{\frac{n_2(n_2+1)}{2} - N(m_2 - n_2) - m_2 n_2 - 1}} \cdot \prod_{i=1}^N \frac{\prod_{j=1}^{n_2} \vartheta_1(b_i d_j^{-1} x_1 x_3^{-1})}{\prod_{j=1}^{m_2} \vartheta_1(b_i f_j^{-1} x_1)} \cdot \frac{\vartheta_1(f_1^2)^{-m_2} \vartheta_1(f_1)^{m_2-1}}{\prod_{i=1}^{m_2} \prod_{j=1}^{n_2} \vartheta_1(f_i d_j x_3)}
 \end{aligned} \tag{4.75}$$

where the fugacities f_1 , e_1 and y are defined as

$$f_1 = x_0^{N_A} x_3^{-n_2}, \quad e_1 = x_0^{-N_A} x_1^N x_2^{-n_1}, \quad y^{N_B} = x_0^{-N_A - N_B} x_1^N. \tag{4.76}$$

The elliptic genus $\mathcal{J}_A$ does not depend on the $SU(n_1)$ fugacity or the $U(1)_2$ fugacity x_2 , while $\mathcal{J}_B$ is independent of the $SU(n_2)$ fugacity and the $U(1)_3$ fugacity x_3 . Moreover, the agreement between $\mathcal{J}_A$ and $\mathcal{J}_B$ has been confirmed, via an expansion in q , up to the rank-five JK integrals.

4.3.7 Dualities with adjoint chiral

The adjoint examples in this subsection are Level A at the level of elliptic genera: the JK residues reduce to closed theta-function products. The Lie-algebra isomorphisms below provide additional consistency checks.

Consider the 4d $\mathcal{N} = 4$ SCFT. Its $(0,2)$ reduction results in the $(0,2)$ gauge theory only with a $(0,2)$ adjoint chiral multiplet ϕ , which is indeed a $(2,2)$ vector multiplet. The IR central charge of the $\mathcal{N} = (2, 2)$ pure Yang-Mills theory with gauge group G is given by

$$c_L = c_R = 3 \text{ rank } G. \tag{4.77}$$

The elliptic-genus computation below shows that the theory is dual, at the level of this protected index, to rank G free $\mathcal{N} = (2, 2)$ chiral multiplets.

Sp(2N)+1Adj: Since the duality for the SU(N) gauge theory with a single adjoint matter field is well-studied in ^[19,71], we now extend our analysis to other classical gauge groups. We begin by considering the Sp(2N) gauge theory with an adjoint chiral field.

A useful observation is that **Adj** $\cong$ Sym for the Sp(2N) gauge group. This allows us to apply the duality depicted in Fig. 4-8. As illustrated in Fig. 4-14, a sequence of duality transformations relates Sp(2N)+1**Adj** to Sp(2N−2)+1**Adj** and SU(2)+1**Adj**, with an extra chiral multiplet X and a Fermi multiplet Ψ .

Furthermore, it is known ^[19,71] that the SU(2)+1**Adj** theory is dual to an $\mathcal{N} = (2, 2)$ twisted chiral multiplet consisting of a free $\mathcal{N} = (0, 2)$ chiral multiplet $\text{Tr } Y^2$ and a Fermi multiplet Γ . We claim that they form a superpotential

$$\mathcal{W} = \Gamma X, \quad (4.78)$$

so that eliminates both the chiral multiplet X and the Fermi multiplet Γ . As a result, the duality transformation establishes an equivalence between Sp(2N)+1**Adj** and Sp(2N−2)+1**Adj** with the additional free chiral multiplet $\text{Tr } Y^2$ and the Fermi multiplet Ψ .

This duality recursively reduces the rank of the Sp(2N) gauge group, and at the end, we obtain the following duality:

$$\text{Sp}(2N) + 1\mathbf{Adj} \iff N \text{ free chirals} + N \text{ free Fermi's} . \quad (4.79)$$

Note that N free chirals correspond to the gauge-invariant operators made out of the adjoint chiral Z :

$$\text{Tr}(Z^{2k}), \quad k = 1, \dots, N, \quad (4.80)$$

and N free Fermi's are necessary to form $\mathcal{N} = (2, 2)$ twisted chiral multiplets. At the level of the protected index, the duality follows from the JK residue computation

$$\begin{aligned} \mathcal{J}_{\text{Sp}(2N)}^{\mathbf{Adj}} &= \frac{\eta(q)^{3N} \vartheta_1(y)^N}{2^N N!} \oint_{\text{JK}} \frac{da}{2\pi i a} \frac{\prod_{i < j}^N \vartheta_1(a_i^\pm a_j^\pm) \cdot \prod_{i=1}^N \vartheta_1(a_i^{2\pm})}{\prod_{i \leq j}^N \vartheta_1(y a_i^\pm a_j^\pm)} \\ &= (-1)^N \prod_{k=1}^N \frac{\vartheta_1(y^{2k-1})}{\vartheta_1(y^{2k})} . \end{aligned} \quad (4.81)$$

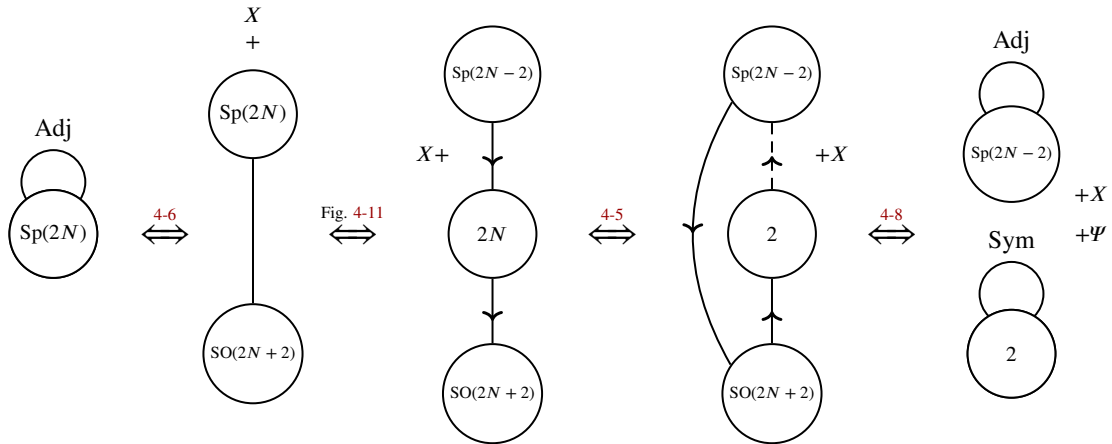


Figure 4-14 Duality chain giving the recursion relation for Sp(2N)+1Adj. The sequence reduces the rank to Sp(2N−2)+1Adj, with the extra chiral X and Fermi Ψ coupled as in (4.78).

SO(n)+1Adj: Now, let us consider the SO(n) gauge group. Note that **Adj** $\cong$ **AS** for the SO(n) gauge group. Depending on whether n is even or odd, we can apply the dualities presented in Fig. 4-6 and Fig. 4-7.

As illustrated in Fig. 4-15, these dualities, when followed by the duality in Fig. 4-8, establish an equivalence between the SO(n) + 1**Adj** theory and the Sp gauge theory with an adjoint matter field, supplemented by additional chiral and Fermi multiplets. This connects the SO(n) case to the previously discussed duality for Sp($2N$) + 1Adj in Eq. (4.79).

In the upper part of Fig. 4-15, the SO($2N$) + 1Adj theory is dual to the Sp($2N - 2$) + 1Adj theory, along with an additional free chiral multiplet X and a Fermi multiplet Ψ . Examining the U(1) charge, we identify that the chiral multiplet X corresponds to the gauge-invariant operator Pf Z , where Z represents the adjoint chiral superfield of the SO($2N$) gauge theory. Consequently, this duality leads to the relation:

$$\text{SO}(2N) + 1\mathbf{Adj} \iff N \text{ free chirals} + N \text{ free Fermi's} , \quad (4.82)$$

where the N free chiral multiplets correspond to gauge-invariant operators constructed from the adjoint chiral superfield Z :

$$\text{Pf}(Z), \quad \text{Tr}(Z^{2k}), \quad k = 1, \dots, N - 1 . \quad (4.83)$$

In the lower part of Fig. 4-15, a similar sequence of dualities establishes that the SO($2N + 1$) + 1Adj theory is dual to the Sp($2N - 2$) + 1Adj theory, again with an additional free chiral multiplet X and Fermi multiplet Ψ . By examining the U(1) charge, we recognize that the chiral multiplet X corresponds to the gauge-invariant operator Tr Z^{2N} , where Z is the adjoint chiral multiplet of the SO($2N + 1$) gauge theory. Incorporating Eq. (4.79), we arrive at the following triality:

$$\begin{array}{ccc} \text{SO}(2N + 1) + 1\mathbf{Adj} & \iff & N \text{ free chirals} + N \text{ free Fermi's} \\ & \swarrow \quad \searrow & \\ & \text{Sp}(2N) + 1\mathbf{Adj} & \end{array} \quad (4.84)$$

where the N free chiral multiplets correspond to gauge-invariant operators constructed from the adjoint chiral multiplet Z :

$$\text{Tr}(Z^{2k}), \quad k = 1, \dots, N . \quad (4.85)$$

In both even and odd cases, N free Fermi multiplets are necessary to form N (2,2) free twisted chiral multiplets.

The explicit computations of the elliptic genera give

$$\begin{aligned}\mathcal{F}_{\text{SO}(2N)}^{\text{Adj}} &= \frac{2\eta(q)^{3N}}{2^{N-1}N! \cdot \vartheta_1(y)^N} \oint_{\text{JK}} \frac{da}{2\pi i a} \prod_{i < j}^N \frac{\vartheta_1(a_i^\pm a_j^\pm)}{\vartheta_1(y a_i^\pm a_j^\pm)} \\ &= (-1)^N \frac{\vartheta_1(y^{N-1})}{\vartheta_1(y^N)} \cdot \prod_{k=1}^{N-1} \frac{\vartheta_1(y^{2k-1})}{\vartheta_1(y^{2k})},\end{aligned}\quad (4.86)$$

$$\begin{aligned}\mathcal{F}_{\text{SO}(2N+1)}^{\text{Adj}} &= \frac{2\eta(q)^{3N}}{2^N N! \cdot \vartheta_1(y)^N} \oint_{\text{JK}} \frac{da}{2\pi i a} \prod_{i < j}^N \frac{\vartheta_1(a_i^\pm a_j^\pm)}{\vartheta_1(y a_i^\pm a_j^\pm)} \cdot \prod_{i=1}^N \frac{\vartheta_1(a_i^\pm)}{\vartheta_1(y a_i^\pm)} \\ &= (-1)^N \prod_{k=1}^N \frac{\vartheta_1(y^{2k-1})}{\vartheta_1(y^{2k})}.\end{aligned}\quad (4.87)$$

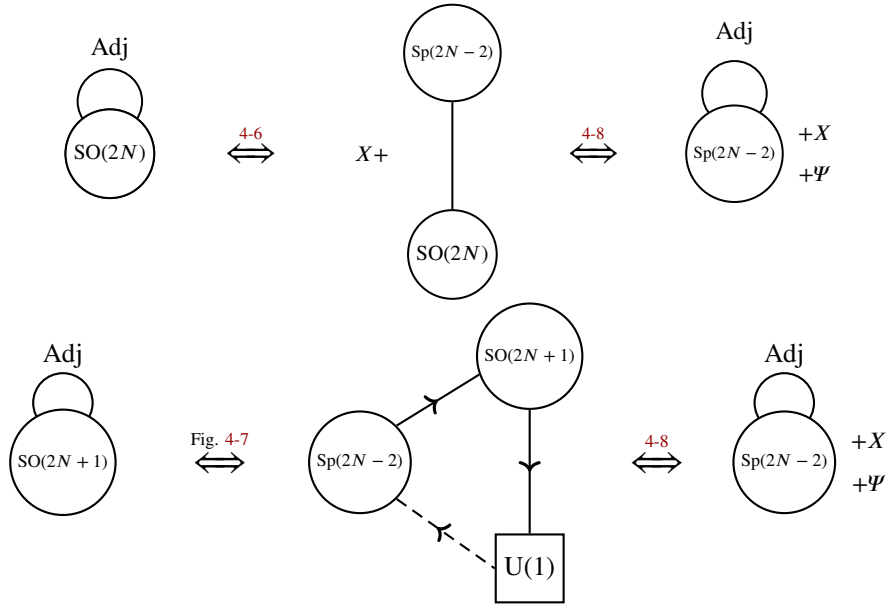


Figure 4-15 Duality chains relating $\text{SO} + 1\text{Adj}$ theories to $\text{Sp} + 1\text{Adj}$ theories. The upper diagram treats the even orthogonal group, and the lower diagram treats the odd orthogonal group.

The Lie-algebra isomorphisms give the following identities of elliptic genera:

$$\mathcal{F}_{\text{SO}(4)}^{\text{Adj}} = (\mathcal{F}_{\text{SU}(2)}^{\text{Adj}})^2, \quad \mathcal{F}_{\text{SO}(5)}^{\text{Adj}} = \mathcal{F}_{\text{Sp}(2)}^{\text{Adj}}, \quad \mathcal{F}_{\text{SO}(6)}^{\text{Adj}} = \mathcal{F}_{\text{SU}(4)}^{\text{Adj}}. \quad (4.88)$$

The chiral algebra of the 4d $\mathcal{N} = 4$ theory with gauge group G admits the free-field realization by the $bc\beta\gamma$ system^[72]. The elliptic genus (4.86, 4.87, 4.81) is the vacuum character of the corresponding $bc\beta\gamma$ system up to a factor^[73]. Consequently, the elliptic genera are reducible module characters of the chiral algebra of the 4d $\mathcal{N} = 4$ theory with the corresponding gauge groups.

4.3.8 SU gauge theories with symmetric and antisymmetric chirals

The combined symmetric-and-antisymmetric recursion has Level B/C/D evidence. The recursion is built from the elementary tensor-matter moves above, and the final lower-

rank or LG descriptions are checked by the finite elliptic-genus comparisons stated in the text.

We now consider the $SU(N)$ gauge theory that includes both symmetric and anti-symmetric chiral multiplets. As discussed earlier, an SU symmetric chiral multiplet can be replaced by an SO gauge group with bifundamental fields, while an SU antisymmetric chiral multiplet can be replaced by an Sp gauge group with bifundamental fields. We then apply a sequence of duality transformations to obtain an LG dual model. This procedure closely parallels the approach in the previous subsection, where we derived a recursion relation to systematically reduce the ranks of the gauge groups.

The replacement of the SU antisymmetric chiral multiplet with an Sp gauge field depends on whether N is even or odd—specifically, via Fig. 4-6 for even N and Fig. 4-7 for odd N . Therefore, we divide the discussion into two cases: even and odd N .

Consider the case that N is even. As shown in Fig. 4-16, a sequence of the known dualities provides a “recursion relation” of $SU+1\text{Sym}+1\text{AS}$ theories in terms of the ranks of gauge groups. Starting with $SU(N)+1\text{Sym}+1\text{AS}$, supplemented by additional 1 chiral multiplets $Z^{(0)}$ and 3 Fermi multiplets $\Psi_{1,2,3}^{(0)}$, with a superpotential

$$\mathcal{W} = \Psi_1^{(0)} \det X^{(0)} + \Psi_2^{(0)} \text{Pf } Y^{(0)} + \Psi_3^{(0)} ((Z^{(0)})^2 + B^{(0)}(2)) \quad (4.89)$$

As shown in Fig. 4-16, a series of duality transformations systematically map the theory to a similar theory: $SU(N-2)+1\text{Sym}+1\text{AS}$, along with additional chiral multiplets $Z^{(1)}$, $\tilde{Z}^{(1)}$ and Fermi multiplets $\Psi_{1,2,3,4}^{(1)}$. (Here, $\Psi_2^{(0)}$ forms a superpotential with a newly generated chiral field $P^{(1)}$, which then decouples from the remaining theory. Since $\Psi_2^{(0)}$ and $P^{(1)}$ do not contribute to either the central charge or the elliptic genus, they are omitted in the subsequent discussion.) The superpotential of the dual side takes a similar form:

$$\mathcal{W} = \Psi_1^{(1)} (\det X^{(1)} + \tilde{Z}^{(1)}) + \Psi_2^{(1)} \text{Pf } Y^{(1)} + \Psi_3^{(1)} ((Z^{(1)})^2 + B^{(1)}(2)) + \Psi_3^{(0)} ((Z^{(0)})^2 + \text{Tr}[(A^{(0)})^2]) \quad (4.90)$$

where $B^{(i)}(n)$ is a neutral baryon with $U(1)_{x^{(i)}} \times U(1)_{y^{(i)}}$ charge $(2n, 2(N-2i-n))$ that defined as:

$$B^{(i)}(n) \equiv \epsilon^{\alpha_1 \dots \alpha_{N-2i}} \epsilon^{\beta_1 \dots \beta_{N-2i}} Y_{\alpha_1 \beta_1}^{(i)} \dots Y_{\alpha_{N-2i} \beta_{N-2i}}^{(i)} X_{\alpha_{N-2i+1} \beta_{N-2i+1}}^{(i)} \dots X_{\alpha_{N-2i} \beta_{N-2i}}^{(i)} \quad (4.91)$$

Moreover, the dual side also has an $SU(2)+1\text{Sym}$ theory with the symmetric tensor $A^{(0)}$, and couples, via (4.90), with the chiral $Z^{(0)}$ and Fermi $\Psi_3^{(0)}$ from the original theory, as shown in the last quiver of Fig. 4-16. Again, the $SU(2)+1\text{Sym}$ theory is dual to one chiral $\text{Tr}(A^{(0)})^2$ and Fermi with opposite charges with the chiral $Z^{(0)}$ and Fermi $\Psi_3^{(0)}$. As a result, their combined contribution to the elliptic genus cancels out:

$$\underbrace{\frac{\vartheta_1(x^{-2}y^{-N+2})}{\vartheta_1(x^4y^{2N-4})}}_{SU(2)+1\text{Sym}} \cdot \underbrace{\frac{\overbrace{\vartheta_1(x^{-4}y^{-2N+4})}^{\Psi_3^{(0)}}}{\vartheta_1(x^2y^{N-2})}}_{Z^{(0)}} = 1. \quad (4.92)$$

Although their contributions to the elliptic genus cancel out, these fields are not entirely negligible, as their combined central charge contribution from $SU(2)+1\text{Sym}$, $\Psi_3^{(0)}$, and

$Z^{(0)}$ is 3. This establishes a recursive pattern of SU+1Sym+1AS demonstrating how an $SU(N)$ gauge group systematically reduces to $SU(N-2)$ through a series of duality transformations, as emphasized in Fig. 4-16. We summarize the charges of the fields in the $SU(N)+1AS+1Sym$ theory in Tab. 4-3, as well as the charges of the fields in the $SU(N-2)+1AS+1Sym$ theory obtained after performing this series of duality transformations.

By repeatedly applying this process, the gauge group rank is systematically reduced, allowing the theory to be expressed in terms of lower-rank cases. The duality recursion reaches its lowest order in the case of $SU(2)+1Sym+1AS$ along with additional Fermi and chiral fields.

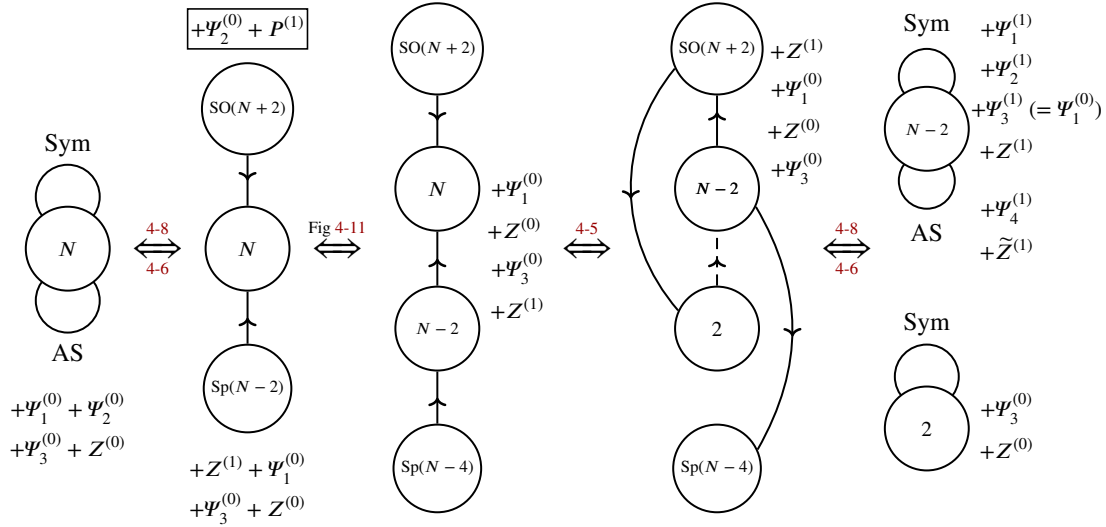


Figure 4-16 Recursive duality step reducing the $SU(N)+1AS+1Sym$ theory to lower-rank quiver data.

With this definition of $B^{(i)}(n)$, the endpoint values are

$$B^{(i)}(0) \propto \det Y^{(i)}, \quad B^{(i)}(N-2i) \propto \det X^{(i)}. \quad (4.93)$$

This allows us to identify part of the free field relations between the two dual theories:

$$\Psi_3^{(i)} = \Psi_1^{(i-1)}, \quad Z^{(i)} = \text{Tr}[(X^{(i-1)})^{N/2}], \quad \tilde{Z}^{(i)} = B^{(i-1)}(4). \quad (4.94)$$

At the final stage of the duality sequence, we obtain $\frac{N-2}{2}$ copies of $SU(2) + 1Sym$, along with $N+1$ Fermi multiplets, $N-1$ chiral multiplets, and an $SU(2)+1AS+1Sym$ theory, as illustrated in Fig. 4-17. The corresponding superpotential takes the form:

$$\begin{aligned} \mathcal{W} = & \Psi_1^{(\frac{N-2}{2})} \left(\det X^{(\frac{N-2}{2})} + \tilde{Z}^{(\frac{N-2}{2})} \right) + \Psi_2^{(\frac{N-2}{2})} \text{Pf } Y^{(\frac{N-2}{2})} + \Psi_3^{(\frac{N-2}{2})} \det X^{(\frac{N-2}{2})} \\ & + \sum_{i=0}^1 \Psi_3^{(i)} \left((Z^{(i)})^2 + \text{Tr}[(A^{(i)})^2] \right) + \sum_{i=2}^{\frac{N-2}{2}} \Psi_3^{(i)} \left(\tilde{Z}^{(i-1)} + (Z^{(i)})^2 + \text{Tr}[(A^{(i)})^2] \right). \end{aligned} \quad (4.95)$$

For $SU(2)$, the antisymmetric representation is trivial. As discussed earlier, $SU(2)+1Sym$ is dual to a pair of chiral and Fermi fields. Therefore, the recursive duality takes

	SU(N)	U(1) _{x(0)}	U(1) _{y(0)}
$X_{\alpha\beta}^{(0)}$	$\square \square$	2	0
$Y_{\alpha\beta}^{(0)}$	$\square$	0	2
$\Psi_1^{(0)}$	1	$-2N$	0
$\Psi_2^{(0)}$	1	0	$-N$
$\Psi_3^{(0)}$	1	-4	$-2N + 4$
$Z^{(0)}$	1	2	$N - 2$
	SU(N - 2)	U(1) _{x(1)}	U(1) _{y(1)}
$X_{\alpha\beta}^{(1)}$	$\square \square$	2	0
$Y_{\alpha\beta}^{(1)}$	$\square$	0	2
$\Psi_1^{(1)}$	1	$-2(N - 2)$	0
$\Psi_2^{(1)}$	1	0	$-(N - 2)$
$\Psi_3^{(1)}$	1	-4	$-2(N - 2) + 4$
$\Psi_4^{(1)}$	1	$-2N + 2$	2
$Z^{(1)}$	1	2	$(N - 2) - 2$
$\tilde{Z}^{(1)}$	1	$2N - 4$	0

Table 4-3 Charge assignments before and after one even- N recursion step.

the following form

$$\begin{array}{ccc}
 \text{SU}(N) + 1\text{Sym} + 1\mathbf{AS} + 3 \text{ Fermi's} + 1 \text{ chiral} & \left(\frac{3N}{2} \right) \text{ chirals} + \left(\frac{3N}{2} + 1 \right) \text{ Fermi's} & \\
 \text{with superpotential (4.89)} & \text{with superpotential (4.95)} & \\
 \Downarrow & \nearrow & \\
 \text{SU}(N - 2) + 1\text{Sym} + 1\mathbf{AS} + 6 \text{ Fermi's} + 4 \text{ chirals} & & \\
 \text{with superpotential (4.90)} & &
 \end{array} \tag{4.96}$$

At each step of this iterative procedure, we neglect the fields $\Psi_2^{(i)}$ and $P^{(i+1)}$, as they decouple from the theory due to the superpotential term $\mathcal{W} = \Psi_2^{(i)} P^{(i+1)}$. The matter content on the right side of the first row is obtained by applying this duality iteration to the final stage, as illustrated in Fig. 4-17.

For free Fermi and free chiral fields, we assign an R-charge of 0. For interacting Fermi and chiral fields, we assign R-charges of 1 and 0, respectively. With these assignments, we compute the central charge of the theory as:

$$c_L = 2N - 4, \quad c_R = 3N - 6. \tag{4.97}$$

The elliptic genus for even N is given by:

$$\begin{aligned} \mathcal{J} &= \frac{\eta(q)^{3N-4}}{N!} \frac{\vartheta_1(x^{-2N})\vartheta_1(y^{-N})\vartheta_1(x^{-4}y^{4-2N})}{\vartheta_1(x^2y^{N-2})} \oint_{JK} \frac{d\mathbf{a}}{2\pi i \mathbf{a}} \prod_{i=1}^N \frac{\prod_{j \neq i} \vartheta_1(a_i/a_j)}{\prod_{j \leq i} \vartheta_1(x^2 a_i a_j) \cdot \prod_{j < i} \vartheta_1(y^2 a_i a_j)} \\ &= (-1)^{\frac{N}{2}+1} \frac{\vartheta_1(x^{2N})}{\eta(q)} \prod_{i=1}^{\frac{N-2}{2}} \frac{\vartheta_1(x^{4i+2}y^{2N-4i-2})}{\vartheta_1(x^{4i+4}y^{2N-4i-4})} \end{aligned} \quad (4.98)$$

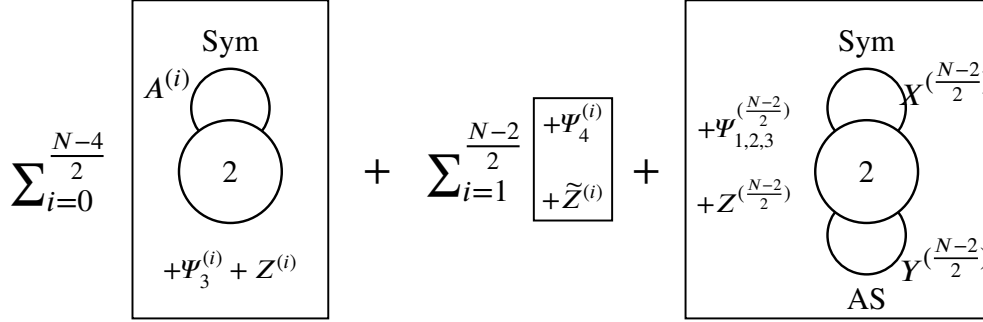


Figure 4-17 Final even- N stage of the recursive elliptic-genus computation.

When N is odd, we can start with the theory $SU(N)+1AS+1Sym$, along with three Fermi's $\Psi_{1,2,3}$ and a free chiral Z . By applying a series of dualities as illustrated in Fig. 4-18, we can transform it into the even case $(N-1)$, which corresponds to the theory $SU(N-1)+1AS+1Sym$, along with three Fermi's $\Psi_{1,2,3}^{(0)}$ and one chiral $Z^{(0)}$. This allows us to use the aforementioned formulas and dualities for iterative derivation. The corresponding charges for these two theories are provided in Tab 4-4. Through this duality, we can apply the iterative dualities illustrated in Fig. 4-16, ultimately resulting in $\frac{N-1}{2} + 3$ free Fermi's and $\frac{N-1}{2} + 2$ free chirals.

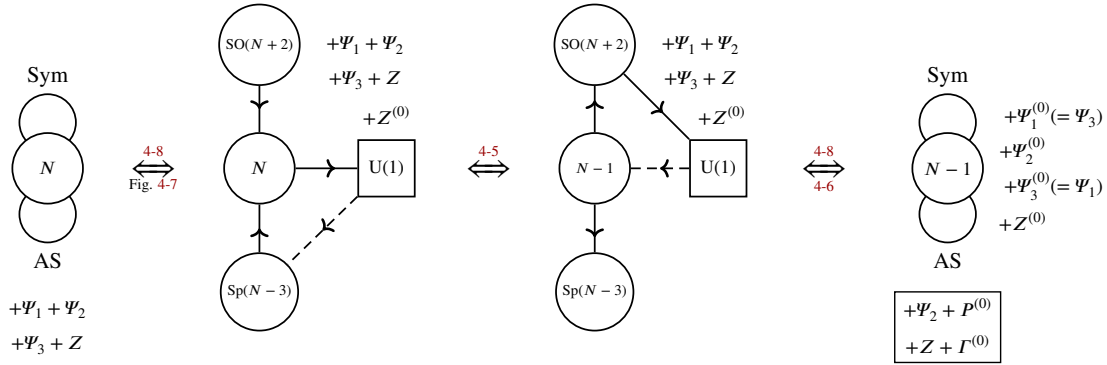


Figure 4-18 A chain of duality transformations relate $SU(N)+1Sym+1AS$ to $SU(N-1)+1Sym+1AS$ where N is odd. In the final quiver, $Z^{(0)}$ and $P^{(0)}$ are chiral fields generated during the duality process, while $\Psi_2^{(0)}$ and $\Gamma^{(0)}$ are newly generated Fermions. The boxed components do not contribute to the elliptic genus. Furthermore, Ψ_2 couples to $P^{(0)}$, Z is a free chiral field, and $\Gamma^{(0)}$ is a free Fermion.

The superpotential for the odd N case is given by:

$$\mathcal{W} = \Psi_1 \det X + \Psi_2 B(2) + \Psi_3 B(3) \quad (4.99)$$

We identify some of the relations of the two sides as:

$$\Psi_1^{(0)} = \Psi_3, \quad \Psi_3^{(0)} = \Psi_1, \quad Z^{(0)} = \text{Tr}[X^{N/2}]. \quad (4.100)$$

	SU(N)	U(1) _x	U(1) _y
$X_{\alpha\beta}$	$\begin{array}{ c c } \hline & \\ \hline \end{array}$	2	0
$Y_{\alpha\beta}$	$\begin{array}{ c } \hline \\ \hline \end{array}$	0	2
Ψ_1	1	$-2N$	0
Ψ_2	1	-2	$-2N + 2$
Ψ_3	1	-6	$-2N + 6$
Z	1	4	$2N - 4$
	SU(N - 1)	U(1) _{x⁽⁰⁾}	U(1) _{y⁽⁰⁾}
$X_{\alpha\beta}^{(0)}$	$\begin{array}{ c c } \hline & \\ \hline \end{array}$	2	0
$Y_{\alpha\beta}^{(0)}$	$\begin{array}{ c } \hline \\ \hline \end{array}$	0	2
$\Psi_1^{(0)}$	1	$-2(N - 1)$	0
$\Psi_2^{(0)}$	1	0	$-(N - 1)$
$\Psi_3^{(0)}$	1	-4	$-2(N - 1) + 4$
$Z^{(0)}$	1	2	$(N - 1) - 2$

Table 4-4 Charge assignments for the odd- N reduction to the even-rank recursion.

The corresponding elliptic genus when N is odd is:

$$\begin{aligned}
\mathcal{F} &= \frac{\eta(q)^{3N-4}}{N!} \frac{\vartheta(x^{-2N})\vartheta(x^{-2}y^{-2N+2})\vartheta(x^{-6}y^{-2N+6})}{\vartheta(x^4y^{2N-4})} \oint_{\text{JK}} \frac{da}{2\pi i a} \prod_{i=1}^N \frac{\prod_{j \neq i} \vartheta_1(a_i/a_j)}{\prod_{j \leq i} \vartheta_1(x^2 a_i a_j) \cdot \prod_{j < i} \vartheta_1(y^2 a_i a_j)} \\
&= (-1)^{\frac{N+1}{2}} \frac{\vartheta(x^6 y^{2N-6})}{\eta(q)} \prod_{i=1}^{\frac{N-3}{2}} \frac{\vartheta(x^{4i+4} y^{2N-4i-4})}{\vartheta(x^{4i+2} y^{2N-4i-2})}.
\end{aligned} \tag{4.101}$$

Chapter 5

Shell Formulas for Instantons

This chapter is the fixed-point research chapter of the thesis and follows the main body of *Shell formulas for instantons and gauge origami*^[74]. The ADHM and brane input was prepared in Section 3.2; here the starting point is the localized Witten index

$$\mathcal{Z}_k = \oint_{\text{JK}} \mathcal{J}_k = \sum_{\text{fixed points}} \mathcal{Z}_{\text{fixed}}.$$

For D0–D4 instantons the fixed points are ordinary two-dimensional Young diagrams. Gauge origami extends the same mechanism to D0–D6, D0–D8, spiked-instanton, DT3, and DT4-type systems^[4,9–11,20–21].

The purpose of the shell formula is to replace case-by-case arm-leg products by a dimension-independent local rule. A Young diagram $\mathbf{Y}$ contributes through its shell $\mathcal{S}(\mathbf{Y})$, an integer charge on each shell box, and the corresponding $\mathcal{J}$ -factor. This has three roles in the chapter:

- (i) it gives uniform closed expressions for 2d, 3d, and 4d Young-diagram fixed points;
- (ii) it produces local recursion relations for adding one box;
- (iii) in special limits, such as the unrefined $\text{Sp}(2N)$ instanton limit, it absorbs factors that otherwise appear as Young-diagram-dependent jumping coefficients.

The common quantum-mechanical input is a set of chiral fields B_a and Fermi constraints A_{ab} . In the ordinary D0–D4 case,

$$J_{A_1} = [B_1, B_2] + IJ, \quad E_{A_1} = 0, \quad (5.1)$$

while gauge-origami systems satisfy

$$\sum_{a < b} \text{Tr } E_{ab} J_{ab} = 0. \quad (5.2)$$

The formulas below are exact in the chambers and limits stated in each example. For refined DT4-type problems, orientation signs and multiplicities may require extra input beyond shell charges. Section 5.1 defines the shell language, Section 5.2 applies it to ordinary 5d instantons, and Section 5.3 treats the higher-dimensional examples.

5.1 Young diagrams and shell formulas

In this section, we introduce Young diagrams in arbitrary dimensions, define the shell formula and its central ingredient—the $\mathcal{J}$ -factor—and establish several algebraic properties that will be used throughout this thesis.

5.1.1 Young diagrams and poles

A Young diagram is a combinatorial object defined by a simple monotonicity rule. In two dimensions, the distinct ways to write a positive integer as an ordered sum of non-increasing positive integers correspond bijectively to 2d Young diagrams, also known as integer partitions. This concept generalizes directly to higher dimensions: 3d Young diagrams (plane partitions) and 4d Young diagrams (solid partitions), and so on. The general definition is as follows.

Definition 5.1 (Young diagram) A d -dimensional Young diagram is a finite subset $Y \subseteq \mathbb{Z}^d$ for which there exists a point $\mathbf{z} = (z_1, \dots, z_d) \in \mathbb{Z}^d$, called the origin of Y , such that the following monotonicity condition holds:

$$\begin{aligned} &\text{if } \mathbf{x} = (x_1, \dots, x_d) \in Y \text{ and } z_i \leq y_i \leq x_i \text{ for all } i = 1, \dots, d, \\ &\text{then } \mathbf{y} = (y_1, \dots, y_d) \in Y. \end{aligned}$$

The empty set $\emptyset$ is also regarded as a Young diagram (with no origin specified).

throughout this thesis, the origin of every Young diagram is taken to be $(1, 1, 1, \dots)$ unless stated otherwise.

Young diagrams provide a natural language for the partition functions discussed in Sec. 5.2 and Sec. 5.3. The key point is that the poles of an instanton partition function are in one-to-one correspondence with the box coordinates of Young diagrams. More precisely, given an integrand $\mathcal{J}(\boldsymbol{\phi})$ depending on integration variables $\boldsymbol{\phi} = (\phi_1, \dots, \phi_k)$, the JK-residue (reviewed in Chapter 2) evaluates the integral as a sum over poles:

$$\oint_{\text{JK}} \prod_{I=1}^k \frac{d\phi_I}{2\pi i} \mathcal{J}(\boldsymbol{\phi}) = \sum_{\boldsymbol{\phi}=\boldsymbol{\phi}_*} \text{JK-Res}(\eta) \mathcal{J}(\boldsymbol{\phi}). \quad (5.3)$$

In all cases considered here, the JK-residue selects poles indexed by a list of Young diagrams $\mathbf{Y} = (\mathbf{Y}_{\mathcal{A}}, \mathbf{Y}_{\mathcal{B}}, \dots)$. Each Young diagram $\mathbf{Y}_{\mathcal{A}}$ carries a label $\mathcal{A} \equiv (A, \alpha) = (a_1 a_2 \dots a_d, \alpha)$, specifying its basis directions $\epsilon_A = (\epsilon_{a_1}, \dots, \epsilon_{a_d})$ and a color α . The pole corresponding to this Young diagram list is:

$$(\phi_{1*}, \dots, \phi_{k*}) = (\mathcal{X}_{\mathcal{A}}(\mathbf{x}_{\mathcal{A},1}), \dots, \mathcal{X}_{\mathcal{A}}(\mathbf{x}_{\mathcal{A},n_{\mathcal{A}}}), \mathcal{X}_{\mathcal{B}}(\mathbf{x}_{\mathcal{B},1}), \dots, \mathcal{X}_{\mathcal{B}}(\mathbf{x}_{\mathcal{B},n_{\mathcal{B}}}), \dots), \quad (5.4)$$

where $n_{\mathcal{A}} = |\mathbf{Y}_{\mathcal{A}}|$ is the number of boxes in $\mathbf{Y}_{\mathcal{A}}$, and the total box count satisfies $\sum_{\mathcal{A}} n_{\mathcal{A}} = k$. The coordinate function $\mathcal{X}_{\mathcal{A}}$ is defined in terms of the Coulomb branch parameter $v_{\mathcal{A}}$ by:

$$\mathcal{X}_{\mathcal{A}}(\mathbf{x}) \equiv v_{\mathcal{A}} + (\mathbf{x} - \mathbf{1}) \cdot \epsilon_A = v_{\mathcal{A}} + \sum_{i=1}^d (x_i - 1) \epsilon_{a_i}. \quad (5.5)$$

A crucial observation is that all partition functions considered in this thesis share a common structure at each pole:

$$\text{JK-Res}(\eta) \mathcal{J}(\boldsymbol{\phi}) \supset \frac{1}{\text{sh}(\phi_{i*} - \mathcal{X}_{\mathcal{A}}(\mathbf{1}))} \prod_{\mathbf{y} \in \mathbf{Y}_{\mathcal{A}}} \frac{\prod_{\mathbf{b} \in \mathbf{B}_{\text{even}}} \text{sh}(\phi_{i*} - \mathcal{X}_{\mathcal{A}}(\mathbf{y}) - \mathbf{b} \cdot \epsilon_A)}{\prod_{\mathbf{b} \in \mathbf{B}_{\text{odd}}} \text{sh}(\phi_{i*} - \mathcal{X}_{\mathcal{A}}(\mathbf{y}) - \mathbf{b} \cdot \epsilon_A)}, \quad (5.6)$$

where $\text{sh}(x) = e^{x/2} - e^{-x/2}$. Here $\mathbf{B}_d = \{\mathbf{b} = (b_1, \dots, b_d) \mid b_i \in \{0, 1\}\}$ is the set of all d -tuples of binary digits, while $\mathbf{B}_{\text{even}} \subset \mathbf{B}_d$ consists of those tuples with $|\mathbf{b}| = \sum_i b_i \in 2\mathbb{Z}$, and $\mathbf{B}_{\text{odd}}$ consists of those with $|\mathbf{b}| \in 2\mathbb{Z} + 1$. It is precisely this universal structure that motivates the definition of the shell formula.

5.1.2 Shell formulas and $\mathcal{J}$ -Factor

The shell formula is built from two geometric ingredients attached to a Young diagram: its *shell* (the set of boxes on its outer boundary) and a *charge* assigned to each shell box. We define these in turn.

Definition 5.2 (Shell of a Young diagram) Given a d -dimensional Young diagram $\mathbf{Y}$ and the set of binary tuples $\mathbf{B}_d$, the shell $\mathcal{S}(\mathbf{Y})$ of $\mathbf{Y}$ is:

$$\mathcal{S}(\mathbf{Y}) \equiv (\mathbf{Y} + \mathbf{B}_d) \setminus \mathbf{Y}. \quad (5.7)$$

In words, $\mathcal{S}(\mathbf{Y})$ consists of all boxes that are not in $\mathbf{Y}$ but can be reached from some box of $\mathbf{Y}$ by a unit step in any combination of coordinate directions.

Definition 5.3 (Charge of a shell box) For each box $\mathbf{x} = (x_1, \dots, x_d)$ in the shell $\mathcal{S}(\mathbf{Y})$, its charge is defined by the inclusion-exclusion sum:

$$Q_{\mathbf{Y}}(\mathbf{x}) = \sum_{\substack{\mathbf{b} \in \mathbf{B}_d \\ \mathbf{x} - \mathbf{b} \in \mathbf{Y}}} (-1)^{|\mathbf{b}|}, \quad (5.8)$$

where $|\mathbf{b}| \equiv \sum_{i=1}^d b_i$ counts the number of 1's in $\mathbf{b}$. Intuitively, the charge measures how many corners of the unit hypercube centered at $\mathbf{x}$ already belong to $\mathbf{Y}$, weighted by sign.

Explicit examples of shells and charges are collected in Appendix A.2.1. For a generic d -dimensional Young diagram, shell box charges take integer values between $-d$ and d .

With these ingredients, the central object of the shell formula can now be defined.

Definition 5.4 ($\mathcal{J}$ -factor) Given a d -dimensional Young diagram $\mathbf{Y}_{\mathcal{A}}$ with label $\mathcal{A} = (A, \alpha) = (a_1 a_2 \dots a_d, \alpha)$, the $\mathcal{J}$ -factor is the product over shell boxes:

$$\begin{aligned} \mathcal{J}(\mathbf{x} | \mathbf{Y}_{\mathcal{A}}) &\equiv \prod_{\mathbf{y} \in \mathcal{S}(\mathbf{Y}_{\mathcal{A}})} \text{sh}(\mathbf{x} - \mathcal{X}_{\mathcal{A}}(\mathbf{y}))^{Q_{\mathbf{Y}_{\mathcal{A}}}(\mathbf{y})}, \\ \mathcal{J}(\mathbf{x} | \emptyset_{\mathcal{A}}) &\equiv \frac{1}{\text{sh}(\mathbf{x} - \mathcal{X}_{\mathcal{A}}(\mathbf{1}))}. \end{aligned} \quad (5.9)$$

That is, each shell box $\mathbf{y}$ contributes a factor of $\text{sh}(\mathbf{x} - \mathcal{X}_{\mathcal{A}}(\mathbf{y}))$ raised to its charge.

Detailed computational examples of the $\mathcal{J}$ -factor are provided in Appendix A.2.1.

We now establish four algebraic properties of the $\mathcal{J}$ -factor that will be used in subsequent sections. This discussion is self-contained and may be skipped on a first reading.

Expansion. The $\mathcal{J}$ -factor admits a box-by-box expansion:

$$\mathcal{J}(\mathbf{x} | \mathbf{Y}_{\mathcal{A}}) = \frac{1}{\text{sh}(\mathbf{x} - \mathcal{X}_{\mathcal{A}}(\mathbf{1}))} \prod_{\mathbf{y} \in \mathbf{Y}_{\mathcal{A}}} \frac{\prod_{\mathbf{b} \in \mathbf{B}_{\text{even}}} \text{sh}(\mathbf{x} - \mathcal{X}_{\mathcal{A}}(\mathbf{y}) - \mathbf{b} \cdot \epsilon_A)}{\prod_{\mathbf{b} \in \mathbf{B}_{\text{odd}}} \text{sh}(\mathbf{x} - \mathcal{X}_{\mathcal{A}}(\mathbf{y}) - \mathbf{b} \cdot \epsilon_A)}. \quad (5.10)$$

This expansion simplifies dramatically after cancellations: for any box $\mathbf{y}$ in the interior of $\mathbf{Y}$, the boxes $\mathbf{y} - \mathbf{b}$ (for all $\mathbf{b} \in \mathbf{B}_d$) all lie in $\mathbf{Y}$ by the Young diagram monotonicity condition, and their contributions cancel pairwise. More precisely:

$$\frac{\prod_{\mathbf{b} \in \mathbf{B}_{\text{even}}} \text{sh}(x - \mathcal{X}_{\mathcal{A}}(\mathbf{y} - \mathbf{b}) - \mathbf{b} \cdot \epsilon_A)}{\prod_{\mathbf{b} \in \mathbf{B}_{\text{odd}}} \text{sh}(x - \mathcal{X}_{\mathcal{A}}(\mathbf{y} - \mathbf{b}) - \mathbf{b} \cdot \epsilon_A)} = \frac{\prod_{\mathbf{b} \in \mathbf{B}_{\text{even}}} \text{sh}(x - \mathcal{X}_{\mathcal{A}}(\mathbf{y}))}{\prod_{\mathbf{b} \in \mathbf{B}_{\text{odd}}} \text{sh}(x - \mathcal{X}_{\mathcal{A}}(\mathbf{y}))} = 1. \quad (5.11)$$

For boxes on the boundary of $\mathbf{Y}$, only half of the shifted boxes $\mathbf{y} - \mathbf{b}$ are present, but they still cancel mutually. After all cancellations, only the shell boxes survive, leaving exactly the definition (5.9). This also confirms that the expansion (5.10) coincides with the universal pole structure (5.6), providing the precise justification for the definition of the $\mathcal{J}$ -factor.

As a concrete illustration, for the 2d Young diagram $\lambda_{12,\alpha}$ with label $(12, \alpha)$:

$$\begin{aligned} \mathcal{J}(x | \lambda_{12,\alpha}) &= \frac{1}{\text{sh}(x - \mathcal{X}_{12,\alpha}(\mathbf{1}))} \prod_{\mathbf{y} \in \lambda_{12,\alpha}} \frac{\text{sh}(x - \mathcal{X}_{12,\alpha}(\mathbf{y}) - (0,0) \cdot \epsilon_{12}) \text{sh}(x - \mathcal{X}_{12,\alpha}(\mathbf{y}) - (1,1) \cdot \epsilon_{12})}{\text{sh}(x - \mathcal{X}_{12,\alpha}(\mathbf{y}) - (0,1) \cdot \epsilon_{12}) \text{sh}(x - \mathcal{X}_{12,\alpha}(\mathbf{y}) - (1,0) \cdot \epsilon_{12})} \\ &= \frac{1}{\text{sh}(x - v_{12,\alpha})} \prod_{\mathbf{y} \in \lambda_{12,\alpha}} \frac{\text{sh}(x - \mathcal{X}_{12,\alpha}(\mathbf{y})) \text{sh}(x - \mathcal{X}_{12,\alpha}(\mathbf{y}) - \epsilon_{12})}{\text{sh}(x - \mathcal{X}_{12,\alpha}(\mathbf{y}) - \epsilon_1) \text{sh}(x - \mathcal{X}_{12,\alpha}(\mathbf{y}) - \epsilon_2)}. \end{aligned} \quad (5.12)$$

Translation invariance. The $\mathcal{J}$ -factor depends only on coordinate differences, so shifting both its argument and the Young diagram by the same vector $\mathbf{y} \in \mathbb{Z}^d$ leaves it unchanged:

$$\mathcal{J}(\mathcal{X}_{A,\alpha}(\mathbf{x}) | \mathbf{Y}_{A,\beta}) = \mathcal{J}(\mathcal{X}_{A,\alpha}(\mathbf{x} + \mathbf{y}) | \mathbf{Y}_{A,\beta} + \mathbf{y}), \quad (5.13)$$

where $\mathbf{Y}_{A,\beta} + \mathbf{y}$ denotes the Young diagram obtained by translating every box of $\mathbf{Y}_{A,\beta}$ by $\mathbf{y}$.

Swapping property. For two d -dimensional Young diagrams $\mathbf{Y}_{A,\alpha}$ and $\mathbf{Y}_{A,\beta}$ sharing the same basis A , the following identity exchanges their roles:

$$\begin{aligned} &\prod_{\mathbf{x} \in \mathbf{Y}_{A,\alpha}} \text{sh}(\mathcal{X}_{A,\alpha}(\mathbf{x}) - \mathcal{X}_{A,\beta}(\mathbf{1})) \mathcal{J}(\mathcal{X}_{A,\alpha}(\mathbf{x}) | \mathbf{Y}_{A,\beta}) \\ &= \prod_{\mathbf{y} \in \mathbf{Y}_{A,\beta} + \mathbf{1}} (\text{sh}(\mathcal{X}_{A,\beta}(\mathbf{y}) - \mathcal{X}_{A,\alpha}(\mathbf{1})) \mathcal{J}(\mathcal{X}_{A,\beta}(\mathbf{y}) | \mathbf{Y}_{A,\alpha}))^{(-1)^d}. \end{aligned} \quad (5.14)$$

The sign $(-1)^d$ reflects the parity of the dimension. In particular, for $d = 2$ the right-hand side has the same sign as the left, while for $d = 3$ it is inverted. Concretely, for two 2d Young diagrams $\lambda_{12,1}$ and $\lambda_{12,2}$:

$$\begin{aligned} &\prod_{\mathbf{x} \in \lambda_{12,1}} \text{sh}(\mathcal{X}_{12,1}(\mathbf{x}) - \mathcal{X}_{12,2}(\mathbf{1})) \mathcal{J}(\mathcal{X}_{12,1}(\mathbf{x}) | \lambda_{12,2}) \\ &= \prod_{\mathbf{y} \in \lambda_{12,2} + \mathbf{1}} \text{sh}(\mathcal{X}_{12,2}(\mathbf{y}) - \mathcal{X}_{12,1}(\mathbf{1})) \mathcal{J}(\mathcal{X}_{12,2}(\mathbf{y}) | \lambda_{12,1}), \end{aligned} \quad (5.15)$$

while for two 3d Young diagrams $\pi_{123,1}$ and $\pi_{123,2}$ the inversion gives:

$$\begin{aligned} & \prod_{\mathbf{x} \in \pi_{123,1}} \text{sh}(\mathcal{X}_{123,1}(\mathbf{x}) - \mathcal{X}_{123,2}(\mathbf{1})) \mathcal{J}(\mathcal{X}_{123,1}(\mathbf{x}) | \pi_{123,2}) \\ &= \prod_{\mathbf{y} \in \pi_{123,2} + \mathbf{1}} \frac{1}{\text{sh}(\mathcal{X}_{123,2}(\mathbf{y}) - \mathcal{X}_{123,1}(\mathbf{1})) \mathcal{J}(\mathcal{X}_{123,2}(\mathbf{y}) | \pi_{123,1})}. \end{aligned} \quad (5.16)$$

Recursion relation. Adding a single box to a Young diagram changes the $\mathcal{J}$ -factor in a controlled way. Specifically, let $\mathbf{Y}'_{\mathcal{A}} = \mathbf{Y}_{\mathcal{A}} \cup \{\mathbf{w}\}$ be obtained by adding one box $\mathbf{w}$ to $\mathbf{Y}_{\mathcal{A}}$. Then:

$$\begin{aligned} & \frac{\prod_{\mathbf{x} \in \mathbf{Y}_{\mathcal{A}} \cup \{\mathbf{w}\}} \mathcal{J}(\mathcal{X}_{\mathcal{A}}(\mathbf{x}) | \mathbf{Y}_{\mathcal{A}} \cup \{\mathbf{w}\})}{\prod_{\mathbf{x} \in \mathbf{Y}_{\mathcal{A}}} \mathcal{J}(\mathcal{X}_{\mathcal{A}}(\mathbf{x}) | \mathbf{Y}_{\mathcal{A}})} \\ &= \mathcal{J}(\mathcal{X}_{\mathcal{A}}(\mathbf{w}) | \mathbf{Y}_{\mathcal{A}} \cup \{\mathbf{w}\}) (\text{sh}(\mathcal{X}_{\mathcal{A}}(\mathbf{w}) - \mathcal{X}_{\mathcal{A}}(\mathbf{0})) \mathcal{J}(\mathcal{X}_{\mathcal{A}}(\mathbf{w} + \mathbf{1}) | \mathbf{Y}_{\mathcal{A}}))^{(-1)^d}, \end{aligned} \quad (5.17)$$

where the derivation uses the swapping property (5.14). The key point is that the entire ratio reduces to a local contribution at the new box $\mathbf{w}$ alone: one factor from the $\mathcal{J}$ -factor of the enlarged diagram evaluated at $\mathbf{w}$, and one from the original diagram evaluated at the shifted point $\mathbf{w} + \mathbf{1}$.

The intermediate steps of the derivation are:

$$\begin{aligned} & \frac{\prod_{\mathbf{x} \in \mathbf{Y}_{\mathcal{A}} \cup \{\mathbf{w}\}} \mathcal{J}(\mathcal{X}_{\mathcal{A}}(\mathbf{x}) | \mathbf{Y}_{\mathcal{A}} \cup \{\mathbf{w}\})}{\prod_{\mathbf{x} \in \mathbf{Y}_{\mathcal{A}}} \mathcal{J}(\mathcal{X}_{\mathcal{A}}(\mathbf{x}) | \mathbf{Y}_{\mathcal{A}})} \\ &= \mathcal{J}(\mathcal{X}_{\mathcal{A}}(\mathbf{w}) | \mathbf{Y}_{\mathcal{A}} \cup \{\mathbf{w}\}) \frac{\prod_{\mathbf{x} \in \mathbf{Y}_{\mathcal{A}}} \mathcal{J}(\mathcal{X}_{\mathcal{A}}(\mathbf{x}) | \mathbf{Y}_{\mathcal{A}} \cup \{\mathbf{w}\})}{\prod_{\mathbf{x} \in \mathbf{Y}_{\mathcal{A}}} \mathcal{J}(\mathcal{X}_{\mathcal{A}}(\mathbf{x}) | \mathbf{Y}_{\mathcal{A}})} \\ &= \mathcal{J}(\mathcal{X}_{\mathcal{A}}(\mathbf{w}) | \mathbf{Y}_{\mathcal{A}} \cup \{\mathbf{w}\}) \frac{\prod_{\mathbf{x} \in \mathbf{Y}_{\mathcal{A}} \cup \{\mathbf{w}\}} (\text{sh}(\mathcal{X}_{\mathcal{A}}(\mathbf{x} + \mathbf{1}) - \mathcal{X}_{\mathcal{A}}(\mathbf{1})) \mathcal{J}(\mathcal{X}_{\mathcal{A}}(\mathbf{x} + \mathbf{1}) | \mathbf{Y}_{\mathcal{A}}))^{(-1)^d}}{\prod_{\mathbf{x} \in \mathbf{Y}_{\mathcal{A}}} \mathcal{J}(\mathcal{X}_{\mathcal{A}}(\mathbf{x}) | \mathbf{Y}_{\mathcal{A}}) \text{sh}(\mathcal{X}_{\mathcal{A}}(\mathbf{x}) - \mathcal{X}_{\mathcal{A}}(\mathbf{1}))} \\ &= \mathcal{J}(\mathcal{X}_{\mathcal{A}}(\mathbf{w}) | \mathbf{Y}_{\mathcal{A}} \cup \{\mathbf{w}\}) (\text{sh}(\mathcal{X}_{\mathcal{A}}(\mathbf{w}) - \mathcal{X}_{\mathcal{A}}(\mathbf{0})) \mathcal{J}(\mathcal{X}_{\mathcal{A}}(\mathbf{w} + \mathbf{1}) | \mathbf{Y}_{\mathcal{A}}))^{(-1)^d} \\ &\quad \times \frac{\prod_{\mathbf{x} \in \mathbf{Y}_{\mathcal{A}}} (\text{sh}(\mathcal{X}_{\mathcal{A}}(\mathbf{x}) - \mathcal{X}_{\mathcal{A}}(\mathbf{0})) \mathcal{J}(\mathcal{X}_{\mathcal{A}}(\mathbf{x} + \mathbf{1}) | \mathbf{Y}_{\mathcal{A}}))^{(-1)^d}}{\prod_{\mathbf{x} \in \mathbf{Y}_{\mathcal{A}}} \mathcal{J}(\mathcal{X}_{\mathcal{A}}(\mathbf{x}) | \mathbf{Y}_{\mathcal{A}}) \text{sh}(\mathcal{X}_{\mathcal{A}}(\mathbf{x}) - \mathcal{X}_{\mathcal{A}}(\mathbf{1}))} \\ &= \mathcal{J}(\mathcal{X}_{\mathcal{A}}(\mathbf{w}) | \mathbf{Y}_{\mathcal{A}} \cup \{\mathbf{w}\}) (\text{sh}(\mathcal{X}_{\mathcal{A}}(\mathbf{w}) - \mathcal{X}_{\mathcal{A}}(\mathbf{0})) \mathcal{J}(\mathcal{X}_{\mathcal{A}}(\mathbf{w} + \mathbf{1}) | \mathbf{Y}_{\mathcal{A}}))^{(-1)^d}, \end{aligned} \quad (5.18)$$

where the swapping property (5.14) is used in the penultimate step, and the last step follows because the remaining product over $\mathbf{Y}_{\mathcal{A}}$ telescopes to 1.

As a concrete example, for 3d Young diagrams the recursion reads:

$$\frac{\prod_{\mathbf{x} \in \pi_{123,1} \cup \{\mathbf{w}\}} \mathcal{J}(\mathcal{X}_{123,1}(\mathbf{x}) | \pi_{123,1} \cup \{\mathbf{w}\})}{\prod_{\mathbf{x} \in \pi_{123,1}} \mathcal{J}(\mathcal{X}_{123,1}(\mathbf{x}) | \pi_{123,1})} = \frac{\mathcal{J}(\mathcal{X}_{123,1}(\mathbf{w}) | \pi_{123,1} \cup \{\mathbf{w}\})}{\text{sh}(\mathcal{X}_{123,1}(\mathbf{w}) - \mathcal{X}_{123,1}(\mathbf{0})) \mathcal{J}(\mathcal{X}_{123,1}(\mathbf{w} + \mathbf{1}) | \pi_{123,1})}. \quad (5.19)$$

Splitting property. Suppose a d -dimensional Young diagram $\mathbf{Y}_{\mathcal{A}}$ decomposes as a disjoint union:

$$\mathbf{Y}_{\mathcal{A}} = \mathbf{Y}'_{\mathcal{A}} \cup \mathbf{Y}''_{\mathcal{A}}, \quad (5.20)$$

where $\mathbf{Y}'_{\mathcal{A}}$ begins at the same origin as $\mathbf{Y}_{\mathcal{A}}$, and $\mathbf{Y}''_{\mathcal{A}}$ begins at some box $\mathbf{y} = (y_1, \dots, y_d) \in \mathbf{Y}_{\mathcal{A}}$. Then the $\mathcal{F}$ -factor factorizes as:

$$\mathcal{F}(x|\mathbf{Y}_{\mathcal{A}}) = \mathcal{F}(x|\mathbf{Y}'_{\mathcal{A}}) \mathcal{F}(x|\mathbf{Y}''_{\mathcal{A}}) \text{sh}(x - \mathcal{X}_{\mathcal{A}}(\mathbf{y})). \quad (5.21)$$

The extra sh factor accounts for the interface between $\mathbf{Y}'$ and $\mathbf{Y}''$ at their junction. As a simple example, taking $\{(1, 1), (1, 2)\} = \{(1, 1)\} \cup \{(1, 2)\}$ with $\{(1, 2)\}$ a single box starting at $(1, 2)$:

$$\mathcal{F}(x|\{(1, 1), (1, 2)\}_{12,1}) = \mathcal{F}(x|\{(1, 1)\}_{12,1}) \mathcal{F}(x|\{(1, 2)\}_{12,1}) \text{sh}(x - \mathcal{X}_{12,1}(1, 2)). \quad (5.22)$$

5.2 Instantons in 5d pure SYM

In this section, we employ the shell formula defined above to rewrite the instanton partition function for 5d $\mathcal{N} = 1$ pure SYM with classical gauge group^[4,20–21]. Although these instanton partition functions can be expressed in terms of the Nekrasov factor, the shell formula representation makes the formulas more intuitive for visualizing the interactions between instantons and various D-branes. Since all space directions lie in $\mathbb{C}_1 \times \mathbb{C}_2 = (x^0, x^1, x^2, x^3)$, all the Young diagrams are oriented in the 1, 2-direction, and the basis specification is temporarily omitted in this section.

5.2.1 5d pure $U(N)$ SYM

Section 3.2 explained why the ADHM quotient solves the four-dimensional pure $U(N)$ instanton equation. In five dimensions the same moduli space is the phase space of k BPS instanton particles on $\mathbb{C}_1 \times \mathbb{C}_2 \times S^1$. Thus the k -instanton sector of pure 5d $\mathcal{N} = 1$ $U(N)$ SYM is the K-theoretic Witten index of the ADHM quiver in Fig. 3-1.

With $K \simeq \mathbb{C}^k$ and $N \simeq \mathbb{C}^N$,

$$\mathcal{M}_{U(N),k}^{\text{inst}} = \{(B_1, B_2, I, J) \mid \mu_{\mathbb{C}} = 0, \mu_{\mathbb{R}} = \zeta \mathbf{1}_k\} / U(k), \quad (5.23)$$

where

$$\mu_{\mathbb{C}} = [B_1, B_2] + IJ, \quad \mu_{\mathbb{R}} = \sum_{a=1}^2 [B_a, B_a^\dagger] + II^\dagger - J^\dagger J. \quad (5.24)$$

The FI parameter ζ is the small-instanton resolution already used in (3.25). The Ω -background acts on the ADHM multiplets by

$$(B_1, B_2, I, J, \Lambda_1) \xrightarrow{U(1)_{\epsilon_1} \times U(1)_{\epsilon_2}} (q_1^{-1} B_1, q_2^{-1} B_2, I, q_{12}^{-1} J, q_{12}^{-1} \Lambda_1) \quad (5.25)$$

where $q_i \equiv e^{-\epsilon_i}$ and $q_{ij} = q_i q_j$.

Thus the Ω -background localizes the two complex planes and leaves a one-dimensional supersymmetric quantum mechanics on S^1 . In the Losev–Moore–Nekrasov–Shatashvili formalism^[75–77], the Nekrasov partition function is

$$\mathcal{Z}_{\text{inst}}^{U(N)}(v_1, \dots, v_N) = \sum_{k=0}^{\infty} q^k \mathcal{Z}_{N,k}^U(v_1, \dots, v_N) \quad (5.26)$$

where k is the instanton number, and $\mathcal{Z}_k$ is:

$$\mathcal{Z}_{N,k}^U(v_1, \dots, v_N) = \oint_{\text{JK}} \prod_{i=1}^k \frac{d\phi_i}{2\pi i} \mathcal{J}_{N,k}^U,$$

$$\mathcal{J}_{N,k}^U = \frac{1}{k!} \prod_{i \neq j}^k \text{sh}(\phi_i - \phi_j) \prod_{i,j=1}^k \frac{\text{sh}(\phi_i - \phi_j - \epsilon_{12})}{\text{sh}(\phi_i - \phi_j - \epsilon_{1,2})} \prod_{i=1}^k \prod_{\alpha=1}^N \frac{1}{\text{sh}(\phi_i - v_\alpha) \text{sh}(v_\alpha - \epsilon_{12} - \phi_i)}$$

where $\text{sh}(x - \epsilon_{1,2}) = \text{sh}(x - \epsilon_1) \text{sh}(x - \epsilon_2)$. The Coulomb branch parameters v_α effectively indicate the location of the α -th D4-brane along the complex planes $\mathbb{C}^2 = \mathbb{C}_1 \times \mathbb{C}_2$. After applying the JK-residue, the poles can be classified by a set of N 2d Young diagrams $\vec{\lambda} = (\lambda_1, \dots, \lambda_N)$. For the α -th Young diagram, the box at position $\mathbf{x} = (i, j)$ contributes a pole at:

$$\mathcal{X}_\alpha(\mathbf{x}) = v_\alpha + (\mathbf{x} - \mathbf{1}) \cdot \epsilon_{12} = v_\alpha + i\epsilon_1 + j\epsilon_2 - \epsilon_{12}. \quad (5.27)$$

To obtain the closed-form expression for the k -instanton partition function, we introduce the *Nekrasov factor*^[4]:

$$N_{\alpha,\beta}^{\vec{\lambda}}(\mathbf{x}) \equiv v_\alpha - v_\beta + L_{\lambda_\alpha}(\mathbf{x})\epsilon_1 - A_{\lambda_\beta}(\mathbf{x})\epsilon_2 - \epsilon_2 \quad (5.28)$$

where $L_{\lambda_\alpha}(\mathbf{x})$ and $A_{\lambda_\alpha}(\mathbf{x})$ are the leg and arm of the box $\mathbf{x}$ in λ_α respectively (see Appendix A.2 for definitions). The instanton partition function is then:

$$\mathcal{Z}_{N,k}^U(v_1, \dots, v_N) = \sum_{||\vec{\lambda}||=k} \mathcal{Z}^U(\vec{\lambda}) = \sum_{||\vec{\lambda}||=k} \prod_{\alpha,\beta=1}^N \prod_{\mathbf{x} \in \lambda_\alpha} \frac{1}{\text{sh}(-N_{\alpha,\beta}^{\vec{\lambda}}(\mathbf{x})) \text{sh}(N_{\alpha,\beta}^{\vec{\lambda}}(\mathbf{x}) + \epsilon_{12})} \quad (5.29)$$

where $||\vec{\lambda}|| \equiv \sum_\alpha |\lambda_\alpha|$ is the total number of boxes.

While the Nekrasov factor provides a compact encoding of Young diagram data and yields a concise expression for the instanton partition function, it relies on arm and leg lengths that are intrinsic to 2d Young diagrams and do not extend naturally to higher dimensions, and it makes the derivation of algebraic relations—such as recursion formulas—rather cumbersome. For this reason, we introduce the $\mathcal{J}$ -factor defined in (5.9) and rewrite the instanton partition function as:

$$\mathcal{Z}_{N,k}^U(v_1, \dots, v_N) = \sum_{||\vec{\lambda}||=k} \prod_{\alpha,\beta=1}^N \prod_{\mathbf{x} \in \lambda_\alpha} \frac{\mathcal{J}(\mathcal{X}_\alpha(\mathbf{x}) | \lambda_\beta)}{\text{sh}(-\mathcal{X}_\alpha(\mathbf{x}) + \mathcal{X}_\beta(\mathbf{0}))}. \quad (5.30)$$

That (5.30) is equal to (5.29) follows from the identity:

$$\prod_{\mathbf{x} \in \lambda_\alpha} \frac{\mathcal{J}(\mathcal{X}_\alpha(\mathbf{x}) | \lambda_\beta)}{\text{sh}(\mathcal{X}_\alpha(\mathbf{x}) - \mathcal{X}_\beta(\mathbf{0}))} = \prod_{\mathbf{x} \in \lambda_\alpha} \frac{1}{\text{sh}(N_{\alpha,\beta}^{\vec{\lambda}}(\mathbf{x})) \text{sh}(N_{\alpha,\beta}^{\vec{\lambda}}(\mathbf{x}) + \epsilon_{12})}, \quad (5.31)$$

whose proof is given in Appendix C.1.1. Thus (5.30), (5.31), and (5.29) form a commutative triangle: the shell formula on the left and the Nekrasov factor on the right are two equivalent representations of the same quantity, related by the identity (5.31).

The shell formula representation makes the D0-D4 interaction structure manifest. Physically, v_α is the position of the α -th D4-brane and $\mathcal{X}_\alpha(\mathbf{x})$ is the position of the D0-brane (instanton) within $D4_\alpha$ at coordinate $\mathbf{x}$. The open strings stretching from $D0_{\alpha,\mathbf{x}}$ to $D4_\beta$ probe the vacuum configuration encoded in λ_β ; the expansion property (5.10) shows that the $\mathcal{J}$ -factor is precisely their Witten index contribution. Schematically, as illustrated in Fig. 5-1, in the vacuum corresponding to Young diagram λ_β , the contribution from the D0-D4 strings is:

$$\frac{\mathcal{J}(\mathcal{X}_\alpha(\mathbf{x})|\lambda_\beta)}{\text{sh}(-\mathcal{X}_\alpha(\mathbf{x}) + \mathcal{X}_\beta(\mathbf{0}))}. \quad (5.32)$$

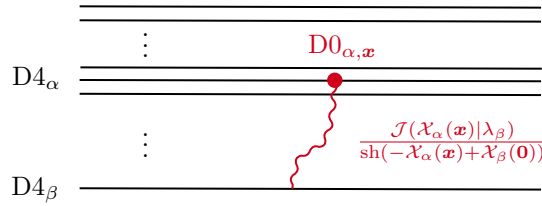


Figure 5-1 The instanton configuration of 5d pure $U(N)$ SYM can be constructed from the type IIA D0-D4 brane system after integrating out the adjoint hypermultiplets. The D4-branes extend along two complex directions $\mathbb{C}_1, \mathbb{C}_2$, and the time direction x^0 . $D4_\alpha$ refers to the α -th D4-brane, while $D0_{\alpha,\mathbf{x}}$ denotes the D0-brane within $D4_\alpha$ corresponding to the instanton at coordinate $\mathbf{x}$ in the Young diagram. The red wavy line represents strings connecting $D0_{\alpha,\mathbf{x}}$ and $D4_\beta$; their contribution to the index is $\mathcal{J}(\mathcal{X}_\alpha|\lambda_\beta)/\text{sh}(-\mathcal{X}_\alpha + \mathcal{X}_\beta(\mathbf{0}))$, as given in (5.32).

The instanton partition function $\mathcal{Z}_{N,k}^{\text{SU}}$ of $SU(N)$ SYM is obtained by imposing the traceless condition $\sum_{\alpha=1}^N v_\alpha = 0$.

Property (5.17) also yields recursion relations for the Nekrasov partition function. When adding the contribution of an instanton within a D4-brane, the ratio of the partition function contribution from the Young diagram $\lambda_\alpha \cup \square$ to that from λ_α is:

$$\begin{aligned} \frac{\mathcal{Z}^U(\lambda_\alpha \cup \square)}{\mathcal{Z}^U(\lambda_\alpha)} &= \left(\prod_{\mathbf{x} \in \lambda_\alpha \cup \square} \frac{\mathcal{J}(\mathcal{X}_\alpha(\mathbf{x})|\lambda_\alpha \cup \square)}{\text{sh}(-\mathcal{X}_\alpha(\mathbf{x}) + \mathcal{X}_\alpha(\mathbf{0}))} \right) / \left(\prod_{\mathbf{y} \in \lambda_\alpha} \frac{\mathcal{J}(\mathcal{X}_\alpha(\mathbf{y})|\lambda_\alpha)}{\text{sh}(-\mathcal{X}_\alpha(\mathbf{y}) + \mathcal{X}_\alpha(\mathbf{0}))} \right) \\ &= -\mathcal{J}(\mathcal{X}_\alpha(\square)|\lambda_\alpha \cup \square) \times \mathcal{J}(\mathcal{X}_\alpha(\square + \mathbf{1})|\lambda_\alpha) \end{aligned} \quad (5.33)$$

where $\square$ denotes the coordinate of the added box in the Young diagram. This recursion relation is directly connected to the quantum toroidal algebra and qq -characters^[78–84], as we now make explicit.

Consider the Gaiotto state, defined as a linear combination of 2d Young diagram basis states corresponding to the Fock representation of the quantum toroidal algebra:

$$|\mathfrak{G}\rangle \equiv \sum_{\vec{\lambda}} \left(\mathbf{q}^{||\vec{\lambda}||} \mathcal{Z}^U(\vec{\lambda}) \right)^{1/2} |\vec{\lambda}\rangle. \quad (5.34)$$

By the following identities:

$$\begin{aligned}
& \mathcal{F}(\mathcal{X}_\alpha(\square) | \lambda_\alpha \cup \square) \mathcal{F}(\mathcal{X}_\alpha(\square + \mathbf{1}) | \lambda_\alpha) \\
&= \mathcal{F}(\mathcal{X}_\alpha(\square) | \lambda_\alpha) \mathcal{F}(\mathcal{X}_\alpha(\square) | \square_\alpha) \text{sh}(\mathcal{X}_\alpha(\square) - \mathcal{X}_\alpha(\square)) \mathcal{F}(\mathcal{X}_\alpha(\square) + \epsilon_{12} | \lambda_\alpha) \\
&= -\mathcal{F}(\mathcal{X}_\alpha(\square) | \lambda_\alpha) \mathcal{F}(\mathcal{X}_\alpha(\square) + \epsilon_{12} | \square_\alpha) \text{sh}(\mathcal{X}_\alpha(\square) + \epsilon_{12} - \mathcal{X}_\alpha(\square)) \mathcal{F}(\mathcal{X}_\alpha(\square) + \epsilon_{12} | \lambda_\alpha) \\
&= -\mathcal{F}(\mathcal{X}_\alpha(\square) + \epsilon_{12} | \lambda_\alpha \cup \square) \mathcal{F}(\mathcal{X}_\alpha(\square) | \lambda_\alpha),
\end{aligned} \tag{5.35}$$

the recursion relation (5.33) can be rearranged to:

$$\frac{\mathcal{Z}^U(\lambda_\alpha \cup \square)}{\mathcal{F}(\mathcal{X}_\alpha(\square) + \epsilon_{12} | \lambda_\alpha \cup \square)} - \mathcal{Z}^U(\lambda_\alpha) \mathcal{F}(\mathcal{X}_\alpha(\square) | \lambda_\alpha) = 0. \tag{5.36}$$

This is precisely the vanishing condition for the A_N qq -character^[78]:

$$\chi(x) \equiv \frac{1}{\mathcal{F}(x + \epsilon_{12})} - \mathbf{q} \mathcal{F}(x), \tag{5.37}$$

where the operator $\mathcal{F}(x)$ is defined by:

$$\mathcal{F}(x) |\vec{\lambda}\rangle \equiv \prod_{\alpha=1}^N \mathcal{F}(x | \lambda_\alpha) |\vec{\lambda}\rangle. \tag{5.38}$$

One can verify directly that $\langle \chi(x) \rangle = \langle \mathfrak{G} | \chi(x) | \mathfrak{G} \rangle$ is a well-defined polynomial in x , which is the defining property of a qq -character. The operator $1/\mathcal{F}(x)$ corresponds precisely to the $\mathcal{Y}(x)$ observable in the conventions of^[78,82], and the $\mathcal{F}$ -factor therefore provides a natural generating function of the qq -character.

5.2.2 5d pure $SO(N)$ SYM

Next, we turn our attention to pure SYM theories for Lie groups of types B and D^[20]. We can express the instanton moduli space via the ADHM construction. First, since $SO(N)$ involves a symmetric bilinear form, to ensure the moment maps $\mu_{\mathbb{R}}$ and $\mu_{\mathbb{C}}$ are invariant under the gauge group $SO(N)$, the quotient group must incorporate an anti-symmetric bilinear form. That is, the quotient group is of type C: $Sp(2k) \subset U(2k)$ ^[21]. The resulting integral form of the instanton partition function is then given by:

$$\begin{aligned}
\mathcal{Z}_{N,k}^{\text{SO}} &= \frac{1}{k! 2^k} \prod_{i \neq j}^k \text{sh}(\phi_i - \phi_j) \prod_{i,j=1}^k \frac{\text{sh}(\phi_i - \phi_j - \epsilon_{12})}{\text{sh}(\phi_i - \phi_j - \epsilon_{1,2})} \frac{\prod_{i \leq j}^k \text{sh}(\pm(\phi_i + \phi_j)) \text{sh}(\pm(\phi_i + \phi_j) - \epsilon_{12})}{\prod_{i < j}^k \text{sh}(\pm(\phi_i + \phi_j) - \epsilon_{1,2})} \\
&\times \prod_{i=1}^k \frac{1}{\prod_{\alpha=1}^n \text{sh}(\pm\phi_i \pm v_\alpha - \frac{1}{2}\epsilon_{12})} \left(\frac{1}{\text{sh}(\pm\phi_i - \frac{1}{2}\epsilon_{12})} \right)^\chi
\end{aligned} \tag{5.39}$$

where $n = \lfloor \frac{N}{2} \rfloor$ is the rank of $SO(N)$, and $\chi = N \bmod 2$ labels the B and D type Lie group.

Classifying the poles of this integral has been a challenging problem. However, the *unrefined limit* $\epsilon_2 = -\epsilon_1$ simplifies matters considerably. In this limit, as in (5.27), the non-trivial poles simplify and are classified by 2d Young diagrams^[85]. In the unrefined limit, the factor $\mathcal{F}(\pm \mathcal{X}_\alpha | \lambda_\beta) / \text{sh}(\pm \mathcal{X}_\alpha + \mathcal{X}_\beta)$ produces singular terms of the form $\text{sh}(a \epsilon_{12}) / \text{sh}(b \epsilon_{12})$ when $\alpha = \beta$ for diagonal boxes (i.e., boxes $\mathbf{x} = (i, i)$) in λ_α ; the limit

$\lim_{\epsilon_2 \rightarrow -\epsilon_1}$ extracts the coefficient a/b , which plays a critical role in the subsequent analysis of $\text{Sp}(2N)$ SYM theory in Sec. 5.2.3. The partition function then takes the concise shell formula form:

$$\mathcal{Z}_{N=2n+\chi, k}^{\text{SO}}(v_1, \dots, v_n) = \lim_{\epsilon_2 \rightarrow -\epsilon_1} \sum_{||\vec{\lambda}||=k} \prod_{\alpha=1}^n \prod_{\mathbf{x} \in \lambda_\alpha} \frac{\text{sh}(2\mathcal{X}_\alpha(\mathbf{x}) + \epsilon_{1,2}) \text{sh}^2(2\mathcal{X}_\alpha(\mathbf{x}))}{\text{sh}^\chi(\pm \mathcal{X}_\alpha(\mathbf{x}))} \times \prod_{\beta=1}^n \frac{\mathcal{J}(\pm \mathcal{X}_\alpha(\mathbf{x}) | \lambda_\beta)}{\text{sh}(\pm \mathcal{X}_\alpha(\mathbf{x}) + \mathcal{X}_\beta(\mathbf{0}))} \quad (5.40)$$

In order to endow this shell formula with physical meaning, similar to that in Fig. 5-1, we first engineer this system using a 5-brane web in IIB theory^[86–88]. The construction requires $\lfloor \frac{N}{2} \rfloor$ D5-branes, 2 NS5-branes, and an O5-plane, with their orientations given in Tab. 5-1.

	$\mathbb{C}_1$		$\mathbb{C}_2$		x^5	x^6	x^7	x^8	$\mathbb{R} \times \mathbb{S}^1$	
	1	2	3	4	5	6	7	8	9	0
D1	•	•	•	•	-	•	•	•	•	-
D5	-	-	-	-	-	•	•	•	•	-
$\text{O5}^\pm, \widetilde{\text{O5}}^\pm$	-	-	-	-	-	•	•	•	•	-
NS5	-	-	-	-	•	-	•	•	•	-

Table 5-1 Brane configuration for 5d SO and Sp pure SYM.

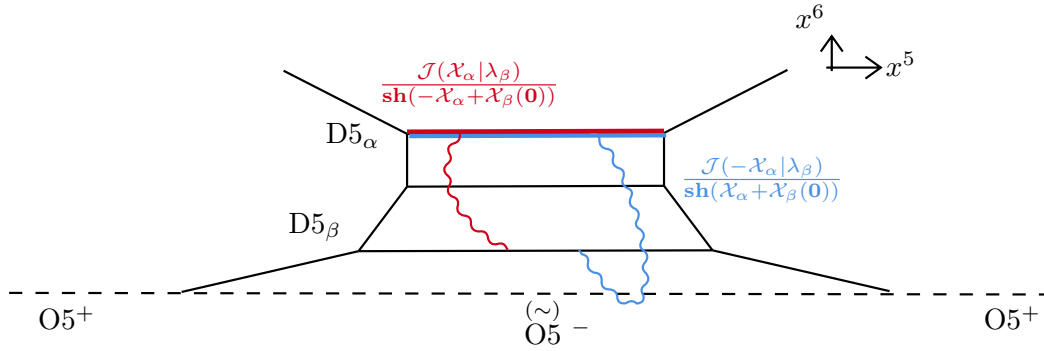


Figure 5-2 Fivebrane-web realization of the SO theory and the O5-plane image-string contribution.

The presence of the O5-plane causes strings winding around it to undergo an orientation reversal. Hence, such strings contribute an additional negative sign: $\mathcal{X}_\alpha \rightarrow -\mathcal{X}_\alpha$ to the partition function, as shown in Fig. 5-2.

5.2.3 5d pure $\text{Sp}(2N)$ SYM

The shell formula is especially powerful in analyzing the *BPS jumping phenomenon*^[69,85]. Let us consider the case of $\text{Sp}(2N)$. According to the ADHM construction, the quotient group for the $\text{Sp}(2N)$ instanton moduli space for k instantons is the orthogonal group $\text{O}(k)$. Furthermore, since $\pi_4(\text{Sp}(2N)) = \mathbb{Z}_2$ for 5d SYM, the 5d Sp SYM partition function naturally includes a discrete topological angle $\theta \in \{0, \pi\}$, corresponding

precisely to the two distinct components of the O group. Based on the ADHM data, the two instanton partition functions for the Sp group are:

$$\begin{aligned}\mathcal{Z}_{2N,k}^{\text{Sp},\theta=0} &= \mathcal{Z}_{2N,k}^{\text{Sp},+} + \mathcal{Z}_{2N,k}^{\text{Sp},-} \\ \mathcal{Z}_{2N,k}^{\text{Sp},\theta=\pi} &= \mathcal{Z}_{2N,k}^{\text{Sp},+} - \mathcal{Z}_{2N,k}^{\text{Sp},-}\end{aligned}\quad (5.41)$$

where the integrand of $\mathcal{Z}^{\text{Sp},\pm}$ are as follow^[21–23]:

$$\begin{aligned}\mathcal{J}_{2N,k}^{\text{Sp},+} &= \frac{1}{l! 2^{l+\chi}} \prod_{i \neq j}^l \text{sh}(\phi_i - \phi_j) \prod_{i,j=1}^l \frac{\text{sh}(\phi_i - \phi_j - \epsilon_{12})}{\text{sh}(\phi_i - \phi_j - \epsilon_{1,2})} \frac{\prod_{i < j}^l \text{sh}(\pm(\phi_i + \phi_j)) \text{sh}(\pm(\phi_i + \phi_j) - \epsilon_{12})}{\prod_{i \leq j}^l \text{sh}(\pm(\phi_i + \phi_j) - \epsilon_{1,2})} \\ &\quad \times \prod_{i=1}^l \frac{1}{\prod_{\alpha=1}^N \text{sh}(\pm\phi_i \pm v_\alpha - \frac{1}{2}\epsilon_{12})} \left(\frac{1}{\text{sh}(\epsilon_{1,2})} \frac{1}{\prod_{\alpha=1}^N \text{sh}(\pm v_\alpha - \frac{1}{2}\epsilon_{12})} \prod_{i=1}^l \frac{\text{sh}(\pm\phi_i) \text{sh}(\pm\phi_i - \epsilon_{12})}{\text{sh}(\pm\phi_i - \epsilon_{1,2})} \right)^\chi\end{aligned}\quad (5.42)$$

$$\begin{aligned}\mathcal{J}_{2N,k}^{\text{Sp},-} &= \frac{1}{(l-1+\chi)! 2^{l+\chi}} \prod_{i \neq j}^{l-1+\chi} \text{sh}(\phi_i - \phi_j) \prod_{i,j=1}^{l-1+\chi} \frac{\text{sh}(\phi_i - \phi_j - \epsilon_{12})}{\text{sh}(\phi_i - \phi_j - \epsilon_{1,2})} \frac{\prod_{i < j}^{l-1+\chi} \text{sh}(\pm(\phi_i + \phi_j)) \text{sh}(\pm(\phi_i + \phi_j) - \epsilon_{12})}{\prod_{i \leq j}^{l-1+\chi} \text{sh}(\pm(\phi_i + \phi_j) - \epsilon_{1,2})} \\ &\quad \times \prod_{i=1}^{l-1+\chi} \frac{1}{\prod_{\alpha=1}^N \text{sh}(\pm\phi_i \pm v_\alpha - \frac{1}{2}\epsilon_{12})} \left(\frac{1}{\text{sh}(\epsilon_{1,2})} \frac{1}{\prod_{\alpha=1}^N \text{ch}(\pm v_\alpha - \frac{1}{2}\epsilon_{12})} \prod_{i=1}^{l-1+\chi} \frac{\text{ch}(\pm\phi_i) \text{ch}(\pm\phi_i - \epsilon_{12})}{\text{ch}(\pm\phi_i - \epsilon_{1,2})} \right) \\ &\quad \times \left(\frac{\text{ch}(\epsilon_{12})}{\text{sh}(2\epsilon_{1,2})} \frac{1}{\prod_{\alpha=1}^N \text{sh}(\pm v_\alpha - \frac{1}{2}\epsilon_{12})} \prod_{i=1}^{l-1+\chi} \frac{\text{sh}(\pm\phi_i) \text{sh}(\pm\phi_i - \epsilon_{12})}{\text{sh}(\pm\phi_i - \epsilon_{1,2})} \right)^{1-\chi}\end{aligned}\quad (5.43)$$

where $l = \lfloor \frac{k}{2} \rfloor$ and $\chi \equiv k \pmod{2}$. Note that in our simplified notation, the expression written as $\mathcal{J}_{2N,k}^{\text{Sp},-}$ is not correct after applying the JK-residue when k is even. Fortunately, however, it is valid in the unrefined limit. Since the classification of poles remains unknown for the refined case, we exclusively focus on the unrefined limit.

In the unrefined limit, the poles are classified by $N+4$ Young diagrams^[85]. Among these, the first N Young diagrams are labeled by the N coulomb branch parameters $v_1, \dots, v_N$ in the partition function as (5.27). The additional four poles require distinct labeling according to the following Tab. 5-2.

	v_{N+1}	v_{N+2}	v_{N+3}	v_{N+4}
$+, k = \text{even}$	$\epsilon_1/2$	$\epsilon_1/2 + \pi i$	$\epsilon_{12}/2$	$\epsilon_{12}/2 + \pi i$
$+, k = \text{odd}$	$\epsilon_1/2$	$\epsilon_1/2 + \pi i$	ϵ_1	$\epsilon_{12}/2 + \pi i$
$-, k = \text{even}$	$\epsilon_1/2$	$\epsilon_1/2 + \pi i$	ϵ_1	$\epsilon_1 + \pi i$
$-, k = \text{odd}$	$\epsilon_1/2$	$\epsilon_1/2 + \pi i$	$\epsilon_{12}/2$	$\epsilon_1 + \pi i$

Table 5-2 The four additional poles required for the Sp instanton partition function.

The closed-form expression for the Sp(2N) plus sector is:

$$\mathcal{Z}_{2N,k=2l+\chi}^{\text{Sp},+}(v_1, \dots, v_N) = \lim_{\epsilon_2 \rightarrow -\epsilon_1} \sum_{||\vec{\lambda}||=l} \left(\prod_{\alpha=1}^{N+4} \prod_{\mathbf{x} \in \lambda_\alpha} \frac{\prod_{\beta=1}^{N+4} \mathcal{F}(\pm \mathcal{X}_\alpha(\mathbf{x}) | \lambda_\beta)}{\prod_{\beta=1}^N \text{sh}(\pm \mathcal{X}_\alpha(\mathbf{x}) + \mathcal{X}_\beta(\mathbf{0}))} \right) \left(\frac{\prod_{\alpha=1}^{N+4} \mathcal{F}(0 | \lambda_\alpha)}{\prod_{\alpha=1}^N \text{sh}(0 + \mathcal{X}_\alpha(\mathbf{0}))} \right)^\chi \quad (5.44)$$

where, as Tab. 5-2 shows, we need to identify $v_{N+1} = \frac{1}{2}\epsilon_1$, $v_{N+2} = \frac{1}{2}\epsilon_1 + \pi i$, $v_{N+3} = \chi \epsilon_1 + \frac{1}{2}(1-\chi)\epsilon_{12}$ and $v_{N+4} = \frac{1}{2}\epsilon_{12} + \pi i$.

We remark that without using the $\lim_{\epsilon_2 \rightarrow -\epsilon_1}$, the BPS jumping coefficients $C_{\vec{\lambda},v}^{\text{Sp}}$ must be manually included in each term of the summation:

$$\mathcal{Z}_{2N,k}^{\text{Sp},\pm} = \sum_{||\vec{\lambda}||=l} C_{\vec{\lambda},v}^{\text{Sp}} \mathcal{Z}^{\text{Sp},\pm}(\vec{\lambda}), \quad C_{\vec{\lambda},v}^{\text{Sp}} = \prod_{\alpha=N+1}^{N+4} C_{\lambda_\alpha, v_\alpha}^{\text{Sp}} \quad (5.45)$$

These coefficients $C_{\vec{\lambda}, \mathbf{v}}^{\text{Sp}}$ depend on the specific shapes of the Young diagrams $\vec{\lambda}$ and the corresponding Coulomb branch parameters $\mathbf{v}$: $C_{\emptyset, \mathbf{v}_\alpha}^{\text{Sp}} = 1$, and

$$\begin{aligned} C_{\lambda_\alpha, \mathbf{v}_\alpha=0, \pi i, \frac{\epsilon_1}{2}, \frac{\epsilon_1}{2} + \pi i}^{\text{Sp}} &= \frac{2^{2j-1}}{\binom{2j-1}{j-1}}, \quad \text{where } j \text{ is number of diagonal boxes } \mathbf{x} = (i, i) \text{ in } \lambda_\alpha, \\ C_{\lambda_\alpha, \mathbf{v}_\alpha=\epsilon_1, \epsilon_1 + \pi i}^{\text{Sp}} &= \frac{2^{2j}}{\binom{2j+1}{j}}, \quad \text{where } j \text{ is number of superdiagonal boxes } \mathbf{x} = (i, i+1) \text{ in } \lambda_\alpha. \end{aligned} \quad (5.46)$$

Consequently, the introduction of these extra coefficients obstructs the computation of the topological vertex for the O^+ -plane^[89] and the derivation of the algebraic properties of the partition functions. Fortunately, by employing the shell formula with the unrefined limit $\lim_{\epsilon_2 \rightarrow -\epsilon_1}$, these coefficients are fully absorbed into the limiting procedure. Detailed calculations in Appendix C.1.2 demonstrate how these coefficients arise.

Similarly, the minus sector for $\text{Sp}(2N)$ can be expressed through analogous formulas:

$$\begin{aligned} \mathcal{Z}_{2N, k=2l+\chi}^{\text{Sp}, -}(v_1, \dots, v_N) &= \lim_{\epsilon_2 \rightarrow -\epsilon_1} \sum_{||\vec{\lambda}||=l-1+\chi} \left(\prod_{\alpha=1}^{N+4} \prod_{\mathbf{x} \in \lambda_\alpha} \frac{\prod_{\beta=1}^{N+4} \mathcal{F}(\pm \mathcal{X}_\alpha(\mathbf{x}) | \lambda_\beta)}{\prod_{\beta=1}^N \text{sh}(\pm \mathcal{X}_\alpha(\mathbf{x}) + \mathcal{X}_\beta(\mathbf{0}))} \right) \\ &\quad \times \left(\frac{\prod_{\alpha=1}^{N+4} \mathcal{F}(\pi i | \lambda_\alpha)}{\prod_{\alpha=1}^N \text{sh}(-\pi i + \mathcal{X}_\alpha(\mathbf{0}))} \right) \left(\frac{\prod_{\alpha=1}^{N+4} \mathcal{F}(0 | \lambda_\alpha)}{\prod_{\alpha=1}^N \text{sh}(0 + \mathcal{X}_\alpha(\mathbf{0}))} \right)^{1-\chi} \end{aligned} \quad (5.47)$$

where we need to impose the extra poles conditions as Tab. 5-2, $v_{N+1} = \frac{1}{2}\epsilon_1$, $v_{N+2} = \frac{1}{2}\epsilon_1 + \pi i$, $v_{N+3} = (1 - \chi)\epsilon_1 + \frac{1}{2}\chi\epsilon_{12}$ and $v_{N+4} = \epsilon_1 + \pi i$.

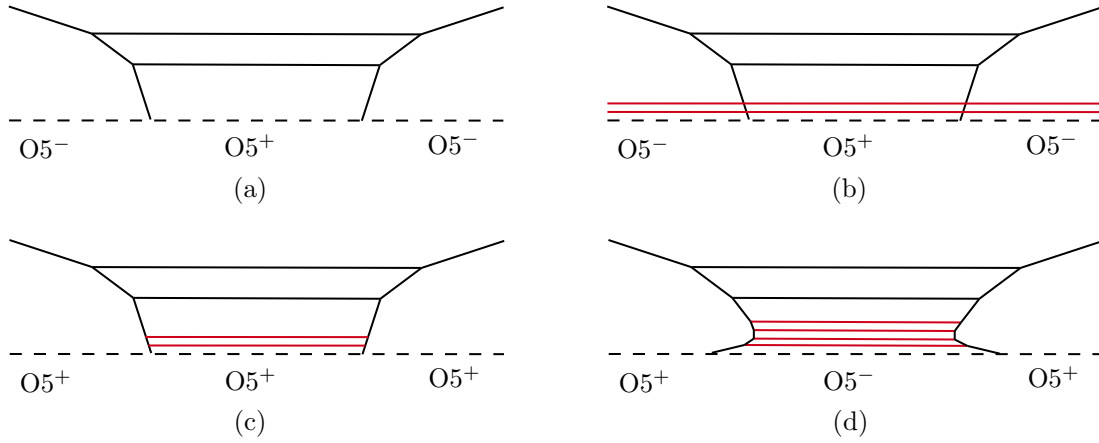
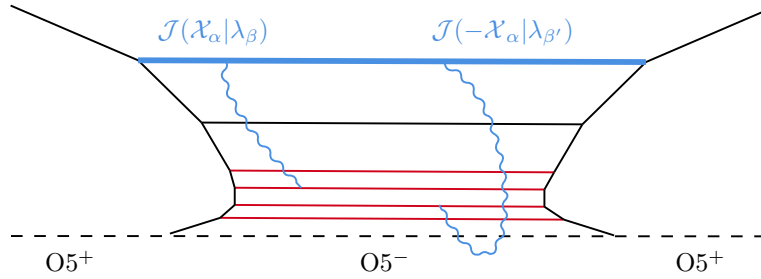
To provide a physical interpretation for these four additional fixed Coulomb branch parameters v_{N+1} to v_{N+4} , we construct the Sp theory using a five-brane web and compare it with the SO theory. As illustrated in Tab. 5-1, the Sp theory is realized with an O5 -plane^[69,88]. Analysis based on RR charge and monodromy reveals that an Op^+ -plane is effectively equivalent to an Op^- -plane plus 2^{p-4} Dp-branes. These Dp-branes are frozen near the orientifold plane and must acquire specific VEVs; hence, an O5^+ -plane corresponds approximately to an O5^- -plane together with two frozen D5-branes. Accordingly, as depicted in Fig. 5-3, the $\text{Sp}(2N)$ theory is related to the $\text{SO}(2N + 8)$ theory by incorporating 4 D5-branes with specific VEVs.

As shown in Fig. 5-4, the effective contribution of the strings connecting the D1-brane to the four frozen D5-branes is $\mathcal{F}(\pm \mathcal{X}_\alpha | \lambda_\beta)$, which differs from the contribution of ordinary strings.

We briefly summarize the contributions of these various types of strings in the Tab. 5-3.

We can check the Lie algebra-theoretic relations of instanton partition functions. The isomorphisms $\text{Sp}(2) \simeq \text{SU}(2)$ and $\text{Sp}(4) \simeq \text{SO}(5)$ of Lie algebras lead to the equality of the partition functions:

$$\begin{aligned} \mathcal{Z}_{2,k}^{\text{SU}}(v_1) &= \mathcal{Z}_{2,k}^{\text{Sp}, \theta=0}(v_1) \\ \mathcal{Z}_{5,k}^{\text{SO}}(v_1 + v_2, v_1 - v_2) &= \mathcal{Z}_{4,k}^{\text{Sp}, \theta=0}(v_1, v_2) \end{aligned} \quad (5.48)$$

Figure 5-3 Brane move relating the $\mathrm{Sp}(2N)$ web to an $\mathrm{SO}(2N+8)$ web with four frozen D5-branes.Figure 5-4 String contributions between D1-branes and the frozen D5-branes in the Sp construction.

5.3 Gauge origami

In this section, we consider a more general setup called gauge origami^[9]. Although the systems treated below appear at first to be distinct, they are hierarchically related through tachyon condensation. A $\mathrm{D8}-\overline{\mathrm{D8}}$ pair condenses into a single D6-brane when the separation between them is tuned to ϵ_4 , restricting the relevant 4d Young diagrams to have at most one layer in the condensed direction and yielding 3d Young diagrams. A further $\mathrm{D6}-\overline{\mathrm{D6}}$ condensation produces a D4-brane and reduces the combinatorics to 2d Young diagrams. Schematically:

$$\mathrm{D0-D8} \xrightarrow{\mathrm{D8}-\overline{\mathrm{D8}} \text{ condensation}} \mathrm{D0-D6} \xrightarrow{\mathrm{D6}-\overline{\mathrm{D6}} \text{ condensation}} \mathrm{D0-D4}. \quad (5.49)$$

The DT3 and DT4 counting problems arise within this hierarchy by placing the D0-brane system on top of a fixed vacuum configuration—a minimal plane or solid partition—determined by D2- and D4-brane boundary conditions. The shell formula provides a unified treatment of all levels of this hierarchy. The following subsections give the shell formula for the gauge origami systems on $\mathbb{C}^4 \times \mathbb{R}^1 \times S^1$, including the D0-D8 system known as the magnificent four^[10–11,52], the D0-D6 system known as tetrahedron instantons^[12,90], the D0-D4 system known as spiked instantons^[9,28,78], the D0-D2-D6 system known as DT3 counting^[82,91–92], and the D0-D2-D4-D8 system known as DT4 counting^[13,93–95]. These systems are further interconnected by introducing appropriate antibranes^[10,96–97], whose tachyon condensation provides a physical mechanism relating different brane configurations within a unified framework. The formulas in this

Type	Contribution
$D1_{\alpha,x}$ – $D5_\beta$ strings	$\frac{\mathcal{F}(\mathcal{X}_\alpha(\mathbf{x}) \lambda_\beta)}{\text{sh}(-\mathcal{X}_\alpha(\mathbf{x}) + \mathcal{X}_\beta(\mathbf{0}))}$
O-plane projected $D1_{\alpha,x}$ – $D5_\beta$ strings	$\frac{\mathcal{F}(-\mathcal{X}_\alpha(\mathbf{x}) \lambda_\beta)}{\text{sh}(\mathcal{X}_\alpha(\mathbf{x}) + \mathcal{X}_\beta(\mathbf{0}))}$
$D1_{\alpha,x}$ –frozen $D5_\beta$ strings	$\mathcal{F}(\mathcal{X}_\alpha(\mathbf{x}) \lambda_\beta)$
O-plane projected $D1_{\alpha,x}$ –frozen $D5_\beta$ strings	$\mathcal{F}(-\mathcal{X}_\alpha(\mathbf{x}) \lambda_\beta)$

Table 5-3 The four types of string configuration that appear with their contributions. The absence of a denominator factor for frozen D5-branes is consistent with the fact that frozen branes do not introduce dynamical Coulomb branch parameters; consequently, they do not contribute fundamental multiplets in the effective quantum mechanics.

section should be read with the following scope in mind: some sign choices are conventions fixed by comparison with the brane index, and some multiplicity constants in refined 4d Young-diagram sums are not known in closed form. Whenever the shell formula absorbs such factors only after the unrefined or Calabi–Yau limit, this limitation is stated explicitly.

5.3.1 Magnificent four

Nekrasov introduced the D0–D8 system on a CY fourfold and dubbed the resulting BPS counting problem the magnificent four^[10]. To investigate the distribution of bound states in SUSY QM on D0-branes, we analyze the energy spectrum of this system. The brane configuration is given in Tab. 5-4; D8-branes and anti-D8-branes wrap the directions $\mathbb{C}_{1,2,3,4}$ and are compactified on a circle S^1 along the x^0 direction.

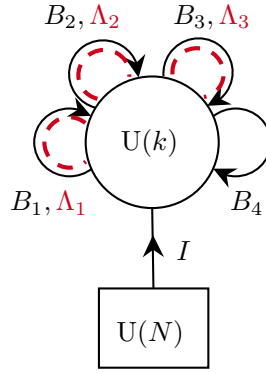
	$\mathbb{C}_1$		$\mathbb{C}_2$		$\mathbb{C}_3$		$\mathbb{C}_4$		$\mathbb{R} \times S^1$	
	1	2	3	4	5	6	7	8	9	0
k D0	•	•	•	•	•	•	•	•	•	–
N D8	–	–	–	–	–	–	–	–	•	–
N $\overline{\text{D8}}$	–	–	–	–	–	–	–	–	•	–

Table 5-4 Brane configuration of the magnificent four. The symbol – denotes an extended direction of the D-branes, whereas • denotes a point-like direction. The D8-branes and anti-D8-branes extend along four complex directions $\mathbb{C}_{1,2,3,4}$ and the time direction x^0 , while the D0-branes, representing instanton probes, extend only along the time direction.

For simplicity, we first consider the case without anti-D8 branes. In the presence of a B-field^[50], the energy spectrum of the D0–D8 system includes four complex adjoint chiral multiplets $B_{1,2,3,4}$ of 1d $\mathcal{N} = 2$ SUSY QM arising from excitations of strings connecting D0–D0, and a fundamental chiral multiplet I arising from excitations of strings connecting D0–D8. The corresponding 1d $\mathcal{N} = 2$ quiver diagram is given in Fig. 5-5.

The moduli space corresponding to this quiver provides the ADHM data for this D0–D8 system as:

$$\mathcal{M}_{N,k}^{\text{D0-D8}} = \{(\mathbf{B}, I) | \mu_{\mathbb{R}} - \zeta \cdot \mathbf{1}_k = \mu_{ab \in \underline{6}} = 0\} / \text{U}(k) \quad (5.50)$$

Figure 5-5 Quiver diagram of the D0-D8 system as $\mathcal{N} = 2$ supersymmetric quantum mechanics.

where the moment maps are defined as:

$$\mu_{\mathbb{R}} = \sum_{a \in \underline{4}} [B_a, B_a^\dagger] + I \cdot I^\dagger$$

$$\mu_{ab} = [B_a, B_b]$$

Similar to the case of pure SYM, where the D0-D4 system is placed in an Ω -background with $U(1)_{\epsilon_1} \times U(1)_{\epsilon_2}$ symmetry, we need to place the entire D0-D8 system in a background with $SU(4)$ symmetry. This is equivalent to considering the system in a space-time with $SU(4)$ holonomy^[98]. Consequently, the charges of the various fields under the $U(1)^3 \subset SU(4)$ transformation are respectively:

$$\begin{pmatrix} B_1, B_2, B_3, B_4 \\ I, \Lambda_1, \Lambda_2, \Lambda_3 \end{pmatrix} \xrightarrow{U(1)_{\epsilon_1} \times U(1)_{\epsilon_2} \times U(1)_{\epsilon_3}} \begin{pmatrix} q_1^{-1} B_1, q_2^{-1} B_2, q_3^{-1} B_3, q_{123} B_4 \\ I, q_{23} \Lambda_1, q_{13} \Lambda_2, q_{12} \Lambda_3 \end{pmatrix} \quad (5.51)$$

The D0-D8 partition function is:

$$\mathcal{Z}_{N,k}^{\text{D0-D8}}(v_{\underline{4},1}, \dots, v_{\underline{4},N}) = \oint_{\text{JK}} \prod_{i=1}^k \frac{d\phi_i}{2\pi i} \mathcal{J}_{N,k}^{\text{D0-D8}}$$

$$\mathcal{J}_{N,k}^{\text{D0-D8}} = \mathcal{J}_k^{\text{D0-D0}} \times \prod_{i=1}^k \prod_{\alpha=1}^N \frac{1}{\text{sh}(\phi_i - v_{\underline{4},\alpha})}$$

$$\mathcal{J}_k^{\text{D0-D0}} = \frac{1}{k!} \prod_{i \neq j}^k \text{sh}(\phi_i - \phi_j) \prod_{i,j}^k \frac{\text{sh}(\phi_i - \phi_j - \epsilon_{1,2,3} - \epsilon_4)}{\text{sh}(\phi_i - \phi_j - \epsilon_{1,2,3,4})} \quad (5.52)$$

where we impose the CY four-fold condition $\epsilon_4 = -\epsilon_{123}$.

The poles of the D0-D8 system are classified by N 4d Young diagrams, and a closed formula follows from the shell formula. Note, however, that this partition function (5.52) differs from the expansion of the $\mathcal{J}$ -factor (5.10) under 4d Young diagrams $\rho_{\underline{4}}$:

$$\begin{aligned} \prod_{x \in \rho_{\underline{4}}} \mathcal{J}(\mathcal{X}_{\underline{4}}(x) | \rho_{\underline{4}}) &= \prod_{x \in \rho_{\underline{4}}} \frac{1}{\text{sh}(\mathcal{X}_{\underline{4}}(x) - \mathcal{X}_{\underline{4}}(\mathbf{1}))} \\ &\quad \times \prod_{y \in \rho_{\underline{4}}} \frac{\text{sh}(\mathcal{X}_{\underline{4}}(x) - \mathcal{X}_{\underline{4}}(y)) \text{sh}(\mathcal{X}_{\underline{4}}(x) - \mathcal{X}_{\underline{4}}(y) - \epsilon_{\underline{4}}) \prod_{ab \in \underline{6}} \text{sh}(\mathcal{X}_{\underline{4}}(x) - \mathcal{X}_{\underline{4}}(y) - \epsilon_{ab})}{\prod_{a \in \underline{4}} \text{sh}(\mathcal{X}_{\underline{4}}(x) - \mathcal{X}_{\underline{4}}(y) - \epsilon_a) \prod_{A \in \underline{4}} \text{sh}(\mathcal{X}_{\underline{4}}(x) - \mathcal{X}_{\underline{4}}(y) - \epsilon_A)} \\ &= \prod_{x \in \rho_{\underline{4}}} \frac{1}{\text{sh}(\mathcal{X}_{\underline{4}}(x) - \mathcal{X}_{\underline{4}}(\mathbf{1}))} \prod_{y \in \rho_{\underline{4}}} \left(\frac{\text{sh}(\mathcal{X}_{\underline{4}}(x) - \mathcal{X}_{\underline{4}}(y)) \text{sh}(\mathcal{X}_{\underline{4}}(x) - \mathcal{X}_{\underline{4}}(y) - \epsilon_{1,2,3} - \epsilon_4)}{\text{sh}(\mathcal{X}_{\underline{4}}(x) - \mathcal{X}_{\underline{4}}(y) - \epsilon_{1,2,3,4})} \right)^2 \end{aligned} \quad (5.53)$$

This expression equals the square of the D0-D8 integrand, as can be seen by comparison with (5.52). Therefore, if we use the original definition (5.9) to express the D0-D8 partition function, it is necessary to take the square root of the $\mathcal{F}$ -factor, and each term in the summation will exhibit an ambiguous sign. Hence, to obtain a canonical sign choice and a well-defined formula, we define a modified $\mathcal{F}$ -factor that selects only those shell boxes whose last coordinate satisfies $x_d \geq y_d$, denoted $\mathcal{F}_{\geq}$:

$$\mathcal{F}_{\geq}(\mathcal{X}_{\mathcal{B}}(\mathbf{x})|\rho_{\mathcal{A}}) \equiv \prod_{\substack{\mathbf{y} \in \mathcal{S}(\rho_{\mathcal{A}}) \\ x_d \geq y_d}} \text{sh}(\mathcal{X}_{\mathcal{B}}(\mathbf{x}) - \mathcal{X}_{\mathcal{A}}(\mathbf{y}))^{Q_{\rho_{\mathcal{A}}}(\mathbf{y})} \quad (5.54)$$

where we compare the last coordinates x_d and y_d of two boxes and only select the contribution from boxes where $x_d \geq y_d$. This definition selects precisely the shell boxes needed and yields the correct sign. The choice of $\mathcal{F}_{\geq}$ is a sign convention adapted to the D0-D8 brane index; other conventions must be accompanied by the corresponding orientation signs. In refined DT4-type problems, additional orientation or multiplicity data may be required. The formula below is the convention used throughout this thesis and is checked explicitly in Appendix C.1.5. The D0-D8 partition function is therefore:

$$\mathcal{Z}_{N,k}^{\text{D0-D8}}(v_{\underline{4},1}, \dots, v_{\underline{4},N}) = \sum_{||\vec{\rho}||=k} \mathcal{Z}^{\text{D0-D8}}(\vec{\rho}) = \sum_{||\vec{\rho}||=k} \prod_{\alpha,\beta=1}^N \prod_{\mathbf{x} \in \rho_{\underline{4},\alpha}} \mathcal{F}_{\geq}(\mathcal{X}_{\underline{4},\alpha}(\mathbf{x})|\rho_{\underline{4},\beta}) \quad (5.55)$$

where $\vec{\rho}$ denotes an N -tuple of 4d Young diagrams with $||\vec{\rho}|| = k$. As in Fig. 5-1, the contribution from strings connecting D0-D8 is:

$$\mathcal{F}_{\geq}(\mathcal{X}_{\underline{4},\alpha}(\mathbf{x})|\rho_{\underline{4},\beta}) \quad (5.56)$$

If we consider an equal number of anti-D8-branes, we need to replace the flavor group with $U(N|N)$ [99]. Effectively, this is equivalent to adding an equal number of Fermi multiplets to the SUSY QM; at the level of the partition function, this amounts to including the corresponding Fermi multiplet contributions:

$$\begin{aligned} \mathcal{Z}_{N,k}^{\text{D0-D8-}\overline{\text{D8}}} &= \mathcal{Z}_{N,k}^{\text{D0-D8}} \times \prod_{i=1}^k \prod_{\alpha=1}^N \text{sh}(\phi_i - w_{\underline{4},\alpha}) \\ \mathcal{Z}_{N,k}^{\text{D0-D8-}\overline{\text{D8}}}(\{v_{\underline{4},\alpha}\}, \{w_{\underline{4},\alpha}\}) &= \sum_{||\vec{\rho}||=k} \prod_{\alpha,\beta=1}^N \prod_{\mathbf{x} \in \rho_{\underline{4},\alpha}} \text{sh}(-\mathcal{X}_{\underline{4},\alpha}(\mathbf{x}) + w_{\underline{4},\beta}) \mathcal{F}_{\geq}(\mathcal{X}_{\underline{4},\alpha}(\mathbf{x})|\rho_{\underline{4},\beta}) \end{aligned} \quad (5.57)$$

The validity of this formula can be verified by computing the plethystic exponent (PE) expression [11]:

$$\mathcal{Z}_N^{\text{D0-D8-}\overline{\text{D8}}}(\{v_{\underline{4},\alpha}\}, \{w_{\underline{4},\alpha}\}) = \sum_{k=0}^{\infty} \mathbf{q}^k \mathcal{Z}_{N,k}^{\text{D0-D8-}\overline{\text{D8}}} = \text{PE} \left(\frac{\text{sh}(\epsilon_{12,13,23})}{\text{sh}(\epsilon_{1,2,3,4})} \frac{\text{sh}(s)}{\text{sh}(p \pm \frac{1}{2}s)} \right) \quad (5.58)$$

where $s \equiv \sum_{i=1}^N (v_{\underline{4},i} - w_{\underline{4},i})$, and $\mathbf{q} \equiv e^{-p}$. The PE operation is defined as:

$$\text{PE } f(x_1, \dots, x_r) \equiv \exp \sum_{m=1}^{\infty} \frac{1}{m} f(m x_1, \dots, m x_r) \quad (5.59)$$

The recursion relation in the D0-D8 system, analogous to (5.33), describes the contribution from adding a 4d box $\square$ to a 4d Young diagram $\rho_{\underline{4},\alpha}$:

$$\frac{\mathcal{Z}^{\text{D0-D8}}(\rho_{\underline{4},\alpha} \cup \square)}{\mathcal{Z}^{\text{D0-D8}}(\rho_{\underline{4},\alpha})} = \mathcal{F}_{\geq}(\mathcal{X}_{\underline{4},\alpha}(\square) | \rho_{\underline{4},\alpha} \cup \square) \mathcal{F}_{<}(\mathcal{X}_{\underline{4},\alpha}(\square) | \rho_{\underline{4},\alpha}) \quad (5.60)$$

Here, $\square$ refers to the coordinate of the 4d box being added. The function $\mathcal{F}_{<}$ is defined analogously to $\mathcal{F}_{\geq}$ in (5.54), but with the inequality $x_4 \geq y_4$ replaced by $x_4 < y_4$. And we use the following identities:

$$\frac{\mathcal{F}_{\geq}(\mathcal{X}_{\beta}(\mathbf{x}) | \rho_{\alpha} \cup \{\mathbf{y}\}_{\alpha})}{\mathcal{F}_{\geq}(\mathcal{X}_{\beta}(\mathbf{x}) | \rho_{\alpha})} = \begin{cases} \frac{\text{sh}(\mathcal{X}_{\beta}(\mathbf{x}) - \mathcal{X}_{\alpha}(\mathbf{y})) \text{sh}(\mathcal{X}_{\beta}(\mathbf{x}) - \mathcal{X}_{\alpha}(\mathbf{y}) - \epsilon_{1234}) \prod_{ab \in \underline{6}} \text{sh}(\mathcal{X}_{\beta}(\mathbf{x}) - \mathcal{X}_{\alpha}(\mathbf{y}) - \epsilon_{ab})}{\text{sh}(\mathcal{X}_{\beta}(\mathbf{x}) - \mathcal{X}_{\alpha}(\mathbf{y}) - \epsilon_{1,2,3,4}) \prod_{A \in \underline{6}} \text{sh}(\mathcal{X}_{\beta}(\mathbf{x}) - \mathcal{X}_{\alpha}(\mathbf{y}) - \epsilon_A)}, & x_4 > y_4 \\ \frac{\text{sh}(\mathcal{X}_{\beta}(\mathbf{x}) - \mathcal{X}_{\alpha}(\mathbf{y})) \text{sh}(\mathcal{X}_{\beta}(\mathbf{x}) - \mathcal{X}_{\alpha}(\mathbf{y}) - \epsilon_{14,24,34})}{\text{sh}(\mathcal{X}_{\beta}(\mathbf{x}) - \mathcal{X}_{\alpha}(\mathbf{y}) - \epsilon_{1,2,3,4})}, & x_4 = y_4 \\ 1, & x_4 < y_4 \end{cases}$$

$$\mathcal{F}_{<}(\mathcal{X}_{\beta}(\mathbf{x}) | \rho_{\alpha}) = \frac{\mathcal{F}(\mathcal{X}_{\beta}(\mathbf{x}) | \rho_{\alpha})}{\mathcal{F}_{\geq}(\mathcal{X}_{\beta}(\mathbf{x}) | \rho_{\alpha})} \quad (5.61)$$

To illustrate its validity, we provide examples in the Appendix C.1.5.

5.3.2 Tetrahedron instanton

We now consider a system whose fixed points are classified by 3d Young diagrams: the D0-D6 system, also known as tetrahedron instantons. This system was first studied in detail by Pomoni, Yan, and Zhang^[12,90] in type IIB string theory using a D1-D7 system. After T-dualizing along x^9 , the corresponding brane configuration is listed in Tab. 5-5. The ADHM data of this system can be represented by an $\mathcal{N} = 2$ SUSY QM

	$\mathbb{C}_1$		$\mathbb{C}_2$		$\mathbb{C}_3$		$\mathbb{C}_4$		$\mathbb{R} \times \mathbb{S}^1$	
	1	2	3	4	5	6	7	8	9	0
k D0	•	•	•	•	•	•	•	•	•	-
N_4^- D6 $_{\bar{4}}$	-	-	-	-	-	-	•	•	•	-
N_3^- D6 $_{\bar{3}}$	-	-	-	-	•	•	-	-	•	-
N_2^- D6 $_{\bar{2}}$	-	-	•	•	-	-	-	-	•	-
N_1^- D6 $_{\bar{1}}$	•	•	-	-	-	-	-	-	•	-

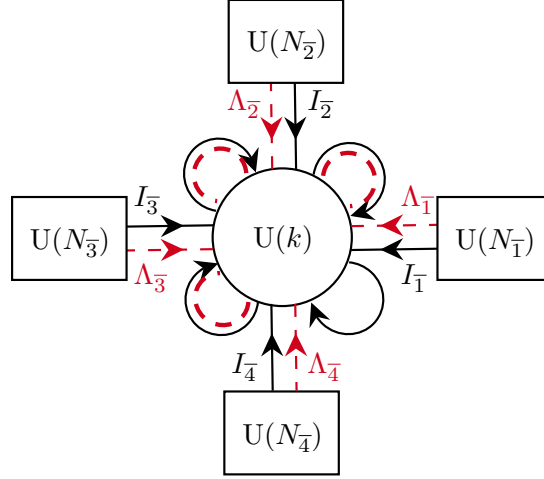
Table 5-5 Brane configuration of tetrahedron instantons.

quiver diagram. It includes all adjoint fields $B_{1,2,3,4}$ from the magnificent four in Fig. 5-5 as well as $\Lambda_{1,2,3}$. It also contains four distinct fundamental chiral $I_{\bar{4},\bar{3},\bar{2},\bar{1}}$ and Fermi multiplets $\Lambda_{\bar{4},\bar{3},\bar{2},\bar{1}}$, corresponding to the four different D6-branes. The transformations of these fields under the $U(1)^3$ symmetries are listed as

$$\begin{pmatrix} I_{\bar{4}}, I_{\bar{3}}, I_{\bar{2}}, I_{\bar{1}} \\ \Lambda_{\bar{1}}, \Lambda_{\bar{2}}, \Lambda_{\bar{3}}, \Lambda_{\bar{4}} \end{pmatrix} \xrightarrow{U(1)_{\epsilon_1} \times U(1)_{\epsilon_2} \times U(1)_{\epsilon_3}} \begin{pmatrix} I_{\bar{4}}, I_{\bar{3}}, I_{\bar{2}}, I_{\bar{1}} \\ q_1^{-1} \Lambda_{\bar{1}}, q_2^{-1} \Lambda_{\bar{2}}, q_3^{-1} \Lambda_{\bar{3}}, q_{123} \Lambda_{\bar{4}} \end{pmatrix} \quad (5.62)$$

The ADHM equations of tetrahedron instantons are:

$$\mathcal{M}_{N,k}^{\text{D0-D6}} = \{(\mathbf{B}, \mathbf{I}) | \mu_{\mathbb{R}} - \zeta \cdot \mathbf{1}_k = \mu_{ab \in \underline{6}} = \sigma_{\bar{a}} = 0\} / U(k) \quad (5.63)$$

Figure 5-6 Quiver diagram of the D0-D6 system as $\mathcal{N} = 2$ supersymmetric quantum mechanics.

where the moment maps are defined as:

$$\begin{aligned}
 \mu_{\mathbb{R}} &= \sum_{a \in \underline{4}} [B_a, B_a^\dagger] + I_{\bar{a}} I_{\bar{a}}^\dagger \\
 \mu_{ab} &= [B_a, B_b] \\
 \sigma_{\bar{a}} &= B_a I_{\bar{a}} \quad (\text{superpotential F-term coupling the adjoint and fundamental fields})
 \end{aligned} \tag{5.64}$$

Thus, the partition function is expressed as:

$$\mathcal{Z}_{N,k}^{\text{D0-D6}} = \oint_{\text{JK}} \prod_{i=1}^k \frac{d\phi_i}{2\pi i} \mathcal{J}_{N,k}^{\text{D0-D6}}, \quad \mathcal{J}_{N,k}^{\text{D0-D6}} = \mathcal{J}_k^{\text{D0-D0}} \times \prod_{i=1}^k \prod_{\mathcal{A}} \frac{\text{sh}(\phi_i - v_{\mathcal{A}} + \epsilon_{\mathcal{A}})}{\text{sh}(\phi_i - v_{\mathcal{A}})} \tag{5.65}$$

where $\mathcal{A} = (A, \alpha) \in \{(\bar{4}, 1), \dots, (\bar{4}, N_{\bar{4}}), (\bar{3}, 1), \dots, (\bar{1}, N_{\bar{1}})\}$ label each individual D6-brane, and $\mathbf{N} = (N_{\bar{4}}, N_{\bar{3}}, N_{\bar{2}}, N_{\bar{1}})$ denote the numbers of D6-branes in the four distinct orientations. $\mathcal{J}_k^{\text{D0-D0}}$ is the contribution from the D0-D0 strings in (5.52). After performing the JK-residue integral, the poles of this system are classified by a set $\{\pi_{\mathcal{A}}\}$ of 3d Young diagrams in the four different directions. The shell formula then gives:

$$\begin{aligned}
 \mathcal{Z}_{N,k}^{\text{D0-D6}} &= \sum_{||\vec{\pi}||=k} \left(\prod_{\mathcal{A}, \mathcal{B}} \prod_{\mathbf{x} \in \pi_{\mathcal{A}}} \text{sh}(\mathcal{X}_{\mathcal{A}}(\mathbf{x}) - \mathcal{X}_{\mathcal{B}}(\mathbf{0})) \mathcal{J}(\mathcal{X}_{\mathcal{A}}(\mathbf{x}) | \pi_{\mathcal{B}}) \right) \\
 &\quad \times \left(\prod_{\mathcal{A}, \mathcal{B}} \prod_{\substack{\mathbf{x} \in \pi_{\mathcal{A}} \\ \mathbf{y} \in \pi_{\mathcal{B}}}} \frac{\text{sh}(\mathcal{X}_{\mathcal{A}}(\mathbf{x}) - \mathcal{X}_{\mathcal{B}}(\mathbf{y}) + \epsilon_B)}{\text{sh}(\mathcal{X}_{\mathcal{A}}(\mathbf{x}) - \mathcal{X}_{\mathcal{B}}(\mathbf{y}) + \epsilon_A)} \right) \left(\prod_{\substack{\mathcal{A} < \mathcal{B} \\ \mathbf{y} \in \pi_{\mathcal{B}}}} \prod_{\substack{a \in A \\ b \notin B}} \frac{\text{sh}(\mathcal{X}_{\mathcal{A}}(\mathbf{x}) - \mathcal{X}_{\mathcal{B}}(\mathbf{y}) + \epsilon_{ab})}{\text{sh}(\mathcal{X}_{\mathcal{A}}(\mathbf{x}) - \mathcal{X}_{\mathcal{B}}(\mathbf{y}) - \epsilon_{ab})} \right)
 \end{aligned} \tag{5.66}$$

where the order $A < B$ is defined by the canonical ordering $123 < 124 < 134 < 234$. The second factor in (5.66) can be interpreted as an additional contribution arising from the interaction between 3d Young diagrams in different directions.

The recursion relation (5.17) directly gives the contribution from adding a D0-brane at location $\square$:

$$\frac{\mathcal{Z}^{\text{D0-D6}}(\pi_{\mathcal{A}} \cup \square)}{\mathcal{Z}^{\text{D0-D6}}(\pi_{\mathcal{A}})} = \frac{\mathcal{J}(\mathcal{X}_{\mathcal{A}}(\square) | \pi_{\mathcal{A}} \cup \square)}{\mathcal{J}(\mathcal{X}_{\mathcal{A}}(\square + \mathbf{1}) | \pi_{\mathcal{A}})} \quad (5.67)$$

Detailed calculations for several illustrative examples are collected in Appendix C.1.3.

The similarity between the D0-D6 system (5.65) and the D0-D8- $\overline{\text{D8}}$ system (5.57) is visible at the level of the integrand. Indeed, D6-branes arise from tachyon condensation between D8 and $\overline{\text{D8}}$ branes^[10,96–97]. For instance, considering a system with only one pair of D8- $\overline{\text{D8}}$ -branes, its integrand is given by:

$$\mathcal{J}_{1,k}^{\text{D0-D8-}\overline{\text{D8}}} = \mathcal{J}_k^{\text{D0-D0}} \times \prod_{i=1}^k \frac{\text{sh}(\phi_i - w_{\underline{4},1})}{\text{sh}(\phi_i - v_{\underline{4},1})} \quad (5.68)$$

Once we identify $v_{\underline{4},1} = v_{123,1}$ and $w_{\underline{4},1} = v_{123,1} - \epsilon_{123}$, this integrand is identical to that of a single D6-brane. Setting $w_{\underline{4},1} = v_{123,1} - \epsilon_{123}$ effectively brings the D8 and $\overline{\text{D8}}$ -branes together. As shown in Fig. 5-7, the open strings connecting them develop a negative mass (tachyon) state, which renders the system unstable and causes it to decay into a single D6-brane.

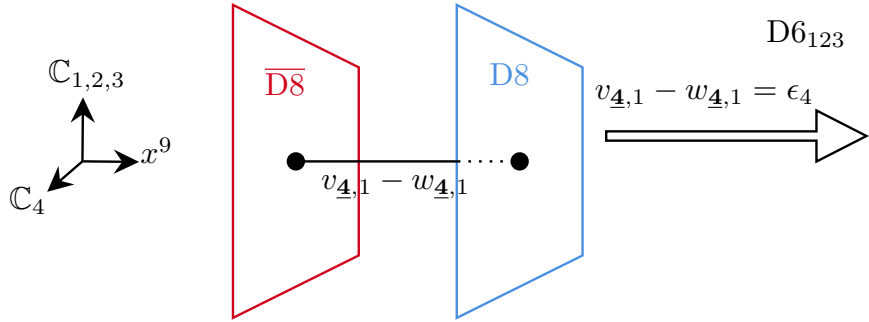


Figure 5-7 The D8- $\overline{\text{D8}}$ pair annihilates into a D6-brane through tachyon condensation. In the figure, the red brane represents the anti-D8-brane, with $w_{\underline{4},1}$ being its corresponding Coulomb branch parameter. The blue brane represents the D8-brane, with $v_{\underline{4},1}$ being its corresponding Coulomb branch parameter. When we adjust the moduli space coordinate to the place $v_{\underline{4},1} - w_{\underline{4},1} = \epsilon_4$, the D8- $\overline{\text{D8}}$ pair becomes equivalent to a single D6-brane oriented along $\mathbb{C}_4$.

From the perspective of the shell formula, if the corresponding 4d Young diagram contains the box located at $(1, 1, 1, 2)$, then the contribution of the $\overline{\text{D8}}$ -brane becomes $\text{sh}(\epsilon_{1234})$ under the condition $w_{\underline{4},1} = v_{123,1} - \epsilon_{123}$ and $v_{\underline{4},1} = v_{123,1}$. Under the CY4 condition $\epsilon_{1234} = 0$, this term vanishes. This implies that the $\overline{\text{D8}}$ -brane contribution selects only those 4d Young diagrams that have exactly one layer in the fourth direction, i.e., 3d Young diagrams. In this case, the condition $x_4 \geq y_4$ in the $\mathcal{J}_{\geq}$ -factor (5.54) becomes trivial.

It follows that the PE expression of the tetrahedron instanton^[11,90], derivable from that of the magnificent four (5.58), is manifestly independent of all the Coulomb branch parameters $\{v_{\mathcal{A}}\}$:

$$\mathcal{Z}_N^{\text{D0-D6}}(\{v_{\mathcal{A}}\}) = \sum_{k=0}^{\infty} \mathbf{q}^k \mathcal{Z}_{N,k}^{\text{D0-D6}} = \text{PE} \left(\frac{\text{sh}(\epsilon_{12,13,23})}{\text{sh}(\epsilon_{1,2,3,4})} \frac{\text{sh}(s)}{\text{sh}(p \pm \frac{1}{2}s)} \right) \quad (5.69)$$

where the parameter $s \equiv \sum_{i=1}^N (v_{\underline{4},i} - w_{\underline{4},i})$ after tachyon condensation becomes:

$$\begin{aligned} s &= N_{\bar{4}}\epsilon_{123} + N_{\bar{3}}\epsilon_{124} + N_{\bar{2}}\epsilon_{134} + N_{\bar{1}}\epsilon_{234} \\ &= (N_{123} - N_{234})\epsilon_1 + (N_{123} - N_{134})\epsilon_2 + (N_{123} - N_{124})\epsilon_3 \end{aligned} \quad (5.70)$$

We briefly discuss the generalized tetrahedron instanton, namely the theory with the addition of anti-D6 branes^[82,100]. Similar to the anti-D8 case, the $\overline{\text{D6}}$ -brane contribution to the index is the reciprocal of that of a D6-brane. The partition function is:

$$\mathcal{J}_{N,\mathbf{M},k}^{\text{D0-D6-}\overline{\text{D6}}} = \mathcal{J}_{N,k}^{\text{D0-D6}} \times \prod_{i=1}^k \prod_{\mathcal{B}} \frac{\text{sh}(-\phi_i + w_{\mathcal{B}})}{\text{sh}(-\phi_i + w_{\mathcal{B}} - \epsilon_{\mathcal{B}})} \quad (5.71)$$

where $w_{\mathcal{B}}$ is the fugacity for the $\overline{\text{D6}}$ -brane with $\mathcal{B} = (B, \beta) \in \{(\bar{4}, 1), \dots, (\bar{4}, M_{\bar{4}}), (\bar{3}, 1), \dots, (\bar{1}, M_{\bar{1}})\}$ label each $\overline{\text{D6}}$ -brane.

5.3.3 Spiked instanton

Spiked instantons arise from D0-branes bound to D4-branes with multiple orientations^[9,28,78]. They can be obtained from the 5d $\mathcal{N} = 1$ theory of Sec. 5.2.1 by incorporating D4-branes with different orientations and turning on adjoint multiplet masses, or alternatively from the tetrahedron instanton of Sec. 5.3.2 by introducing an equal number of anti-D6-branes through tachyon condensation. The corresponding brane configuration involves six types of D4-branes with distinct orientations, as shown in Tab. 5-6. In the low-energy regime, D0-instantons attach to any of these D4-branes in the form of 2d Young diagrams. With the quiver diagram as $\mathcal{N} = 2$ SUSY QM as in Fig. 5-8,

	$\mathbb{C}_1$		$\mathbb{C}_2$		$\mathbb{C}_3$		$\mathbb{C}_4$		$\mathbb{R} \times \mathbb{S}^1$	
	1	2	3	4	5	6	7	8	9	0
D0	•	•	•	•	•	•	•	•	•	-
D4 ₁₂	-	-	-	-	•	•	•	•	•	-
D4 ₁₃	-	-	•	•	-	-	•	•	•	-
D4 ₁₄	-	-	•	•	•	•	-	-	•	-
D4 ₂₃	•	•	-	-	-	-	•	•	•	-
D4 ₂₄	•	•	-	-	•	•	-	-	•	-
D4 ₃₄	•	•	•	•	-	-	-	-	•	-

Table 5-6 Brane configuration of spiked instantons. – represents the direction along which the D-branes extend, while • represents the point-like directions of the D-branes. In the spiked instanton, we consider six types of D4-branes, where the D4 _{$ab \in \underline{6}$} -branes extend along $\mathbb{C}_{ab}$ and the time direction x^0 .

the Instanton moduli space is defined as:

$$\mathcal{M}_{N,k}^{\text{D0-D4}} = \{(\mathbf{B}, \mathbf{I}, \mathbf{J}) | \mu_{\mathbb{R}} - \zeta \cdot \mathbf{1}_k = \mu_{ab} = \sigma_{a;bc} = \tilde{\sigma}_{a;bc} = 0\} / \text{U}(k) \quad (5.72)$$

where $a \in \underline{4}$, $ab, bc \in \underline{6}$. The moment maps in the spiked instantons cases are modified as:

$$\begin{aligned}
\mu_{\mathbb{R}} &= \sum_{a \in \underline{4}} [B_a, B_a^\dagger] + \sum_{ab \in \underline{6}} (I_{ab} I_{ab}^\dagger - J_{ab}^\dagger J_{ab}) \\
\mu_{ab} &= [B_a, B_b] + I_{ab} J_{ab} \\
\sigma_{a;bc} &= B_a I_{bc} \\
\tilde{\sigma}_{a;bc} &= J_{bc} B_a
\end{aligned} \tag{5.73}$$

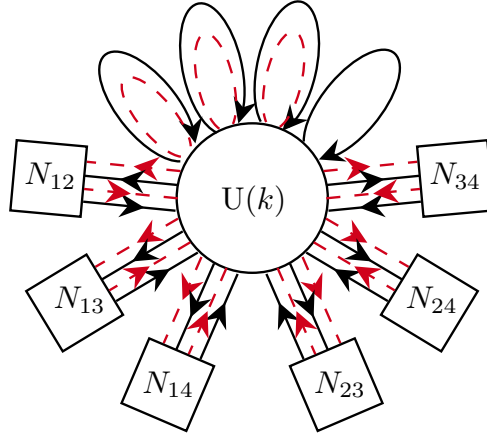


Figure 5-8 Quiver diagram of D0-D4 system as $\mathcal{N} = 2$ SUSY QM. In this quiver, we have the $U(k)$ gauge group, six flavor groups $U(N_{ab \in \underline{6}})$, six types of fundamental chiral multiplets that contain $I_{ab \in \underline{6}}$, six types of anti-fundamental chirals that contain $J_{ab \in \underline{6}}$, twelve types of Fermi multiplets $\Lambda_{I_{ab}}, \Lambda_{J_{ab}}$ correspond to I_{ab} and J_{ab} respectively, four adjoint chiral multiplets $B_{1,2,3,4}$, and three adjoint Fermi multiplets $\Lambda_{1,2,3}$.

The spiked instanton partition function is:

$$\begin{aligned}
\mathcal{Z}_{N,k}^{\text{D0-D4}} &= \oint_{\text{JK}} \prod_{i=1}^k \frac{d\phi_i}{2\pi i} \mathcal{J}_{N,k}^{\text{D0-D4}} \\
\mathcal{J}_{N,k}^{\text{D0-D4}} &= \mathcal{J}_k^{\text{D0-D0}} \times \prod_{i=1}^k \prod_{\mathbf{ab}} \frac{\prod_{c \in \overline{ab}} \text{sh}(\phi_i - v_{\mathbf{ab}} - \epsilon_c)}{\text{sh}(\phi_i - v_{\mathbf{ab}}) \text{sh}(-\phi_i + v_{\mathbf{ab}} - \epsilon_{ab})}
\end{aligned} \tag{5.74}$$

where $\mathbf{ab} = (ab, \alpha) \in \{(12, 1), \dots, (12, N_{12}), (13, 1), \dots, (34, N_{34})\}$ label each D4 brane, $\overline{ab}$ denotes the complement of ab within $\underline{4}$, and $\mathbf{N} = (N_{12}, \dots, N_{34})$ corresponds to the numbers of D4-branes with different orientations. The poles of this residue integral are classified by a set of 2d Young diagrams, and the shell formula gives:

$$\begin{aligned}
\mathcal{Z}_{N,k}^{\text{D4}, \mathbb{C}^4} &= \sum_{||\tilde{\lambda}||=k} (-1)^k \prod_{\mathbf{ab}, \mathbf{ab}'} \left(\prod_{\mathbf{x} \in \lambda_{\mathbf{ab}}} \frac{\mathcal{J}(\mathcal{X}_{\mathbf{ab}}(\mathbf{x}) | \lambda_{\mathbf{ab}'})}{\text{sh}(-\mathcal{X}_{\mathbf{ab}}(\mathbf{x}) + \mathcal{X}_{\mathbf{ab}'}(\mathbf{0}))} \prod_{c \in \overline{ab}} \text{sh}(\mathcal{X}_{\mathbf{ab}}(\mathbf{x}) - \mathcal{X}_{\mathbf{ab}'}(\mathbf{0}) + \epsilon_c) \right) \\
&\times \left(\prod_{\substack{\mathbf{x} \in \lambda_{\mathbf{ab}} \\ \mathbf{y} \in \lambda_{\mathbf{ab}'}}} \frac{\text{sh}(\mathcal{X}_{\mathbf{ab}}(\mathbf{x}) - \mathcal{X}_{\mathbf{ab}'}(\mathbf{y}) - \epsilon_{1,2,3} - \epsilon_4)}{\text{sh}(\mathcal{X}_{\mathbf{ab}}(\mathbf{x} + \mathbf{1}) - \mathcal{X}_{\mathbf{ab}'}(\mathbf{y})) \prod_{c \in \overline{ab}} \text{sh}(\mathcal{X}_{\mathbf{ab}}(\mathbf{x}) - \mathcal{X}_{\mathbf{ab}'}(\mathbf{y}) + \epsilon_c)} \right)
\end{aligned} \tag{5.75}$$

From the perspective of tachyon condensation, this spiked instanton system can arise from a D6-anti-D6 system^[82,100]. For instance, on the integrand level, a $D4_{12}$ -brane can be obtained from a $D6_{123}-\overline{D6}_{123}$ system by taking $v_{123,1} = v_{12,1}$, $w_{123,1} = v_{12,1} + \epsilon_3$, or from a $D6_{124}-\overline{D6}_{124}$ system by taking $v_{123,1} = v_{12,1}$, $w_{123,1} = v_{12,1} + \epsilon_4$ as in (5.71). Furthermore, if we consider only a single type of D4-brane, the resulting theory is precisely the 5d $U(N)$ SYM theory with an additional adjoint hypermultiplet, where ϵ_3 can be interpreted as the mass fugacity for the adjoint hypermultiplet^[69].

5.3.4 Donaldson-Thomas 3 counting

Another natural application of the shell formula is DT3 counting^[82,91–92,100–102]. Mathematically, the DT3 invariants measure the virtual Euler characteristics of moduli spaces of ideal sheaves on a CY threefold, or equivalently, the virtual counts of curve and point subschemes. Physically, they enumerate bound states of D0-D2-D6 branes. Since a single D2-brane corresponds to an infinitely long 3d Young diagram composed of individual boxes, the $\mathcal{F}$ -factor provides a natural tool for computing the partition function of the D0-D2-D6 system.

We consider the simplest DT3 invariant, corresponding to the system placed on $\mathbb{C}^3$. The brane construction data are summarized in Tab. 5-7.

	$\mathbb{C}_1$		$\mathbb{C}_2$		$\mathbb{C}_3$		$\mathbb{C}_4$		$\mathbb{R} \times S^1$	
	1	2	3	4	5	6	7	8	9	0
k D0	•	•	•	•	•	•	•	•	•	–
1 $D6_{123}$	–	–	–	–	–	–	•	•	•	–
N_1 D2 ₁	–	–	•	•	•	•	•	•	•	–
N_2 D2 ₂	•	•	–	–	•	•	•	•	•	–
N_3 D2 ₃	•	•	•	•	–	–	•	•	•	–

Table 5-7 Brane configuration for DT3 counting. The symbol – denotes an extended worldvolume direction, while • denotes a transverse (point-like) direction. Each D2-brane extends along one of the complex directions $\mathbb{C}_{1,2,3}$ and the time direction x^0 . We consider a single D6-brane extending over $\mathbb{C}_{123}^3$.

The D6-brane extends over the full CY threefold $\mathbb{C}_{123}^3$, with D2-branes wrapping curve subschemes of $\mathbb{C}_{123}^3$. Fixing a stable D2-brane configuration (the minimal plane partition, or vacuum) reduces the problem to counting bound states on the worldvolume of the k D0-branes—equivalently, counting placements of k boxes that extend the 3d Young diagram. The corresponding D0-D2-D6 framed quiver, shown in Fig. 5-9, is constructed following^[102].

The vacuum configuration of the D2-branes is identified with the minimal infinite 3d Young diagram $\pi_{\lambda\mu\nu}$ (minimal plane partition) admitting three prescribed asymptotic boundary conditions λ, μ, ν satisfying $|\lambda| = N_1$, $|\mu| = N_2$, and $|\nu| = N_3$. As illustrated in Fig. 5-10, the three asymptotic planes at infinity are precisely the 2d Young diagrams λ, μ , and ν . We refer to the configuration as the 3-leg case when all three boundaries are non-empty, the 2-leg case when exactly two are non-empty, and the 1-leg case when only one is non-empty.

Given a vacuum configuration, the multiplicities and $U(1)^3$ charges of the fields $\{I_x\}$, $\{J_y\}$, $\{A_{I_x}\}$, and $\{A_{J_y}\}$ are determined via the framed quiver and its superpoten-

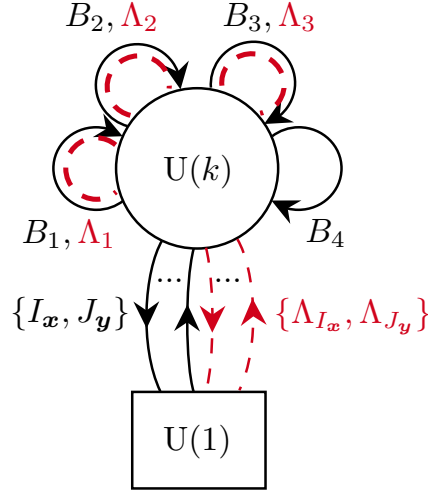


Figure 5-9 Quiver diagram of the D0-D2-D6 system as $\mathcal{N} = 2$ SUSY quantum mechanics. Solid black lines represent chiral multiplets: the adjoint chirals $B_{1,2,3,4}$, fundamental chirals $\{I_x\}$, and anti-fundamental chirals $\{J_y\}$. Red dashed lines denote Fermi multiplets: the adjoint Fermis $\Lambda_{1,2,3,4}$, fundamental Fermis $\{\Lambda_{I_x}\}$, and anti-fundamental Fermis $\{\Lambda_{J_y}\}$. The multiplicities of $\{I_x\}$, $\{J_y\}$, $\{\Lambda_{I_x}\}$, and $\{\Lambda_{J_y}\}$, together with their $U(1)^3$ charges, are determined by the vacuum configuration of the D2-branes.

tial^[92,102] (the construction of the framed quiver and superpotential is omitted here). This in turn yields the integral expression for the partition function.

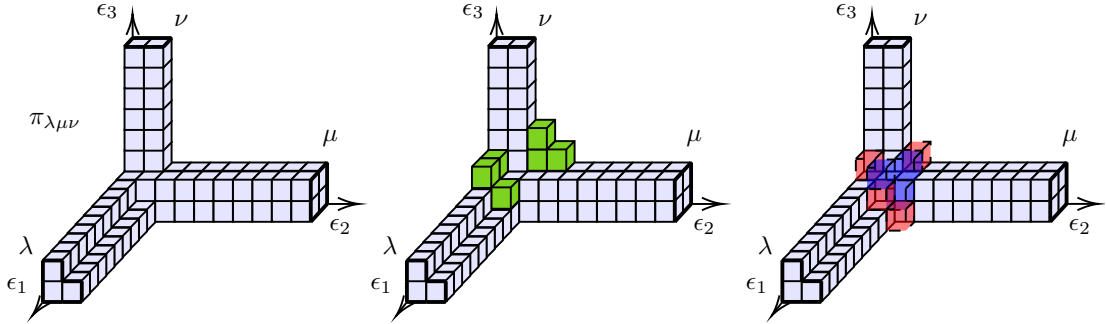


Figure 5-10 Minimal plane partition, DT3 excitation, and charged shell boxes for fixed asymptotic boundaries.

By applying the shell formula together with the recursion relation (5.67) for 3d Young diagrams, the partition function is obtained directly from the minimal plane partition $\pi_{\lambda\mu\nu}$:

$$\mathcal{Z}_{\lambda,\mu,\nu;k}^{\text{D0-D2-D6}} = \mathcal{Z}_k^{\text{D0-D0}} \times \prod_{i=1}^k \frac{\mathcal{F}(\phi_i | \pi_{\lambda\mu\nu})}{\mathcal{F}_-(\phi_i + \epsilon_{123} | \pi_{\lambda\mu\nu})}, \quad (5.76)$$

where the D6-brane label $(123, 1)$ and the Coulomb branch parameter v are suppressed for brevity. To reproduce the correct $U(1)^3$ charges and multiplet content, the signs of the terms in the denominator $\mathcal{F}$ -factor must be flipped via $\text{sh}(x) \rightarrow -\text{sh}(-x)$. Accord-

ingly, $\mathcal{F}_-$ is defined by:

$$\begin{aligned}\mathcal{F}_-(x|\mathbf{Y}_{\mathcal{A}}) &\equiv \prod_{\mathbf{y} \in \mathcal{S}(\mathbf{Y}_{\mathcal{A}})} -\text{sh}(\mathcal{X}_{\mathcal{A}}(\mathbf{y}) - x)^{Q_{\mathbf{Y}_{\mathcal{A}}}(\mathbf{y})}, \\ \mathcal{F}_-(x|\emptyset_{\mathcal{A}}) &\equiv \frac{-1}{\text{sh}(\mathcal{X}_{\mathcal{A}}(\mathbf{1}) - x)}.\end{aligned}\quad (5.77)$$

The $\mathcal{F}$ -factor for $\pi_{\lambda\mu\nu}$ is computed via the following inclusion-exclusion relation:

$$\mathcal{F}(x|\pi_{\lambda\mu\nu}) = \frac{\mathcal{F}(x|\pi_{\lambda\emptyset\emptyset}) \mathcal{F}(x|\pi_{\emptyset\mu\emptyset}) \mathcal{F}(x|\pi_{\emptyset\emptyset\nu}) \mathcal{F}(x|\pi_{\lambda\emptyset\emptyset} \cap \pi_{\emptyset\mu\emptyset} \cap \pi_{\emptyset\emptyset\nu})}{\mathcal{F}(x|\pi_{\lambda\emptyset\emptyset} \cap \pi_{\emptyset\mu\emptyset}) \mathcal{F}(x|\pi_{\lambda\emptyset\emptyset} \cap \pi_{\emptyset\emptyset\nu}) \mathcal{F}(x|\pi_{\emptyset\mu\emptyset} \cap \pi_{\emptyset\emptyset\nu})}, \quad (5.78)$$

where the pairwise and triple intersections $\pi_{\lambda\emptyset\emptyset} \cap \pi_{\emptyset\mu\emptyset}$, $\pi_{\lambda\emptyset\emptyset} \cap \pi_{\emptyset\emptyset\nu}$, $\pi_{\emptyset\mu\emptyset} \cap \pi_{\emptyset\emptyset\nu}$, and $\pi_{\lambda\emptyset\emptyset} \cap \pi_{\emptyset\mu\emptyset} \cap \pi_{\emptyset\emptyset\nu}$ are all finite 3d Young diagrams, so their $\mathcal{F}$ -factors are computed directly from the charges of their shell boxes. For the 1-leg factors $\mathcal{F}(x|\pi_{\lambda\emptyset\emptyset})$, $\mathcal{F}(x|\pi_{\emptyset\mu\emptyset})$, and $\mathcal{F}(x|\pi_{\emptyset\emptyset\nu})$, a recursive argument via (5.21)–illustrated in Fig. 5-11–shows that shell boxes with non-zero charge are confined to the two ends of each leg. Moreover, the numbers of +1 and −1 charged shell boxes at the infinite end are equal, so their net contribution to the $\mathcal{F}$ -factor is:

$$\frac{\text{sh}(\infty\epsilon_2 + \cdots)}{\text{sh}(\infty\epsilon_2 + \cdots)} \frac{\text{sh}(\infty\epsilon_2 + \cdots)}{\text{sh}(\infty\epsilon_2 + \cdots)} \frac{\text{sh}(\infty\epsilon_2 + \cdots)}{\text{sh}(\infty\epsilon_2 + \cdots)} \times \cdots = 1. \quad (5.79)$$

Since the contribution from the infinite end is exactly 1, the $\mathcal{F}$ -factor for each 1-leg diagram reduces to a contribution from the 2d Young diagram defining its asymptotic boundary condition:

$$\begin{aligned}\mathcal{F}(x|\pi_{\lambda\emptyset\emptyset}) &= \mathcal{F}(x + v_{23,1} - v|\lambda_{23,1}), \\ \mathcal{F}(x|\pi_{\emptyset\mu\emptyset}) &= \mathcal{F}(x + v_{13,1} - v|\mu_{13,1}), \\ \mathcal{F}(x|\pi_{\emptyset\emptyset\nu}) &= \mathcal{F}(x + v_{12,1} - v|\nu_{12,1}).\end{aligned}\quad (5.80)$$

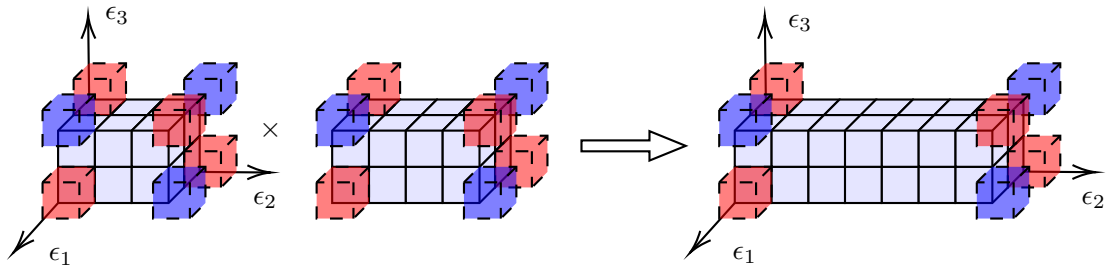


Figure 5-11 Shell boxes for gluing two long legs with identical 2d Young-diagram cross-sections.

Applying the JK residue, the refined DT3 invariant is:

$$\begin{aligned}\mathcal{Z}_{\lambda,\mu,\nu;k}^{\text{D0-D2-D6}} &= \oint_{\text{JK}} \prod_{i=1}^k \frac{d\phi_i}{2\pi i} \mathcal{Z}_{\lambda,\mu,\nu;k}^{\text{D0-D2-D6}} \\ &= \sum_{|\tilde{x}_{\lambda\mu\nu}|=k} \prod_{\mathbf{x} \in \tilde{\pi}_{\lambda\mu\nu}} \left[\frac{\mathcal{F}(\mathcal{X}(\mathbf{x})|\pi_{\lambda\mu\nu})}{\mathcal{F}(\mathcal{X}(\mathbf{x}) + \epsilon_{123}|\pi_{\lambda\mu\nu})} \prod_{\mathbf{y} \in \tilde{\pi}_{\lambda\mu\nu}} \text{sh}(\mathcal{X}(\mathbf{x}) - \mathcal{X}(\mathbf{y})) \mathcal{F}(\mathcal{X}(\mathbf{x})|\{\mathbf{y}\}) \right],\end{aligned}\quad (5.81)$$

where, for any infinite 3d Young diagram $\pi \supset \pi_{\lambda\mu\nu}$ containing the vacuum, we set $\tilde{\pi}_{\lambda\mu\nu} \equiv \pi \setminus \pi_{\lambda\mu\nu}$. Here $\{\mathbf{x}\}$ denotes the 3d Young diagram consisting of the single box at $\mathbf{x}$ (rather than at $\mathbf{1}$), whose $\mathcal{F}$ -factor is computed via (5.13). Note that although $\text{sh}(0)$ appears when $\mathbf{x} = \mathbf{y}$, the overall expression remains finite.

In the special case $\lambda = \mu = \nu = \emptyset$, i.e., $N_1 = N_2 = N_3 = 0$ with no D2-branes present, the system reduces to the simplest sector of the tetrahedron instanton of Sec. 5.3.2—namely, the D0-D6 system with a single D6-brane. The DT3 invariant (5.81) for $\pi_{\emptyset\emptyset\emptyset} = \emptyset$ then reads:

$$\begin{aligned} \mathcal{Z}_{\emptyset,\emptyset,\emptyset;k}^{\text{D0-D2-D6}} &= \sum_{|\pi|=k} \prod_{\mathbf{x} \in \pi} \frac{\text{sh}(\mathcal{X}(\mathbf{x}) - \mathcal{X}(\mathbf{0}))}{\text{sh}(\mathcal{X}(\mathbf{x}) - \mathcal{X}(\mathbf{1}))} \prod_{\mathbf{y} \in \pi} \text{sh}(\mathcal{X}(\mathbf{x}) - \mathcal{X}(\mathbf{y})) \mathcal{F}(\mathcal{X}(\mathbf{x}) | \{\mathbf{y}\}) \\ &= \sum_{|\pi|=k} \prod_{\mathbf{x} \in \pi} \frac{\text{sh}(\mathcal{X}(\mathbf{x}) - \mathcal{X}(\mathbf{0}))}{\text{sh}(\mathcal{X}(\mathbf{x}) - \mathcal{X}(\mathbf{1}))} \prod_{\mathbf{y} \in \pi} \frac{\text{sh}(\mathcal{X}(\mathbf{x}) - \mathcal{X}(\mathbf{y})) \text{sh}(\mathcal{X}(\mathbf{x}) - \mathcal{X}(\mathbf{y}) - \epsilon_{12,13,23})}{\text{sh}(\mathcal{X}(\mathbf{x}) - \mathcal{X}(\mathbf{y}) - \epsilon_{1,2,3}) \text{sh}(\mathcal{X}(\mathbf{x}) - \mathcal{X}(\mathbf{y}) - \epsilon_{123})} \\ &= \sum_{|\pi|=k} \prod_{\mathbf{x} \in \pi} \text{sh}(\mathcal{X}(\mathbf{x}) - \mathcal{X}(\mathbf{0})) \mathcal{F}(\mathcal{X}(\mathbf{x}) | \pi), \end{aligned} \quad (5.82)$$

where the first equality uses $\mathcal{F}(\mathbf{x} | \emptyset) = 1/\text{sh}(\mathbf{x} - \mathcal{X}(\mathbf{1}))$, and the second and third equalities follow from the expansion formula (5.10). Since $\tilde{\pi}_{\emptyset\emptyset\emptyset} = \pi$, the final line coincides precisely with the partition function (5.66) of the tetrahedron instanton with a single D6-brane, giving:

$$\mathcal{Z}_{\emptyset,\emptyset,\emptyset;k}^{\text{D0-D2-D6}} = \mathcal{Z}_{(1,0,0,0),k}^{\text{D0-D6}}. \quad (5.83)$$

Detailed calculations for simple 1-leg and 3-leg examples, together with the derivation of (5.81), are collected in Appendix C.1.4.

5.3.5 Donaldson-Thomas 4 counting

Analogous to DT3 counting, the shell formula extends naturally to a 4d generalization, governing the enumeration of bound states in D8-D2-D0 systems on a CY fourfold. This setup is referred to as DT4 counting with leg boundary conditions. In contrast to DT3 counting, one may also impose surface boundary conditions on the CY fourfold, corresponding to bound states of D8-D4-D0 systems^[13,93–95].

We consider DT4 counting on $\mathbb{C}_{1234}^4$. The setup consists of k D0-branes wrapping S^1 , a single D8-brane extending over $\mathbb{C}_{1234}^4 \times S^1$, N_a D2_{*a*}-branes on $\mathbb{C}_a \times S^1$ for $a \in \underline{4}$, and N_{ab} D4_{*ab*}-branes on $\mathbb{C}_{ab} \times S^1$ for $ab \in \underline{6}$. For brevity, throughout this chapter we suppress the $\underline{4}, 1$ labels on Young diagrams.

DT4 counting is formulated by taking the minimal 4d Young diagram $\rho_{\{\pi_a\}, \{\lambda_{ab}\}}$ —characterized by four leg boundaries π_a and six surface boundaries λ_{ab} —as the vacuum configuration, and enumerating BPS bound states of D0-branes with $|\pi_a| = N_a$ and $|\lambda_{ab}| = N_{ab}$. The worldvolume theory on the D0-branes is described by an $\mathcal{N} = 2$ supersymmetric quantum mechanics quiver analogous to that in Fig. 5-9, with differences arising in the specific content of fundamental and anti-fundamental multiplets. For a given minimal 4d Young diagram $\rho_{\{\pi_a\}, \{\lambda_{ab}\}}$, the integrand associated with the D0-branes is:

$$\mathcal{Z}_{\{\pi_a\}, \{\lambda_{ab}\};k}^{\text{D0-D2-D4-D8}} = \mathcal{Z}_k^{\text{D0-D0}} \times \prod_{i=1}^k \mathcal{F}_{-\mathfrak{A}}(\phi_i | \rho_{\{\pi_a\}, \{\lambda_{ab}\}}). \quad (5.84)$$

To obtain the correct DT4 character within the JK-residue formalism, one must perform the sign reversal $\text{sh}(x) \rightarrow -\text{sh}(-x - \epsilon_{1234})$ on certain terms in the $\mathcal{F}$ -factor, as in (5.76); specifically, the contributions from certain fundamental multiplets are replaced by those from anti-fundamental ones. Accordingly, $\mathcal{F}_{-\mathfrak{A}}$ is defined by:

$$\begin{aligned} \mathcal{F}_{-\mathfrak{A}}(x|\mathbf{Y}_{\mathcal{A}}) &\equiv \left(\prod_{y \in \mathfrak{A}(\mathbf{Y}_{\mathcal{A}})} \text{sh}(x - \mathcal{X}_{\mathcal{A}}(y))^{Q_{Y_{\mathcal{A}}}(y)} \right) \left(\prod_{y \in \mathcal{S}(\mathbf{Y}_{\mathcal{A}}) \setminus \mathfrak{A}(\mathbf{Y}_{\mathcal{A}})} -\text{sh}(\mathcal{X}_{\mathcal{A}}(y) - x - \epsilon_{1234})^{Q_{Y_{\mathcal{A}}}(y)} \right), \\ \mathcal{F}_{-\mathfrak{A}}(x|\emptyset_{\mathcal{A}}) &\equiv \frac{1}{\text{sh}(x - \mathcal{X}_{\mathcal{A}}(\mathbf{1}))}. \end{aligned} \quad (5.85)$$

Here $\mathfrak{A}(\mathbf{Y})$ denotes the set of all *addable boxes* of $\mathbf{Y}$, so that $\mathcal{F}_{-\mathfrak{A}}$ is obtained from $\mathcal{F}$ by flipping the sign of every term except those corresponding to addable boxes. This sign rule is part of the orientation convention for the DT4 index. The Calabi–Yau condition controls the supersymmetric specialization, while the fully refined theory may require extra orientation or multiplicity data not determined by the shell charges alone. This is why the examples in Appendix C.1.6 are presented as checks of the convention rather than as a closed proof of the refined DT4 theory in all chambers.

After performing the JK residue, the partition function takes the form:

$$\begin{aligned} \mathcal{Z}_{\{\pi_a\}, \{\lambda_{ab}\}; k}^{\text{D0-D2-D4-D8}} &= \oint_{\text{JK}} \prod_{i=1}^k \frac{d\phi_i}{2\pi i} \mathcal{Z}_{\{\pi_a\}, \{\lambda_{ab}\}; k}^{\text{D0-D2-D4-D8}} \\ &= \sum_{|\tilde{\rho}_{\{\pi_a\}, \{\lambda_{ab}\}}| = k} \prod_{\mathbf{x} \in \tilde{\rho}_{\{\pi_a\}, \{\lambda_{ab}\}}} \left[\mathcal{F}(\mathcal{X}(\mathbf{x})|\rho_{\{\pi_a\}, \{\lambda_{ab}\}}) \prod_{y \in \tilde{\rho}_{\{\pi_a\}, \{\lambda_{ab}\}}} \text{sh}(\mathcal{X}(\mathbf{x}) - \mathcal{X}(y)) \mathcal{F}_{\geq}(\mathcal{X}(\mathbf{x})|\{y\}) \right], \end{aligned} \quad (5.86)$$

where the notation parallels that of DT3 counting introduced earlier, and for a 4d Young diagram $\rho \supset \rho_{\{\pi_a\}, \{\lambda_{ab}\}}$ we write $\tilde{\rho}_{\{\pi_a\}, \{\lambda_{ab}\}} = \rho \setminus \rho_{\{\pi_a\}, \{\lambda_{ab}\}}$.

Specializing to the D8-D2-D0 system—i.e., DT4 counting with leg boundaries $\pi_a \in \underline{4}$ —the minimal solid partition $\rho_{\{\pi_a\}} = \rho_{\pi_1 \pi_2 \pi_3 \pi_4}$ is schematically illustrated in Fig. 5-12. The associated $\mathcal{F}$ -factor is computed by an inclusion-exclusion analogous to (5.78):

$$\mathcal{F}(x|\rho_{\pi_1 \pi_2 \pi_3 \pi_4}) = \frac{\left(\prod_{a \in \underline{4}} \mathcal{F}(x|\rho_{\pi_a}) \right) \left(\prod_{abc \in \check{\underline{4}}} \mathcal{F}(x|\rho_{\pi_a} \cap \rho_{\pi_b} \cap \rho_{\pi_c}) \right)}{\left(\prod_{ab \in \underline{6}} \mathcal{F}(x|\rho_{\pi_a} \cap \rho_{\pi_b}) \right) \mathcal{F}(x|\rho_{\pi_1} \cap \rho_{\pi_2} \cap \rho_{\pi_3} \cap \rho_{\pi_4})}, \quad (5.87)$$

where ρ_{π_1} is shorthand for $\rho_{\pi_1 \emptyset \emptyset \emptyset}$, and similarly for $\rho_{\pi_{2,3,4}}$. The pairwise, triple, and quadruple intersections $\rho_{\pi_a} \cap \rho_{\pi_b}$, $\rho_{\pi_a} \cap \rho_{\pi_b} \cap \rho_{\pi_c}$, and $\rho_{\pi_1} \cap \rho_{\pi_2} \cap \rho_{\pi_3} \cap \rho_{\pi_4}$ are all finite 4d Young diagrams, so the charges of their boxes are computed in the standard way. For the infinite Young diagram ρ_{π_a} , as in the DT3 case, the $\mathcal{F}$ -factor is determined entirely by the boundary condition π_a :

$$\mathcal{F}(x|\rho_{\pi_a}) = \mathcal{F}(x + v_{\bar{a},1} - v | (\pi_a)_{\bar{a},1}). \quad (5.88)$$

The computation of the $\mathcal{F}$ -factor for the minimal 4d Young diagram $\rho_{\{\lambda_{ab}\}}$ with surface boundaries $\{\lambda_{ab}\}_{ab \in \underline{6}}$ is more involved. Proceeding by the same inclusion-

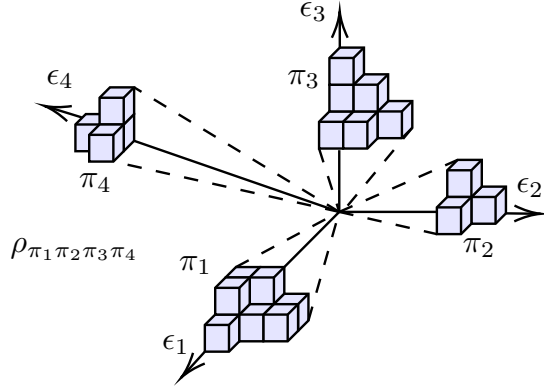


Figure 5-12 A schematic illustration of the minimal 4d Young diagram (minimal solid partition) $\rho_{\pi_1 \pi_2 \pi_3 \pi_4}$. Its basis is spanned by $\epsilon_{1,2,3,4}$, with four distinct 3d Young diagrams $\pi_1, \pi_2, \pi_3, \pi_4$ serving as asymptotic leg boundary conditions.

exclusion method yields:

$$\begin{aligned} \mathcal{F}(x | \rho_{\lambda_{12} \lambda_{13} \lambda_{14} \lambda_{23} \lambda_{24} \lambda_{34}}) &= \frac{\left(\prod_{ab \in \mathbf{6}} \mathcal{F}(x | \rho_{\lambda_{ab}}) \right) \left(\prod_{ab_1 < ab_2 < ab_3} \mathcal{F}(x | \cap_{i=1}^3 \rho_{\lambda_{ab_i}}) \right)}{\left(\prod_{ab_1 < ab_2} \mathcal{F}(x | \cap_{i=1}^2 \rho_{\lambda_{ab_i}}) \right) \left(\prod_{ab_1 < ab_2 < ab_3 < ab_4} \mathcal{F}(x | \cap_{i=1}^4 \rho_{\lambda_{ab_i}}) \right)} \\ &\times \frac{\left(\prod_{ab_1 < ab_2 < ab_3 < ab_4 < ab_5} \mathcal{F}(x | \cap_{i=1}^5 \rho_{\lambda_{ab_i}}) \right)}{\mathcal{F}(x | \cap_{i=1}^6 \rho_{\lambda_{ab_i}})}, \end{aligned} \quad (5.89)$$

where for each surface boundary:

$$\mathcal{F}(x | \rho_{\lambda_{ab}}) = \mathcal{F}(x + v_{\overline{ab},1} - v | (\lambda_{ab})_{\overline{ab},1}). \quad (5.90)$$

In practice, a more convenient computational strategy is available. As established in (5.79), all contributions from asymptotic boundary conditions cancel pairwise. One may therefore impose a cutoff on $\rho_{\{\pi_a\}, \{\lambda_{ab}\}}$ at a sufficiently large distance from the origin, compute the $\mathcal{F}$ -factor directly on the truncated diagram, and then discard all terms that depend on the cutoff. Concretely, for a sufficiently large integer m , we define:

$$\rho_{\{\pi_a\}, \{\lambda_{ab}\}; m} = \{(x_1, x_2, x_3, x_4) \in \rho_{\{\pi_a\}, \{\lambda_{ab}\}} \mid x_i \leq m\}; \quad (5.91)$$

then $\mathcal{F}(x | \rho_{\{\pi_a\}, \{\lambda_{ab}\}})$ is recovered from $\mathcal{F}(x | \rho_{\{\pi_a\}, \{\lambda_{ab}\}; m})$ by discarding all terms of the form $\text{sh}(x + v - m \epsilon_a + \dots)$. Detailed computational examples are collected in Appendix C.1.6.

Finally, the inclusion of D6-branes modifies the D0–D2–D4–D8 system by introducing a nontrivial background on which the 4d Young diagram is supported. Combinatorially, this is equivalent to placing the original configuration on top of a semi-infinite bulk, or, equivalently, shifting the origin of the minimal 4d Young diagram $\rho_{\{\pi_a\}, \{\lambda_{ab}\}}$, so that all boxes are measured relative to a displaced reference corner.

This shift does not alter the local growth rules of the diagram, but changes the vacuum structure and the asymptotic data entering the shell formula, effectively reweighting the contributions of boxes. In this way, the shell formula naturally extends to the D0–D2–D4–D6–D8 system, i.e., the 4G network system^[94]. A detailed analysis of this system is left to future work; the concrete proposal used here is to obtain its partition function from the DT4 system by shifting the reference configuration. For example, for

a DT4 system with $\mathbf{N} = (N_{\bar{4}}, N_{\bar{3}}, N_{\bar{2}}, N_{\bar{1}})$ D6-branes (of types $D6_{\bar{4}}, D6_{\bar{3}}, D6_{\bar{2}}, D6_{\bar{1}}$), the origin of the corresponding 4d Young diagram $\rho_{\{\pi_a\}, \{\lambda_{ab}\}}$ is shifted from $(1, 1, 1, 1)$ to $\mathbf{N} + \mathbf{1}$. Consequently, the integrand is given by:

$$\mathcal{J}_{\{\pi_a\}, \{\lambda_{ab}\}, \mathbf{N}; k}^{\text{D0-D2-D4-D6-D8}} = \mathcal{J}_k^{\text{D0-D0}} \times \prod_{i=1}^k \mathcal{J}_{-\mathfrak{A}}(\phi_i - \mathbf{N} \cdot \epsilon | \rho_{\{\pi_a\}, \{\lambda_{ab}\}}). \quad (5.92)$$

Chapter 6

Conclusion and Outlook

This thesis has studied non-perturbative effects in supersymmetric gauge theories through protected indices and Jeffrey–Kirwan residues. The guiding question is how exact quantities can capture physics that is invisible in ordinary perturbation theory. In the two-dimensional part, elliptic genera test dualities obtained from four-dimensional compactification. In the instanton part, ADHM-type Witten indices compute instanton partition functions and organize gauge-origami fixed points by the shell formula.

Supersymmetry turns these protected observables into exact contour integrals. After localization, the path integral is replaced by a finite-dimensional residue problem, and the JK prescription selects the physical poles:

$$\mathcal{J}_{\text{ell}} = \frac{1}{|W|} \text{JK-Res}_\eta \left[\mathcal{J}_{1\text{-loop}}^{2d}(\mathbf{x}) \prod_{i=1}^r \frac{dx_i}{2\pi i x_i} \right], \quad (6.1)$$

$$\mathcal{Z}_{\text{inst}} = \sum_{k \geq 0} \mathfrak{q}^k \frac{1}{|W_k|} \text{JK-Res}_\eta \left[\mathcal{J}_k^{\text{ADHM}}(\phi) \prod_{I=1}^k \frac{d\phi_I}{2\pi i} \right]. \quad (6.2)$$

This is the technical thread running through the thesis: protected traces are localized to residue integrals, and the residue calculus turns non-perturbative questions into explicit formulas.

6.1 Two-Dimensional Dualities

Chapter 4 concerns two-dimensional $\mathcal{N} = (2, 2)$ and $\mathcal{N} = (0, 2)$ theories obtained from four-dimensional physics by compactification on S^2 . In the $\mathcal{N} = (2, 2)$ part, Lagrangian class- $\mathcal{S}$ theories of type A provide a controlled source of quiver gauge theories. Different pants decompositions of the same parent surface give different ultraviolet quivers, but the elliptic genus is insensitive to the duality frame in the examples considered. The basic building blocks, vector multiplet factors, and gauging operations therefore realize a two-dimensional shadow of the four-dimensional frame-independence.

The compactification inputs and the corresponding two-dimensional structures can be summarized as follows:

Four-dimensional input	Two-dimensional output used in the thesis
Lagrangian class- $\mathcal{S}$ theories of type A	(2, 2) quiver dualities and gluing-independent elliptic genera.
Four-dimensional $\mathcal{N} = 1$ Seiberg dual pairs	(0, 2) SU Seiberg-like dualities with chiral and Fermi multiplets.
Classical-group dualities with tensor matter	Sp/SO trialities, SU/Sp quiver dualities, and symmetric or antisymmetric tensor-matter dualities.
Adjoint and enhanced-supersymmetry examples	Gauge/Landau–Ginzburg or free-field dual descriptions tested by elliptic genera.

In this part, the residue computation is used to test identities of the form

$$\mathcal{J}_{\text{ell}}^{\text{UV}_1}(q, \mathbf{z}) = \mathcal{J}_{\text{ell}}^{\text{UV}_2}(q, \mathbf{z}), \quad (6.3)$$

where the two sides can have different gauge groups, matter fields, and superpotentials. For (2, 2) class- $\mathcal{S}$ examples this equality expresses gluing independence; for (0, 2) examples it is one component of a broader duality check.

The $\mathcal{N} = (0, 2)$ part is more chiral and more delicate. Twisted compactification of four-dimensional dualities produces two-dimensional Seiberg-like dualities, trialities, and tensor-matter dualities. The thesis treats the SU Seiberg-like duality as a basic input, then uses it to study Sp/SO trialities, SU/Sp quiver dualities, dualities with symmetric or antisymmetric chiral multiplets, adjoint examples, and recursive dualities involving both symmetric and antisymmetric matter. In these theories, equality of elliptic genera is only one test. It must be accompanied by gauge anomaly cancellation, flavor and gravitational anomaly matching, compatible R-charge assignments, fugacity maps, and E, J -term consistency.

The resulting conclusion is that compactification from four dimensions is a productive way to generate two-dimensional dualities, but it does not remove the need for two-dimensional checks. Some elliptic-genus identities reduce to analytic theta-function or residue identities. Others are supported by finite-rank JK-residue computations or finite q -expansions, with the range of verification stated explicitly in the main text. This distinction is part of the result: it separates identities established at the level of protected indices from broader conjectural infrared equivalences.

These results also point to several open problems. One would like analytic proofs of the elliptic-genus identities that are currently supported by finite checks. It would also be valuable to understand whether the relation between (0, 2) elliptic genera and four-dimensional chiral-algebra characters, suggested in the class- $\mathcal{S}$ setting^[19,62], extends to the broader reductions studied here. Brane constructions, integrable structures, boundary conditions, and non-Lagrangian two-dimensional theories provide further natural directions.

6.2 Instantons and Gauge Origami

Chapter 5 starts from instanton partition functions. In the Ω -background, the ADHM quantum mechanics for k instantons has a Witten index that localizes to a JK-residue integral. For ordinary five-dimensional gauge theories, the selected poles are labelled by tuples of two-dimensional Young diagrams, and the fixed-point contribution is the usual Nekrasov product written in arm-leg variables.

The main point of the thesis is that this two-dimensional arm-leg language is not flexible enough for the systems encountered in gauge origami. Orientifold projections, non-fundamental matter, Higgsing identities, D0–D6 and D0–D8 systems, spiked instantons, and DT-type counting naturally lead to three-dimensional or four-dimensional Young diagrams, as well as diagrams with asymptotic boundary conditions. The shell formula replaces arm and leg data by the shell of a Young diagram, an integer shell charge, and the associated $\mathcal{J}$ -factor. In this language, the same type of local fixed-point factor appears across several systems.

The basic shell factor used throughout Chapter 5 is

$$\mathcal{J}(x|\mathbf{Y}_{\mathcal{A}}) = \prod_{\mathbf{y} \in \mathcal{S}(\mathbf{Y}_{\mathcal{A}})} \text{sh}(x - \mathcal{X}_{\mathcal{A}}(\mathbf{y}))^{Q_{\mathbf{Y}_{\mathcal{A}}}(\mathbf{y})}. \quad (6.4)$$

Here $\mathcal{S}(\mathbf{Y}_{\mathcal{A}})$ is the shell of the diagram and $Q_{\mathbf{Y}_{\mathcal{A}}}$ is the inclusion-exclusion charge on shell boxes. This formula is the compact replacement for the many arm-leg products that appear in ordinary instanton counting.

The gauge-origami discussion can be summarized by two structural features. First, the D0–D0 sector has a universal form: its determinant is the box-by-box expansion of the $\mathcal{J}$ -factor. Second, the BPS bound states are classified by Young diagrams of the dimension appropriate to the brane system: two-dimensional diagrams for ordinary five-dimensional instantons and spiked instantons, three-dimensional diagrams for tetrahedron instantons and DT3 counting, and four-dimensional diagrams for the magnificent four and DT4 counting. Whenever these two features are present, the shell formula gives a compact way to write the fixed-point contribution.

Within the scope of the thesis, the shell formula is therefore an organizing principle for instanton and gauge-origami partition functions. It rewrites ordinary classical-group instanton factors, D0–D6 tetrahedron-instanton factors, D0–D8 magnificent-four factors, spiked-instanton factors, and DT3/DT4-type factors in a common notation. The cutoff prescriptions for asymptotic diagrams and the special $\mathcal{J}_{\geq}$ -type factors in four dimensions are part of this organization. At the same time, refined DT4 orientations, multiplicities, and sign choices remain genuine extra data and are not hidden inside the definition of the shell charge.

Several open questions remain for instanton counting and gauge origami. It would be useful to clarify the precise relation between the shell formula and topological-vertex formalisms, including orientifold versions. Another direction is to extend the formula to more general matter representations, exceptional gauge groups, orbifold or crystal versions of gauge origami, and the full D0–D2–D4–D6–D8 system. The relation to qq -characters and quantum toroidal algebras also deserves further study^[78,83]. From the viewpoint of enumerative geometry, one expects analogous structures for Donaldson–Thomas theory on more general Calabi–Yau threefolds and fourfolds, where orientation and obstruction-theory issues become essential.

6.3 Final Perspective

The main lesson of the thesis is that supersymmetry does not eliminate non-perturbative phenomena, but it can make protected pieces of them exactly computable. Elliptic genera reduce infrared duality tests to meromorphic identities supplemented by anomaly, R-charge, and superpotential checks. ADHM Witten indices reduce instanton counting

to fixed-point sums, whose combinatorics can be reorganized by Young diagrams and shell factors.

The resulting formulas have several interpretations. As physical observables, they test infrared equivalences and count BPS sectors. As localization integrals, they are JK-residue problems with chamber-dependent pole selection. As mathematical expressions, they produce theta-function identities, equivariant fixed-point contributions, and DT-type counting functions. This interplay between field theory, residue calculus, and combinatorics is the central organizing theme of the thesis.

Appendix A

Conventions, Special Functions, and Basic Shell Calculations

This appendix collects conventions used throughout the thesis. The notation is chosen to stay close to the three source projects while making the common index structure explicit.

A.1 Basic Conventions

Groups, parameters, and exponential variables. Classical gauge groups are denoted by $U(N)$, $SU(N)$, $SO(N)$, and $Sp(N)$. Coulomb branch parameters are written as a_s or v_s depending on the source project. Equivariant parameters are denoted by ϵ_i ; for ordinary instantons on $\mathbb{C}^2$ we use

$$\epsilon_{12} = \epsilon_1 + \epsilon_2, \quad \epsilon_{\pm} = \frac{\epsilon_1 \pm \epsilon_2}{2}. \quad (\text{A.1})$$

For elliptic genera we use $q = e^{2\pi i \tau}$, and for an additive chemical potential u we write the corresponding multiplicative fugacity as $X = e^{2\pi i u}$. In concrete duality formulas this fugacity may be denoted by $x, a_i, b_j, \dots$. For instanton sums we often use $A_s = e^{-a_s}$ and $M = e^{-m}$.

Hyperbolic functions. The 1d index conventions use

$$\text{sh}(x) = e^{x/2} - e^{-x/2}, \quad \text{ch}(x) = e^{x/2} + e^{-x/2}. \quad (\text{A.2})$$

Thus $\text{sh}(-x) = -\text{sh}(x)$ and $\text{ch}(-x) = \text{ch}(x)$. We also use the shorthand

$$\text{sh}(x \pm y) = \text{sh}(x + y) \text{sh}(x - y), \quad (\text{A.3})$$

and similarly for multiple signs. With this convention a 1d chiral multiplet of weight x contributes $1/\text{sh}(x)$ and a 1d Fermi multiplet contributes $\text{sh}(x)$.

Theta functions. The odd theta function and Dedekind eta function are

$$\vartheta_1(X) = -iq^{1/8} X^{1/2} \prod_{n=1}^{\infty} (1 - q^n)(1 - X^{-1}q^n)(1 - Xq^{n-1}), \quad (\text{A.4})$$

$$\eta(q) = q^{1/24} \prod_{n=1}^{\infty} (1 - q^n), \quad (\text{A.5})$$

where $X = e^{2\pi i u}$. The useful multiplicative quasi-periodicities are

$$\vartheta_1(qX) = -q^{-1/2} X^{-1} \vartheta_1(X), \quad (\text{A.6})$$

$$\vartheta_1(q^n X) = (-1)^n q^{-n^2/2} X^{-n} \vartheta_1(X), \quad n \in \mathbb{Z}. \quad (\text{A.7})$$

In 2d (0, 2) elliptic genera, a chiral multiplet of total fugacity X gives $i\eta(q)/\vartheta_1(X)$, and a Fermi multiplet gives $\vartheta_1(X)/(i\eta(q))$.

Young diagrams and pole coordinates. A two-dimensional Young diagram is denoted by λ , and a tuple by $\vec{\lambda} = (\lambda^{(1)}, \dots, \lambda^{(N)})$. A box $x = (x_1, x_2)$ in color s gives the unrefined pole

$$\phi_s(x) = a_s + (x_1 - x_2)\hbar. \quad (\text{A.8})$$

For a Young diagram of arbitrary dimension we write $\mathbf{Y}_{\mathcal{A}}$, where $\mathcal{A} = (A, \alpha)$ records the chosen coordinate directions and a color. Its pole map is

$$\mathcal{X}_{\mathcal{A}}(x) = v_{\mathcal{A}} + \sum_{i=1}^d (x_i - 1)\epsilon_{a_i}. \quad (\text{A.9})$$

A.2 Examples of Shellbox Charges

This section refines the Young-diagram notation above for the gauge-origami examples and presents the basic shell computations used in Sec. 5.1.

Labels of Young diagrams and notation. To concisely indicate the required ϵ_i parameter and the basis of Young diagrams, we adopt the following simplified notation:

- $\underline{\mathbf{4}} = \{1, 2, 3, 4\}$, $a, b \in \underline{\mathbf{4}}$. Elements of $\underline{\mathbf{4}}$ index the four complex directions of $\mathbb{C}^4$; they label individual D-brane worldvolume directions and appear as subscripts on the Ω -background parameters ϵ_a .
- $\check{\underline{\mathbf{4}}} = \{123, 124, 134, 234\} = \{\bar{4}, \bar{3}, \bar{2}, \bar{1}\}$, $A, B \in \check{\underline{\mathbf{4}}}$. Elements of $\check{\underline{\mathbf{4}}}$ index the four coordinate hyperplanes (complements of each direction in $\underline{\mathbf{4}}$); $A \in \check{\underline{\mathbf{4}}}$ specifies the basis directions of a 3d or 4d Young diagram, and appears in the labels of tetrahedron instanton and D0-D8 systems (Sec. 5.3.2–5.3.5).
- $\underline{\mathbf{6}} = \{12, 13, 23, 14, 24, 34\}$, $ab \in \underline{\mathbf{6}}$. Elements of $\underline{\mathbf{6}}$ index the six coordinate 2-planes; $ab \in \underline{\mathbf{6}}$ labels D4-brane orientations in the spiked instanton and DT3 systems (Sec. 5.3.3–5.3.4), and specifies the basis of a 2d Young diagram.

The Ω -background parameters $\epsilon_{1,2,3,4}$ are also denoted as:

$$q_i \equiv e^{-\epsilon_i}, \quad q_{i_1 \dots i_s} = q_{i_1} \dots q_{i_s} = e^{-(\epsilon_{i_1} + \dots + \epsilon_{i_s})} \quad (\text{A.10})$$

in this thesis, we adopt the CY fourfold condition $\epsilon_4 = -\epsilon_{123}$ as a default assumption.

Each label $\mathcal{A} = (A, \alpha)$ consists of a basis specification $A \in \check{\underline{\mathbf{4}}} \cup \underline{\mathbf{6}} \cup \underline{\mathbf{4}}$ and a color index α counting Young diagrams with the same basis. We employ $\mathcal{A}$, $\mathcal{B}$, $\mathbf{ab}$ and $\mathbf{ab}'$

to denote such labels; for instance, $\mathcal{A} = (234, 7)$ means $\mathbf{Y}_{\mathcal{A}}$ is the 7th Young diagram in the basis $\epsilon_2, \epsilon_3, \epsilon_4$.

The coordinate function converting box positions to integration variables is:

$$\mathcal{X}_{\mathcal{A}}(\mathbf{x}) \equiv v_{\mathcal{A}} + (\mathbf{x} - \mathbf{1}) \cdot \epsilon_A = v_{\mathcal{A}} + \sum_{i=1}^d (x_i - 1) \epsilon_{a_i} \quad (\text{A.11})$$

This is the function appearing in the pole classification of Sec. 5.1: a JK-selected pole at ϕ_* corresponds to setting $\phi_{i*} = \mathcal{X}_{\mathcal{A}}(\mathbf{x})$ for each box $\mathbf{x}$ in the corresponding Young diagram.

Given a 2d Young diagram λ , we can define, for each box $\mathbf{x}$ in λ , its leg $L_{\lambda}(\mathbf{x})$ and arm $A_{\lambda}(\mathbf{x})$ as illustrated in Fig. A-1. The leg is the number of boxes in direction 1 from $\mathbf{x}$ within λ , while the arm is the number of boxes in direction 2.

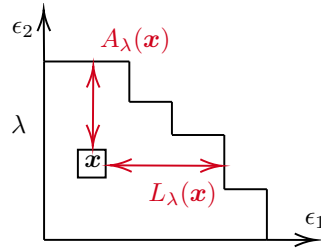


Figure A-1 Leg length $L_{\lambda}(\mathbf{x})$ and arm length $A_{\lambda}(\mathbf{x})$ of a box $\mathbf{x}$ in a Young diagram λ .

Given a tuple of Young diagrams $\vec{\mathbf{Y}} = (\mathbf{Y}_{\mathcal{A}}, \dots)$, the total number of boxes is defined as follows. For a single Young diagram $\mathbf{Y}_{\mathcal{A}}$, we denote its number of boxes by $|\mathbf{Y}_{\mathcal{A}}|$. Then, the total number of boxes for the tuple is defined as $||\vec{\mathbf{Y}}|| \equiv \sum_{\mathcal{A}} |\mathbf{Y}_{\mathcal{A}}|$.

The sets $\underline{4}$, $\check{4}$, and $\underline{6}$ together cover all the index types needed in this thesis: $\check{4}$ appears in the tetrahedron instanton and D0-D8 system (Sec. 5.3.2–C.1.5), $\underline{6}$ in the spiked instanton and DT3 (Sec. 5.3.3–C.1.4), and $\underline{4}$ in the magnificent four and DT4 (Sec. 5.3.1–C.1.6).

A.2.1 Shell and $\mathcal{J}$ -factor

This subsection presents three progressively higher-dimensional examples of shell and $\mathcal{J}$ -factor computations: a single-box 2d diagram worked out fully from Definition 5.7, a general 2d diagram showing the charge-to-box correspondence, and a single-box 3d diagram that is the fundamental building block of all 3d partition functions in this thesis.

- **2d single box:** $\lambda_{12,1} = \{(1, 1)\}$. With $\mathbf{B}_2 = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$ from Definition 5.7, the shell is:

$$\begin{aligned} \mathcal{S}(\lambda_{12,1}) &= (\lambda_{12,1} + \mathbf{B}_2) \setminus \lambda_{12,1} \\ &= \{(1, 1), (1, 2), (2, 1), (2, 2)\} \setminus \{(1, 1)\} \\ &= \{(1, 2), (2, 1), (2, 2)\} \end{aligned} \quad (\text{A.12})$$

Therefore, the charge of each shellbox is defined as (5.8):

$$\begin{aligned} Q_{\lambda_{12,1}}(1, 2) &= (-1)^{|(0,1)|} = -1 \\ Q_{\lambda_{12,1}}(2, 1) &= (-1)^{|(1,0)|} = -1 \\ Q_{\lambda_{12,1}}(2, 2) &= (-1)^{|(1,1)|} = 1 \end{aligned} \quad (\text{A.13})$$

The $\mathcal{F}$ -factor (5.9) is therefore:

$$\begin{aligned} \mathcal{F}(x|\lambda_{12,1}) &= \frac{\text{sh}(x - \mathcal{X}_{12,1}(2, 2))}{\text{sh}(x - \mathcal{X}_{12,1}(1, 2)) \text{sh}(x - \mathcal{X}_{12,1}(2, 1))} \\ &= \frac{\text{sh}(x - v_{12,1} - \epsilon_{12})}{\text{sh}(x - v_{12,1} - \epsilon_1) \text{sh}(x - v_{12,1} - \epsilon_2)} \end{aligned} \quad (\text{A.14})$$

The Young diagram, its shell, and the corresponding charges are illustrated in Fig. A-2.

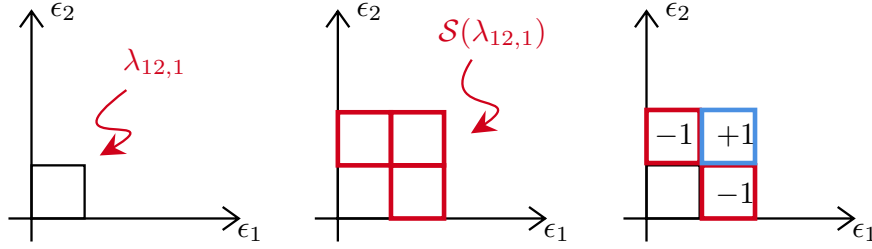


Figure A-2 The leftmost diagram shows a 2d Young diagram $\lambda_{12,1}$ with only one box, labeled $\{12, 1\}$ indicating it is the first Young diagram in the 12-plane. In the middle diagram, the red boxes represent the shell $\mathcal{S}(\lambda_{12,1})$ of $\lambda_{12,1}$; for a 2d Young diagram with only one box, its shell $\mathcal{S}(\lambda_{12,1})$ consists of only 3 shellboxes. In the rightmost diagram, the charge of each shellbox is shown, where the $+1$ shellboxes are marked in blue and the -1 shellboxes are marked in red.

- For a more general 2d Young diagram λ , Fig. A-3 shows the shell and the charges of each shellbox. The charge pattern reveals a precise combinatorial correspondence:

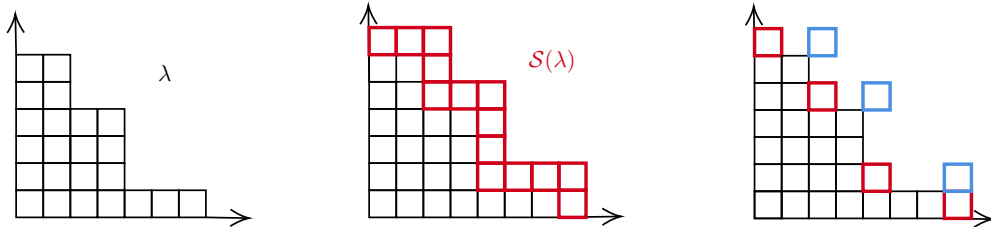


Figure A-3 For a general 2d Young diagram λ (leftmost), the shellboxes $\mathcal{S}(\lambda)$ are marked in the middle figure. In the rightmost figure, red boxes have -1 charge, blue boxes have $+1$ charge, and unmarked shellboxes have 0 charge.

Property (2d charge-box correspondence). For any 2d Young diagram λ , the shellboxes with charge -1 are exactly the addable boxes $\mathfrak{A}(\lambda)$ (positions where a new box may be placed while preserving the Young diagram property), and each shellbox with charge $+1$ at position (i, j) corresponds to a removable box $\mathfrak{R}(\lambda)$ at position $(i - 1, j - 1)$.

This correspondence is a consequence of the inclusion-exclusion definition (5.8): for a 2d diagram, each shell box has at most one binary neighbor inside λ , so only charges ± 1 and 0 appear, and the geometric roles of ± 1 boxes are exactly as stated. As a result, the $\mathcal{F}$ -factor for a 2d Young diagram can be expressed directly

in terms of addable and removable boxes:

$$\mathcal{J}(x|\lambda_{\mathbf{ab}}) = \frac{\prod_{y \in \mathfrak{R}} \text{sh}(x - \mathcal{X}_{\mathbf{ab}}(y + \mathbf{1}))}{\prod_{y \in \mathfrak{A}} \text{sh}(x - \mathcal{X}_{\mathbf{ab}}(y))} \quad (\text{A.15})$$

- For 3d Young diagrams, the charge-to-box correspondence of the previous item breaks down. As illustrated in Fig. A-4, a 3d shellbox may carry charges 0, ± 1 , or even ± 2 —values that do not correspond to single addable or removable boxes. Consequently, the $\mathcal{J}$ -factor for a 3d Young diagram cannot be expressed in the same form as (A.15).

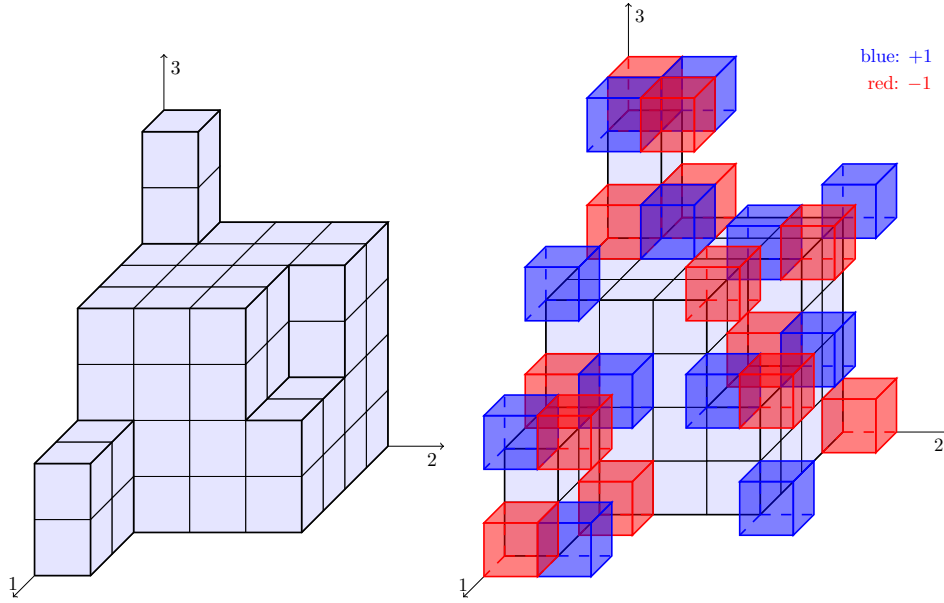


Figure A-4 For an arbitrary 3d Young diagram (left), the shellbox charges are shown on the right: red boxes carry charge -1 and blue boxes carry charge $+1$.

Example (3d single box). The 3d Young diagram $\pi_{\bar{4},1} = \{(1, 1, 1)\}$ has binary set $\mathbf{B}_3$ of size $2^3 = 8$. Applying (5.7) and (5.8) gives:

$$\begin{aligned} Q = +1 : & \quad (1, 2, 2), (2, 1, 2), (2, 2, 1) \\ Q = -1 : & \quad (1, 1, 2), (1, 2, 1), (2, 1, 1), (2, 2, 2) \end{aligned}$$

The $\mathcal{J}$ -factor (5.9) is therefore:

$$\mathcal{J}(x|\{(1, 1, 1)\}_{\bar{4},1}) = \frac{\text{sh}(x - v_{\bar{4},1} - \epsilon_{12}) \text{sh}(x - v_{\bar{4},1} - \epsilon_{13}) \text{sh}(x - v_{\bar{4},1} - \epsilon_{23})}{\text{sh}(x - v_{\bar{4},1} - \epsilon_{1,2,3}) \text{sh}(x - v_{\bar{4},1} - \epsilon_{123})} \quad (\text{A.16})$$

This is the building block of the tetrahedron instanton and DT3 partition functions in Sec. 5.3.2–C.1.4.

- For a d -dimensional Young diagram, the charge of a shellbox can take any integer value from $-d$ to d . The extreme values $\pm d$ arise only at the corner of the Young diagram, where the box $(1, 1, \dots, 1)$ is adjacent to all 2^d binary neighbors simultaneously: $Q = -d$ occurs when none of those neighbors is in $\mathbf{Y}$ (the box

is addable in all d directions at once), and $Q = +d$ when all are in $\mathbf{Y}$. In practice, charges beyond ± 1 first appear in 3d diagrams, and for 4d diagrams the charges $Q = +2$ and $Q = -3$ visible in Tab. C-8 reflect the geometry of four infinite legs meeting at a common origin.

Appendix B

Additional Two-Dimensional Duality Checks

This appendix collects the longer checks and derivations for the two-dimensional dualities used in Chapter 4.

B.1 Some details on 2d $\mathcal{N} = (0, 2)$ dualities

This appendix presents the detailed and technical aspects of the 2d $\mathcal{N} = (0, 2)$ dualities discussed in §4.3. While the main chapters focuses on key results, some derivations, additional dualities, and further supporting evidence are presented here. Although these details are too technical for the main chapters, they are essential for an understanding of the underlying structure of $\mathcal{N} = (0, 2)$ dualities.

B.1.1 Twisted compactification of 4d Intriligator-Seiberg duality on S^2

We consider the twisted compactification of 4d $\mathcal{N} = 1$ Intriligator-Seiberg duality^[68] on S^2 , resulting in a 2d $(0, 2)$ SO gauge theory dual to Landau-Ginzburg model. Our approach follows a method similar to that used in^[18]. Tab. B-1 summarizes the matter contents of the 4d $\mathcal{N} = 1$ Intriligator-Seiberg duality where the theory in the right has a superpotential

$$\mathcal{W} = \hat{Q}^i \hat{Q}^j M_{ij} . \quad (\text{B.1})$$

	$\text{SO}(N_c)$	$\text{SU}(N_f)$	$\text{U}(1)_R$
Q	$\square$	$\square$	$\frac{N_f - N_c + 2}{N_f}$
	$\text{SO}(N_f - N_c + 4)$	$\text{SU}(N_f)$	$\text{U}(1)_R$
$\hat{Q}$	$\square$	$\overline{\square}$	$\frac{N_c - 2}{N_f}$
M	$\mathbf{1}$	$\mathbf{Sym}$	$\frac{2(N_f - N_c + 2)}{N_f}$

Table B-1 Matter contents of 4d $\mathcal{N} = 1$ SO Intriligator-Seiberg duality

Upon compactification on S^2 , we reassign non-negative integer R -charges r_a to each fundamental chiral multiplet in the left theory and $\hat{r}_a$ to those in the right theory^[54]. In

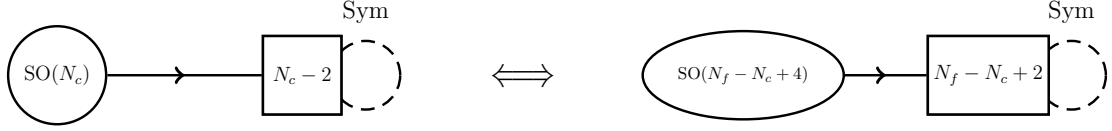


Figure B-1 A trivial 2d $\mathcal{N} = (0, 2)$ duality obtained from the compactification of 4d $\mathcal{N} = 1$ Intriligator-Seiberg duality on S^2 .

4d, the $U(1)_R SO(N_c) SO(N_c)$ mixed gauge anomaly can be calculated as

$$\text{Tr } RGG = T_{SO(\square)} \sum_{a=1}^{N_f} (r_a - 1) + T_{SO(\mathbf{Adj})} = 0. \quad (\text{B.2})$$

This anomaly cancellation imposes the following constraint on the R -charges of the fundamental chirals:

$$\sum_{a=1}^{N_f} r_a = N_f - N_c + 2. \quad (\text{B.3})$$

Similarly, in the dual theory, the anomaly condition leads to:

$$\sum_{a=1}^{N_f} \hat{r}_a = N_c - 2. \quad (\text{B.4})$$

To satisfy these conditions, we assign $r_a = 1$ for $a = 1, \dots, N_f - N_c + 2$ and $r_a = 0$ for the remaining indices. The dual R -charges are then given by $\hat{r}_a = 1 - r_a$. Consequently, the R -charges of the mesons in the symmetric representation of the dual theory are determined as:

$$r(M_{ab}) = 2 - \hat{r}_a - \hat{r}_b, \quad a \leq b. \quad (\text{B.5})$$

Following^[54], upon compactification on S^2 , the 2d $(0, 2)$ chiral multiplets arise from 4d fields with integral R -charges less than 1, while 2d $(0, 2)$ Fermi multiplets originate from 4d fields with integral R -charges greater than 1. 4d fields with R -charge exactly equal to 1 do not contribute to the 2d theory.

Thus, we arrive at the conclusion that the 2d $(0, 2)$ $SO(N_c)$ gauge theory with $N_c - 2$ fundamental chirals is dual to the $SO(N_f - N_c + 4)$ gauge theory, which includes $N_f - N_c + 2$ fundamental chirals, $(N_c - 2)(N_c - 1)/2$ chiral mesons, and $(N_f - N_c + 2)(N_f - N_c + 3)/2$ Fermi mesons. After transferring the chiral mesons to the other side, we obtain the duality depicted in Fig. B-1.

It is evident from Fig. B-1 that the dual theories are independent of the rank of the gauge groups. By evaluating the elliptic genus

$$\begin{aligned} \mathcal{F} &= \frac{2\eta(q)^{\frac{3N-3\chi-2}{2}} \prod_{i \leq j} \vartheta_1(x^2 b_i b_j)}{2^{[(N-1)/2]} [N/2]!} \oint_{JK} \frac{d\mathbf{a}}{2\pi i \mathbf{a}} \frac{\prod_{i < j}^{[N/2]} \vartheta_1(a_i^\pm a_j^\pm)}{\prod_{i=1}^{[N/2]} \prod_{j=1}^{N-2} \vartheta_1(x a_i^\pm b_j)} \left(\frac{\prod_{i=1}^{[N/2]} \vartheta_1(a_i^\pm)}{\prod_{j=1}^{N-2} \vartheta_1(x b_j)} \right)^\chi \\ &= \frac{\vartheta_1(x^{N-2})}{\eta(q)}, \end{aligned} \quad (\text{B.6})$$

we conclude that the theory is dual to one free Fermi field.

By reorganizing the field content, we obtain the 2d $(0, 2)$ version of the Intriligator-Seiberg duality between Sym-1 and Sym-3 in Fig. 4-8.

B.1.2 Triality with $SU(N)$ flavor symmetry for odd N

We have identified a new triality, as illustrated in Fig. 4-6, in which the flavor symmetry is $SU(N)$ with N being even. Building on this discovery, we now present a closely related triality that arises when the flavor symmetry is $SU(N)$ with N being odd. This new case exhibits analogous structures while introducing subtle differences in superpotentials due to the parity of N . We propose that the following three theories become equivalent in the infrared.

AS-1'. $Sp(N-1)$ gauge theory with N fundamental chirals P , one fundamental chiral Q , and no superpotential.

AS-2'. $SU(N-2)$ gauge theory with one antisymmetric chiral X , N fundamental chirals Y and N chiral mesons Z . Additionally, there is a neutral Fermi multiplet Ψ , forming a superpotential

$$\mathcal{W} = \Psi(\epsilon^{\alpha_1 \dots \alpha_{N-2}} X_{\alpha_1 \alpha_2} \dots X_{\alpha_{N-4} \alpha_{N-3}} Y_{\alpha_{N-2}}^i Z_i). \quad (\text{B.7})$$

AS-3'. LG model of one Fermi Ψ , N chiral mesons Z and $\frac{1}{2}N(N-1)$ chirals, forming an antisymmetric $N \times N$ matrix A with a superpotential

$$\mathcal{W} = \Psi(\epsilon^{i_1 \dots i_N} A_{i_1 i_2} \dots A_{i_{N-2} i_{N-1}} Z_{i_N}). \quad (\text{B.8})$$

The $U(1)_x$ and the $U(1)_y$ charges of the fields are given as follows:

	P	Q	Ψ	A	X	Y	Z
$U(1)_x$	1	0	$-N$	2	$\frac{2N}{N-2}$	$\frac{2}{N-2}$	1
$U(1)_y$	0	-1	1	0	0	0	-1

(B.9)

The central charges of these theories agree:

$$c_L = (N+2)(N-1) \quad c_R = \frac{3}{2}(N+2)(N-1). \quad (\text{B.10})$$

Their elliptic genera agree in the checked JK-residue computation:

$$\begin{aligned} \mathcal{J} &= -\frac{\eta(q)^{(N^2+4N-5)/2}}{2^{\frac{N-1}{2}} \frac{N-1}{2}!} \oint_{\text{JK}} \frac{d\mathbf{a}}{2\pi i \mathbf{a}} \prod_{i=1}^{\frac{N-1}{2}} \frac{\vartheta_1(a_i^{\pm 2}) \prod_{j < i} \vartheta_1(a_i^{\pm} a_j^{\pm})}{\prod_{j=1}^N \vartheta_1(a_i^{\pm} b_j x) \cdot \vartheta_1(a_i^{\pm} y^{-1})}, \\ &= \frac{\eta(q)^{\frac{N^2+7N-20}{2}} \vartheta_1(x^{-N} y)}{(N-2)! \prod_{i=1}^N \vartheta_1(b_i y^{-1} x)} \oint_{\text{JK}} \frac{d\mathbf{a}}{2\pi i \mathbf{a}} \prod_{i=1}^{N-2} \frac{\prod_{j \neq i} \vartheta_1(a_i/a_j)}{\prod_{j < i} \vartheta_1(a_i a_j x^{\frac{2N}{N-2}}) \cdot \prod_{j=1}^N \vartheta_1(a_i b_j^{-1} x^{\frac{2}{N-2}})}, \\ &= \frac{\eta(q)^{(N^2+N-2)/2} \vartheta_1(x^{-N} y)}{\prod_{i < j} \vartheta_1(x^2 b_i b_j) \prod_{i=1}^N \vartheta_1(b_i y^{-1} x)}. \end{aligned} \quad (\text{B.11})$$

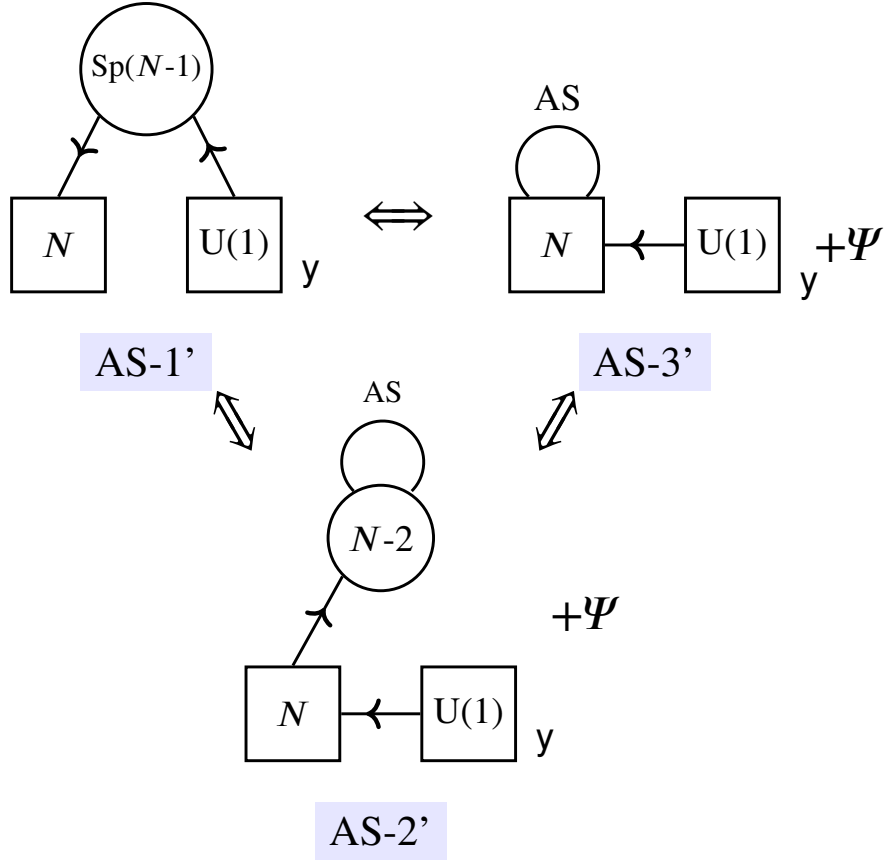


Figure B-2 A triality among three distinct theories: (AS-1') an $Sp(N-1)$ gauge theory of SQCD type with $N+1$ fundamental chirals, (AS-2') an $SU(N-2)$ gauge theory with one antisymmetric, N fundamental chirals and N chiral mesons, coupled to a neutral Fermi multiplet Ψ , and (AS-3') a Landau-Ginzburg model consisting of $\frac{1}{2}N(N-1)$ chiral multiplets and N chiral mesons coupled to a Fermi multiplet Ψ .

B.1.3 SU gauge theory with one antisymmetric and one fundamental chiral

In §4.3.5, we studied the dualities of $SU(n)$ gauge theories with one antisymmetric chiral multiplet. For the duality in Fig. 4-12 to hold, the number of fundamental chirals must satisfy $N \geq 2$, as required by condition (4.70). Here, we consider the special case where the number of fundamental chirals is reduced to one, i.e., $N = 1$.

As illustrated in Fig. B-3, a sequence of dualities leads to an equivalence between an $SU(n)$ gauge theory and an $Sp(n+m-3)$ gauge theory, where m is either 0 or 1 such that $n+m$ is odd. The matter content and superpotentials for both theories are detailed below.

- **Theory A:** an $SU(n-1)$ gauge theory with the following matters:
 - One antisymmetric chiral Y
 - One fundamental chiral X_2
 - n anti-fundamental chirals X_1
 - One free Fermi Ψ that transforms as $(1-n, m)$ under the symmetries $U(1)_0 \times U(1)_1$.

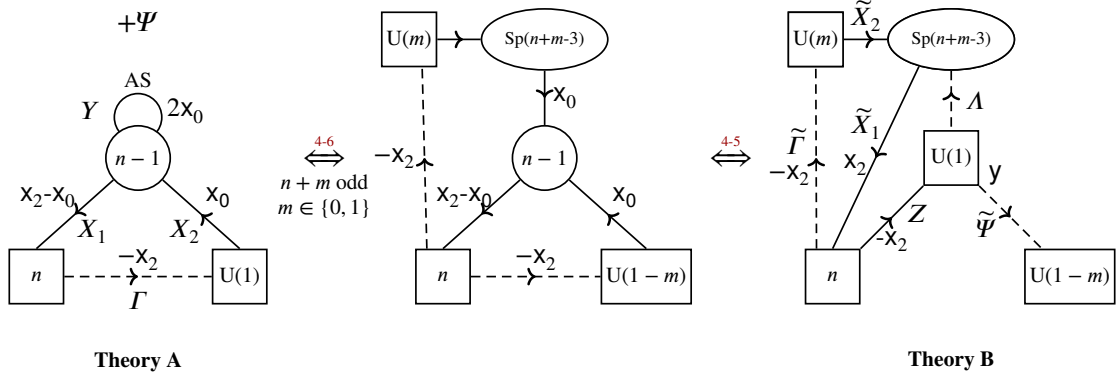


Figure B-3 A duality between $SU(n)$ and $Sp(n + m - 3)$ gauge theories, where m is either 0 or 1 such that $n + m$ is odd. The transformation between Theory A and Theory B follows 4-6 and 4-8, as indicated. The chemical potential is given by $y = nx_2 - (n - 1)x_1$.

- A Fermi meson Γ

The superpotential of the theory takes the following form:

$$\mathcal{W} = \begin{cases} \Psi \text{Pf } Y + \text{Tr } \Gamma X_1 X_2 & \text{when } n \text{ is odd} \\ \Psi \epsilon^{\alpha_1 \dots \alpha_{n-1}} Y_{\alpha_1 \alpha_2} \dots Y_{\alpha_{n-3} \alpha_{n-2}} X_{2, \alpha_{n-1}} + \text{Tr } \Gamma X_1 X_2 & \text{when } n \text{ is even} \end{cases} \quad (\text{B.12})$$

- **Theory B:** an $Sp(n + m - 3)$ gauge theory, where m is either zero or one such that $n + m$ is odd, with the following matters:
 - One fundamental chiral $\tilde{X}_1$ with charge $(\square, \square, 1)$ under the flavor symmetries $Sp(n + m - 3) \times SU(n) \times U(1)_2$.
 - One fundamental chiral $\tilde{X}_2$ with charge $(\square, -m)$ under the symmetries $Sp(n + m - 3) \times U(m)_1$.
 - One chiral meson Z that charges $(\overline{\square}, 1 - n, n - 1)$ under $SU(n) \times U(1)_0 \times U(1)_2$.
 - One fundamental Fermi Λ with charge $(\square, n - 1, -n)$ under $Sp(n + m - 3) \times U(1)_0 \times U(1)_2$.
 - One Fermi meson $\tilde{\Gamma}$ with charge $(\overline{\square}, 1, -1)$ under $SU(n) \times U(m)_1 \times U(1)_2$.
 - One single free Fermi $\tilde{\Psi}$ with charge $(n - 1, 1, -n)$ under $U(1)_0 \times U(1 - m)_1 \times U(1)_2$.

The superpotential of theory B is given by:

$$\mathcal{W} = \text{Tr } \tilde{\Gamma} \tilde{X}_1 \tilde{X}_2 + \text{Tr } \Lambda \tilde{X}_1 Z \quad (\text{B.13})$$

The central charges agree as

$$c_L = n^2 - n - 2, \quad c_R = \frac{3}{2}(n^2 - n - 2). \quad (\text{B.14})$$

Moreover, the elliptic genera agree and are expressed as follows:

$$\mathcal{J}_A = \frac{\eta(q)^{\frac{n^2+5n-14}{2}}}{(n-1)!} \oint_{\text{JK}} \frac{da}{2\pi i a} \prod_{i=1}^{n-1} \frac{\prod_{j \neq i} \vartheta_1(a_i/a_j)}{\prod_{j < i} \vartheta_1(x_0^2 a_i a_j) \cdot \prod_{j=1}^n \vartheta_1(a_i^{-1} c_j x_2 x_0^{-1}) \cdot \vartheta_1(a_i x_1^{-1} x_0)} \\ \times \vartheta_1(x_0^{1-n} x_1^m) \prod_{i=1}^n \vartheta_1(c_i^{-1} x_1 x_2^{-1}), \quad (\text{B.15})$$

$$\mathcal{J}_B = \frac{\eta(q)^{N_B(N_B+6)/2}}{2^{\frac{N_B}{2}} \frac{N_B}{2}!} \oint_{\text{JK}} \frac{dg}{2\pi i g} \prod_{i=1}^{\frac{N_B}{2}} \frac{\vartheta_1(g_i^{\pm 2}) \prod_{j < i} \vartheta_1(g_i^{\pm} g_j^{\pm}) \cdot \vartheta_1(g_i^{\pm} y^{-1})}{\prod_{j=1}^n \vartheta_1(x_2 g_i^{\pm} c_j) \cdot \vartheta_1(g_i^{\pm} x_1^{-1})^m} \\ \times \frac{\vartheta_1(y^{-1} x_1)^{1-m} \prod_{i=1}^n \vartheta_1(x_1 x_2^{-1} c_i^{-1})^m}{\eta(q)^{(m-1)(n-1)} \prod_{i=1}^n \vartheta_1(y c_i^{-1} x_2^{-1})}, \quad (\text{B.16})$$

where $N_B = n + m - 3$ and $y = x_1^{-(n-1)} x_2^n$.

B.1.4 SU gauge theory with one antisymmetric and no fundamental chiral

Finally, we examine the case where there is no fundamental chiral multiplet, i.e., $N = 0$. This corresponds to an $\text{SU}(n-2)$ gauge theory with one antisymmetric chiral multiplet and n anti-fundamental chirals. One approach to studying this case is to directly compute the elliptic genus using the JK residue integrals. Alternatively, we can derive the result through a sequence of known dualities and operations.

In this case, we consider transformations that provide the identities of the elliptic genus, focusing only on the field contents while ignoring detailed aspects of the theory, such as the superpotential. As before, this depends on whether n is even or odd: we apply the duality in Fig. 4-6 or Fig. B-2 to the antisymmetric chiral multiplet, followed by a sequence of additional dualities and transformations.

Let us consider the case of even n , and Fig. B-4 illustrates a series of transformations that provides the identity of elliptic genera. Based on this identity, we propose the duality between the following two theories:

- An $\text{SU}(n-2)$ gauge theory with one antisymmetric chiral X , n anti-fundamental chirals Y , and a neutral Fermi multiplet Ψ . The superpotential is given by:

$$\mathcal{W} = \Psi \text{Pf } X. \quad (\text{B.17})$$

- A disjoint union of $\frac{n-2}{2}$ copies of a theory with $\frac{1}{2}n(n-1)$ chirals, forming an antisymmetric $n \times n$ matrix A , and a Fermi multiplet Ψ , with the superpotential:

$$\mathcal{W} = \Psi \text{Pf } A. \quad (\text{B.18})$$

The $\text{U}(1)_x$ charges of the fields are summarized as follows,

$$\begin{array}{c|cccc} & X & Y & \Psi & A \\ \hline \text{U}(1)_x & \frac{2n}{n-2} & -\frac{2}{n-2} & -n & 2 \end{array}. \quad (\text{B.19})$$

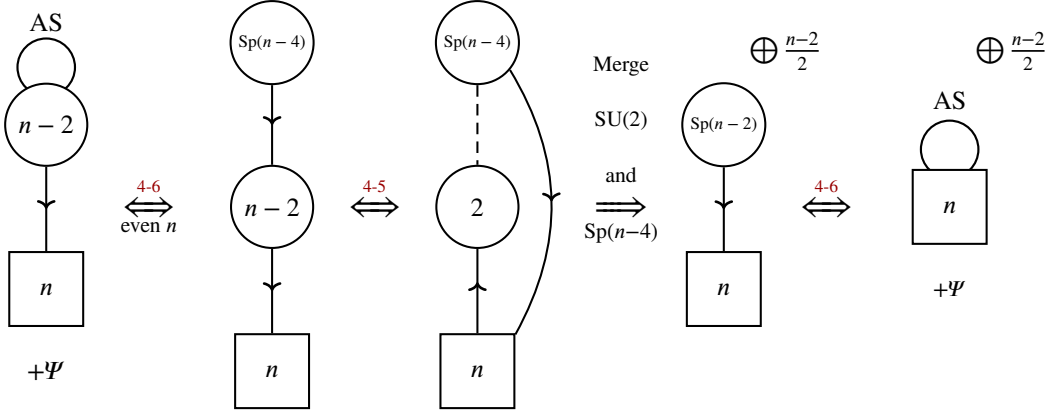


Figure B-4 A sequence of transformations applied to the $SU(n-2)$ gauge theory with one antisymmetric chiral and n anti-fundamental chirals where n is even. In the transition from the third to the fourth diagram, the $Sp(n-4)$ and $SU(2)$ gauge nodes are merged into a single $Sp(n-2)$ node. This merging introduces an additional factor of $(n-2)/2 = |W(Sp(n-2))|/|W(Sp(n-4)) \times W(SU(2))|$, reflecting the ratio of Weyl group orders in the elliptic genus. Additionally, at this step, the direction of the arrow between nodes 2 and n is reversed from $(2 \leftarrow n)$ to $(2 \rightarrow n)$, utilizing the identity $\vartheta_1(z) = -\vartheta_1(1/z)$. For the 4th and 5th quiver diagrams, we consider a disjoint union of $(n-2)/2$ copies of the theory. Through these transformations, we establish an identity between the elliptic genera at each step of the sequence.

The elliptic genera are given by

$$\begin{aligned} \mathcal{J} &= -\frac{\eta(q)^{\frac{n^2+5n-20}{2}} \vartheta_1(x^{-n})}{(n-2)!} \oint_{JK} \frac{da}{2\pi i a} \prod_{i=1}^{n-2} \frac{\prod_{j \neq i} \vartheta_1(a_i/a_j)}{\prod_{j < i} \vartheta_1(x^{\frac{2n}{n-2}} a_i a_j) \cdot \prod_{j=1}^n \vartheta_1(a_i^{-1} b_j x^{-\frac{2}{n-2}})} \\ &= \frac{n-2}{2} \frac{\vartheta_1(x^{-n})}{\eta(q)} \prod_{i < j} \frac{\eta(q)}{\vartheta_1(x^2 b_i b_j)}. \end{aligned} \quad (B.20)$$

Now we turn to the case of odd n , and Fig. B-5 illustrates a sequence of transformations that provide the identities of the elliptic genus. Based on this, we propose a duality between the following two theories

- An $SU(n-2)$ gauge theory with one antisymmetric chiral X , n anti-fundamental chirals Y , and one Fermi multiplet Ψ . The superpotential is given by:

$$\mathcal{W} = \Psi \det X. \quad (B.21)$$

- A disjoint union of $\frac{n-1}{2}$ copies of a theory with $\frac{n(n-1)}{2}$ chiral multiplets A , and one Fermi multiplet Ψ . The superpotential is given by:

$$\mathcal{W} = \Psi \det A. \quad (B.22)$$

The $U(1)_x$ and $U(1)_d$ charges of the fields are summarized as follows:

	Ψ	X	Y	A
$U(1)_x$	$-2n$	$\frac{2n}{n-2}$	$-\frac{2}{n-2}$	2

(B.23)

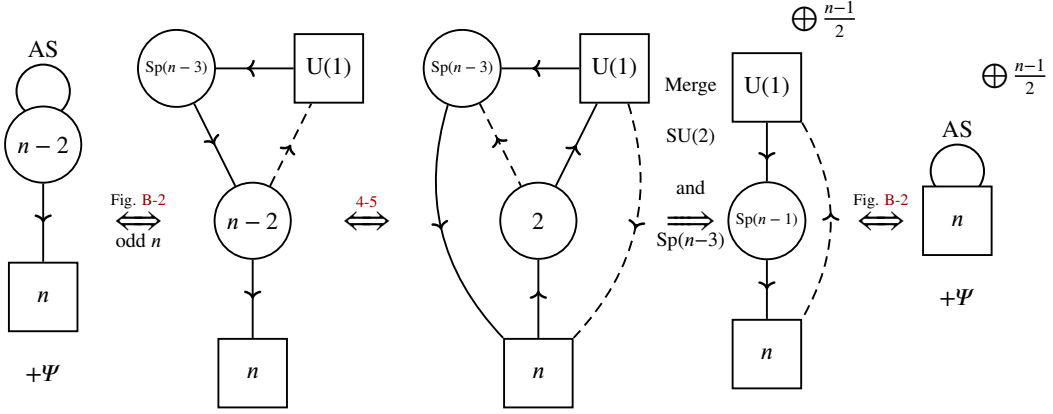


Figure B-5 A sequence of transformations applied to the $SU(n-2)$ gauge theory with one antisymmetric chiral and n anti-fundamental chirals where n is odd. In the transition from the third to the fourth diagram, the $Sp(n-3)$ and $SU(2)$ gauge nodes are merged into a single $Sp(n-1)$ node. This merging introduces an additional factor of $(n-1)/2 = |W(Sp(n-1))|/|W(Sp(n-3)) \times W(SU(2))|$, reflecting the ratio of Weyl group orders in the elliptic genus. Additionally, at this step, the direction of the arrow between nodes 2 and n is reversed from $(2 \leftarrow n)$ to $(2 \rightarrow n)$, utilizing the identity $\vartheta_1(z) = -\vartheta_1(1/z)$. For the 4th and 5th quiver diagrams, we consider a disjoint union of $(n-1)/2$ copies of the theory. Through these transformations, we establish an identity between the elliptic genera at each step of the sequence.

Therefore, under the duality, there is a correspondence between the operator $\Psi(\det B)^2$ and Ψ . The elliptic genera are given by:

$$\begin{aligned}
 \mathcal{J} &= -\frac{\eta(q)^{\frac{n^2+5n-20}{2}} \vartheta_1(x^{-2n})}{(n-2)!} \oint_{JK} \frac{da}{2\pi i a} \prod_{i=1}^{n-2} \frac{\prod_{j \neq i} \vartheta_1(a_i/a_j)}{\prod_{j < i} \vartheta_1(a_i a_j x^{\frac{2n}{n-2}}) \cdot \prod_{j=1}^n \vartheta_1(a_i^{-1} b_j x^{-\frac{2}{n-2}})}, \\
 &= \frac{n-1}{2} \frac{\vartheta_1(x^{-2n})}{\eta(q)} \prod_{i < j} \frac{\eta(q)}{\vartheta_1(x^2 b_i b_j)}.
 \end{aligned} \tag{B.24}$$

Appendix C

Detailed Shell-Formula Computations

This appendix collects explicit low-instanton and low-box computations illustrating the shell formula. Definition-level examples of shell boxes and shell charges are kept in Appendix A; here they are used only inside fixed-point calculations.

C.1 Detailed computations for various cases

This appendix provides explicit low-instanton calculations that confirm the main-text formulas and illustrate how the shell formula operates in practice. The subsections are organized in order of increasing Young diagram dimension, and each is self-contained.

- Appendix C.1.1 ($U(N)$ SYM, 2d Young diagrams) provides a proof of the equivalence (5.31) between the shell formula and the Nekrasov factor.
- Appendix C.1.2 ($Sp(2)$ SYM, 2d Young diagrams) verifies the closed-form expressions (5.44)–(5.47) at $k = 1, 2$ and demonstrates how the BPS jumping coefficients (5.45) are absorbed by the unrefined limit.
- Appendix C.1.3 (D0-D6, 3d Young diagrams) confirms the $k = 1, 2$ contributions match the MacMahon function under the CY3 condition, and verifies the recursion relation (5.67).
- Appendix C.1.4 (DT3, 3d Young diagrams with boundary) presents explicit 1-leg residue computations, derives the general DT3 integrand (5.81) via the recursion relation, and computes the simplest 3-leg example.
- Appendix C.1.5 (D0-D8, 4d Young diagrams) demonstrates the $\mathcal{J}_{\geq}$ -factor computation, verifies the 4d recursion relation (5.60), and explains the sign discrepancy between the partition function convention of^[11] and ours.
- Appendix C.1.6 (DT4, 4d Young diagrams with boundary) illustrates the cutoff method for two boundary configurations and derives the DT4 integrand (5.84).

C.1.1 Shell formula and Nekrasov factor

We want to show that the shell formula (5.31) is equivalent to the Nekrasov factor. The idea is simple: we first handle a small rectangular piece of a Young diagram, then tile the whole diagram with such pieces, and finally check that leftover terms cancel. We treat the single-diagram case $\alpha = \beta$ first, then explain how the argument carries over to $\alpha \neq \beta$.

Step 1: A rectangle case

Fix a rectangular subdiagram $\mu \subset \lambda_\alpha$ and two shell boxes $\mathbf{x}_1$ (addable) and $\mathbf{x}_2$ (removable) as in Fig. C-1: $\mathbf{x}_2$ sits flush with the bottom of μ , and $\mathbf{x}_1$ sits directly above it. Split μ into three regions A, B, C (with $A \simeq C$, and C being the strip between $\mathbf{x}_1$ and $\mathbf{x}_2$). Two things can be checked directly:

$$\prod_{\mathbf{y} \in A} \frac{1}{\text{sh}(\mathcal{X}_\alpha(\mathbf{y}) - \mathcal{X}_\alpha(\mathbf{x}_2))} = \prod_{\mathbf{y} \in C} \frac{1}{\text{sh}((L_\lambda(\mathbf{y}) + 1)\epsilon_1 - A_\lambda(\mathbf{y})\epsilon_2)}, \quad (\text{C.1})$$

$$\frac{\prod_{\mathbf{y} \in A \cup B} \text{sh}(\mathcal{X}_\alpha(\mathbf{y}) - \mathcal{X}_\alpha(\mathbf{x}_1))}{\prod_{\mathbf{y} \in B \cup C} \text{sh}(\mathcal{X}_\alpha(\mathbf{y}) - \mathcal{X}_\alpha(\mathbf{x}_2))} = 1. \quad (\text{C.2})$$

The first says that the contribution of A can be rewritten using arm/leg lengths over C. The second says that B drops out. Putting them together, the ratio over all of μ reduces to just a product over C:

$$\prod_{\mathbf{y} \in \mu} \frac{\text{sh}(\mathcal{X}_\alpha(\mathbf{y}) - \mathcal{X}_\alpha(\mathbf{x}_1))}{\text{sh}(\mathcal{X}_\alpha(\mathbf{y}) - \mathcal{X}_\alpha(\mathbf{x}_2))} = \prod_{\mathbf{y} \in C} \frac{\text{sh}(\mathcal{X}_\alpha(\mathbf{y}) - \mathcal{X}_\alpha(\mathbf{x}_1))}{\text{sh}((L_\lambda(\mathbf{y}) + 1)\epsilon_1 - A_\lambda(\mathbf{y})\epsilon_2)}. \quad (\text{C.3})$$

This equality still holds even when A and C intersect, and it is the repeated reduction used in the next step.

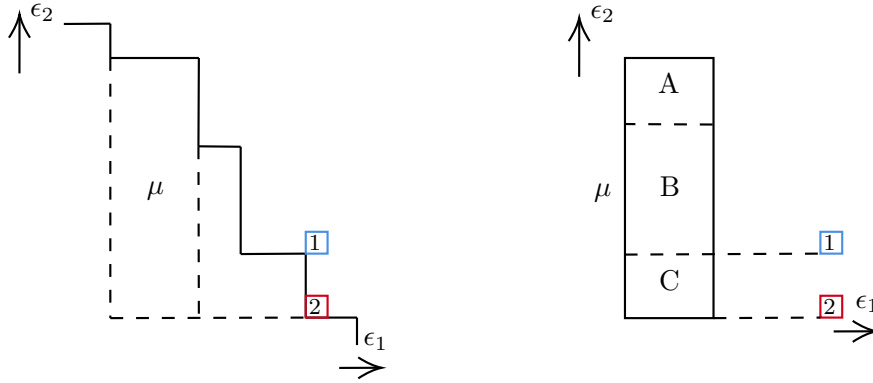


Figure C-1 Left: the Young diagram λ with a rectangular subdiagram μ , an addable box $\mathbf{x}_2$ flush with the bottom of μ , and an addable box $\mathbf{x}_1$ directly above $\mathbf{x}_2$. Right: μ split into regions A, B, C (with $A \simeq C$), where C is the strip between $\mathbf{x}_1$ and $\mathbf{x}_2$.

Step 2: Tiling the full Young diagram

Now take an arbitrary Young diagram λ_α as in Fig. C-2. Label its addable boxes $\mathbf{x}_{1,3,5,7}$ (red) and removable boxes $\mathbf{x}_{2,4,6}$ (blue), with $\mathbf{x}_8$ keeping track of the factor $1/\text{sh}(x - \mathcal{X}(\mathbf{0}))$ in (5.31). Partition λ_α into regions A, ..., F using the addable boxes as dividers.

Apply (C.3) to each rectangle in λ_α —for instance, to $A \cup B \cup C$ paired with $\mathbf{x}_{6,7}$,

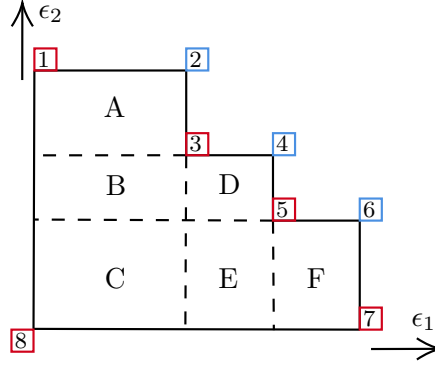


Figure C-2 The 2d Young diagram λ_α split into regions A–F. Red boxes $\mathbf{x}_{1,3,5,7}$: addable; blue boxes $\mathbf{x}_{2,4,6}$: removable; $\mathbf{x}_8$: accounts for $1/\text{sh}(x - \mathcal{X}(\mathbf{0}))$ in (5.31).

and to $E \cup F$ paired with $\mathbf{x}_{3,4}$, and so on. Writing

$$(A - \mathbf{x}_2) \equiv \prod_{y \in A} \text{sh}(\mathcal{X}_\alpha(y) - \mathcal{X}_\alpha(\mathbf{x}_2)), \quad (\text{C.4})$$

$$(B - \mathbf{x}_{2,4}) \equiv \prod_{y \in B} \text{sh}(\mathcal{X}_\alpha(y) - \mathcal{X}_\alpha(\mathbf{x}_2)) \text{sh}(\mathcal{X}_\alpha(y) - \mathcal{X}_\alpha(\mathbf{x}_4)), \quad (\text{C.5})$$

and so on for the other regions, multiplying all the rectangle contributions together gives:

$$\frac{(A - \mathbf{x}_2)(D - \mathbf{x}_4)(F - \mathbf{x}_6)(B - \mathbf{x}_{2,4})(C - \mathbf{x}_{2,6})(E - \mathbf{x}_{4,6})}{\prod_{y \in \lambda_\alpha} \text{sh}((L_{\lambda_\alpha}(y) + 1)\epsilon_1 - A_{\lambda_\alpha}(y)\epsilon_2) \text{sh}((A_{\lambda_\alpha}(y) + 1)\epsilon_2 - L_{\lambda_\alpha}(y)\epsilon_1)}. \quad (\text{C.6})$$

The denominator is exactly the Nekrasov factor in (5.31).

Step 3: The remaining terms cancel

Comparing (C.6) with the full right-hand side of (5.31), there are still terms not yet accounted for:

$$\frac{(C - \mathbf{x}_4)}{(A - \mathbf{x}_8)(B - \mathbf{x}_{3,8})(C - \mathbf{x}_{3,5,8})(D - \mathbf{x}_8)(E - \mathbf{x}_{5,8})(F - \mathbf{x}_8)}. \quad (\text{C.7})$$

One can check directly that when all these terms are put together, they cancel:

$$\frac{(A - \mathbf{x}_2)(B - \mathbf{x}_{2,4})(C - \mathbf{x}_{2,4,6})(D - \mathbf{x}_4)(E - \mathbf{x}_{4,6})(F - \mathbf{x}_6)}{(A - \mathbf{x}_8)(B - \mathbf{x}_{3,8})(C - \mathbf{x}_{3,5,8})(D - \mathbf{x}_8)(E - \mathbf{x}_{5,8})(F - \mathbf{x}_8)} = 1, \quad (\text{C.8})$$

which follows from five identities, each of which is just (C.3) applied to a different sub-region:

$$\frac{(A \cup B \cup C - \mathbf{x}_2)}{(A \cup B \cup C - \mathbf{x}_8)} = \frac{(B \cup C - \mathbf{x}_8)}{(B \cup C - \mathbf{x}_3)} = \frac{(C \cup E \cup F - \mathbf{x}_6)}{(C \cup E \cup F - \mathbf{x}_8)} = \frac{(C \cup E - \mathbf{x}_8)}{(C \cup E - \mathbf{x}_5)} = \frac{(B \cup C \cup D \cup E - \mathbf{x}_4)}{(B \cup C \cup D \cup E - \mathbf{x}_8)} = 1. \quad (\text{C.9})$$

This finishes the proof for $\alpha = \beta$. The same argument works for Young diagrams of any shape.

The case of two distinct Young diagrams ($\alpha \neq \beta$)

When $\lambda_\alpha \neq \lambda_\beta$, we use the same rectangle-tiling idea but now applied to both diagrams simultaneously, as shown in Fig. C-3. The relevant identities become:

$$\prod_{y \in B} \frac{1}{\text{sh}(\mathcal{X}_\beta(y) - \mathcal{X}_\alpha(x_1))} = \prod_{y \in A} \frac{1}{\text{sh}((A_{\lambda_\alpha}(y) + 1)\epsilon_2 - L_{\lambda_\beta}(y)\epsilon_1)}, \quad (\text{C.10})$$

$$\prod_{y \in D} \frac{1}{\text{sh}(\mathcal{X}_\beta(y) - \mathcal{X}_\alpha(x_4))} = \prod_{y \in C} \frac{1}{\text{sh}((L_{\lambda_\alpha}(y) + 1)\epsilon_1 - A_{\lambda_\beta}(y)\epsilon_2)}. \quad (\text{C.11})$$

The remaining terms cancel by the same kind of check as in Step 3. Since the calculation is entirely parallel and adds nothing new, we omit it.

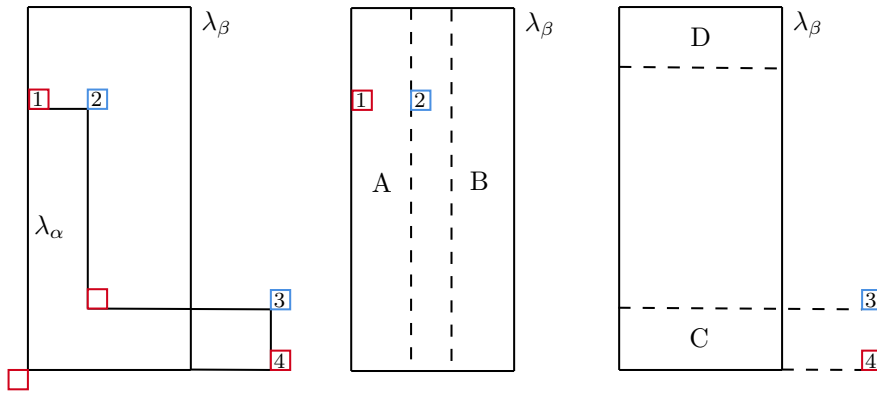


Figure C-3 For two distinct Young diagrams λ_α and λ_β , we apply the same rectangle-tiling strategy to both diagrams at once.

C.1.2 5d Sp(2) SYM

We verify the shell formula expressions (5.44)–(5.47) for Sp(2) at $k = 1$ and $k = 2$, and demonstrate how the BPS jumping coefficient (5.45) arises and is absorbed by the limiting procedure $\lim_{\epsilon_2 \rightarrow -\epsilon_1}$. As a consistency check, the $k = 1$ result should reproduce the SU(2) partition function by the Lie algebra isomorphism $\text{Sp}(2) \simeq \text{SU}(2)$.

$k = 1$. Since the corresponding auxiliary gauge group has rank 0 at level $k = 1$, this reduces to a single term with no integration. The partition function follows directly

from (5.44) and (5.47):

$$\begin{aligned}
 \mathcal{Z}_{2,k=1}^{\text{Sp},+} &= \lim_{\epsilon_2 \rightarrow -\epsilon_1} \frac{\prod_{\alpha=1}^5 \mathcal{J}(0|\emptyset_\alpha)}{\text{sh}(0 + v_1 - \epsilon_{12})} \\
 &= \lim_{\epsilon_2 \rightarrow -\epsilon_1} \frac{1}{\text{sh}(v_1 - \epsilon_{12}) \prod_{\alpha=1}^5 \text{sh}(-v_\alpha)} \\
 &= \lim_{\epsilon_2 \rightarrow -\epsilon_1} \frac{-1}{\text{sh}(\frac{\epsilon_1}{2}) \text{sh}(\frac{\epsilon_1}{2} + \pi i) \text{sh}(\epsilon_1) \text{sh}(\frac{\epsilon_{12}}{2} + \pi i) \text{sh}(v_1) \text{sh}(v_1 - \epsilon_{12})} \\
 &= \frac{1}{2 \text{sh}(v_1)^2 \text{sh}(\epsilon_1)^2} \\
 \mathcal{Z}_{2,k=1}^{\text{Sp},-} &= \lim_{\epsilon_2 \rightarrow -\epsilon_1} \frac{\prod_{\alpha=1}^5 \mathcal{J}(\pi i|\emptyset_\alpha)}{\text{sh}(-\pi i + v_1 - \epsilon_{12})} \\
 &= \lim_{\epsilon_2 \rightarrow -\epsilon_1} \frac{-1}{\text{sh}(\frac{\epsilon_1}{2}) \text{sh}(\frac{\epsilon_1}{2} + \pi i) \text{sh}(\epsilon_1) \text{sh}(\frac{\epsilon_{12}}{2} + \pi i) \text{sh}(v_1 + \pi i) \text{sh}(v_1 - \epsilon_{12} + \pi i)} \\
 &= -\frac{\text{sh}(v_1)^2}{2 \text{sh}(2v_1)^2 \text{sh}(\epsilon_1)^2} \tag{C.12}
 \end{aligned}$$

where we substituted the frozen brane values $v_2, \dots, v_5$ from Tab. 5-2 and used the identity $\text{sh}(x + \pi i) = i \text{sh}(2x)/\text{sh}(x)$. A further identity $\text{sh}(x + \pi i)^2 + \text{sh}(x)^2 = -4$ then gives the full $k = 1$ partition function:

$$\mathcal{Z}_{2,k=1}^{\text{Sp}} = \mathcal{Z}_{2,k=1}^{\text{Sp},+} + \mathcal{Z}_{2,k=1}^{\text{Sp},-} = \frac{2}{\text{sh}(2v_1)^2 \text{sh}(\epsilon_1)^2} \tag{C.13}$$

This agrees exactly with the $\text{SU}(2)$ unrefined instanton partition function, as required by the Lie algebra isomorphism $\text{Sp}(2) \simeq \text{SU}(2)$.

$k = 2$. For the minus sector, $k = 2$ admits no non-trivial Young diagrams, so the result is immediate:

$$\begin{aligned}
 \mathcal{Z}_{2,k=2}^{\text{Sp},-} &= \lim_{\epsilon_2 \rightarrow -\epsilon_1} \frac{\prod_{\alpha=1}^5 \mathcal{J}(0|\emptyset_\alpha)}{\text{sh}(0 + v_1 - \epsilon_{12})} \frac{\prod_{\alpha=1}^5 \mathcal{J}(\pi i|\emptyset_\alpha)}{\text{sh}(-\pi i + v_1 - \epsilon_{12})} \\
 &= \frac{-1}{\text{sh}(2v_1)^2 \text{sh}(\epsilon_1)^2 \text{sh}(2\epsilon_1)^2} \tag{C.14}
 \end{aligned}$$

However, the plus sector at level $k = 2$ possesses one degree of freedom corresponding to a single box. This box can be placed on any of the five empty Young diagrams, resulting in a total of five distinct Young diagram configurations λ :

$$\begin{aligned}
 &((1, 1)_1, \emptyset_2, \emptyset_3, \emptyset_4, \emptyset_5), \quad (\emptyset_1, \{(1, 1)_2, \emptyset_3, \emptyset_4, \emptyset_5\}, \quad (\emptyset_1, \emptyset_2, \{(1, 1)_3, \emptyset_4, \emptyset_5\}) \\
 &(\emptyset_1, \emptyset_2, \emptyset_3, \{(1, 1)_4, \emptyset_5\}), \quad (\emptyset_1, \emptyset_2, \emptyset_3, \emptyset_4, \{(1, 1)_5\}) \tag{C.15}
 \end{aligned}$$

We also know the $\mathcal{J}$ -factor for a single 2d box from (A.14). The plus sector partition function is then (first equality uses $\mathcal{J}(x|\emptyset_\alpha) = 1/\text{sh}(x - v_\alpha)$; second equality substitutes the frozen brane values from Tab. 5-2; the limit $\epsilon_2 \rightarrow -\epsilon_1$ reduces the five contributions

to three distinct types):

$$\begin{aligned}
 \mathcal{Z}_{2,k=2}^{\text{Sp},+} &= \lim_{\epsilon_2 \rightarrow -\epsilon_1} \sum_{i=1}^5 \frac{\mathcal{J}(\pm v_i | \{(1, 1)\}_i)}{\text{sh}(v_i \pm v_i - \epsilon_{12}) \prod_{j \neq i}^5 \text{sh}(\pm v_i - v_j)} \\
 &= \lim_{\epsilon_2 \rightarrow -\epsilon_1} \left(-\frac{1}{\text{sh}(2v_1 \pm \epsilon_1) \text{sh}(2v_1 + \epsilon_{1,2}) \text{sh}(\epsilon_{1,2}) \text{sh}(2v_1 - \epsilon_{12})^2} \right. \\
 &\quad + \frac{1}{2 \text{sh}(v_1 \pm \frac{\epsilon_1}{2}) \text{sh}(v_1 - \epsilon_{12} \pm \frac{\epsilon_1}{2}) \text{sh}(2\epsilon_1)^2 \text{sh}(\epsilon_2)^2} \\
 &\quad + \frac{\text{sh}(v_1 \pm \frac{\epsilon_1}{2}) \text{sh}(v_1 - \epsilon_{12} \pm \frac{\epsilon_1}{2})}{2 \text{sh}(2v_1 \pm \epsilon_1) \text{sh}(2v_1 - 2\epsilon_{12} \pm \epsilon_1) \text{sh}(2\epsilon_1)^2 \text{sh}(\epsilon_2)^2} \\
 &\quad - \frac{\text{sh}(\epsilon_{12}^2)}{2 \text{sh}(v_1 \pm \frac{\epsilon_{12}}{2}) \text{sh}(v_1 - \epsilon_{12} \pm \frac{\epsilon_{12}}{2}) \text{sh}(\epsilon_1) \text{sh}(\epsilon_2)^2 \text{sh}(2\epsilon_1 + \epsilon_2)^2 \text{sh}(\epsilon_1 + 2\epsilon_2)} \\
 &\quad \left. - \frac{\text{sh}(\epsilon_{12})^2 \text{sh}(v_1 \pm \frac{\epsilon_{12}}{2}) \text{sh}(v_1 - \epsilon_{12} \pm \frac{\epsilon_{12}}{2})}{2 \text{sh}(2v_1 \pm \epsilon_{12}) \text{sh}(2v_1 - 2\epsilon_{12} \pm \epsilon_{12}) \text{sh}(\epsilon_1) \text{sh}(\epsilon_2)^2 \text{sh}(\epsilon_1 + 2\epsilon_2) \text{sh}(2\epsilon_1 + \epsilon_2)^2} \right) \\
 &= \frac{1}{\text{sh}(2v_1)^2 \text{sh}(2v_1 \pm \epsilon_1)^2 \text{sh}(\epsilon_1)^2} + \frac{1}{2 \text{sh}(v_1 \pm \frac{\epsilon_1}{2})^2 \text{sh}(\epsilon_1)^2 \text{sh}(2\epsilon_1)^2} + \frac{\text{sh}(v_1 \pm \frac{\epsilon_1}{2})^2}{2 \text{sh}(2v_1 \pm \epsilon_1)^2 \text{sh}(\epsilon_1)^2 \text{sh}(2\epsilon_1)^2} \\
 &\quad (C.16)
 \end{aligned}$$

Here, at the first equality, we have used $\mathcal{J}(x | \emptyset_\alpha) = 1/\text{sh}(x - v_\alpha)$. At the second equality, we substituted the specific values of the frozen branes as Tab. 5-2. The complete partition function for the Sp(2) $k = 2$ theory is:

$$\begin{aligned}
 \mathcal{Z}_{2,k=2}^{\text{Sp}} &= \frac{1}{\text{sh}(2v_1)^2 \text{sh}(2v_1 \pm \epsilon_1)^2 \text{sh}(\epsilon_1)^2} + \frac{1}{2 \text{sh}(v_1 \pm \frac{\epsilon_1}{2})^2 \text{sh}(\epsilon_1)^2 \text{sh}(2\epsilon_1)^2} \\
 &\quad + \frac{\text{sh}(v_1 \pm \frac{\epsilon_1}{2})^2}{2 \text{sh}(2v_1 \pm \epsilon_1)^2 \text{sh}(\epsilon_1)^2 \text{sh}(2\epsilon_1)^2} - \frac{1}{\text{sh}(2v_1)^2 \text{sh}(\epsilon_1)^2 \text{sh}(2\epsilon_1)^2} \quad (C.17)
 \end{aligned}$$

BPS jumping coefficient. To illustrate the emergence of the BPS jumping coefficients (5.45) and their absorption into the shell formula, we consider the configuration $\vec{\lambda} = (\emptyset_1, \lambda_2, \emptyset_3, \emptyset_4, \emptyset_5)$, where λ_2 is the first Young diagram shape that produces a non-trivial coefficient. It is the L-shaped diagram:

$$\lambda_2 = \{(1, 1), (1, 2), (1, 3), (2, 1), (3, 1), (2, 2)\} \quad (C.18)$$

The corresponding $\mathcal{J}$ -factor is given by:

$$\mathcal{J}(x | \lambda_2) = \frac{\text{sh}(x - v_2 - \epsilon_1 - 3\epsilon_2) \text{sh}(x - v_2 - 2\epsilon_{12}) \text{sh}(x - v_2 - 3\epsilon_1 - \epsilon_2)}{\text{sh}(x - v_2 - 3\epsilon_{1,2}) \text{sh}(x - v_2 - \epsilon_1 - 2\epsilon_2) \text{sh}(x - v_2 - 2\epsilon_1 - \epsilon_2)} \quad (C.19)$$

The corresponding contribution of $\vec{\lambda}$ is then:

$$\mathcal{Z}^{\text{Sp},+}(\vec{\lambda}) \propto \lim_{\epsilon_2 \rightarrow -\epsilon_1} \frac{\text{sh}(\epsilon_{12})^3}{2 \text{sh}(2\epsilon_{12})^2 \text{sh}(3\epsilon_{12})} = \frac{1}{24} \quad (C.20)$$

The term in $\mathcal{Z}^{\text{Sp},+}$ corresponding to $\vec{\lambda}$ thus carries an extra coefficient $\frac{1}{24}$ at the unrefined limit. Indeed, the exact result is:

$$\begin{aligned}
 \mathcal{Z}^{\text{Sp},+}(\vec{\lambda}) &= \frac{1}{24 \text{sh}(\epsilon_1)^2 \text{sh}(2\epsilon_1)^4 \text{sh}(3\epsilon_1)^6 \text{sh}(4\epsilon_1)^6 \text{sh}(5\epsilon_1)^5 \text{sh}(6\epsilon_1)^2} \\
 &\quad \times \frac{1}{\text{sh}(v_1 \pm \frac{5}{2}\epsilon_1)^2 \text{sh}(v_1 \pm \frac{3}{2}\epsilon_1)^4 \text{sh}(v_1 \pm \frac{1}{2}\epsilon_1)^6} \quad (C.21)
 \end{aligned}$$

By comparing the Eq.(2.12) of^[85], the expression for the $\text{Sp}(2)$ plus sector with the configuration $\vec{\lambda} = (\emptyset_1, \lambda_2, \emptyset_3, \emptyset_4, \emptyset_5)$, is given by:

$$\begin{aligned} \mathcal{Z}^{\text{Sp},+}(\vec{\lambda}) = & C_{\vec{\lambda},\nu}^{\text{Sp}} \times \frac{1}{64 \text{sh}(\epsilon_1)^2 \text{sh}(2\epsilon_1)^4 \text{sh}(3\epsilon_1)^6 \text{sh}(4\epsilon_1)^6 \text{sh}(5\epsilon_1)^5 \text{sh}(6\epsilon_1)^2} \\ & \times \frac{1}{\text{sh}(v_1 \pm \frac{5}{2}\epsilon_1)^2 \text{sh}(v_1 \pm \frac{3}{2}\epsilon_1)^4 \text{sh}(v_1 \pm \frac{1}{2}\epsilon_1)^6} \end{aligned} \quad (\text{C.22})$$

In this configuration, the number of diagonal boxes in λ_2 is $j = 2$. Hence, by (5.45), the coefficient $C_{\vec{\lambda},\nu}^{\text{Sp}}$ reads:

$$C_{\vec{\lambda},\nu}^{\text{Sp}} = C_{\emptyset, \frac{\epsilon_1}{2}}^{\text{Sp}} C_{\lambda_2, \frac{\epsilon_1}{2} + \pi i}^{\text{Sp}} C_{\emptyset, 0}^{\text{Sp}} C_{\emptyset, \pi i}^{\text{Sp}} = \frac{2^{2j-1}}{\binom{2j-1}{j-1}} = \frac{8}{3} \quad (\text{C.23})$$

Thus, the factor $\frac{1}{64}$ contained in (C.22), combined with the prefactor $C_{\vec{\lambda},\nu}^{\text{Sp}} = \frac{8}{3}$, yields the complete coefficient $\frac{1}{24}$, which matches exactly the result in (C.21). This example demonstrates that, through careful implementation of the limiting procedure $\lim_{\epsilon_2 \rightarrow -\epsilon_1}$, the coefficient $C_{\vec{\lambda},\nu}^{\text{Sp}}$ can be fully absorbed into the shell formula.

C.1.3 D0-D6 partition function

We compute the $k = 1$ and $k = 2$ instanton contributions for a single D6_4 -brane using the shell formula (5.66), verify that the results match the MacMahon function coefficients under the CY3 condition $\epsilon_{123} = 0$, and confirm the recursion relation (5.67) explicitly. We focus on the configuration $\{(1, 1, 1), (1, 1, 2)\}$ at $k = 2$ as a representative; the other two $k = 2$ configurations follow by symmetry.

$k = 1$. The partition function (5.65) involves only a first-order contour integral, and there is a unique 3d Young diagram configuration $\vec{\pi} = \{(1, 1, 1)\}$. Computing the shellboxes and their charges (Tab. C-1) gives:

$Q = +1$	(1, 2, 2)	(2, 1, 2)	(2, 2, 1)
$Q = -1$	(1, 1, 2)	(1, 2, 1)	(2, 1, 1) (2, 2, 2)

Table C-1 The charges of the shellboxes of the 3d Young diagram $\{(1, 1, 1)\}$. The second column displays the coordinates of each shellbox.

Therefore the $\mathcal{J}$ -factor of $\{(1, 1, 1)\}$ is:

$$\mathcal{J}(x | \{(1, 1, 1)\}_{\vec{4},1}) = \frac{\text{sh}(x - v_{\vec{4},1} - \epsilon_{12}) \text{sh}(x - v_{\vec{4},1} - \epsilon_{13}) \text{sh}(x - v_{\vec{4},1} - \epsilon_{23})}{\text{sh}(x - v_{\vec{4},1} - \epsilon_{1,2,3}) \text{sh}(x - v_{\vec{4},1} - \epsilon_{123})} \quad (\text{C.24})$$

The partition function (5.66) becomes:

$$\begin{aligned} \mathcal{Z}_{(1,0,0,0),k=1}^{\text{D0-D6}} &= \text{sh}(\mathcal{X}_{\vec{4},1}(1, 1, 1) - \mathcal{X}_{\vec{4},1}(0, 0, 0)) \mathcal{J}(\mathcal{X}_{\vec{4},1}(1, 1, 1) | \{(1, 1, 1)\}_{\vec{4},1}) \\ &= - \frac{\text{sh}(\epsilon_{12}) \text{sh}(\epsilon_{13}) \text{sh}(\epsilon_{23})}{\text{sh}(\epsilon_1) \text{sh}(\epsilon_2) \text{sh}(\epsilon_3)} \end{aligned} \quad (\text{C.25})$$

Under the CY threefold condition $\epsilon_{123} = 0^{[103]}$, this reduces to 1, matching the first-order coefficient of the MacMahon function:

$$\prod_{k=1}^{\infty} \frac{1}{(1 - q^k)^k} = 1 + q + 3q^2 + 6q^3 + 13q^4 + \dots \quad (\text{C.26})$$

$k = 2$. There are three 3d Young diagram configurations:



$$\{(1, 1, 1), (2, 1, 1)\}$$



$$\{(1, 1, 1), (1, 2, 1)\}$$



$$\{(1, 1, 1), (1, 1, 2)\}$$

Let us focus on the configuration $\{(1, 1, 1), (1, 1, 2)\}$, the charges of the shellboxes are listed in Tab. C-2.

$Q = +1$	(1, 2, 3)	(2, 1, 3)	(2, 2, 1)
$Q = -1$	(1, 1, 3)	(1, 2, 1)	(2, 1, 1) (2, 2, 3)

Table C-2 The charges of the shellboxes of the 3d Young diagram $\{(1, 1, 1), (1, 1, 2)\}$.

Therefore the corresponding $\mathcal{F}$ -factor is:

$$\mathcal{F}(x | \{(1, 1, 1), (1, 1, 2)\}_{\bar{4},1}) = \frac{\text{sh}(x - v_{\bar{4},1} - \epsilon_{12}) \text{sh}(x - v_{\bar{4},1} - \epsilon_1 - 2\epsilon_3) \text{sh}(x - v_{\bar{4},1} - \epsilon_2 - 2\epsilon_3)}{\text{sh}(x - v_{\bar{4},1} - \epsilon_{1,2}) \text{sh}(x - v_{\bar{4},1} - 2\epsilon_3) \text{sh}(x - v_{\bar{4},1} - \epsilon_{12} - 2\epsilon_3)} \quad (\text{C.27})$$

The contribution from $\{(1, 1, 1), (1, 1, 2)\}$ is:

$$\mathcal{Z}_{(1,0,0)}^{\text{D0-D6}}(\{(1, 1, 1), (1, 1, 2)\}_{\bar{4},1}) = \frac{\text{sh}(\epsilon_{12}) \text{sh}(\epsilon_{13}) \text{sh}(\epsilon_{23}) \text{sh}(\epsilon_{12} - \epsilon_3) \text{sh}(\epsilon_1 + 2\epsilon_3) \text{sh}(\epsilon_2 + 2\epsilon_3)}{\text{sh}(\epsilon_1) \text{sh}(\epsilon_2) \text{sh}(\epsilon_3) \text{sh}(2\epsilon_3) \text{sh}(\epsilon_1 - \epsilon_3) \text{sh}(\epsilon_2 - \epsilon_3)} \quad (\text{C.28})$$

By symmetry among $\epsilon_1, \epsilon_2, \epsilon_3$, the contributions from $\{(1, 1, 1), (2, 1, 1)\}$ and $\{(1, 1, 1), (1, 2, 1)\}$ follow immediately. Under the CY3 condition $\epsilon_{123} = 0$ each of the three terms reduces to 1, giving $\mathcal{Z}_{(1,0,0),k=2}^{\text{D0-D6}} = 3$, which matches the second-order MacMahon coefficient (C.26). This confirms that the shell formula reproduces the known enumerative-geometry result at low instanton number.

Recursion relation (5.67). The ratio of consecutive instanton contributions is:

$$\frac{\mathcal{Z}^{\text{D0-D6}}(\{(1, 1, 1), (1, 1, 2)\}_{\bar{4},1})}{\mathcal{Z}^{\text{D0-D6}}(\{(1, 1, 1)\}_{\bar{4},1})} = - \frac{\text{sh}(\epsilon_{12} - \epsilon_3) \text{sh}(\epsilon_1 + 2\epsilon_3) \text{sh}(\epsilon_2 + 2\epsilon_3)}{\text{sh}(\epsilon_1 - \epsilon_3) \text{sh}(\epsilon_2 - \epsilon_3) \text{sh}(2\epsilon_3)} \quad (\text{C.29})$$

The recursion relation (5.67) is verified computationally: evaluating the $\mathcal{F}$ -factors at the new box location (1, 1, 2) in the enlarged diagram, and at the shifted point (2, 2, 3) in the original diagram, gives:

$$\begin{aligned} \mathcal{F}(\mathcal{X}_{\bar{4},1}(1, 1, 2) | \{(1, 1, 1), (1, 1, 2)\}_{\bar{4},1}) &= - \frac{\text{sh}(\epsilon_{13}) \text{sh}(\epsilon_{23}) \text{sh}(\epsilon_{12} - \epsilon_3)}{\text{sh}(\epsilon_3) \text{sh}(\epsilon_{123}) \text{sh}(\epsilon_1 - \epsilon_3) \text{sh}(\epsilon_2 - \epsilon_3)} \\ \mathcal{F}(\mathcal{X}_{\bar{4},1}(2, 2, 3) | \{(1, 1, 1)\}_{\bar{4},1}) &= \frac{\text{sh}(2\epsilon_3) \text{sh}(\epsilon_{13}) \text{sh}(\epsilon_{23})}{\text{sh}(\epsilon_3) \text{sh}(\epsilon_{123}) \text{sh}(\epsilon_1 + 2\epsilon_3) \text{sh}(\epsilon_2 + 2\epsilon_3)} \end{aligned} \quad (\text{C.30})$$

Their ratio reproduces the partition function ratio, confirming (5.67):

$$\frac{\mathcal{Z}^{\text{D0-D6}}(\{(1, 1, 1), (1, 1, 2)\}_{\bar{4},1})}{\mathcal{Z}^{\text{D0-D6}}(\{(1, 1, 1)\}_{\bar{4},1})} = \frac{\mathcal{F}(\mathcal{X}_{\bar{4},1}(1, 1, 2) | \{(1, 1, 1), (1, 1, 2)\}_{\bar{4},1})}{\mathcal{F}(\mathcal{X}_{\bar{4},1}(2, 2, 3) | \{(1, 1, 1)\}_{\bar{4},1})} \quad (\text{C.31})$$

C.1.4 DT3 counting

This subsection presents three calculations: (i) an explicit $k = 1$ residue computation for a 1-leg vacuum with boundary $\lambda = \{(1, 1), (1, 2), (2, 1)\}$, demonstrating that the asymptotic contribution cancels and the $\mathcal{F}$ -factor reduces to a finite 2d contribution as in (5.80); (ii) a recursive derivation of the same result using (5.67), which generalizes to arbitrary k and yields the DT3 integrand (5.81); and (iii) the simplest 3-leg case $\pi_{\lambda\lambda\lambda}$ with $\lambda = \{(1, 1)\}$. Since there is only a single D6-brane throughout, we omit the Young diagram label $(123, 1)$.

1-leg case: direct residue computation. We consider the partition function of three D2_1 -branes inside a D6_{123} -brane. The vacuum is a minimal 3d Young diagram $\pi_{\lambda\emptyset\emptyset}$ extending infinitely in the $\mathbb{C}_1$ direction with boundary condition $\lambda_{23,1} = \{(1, 1), (1, 2), (2, 1)\}$, as shown in Fig. C-4. The charges of the shellboxes of $\pi_{\lambda\emptyset\emptyset}$ are listed in Tab. C-3.

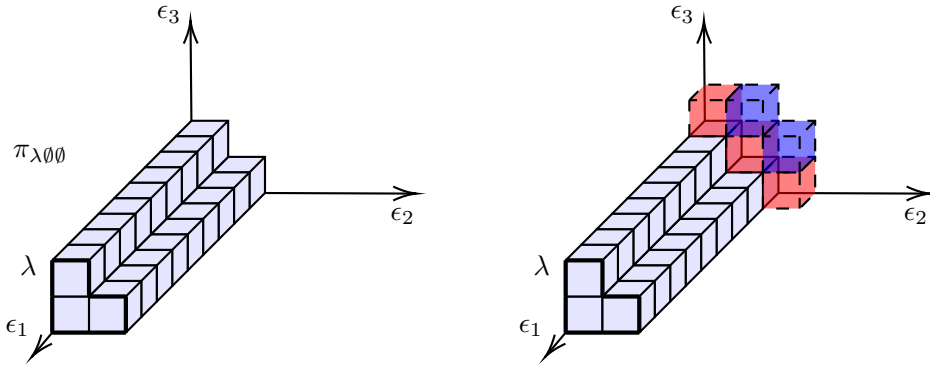


Figure C-4 The left figure shows a one-leg 3d Young diagram $\pi_{\lambda\emptyset\emptyset}$ with boundaries, where the boundary in the ϵ_1 direction is a 2d Young diagram λ in the 23-plane, and the boundaries in the other two directions are empty sets $\emptyset$. The right figure shows the shellboxes with non-zero charges for $\pi_{\lambda\emptyset\emptyset}$, where red represents -1 charge and blue represents $+1$ charge. Note that the contributions from all shellboxes at the infinite boundary are canceled out; therefore, only the finite-distance shellboxes provide non-trivial contributions to the partition function.

$Q = +1$	$(1, 2, 3)$	$(1, 3, 2)$	$(\infty, 1, 3)$	$(\infty, 2, 2)$	$(\infty, 3, 1)$
$Q = -1$	$(1, 1, 3)$	$(1, 2, 2)$	$(1, 3, 1)$	$(\infty, 1, 1)$	$(\infty, 2, 3)$ $(\infty, 3, 2)$

Table C-3 The charges of the shellboxes of the 3d Young diagram $\pi_{\lambda\emptyset\emptyset}$.

As (5.79) mentioned, the contribution to the $\mathcal{F}$ -factor from the shellboxes at the infinite end reads:

$$\frac{\text{sh}(x - \infty\epsilon_1 - 2\epsilon_3) \text{sh}(x - \infty\epsilon_1 - \epsilon_{23}) \text{sh}(x - \infty\epsilon_1 - 2\epsilon_2)}{\text{sh}(x - \infty\epsilon_1) \text{sh}(x - \infty\epsilon_1 - \epsilon_2 - 2\epsilon_3) \text{sh}(x - \infty\epsilon_1 - 2\epsilon_2 - \epsilon_3)} = 1 \quad (\text{C.32})$$

Therefore, the complete $\mathcal{F}$ -factor as (5.80) is:

$$\begin{aligned}\mathcal{F}(x|\pi_{\lambda\emptyset\emptyset}) &= \frac{\text{sh}(x-v-\epsilon_2-2\epsilon_3)\text{sh}(x-v-2\epsilon_2-\epsilon_3)}{\text{sh}(x-v-2\epsilon_2)\text{sh}(x-v-2\epsilon_3)\text{sh}(x-v-\epsilon_{23})} \\ &= \mathcal{F}(x+v_{23,1}-v|\lambda_{23,1})\end{aligned}\quad (\text{C.33})$$

The partition function before integration reads:

$$\mathcal{Z}_{\lambda,\emptyset,\emptyset;k}^{\text{D0-D2-D6}} = (-1)^k \mathcal{Z}_k^{\text{D0-D0}} \times \prod_{i=1}^k \frac{\text{sh}(\phi_i-v-\epsilon_{234})}{\text{sh}(\phi_i-v-\epsilon_{23})} \frac{\text{sh}(\phi_i-v-2\epsilon_{2,3}-\epsilon_4)}{\text{sh}(\phi_i-v-2\epsilon_{2,3})} \frac{\text{sh}(-\phi_i+v+\epsilon_{2,3}-\epsilon_{14})}{\text{sh}(-\phi_i+v+\epsilon_{2,3}-\epsilon_1)} \quad (\text{C.34})$$

For the vacuum configuration $\pi_{\lambda\emptyset\emptyset}$ at $k=1$, there are three poles, which correspond to three distinct box locations:

$$\tilde{\pi}_{\lambda\emptyset\emptyset} = \pi \setminus \pi_{\lambda\emptyset\emptyset} = \{(1,2,2)\}, \quad \{(1,3,1)\}, \quad \text{and} \quad \{(1,1,3)\} \quad (\text{C.35})$$

where $|\tilde{\pi}_{\lambda\emptyset\emptyset}| = k = 1$. Using (5.81), the partition functions read:

$$\begin{aligned}\mathcal{Z}_{\lambda,\emptyset,\emptyset}^{\text{D0-D2-D6}}(1,2,2) &= \frac{\text{sh}(\epsilon_{23})\text{sh}(\epsilon_1+2\epsilon_{2,3})}{\text{sh}(\epsilon_1)\text{sh}(\epsilon_2-\epsilon_3)^2} \\ \mathcal{Z}_{\lambda,\emptyset,\emptyset}^{\text{D0-D2-D6}}(1,3,1) &= \frac{\text{sh}(\epsilon_{13})\text{sh}(\epsilon_{23})\text{sh}(\epsilon_1+2\epsilon_2)\text{sh}(\epsilon_2-2\epsilon_3)\text{sh}(\epsilon_1+3\epsilon_2-\epsilon_3)}{\text{sh}(\epsilon_{1,2})\text{sh}(\epsilon_2-\epsilon_3)\text{sh}(2\epsilon_2-2\epsilon_3)\text{sh}(\epsilon_1+2\epsilon_2-\epsilon_3)} \\ \mathcal{Z}_{\lambda,\emptyset,\emptyset}^{\text{D0-D2-D6}}(1,1,3) &= -\frac{\text{sh}(\epsilon_{12})\text{sh}(\epsilon_{23})\text{sh}(2\epsilon_2-\epsilon_3)\text{sh}(\epsilon_1+2\epsilon_3)\text{sh}(\epsilon_1-\epsilon_2+3\epsilon_3)}{\text{sh}(\epsilon_{1,3})\text{sh}(\epsilon_2-\epsilon_3)\text{sh}(2\epsilon_2-2\epsilon_3)\text{sh}(\epsilon_1-\epsilon_2+2\epsilon_3)} \\ \mathcal{Z}_{\lambda,\emptyset,\emptyset;k=1}^{\text{D0-D2-D6}} &= \mathcal{Z}_{\lambda,\emptyset,\emptyset}^{\text{D0-D2-D6}}(1,2,2) + \mathcal{Z}_{\lambda,\emptyset,\emptyset}^{\text{D0-D2-D6}}(1,3,1) + \mathcal{Z}_{\lambda,\emptyset,\emptyset}^{\text{D0-D2-D6}}(1,1,3)\end{aligned}\quad (\text{C.36})$$

Derivation via recursion relation (5.67). An equivalent approach identifies the D2-brane with an infinite row of D0-branes. Interpreted as additional boxes placed on the vacuum 3d Young diagram $\pi_{\lambda\emptyset\emptyset}$ with $\lambda = \{(1,1), (1,2), (2,1)\}$, the DT3 invariant follows from the D0-D6 recursion relation (5.67):

$$\begin{aligned}\mathcal{Z}_{\lambda,\emptyset,\emptyset}^{\text{D0-D2-D6}}(1,2,2) &= \frac{\mathcal{F}(\mathcal{X}(1,2,2)|\pi_{\lambda\emptyset\emptyset} \cup \{(1,2,2)\})}{\mathcal{F}(\mathcal{X}(1,2,2) + \epsilon_{123}|\pi_{\lambda\emptyset\emptyset})} \\ &= \frac{\mathcal{F}(\mathcal{X}(1,2,2)|\pi_{\lambda\emptyset\emptyset})}{\mathcal{F}(\mathcal{X}(1,2,2) + \epsilon_{123}|\pi_{\lambda\emptyset\emptyset})} \text{sh}(0) \mathcal{F}(\mathcal{X}(1,2,2)|\{(1,2,2)\}) \\ &= \frac{\text{sh}(\epsilon_{23})\text{sh}(\epsilon_1+2\epsilon_{2,3})}{\text{sh}(\epsilon_1)\text{sh}(\epsilon_2-\epsilon_3)^2}\end{aligned}\quad (\text{C.37})$$

where the second equality is obtained by applying the splitting property (5.21): the $\mathcal{F}$ -factor on the enlarged diagram $\pi_{\lambda\emptyset\emptyset} \cup \{(1,2,2)\}$ factorizes into the $\mathcal{F}$ -factor on the vacuum times a single-box contribution at the new box location, with the interface $\text{sh}(0)$ factor canceling trivially. The contributions $\mathcal{Z}_{\lambda,\emptyset,\emptyset}^{\text{D0-D2-D6}}(1,1,3)$ and $\mathcal{Z}_{\lambda,\emptyset,\emptyset}^{\text{D0-D2-D6}}(1,3,1)$ from the other two poles follow similarly.

Derivation of the general DT3 integrand (5.81). The recursive approach generalizes to arbitrary k . For a given pole corresponding to a set $\tilde{\pi}_{\lambda\mu\nu} = \{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ arranged so that each step $\pi_{\lambda\mu\nu} \cup \{\mathbf{x}_1\}$, $\pi_{\lambda\mu\nu} \cup \{\mathbf{x}_1\} \cup \{\mathbf{x}_2\}$, $\dots$ remains a valid 3d Young diagram, iterating the recursion relation (5.67) gives:

$$\begin{aligned} \mathcal{F}_{\lambda,\mu,\nu}^{\text{D0-D2-D6}}(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k) &= \frac{\mathcal{F}(\mathcal{X}(\mathbf{x}_1) | \pi_{\lambda\mu\nu} \cup \{\mathbf{x}_1\})}{\mathcal{F}(\mathcal{X}(\mathbf{x}_1) + \epsilon_{123} | \pi_{\lambda\mu\nu})} \frac{\mathcal{F}(\mathcal{X}(\mathbf{x}_2) | \pi_{\lambda\mu\nu} \cup \{\mathbf{x}_1\} \cup \{\mathbf{x}_2\})}{\mathcal{F}(\mathcal{X}(\mathbf{x}_2) + \epsilon_{123} | \pi_{\lambda\mu\nu} \cup \{\mathbf{x}_1\})} \times \dots \\ &= \left(\prod_{i=1}^k \frac{\mathcal{F}(\mathcal{X}(\mathbf{x}_i) | \pi_{\lambda\mu\nu})}{\mathcal{F}(\mathcal{X}(\mathbf{x}_i) + \epsilon_{123} | \pi_{\lambda\mu\nu})} \right) \prod_{i,j} \text{sh}(\mathcal{X}(\mathbf{x}_i) - \mathcal{X}(\mathbf{x}_j)) \mathcal{F}(\mathcal{X}(\mathbf{x}_i) | \mathbf{x}_j) \\ &= \left(\prod_{i=1}^k \frac{\mathcal{F}(\mathcal{X}(\mathbf{x}_i) | \pi_{\lambda\mu\nu})}{\mathcal{F}(\mathcal{X}(\mathbf{x}_i) + \epsilon_{123} | \pi_{\lambda\mu\nu})} \right) \\ &\quad \times \left(\prod_{i,j} \frac{\text{sh}(\mathcal{X}(\mathbf{x}_i) - \mathcal{X}(\mathbf{x}_j)) \text{sh}(\mathcal{X}(\mathbf{x}_i) - \mathcal{X}(\mathbf{x}_j) - \epsilon_{12,13,23})}{\text{sh}(\mathcal{X}(\mathbf{x}_i) - \mathcal{X}(\mathbf{x}_j) - \epsilon_{1,2,3}) \text{sh}(\mathcal{X}(\mathbf{x}_i) - \mathcal{X}(\mathbf{x}_j) - \epsilon_{123})} \right) \end{aligned} \quad (\text{C.38})$$

The passage from the first to the second line applies the splitting property (5.21) to factor the $\mathcal{F}$ -factor on the enlarged diagram into a vacuum contribution and a single-box contribution at each new box location. The swapping property (5.14) then converts the resulting double product into a symmetric pairwise form. Note that although $\text{sh}(0)$ appears when $i = j$ in the second line, the overall expression remains finite.

Substituting the integration variables ϕ_i for $\mathcal{X}(\mathbf{x}_i)$ and replacing the denominator $\mathcal{F}$ -factor with $\mathcal{F}_-$ to enforce the correct pole structure, one arrives at the integral form of the DT3 invariant (5.81).

3-leg case: $\pi_{\lambda\lambda\lambda}$ with $\lambda = \{(1, 1)\}$. The simplest 3-leg vacuum arises when all three legs carry a single-box boundary. A key simplification occurs because all pairwise and triple intersections in the inclusion-exclusion formula (5.78) coincide with a single box: $\pi_{\lambda\emptyset\emptyset} \cap \pi_{\emptyset\lambda\emptyset} = \pi_{\lambda\emptyset\emptyset} \cap \pi_{\emptyset\emptyset\lambda} = \pi_{\emptyset\lambda\emptyset} \cap \pi_{\emptyset\emptyset\lambda} = \pi_{\lambda\emptyset\emptyset} \cap \pi_{\emptyset\lambda\emptyset} \cap \pi_{\emptyset\emptyset\lambda} = \{(1, 1, 1)\}$, so all intersection $\mathcal{F}$ -factors are equal and the formula reduces considerably. Using (5.78) and Fig. C-5:

$$\begin{aligned} \mathcal{F}(x | \pi_{\lambda\lambda\lambda}) &= \frac{\mathcal{F}(x | \pi_{\lambda\emptyset\emptyset}) \mathcal{F}(x | \pi_{\emptyset\lambda\emptyset}) \mathcal{F}(x | \pi_{\emptyset\emptyset\lambda}) \mathcal{F}(x | \pi_{\lambda\emptyset\emptyset} \cap \pi_{\emptyset\lambda\emptyset} \cap \pi_{\emptyset\emptyset\lambda})}{\mathcal{F}(x | \pi_{\lambda\emptyset\emptyset} \cap \pi_{\emptyset\lambda\emptyset}) \mathcal{F}(x | \pi_{\lambda\emptyset\emptyset} \cap \pi_{\emptyset\emptyset\lambda}) \mathcal{F}(x | \pi_{\emptyset\lambda\emptyset} \cap \pi_{\emptyset\emptyset\lambda})} \\ &= \frac{\text{sh}(x - v - \epsilon_{123})^2}{\text{sh}(x - v - \epsilon_{12}) \text{sh}(x - v - \epsilon_{13}) \text{sh}(x - v - \epsilon_{23})} \end{aligned} \quad (\text{C.39})$$

where in this configuration:

$$\pi_{\lambda\emptyset\emptyset} \cap \pi_{\emptyset\lambda\emptyset} \cap \pi_{\emptyset\emptyset\lambda} = \pi_{\lambda\emptyset\emptyset} \cap \pi_{\emptyset\lambda\emptyset} = \pi_{\lambda\emptyset\emptyset} \cap \pi_{\emptyset\emptyset\lambda} = \pi_{\emptyset\lambda\emptyset} \cap \pi_{\emptyset\emptyset\lambda} = \{(1, 1, 1)\} \quad (\text{C.40})$$

The integrand is:

$$\mathcal{F}_{\lambda,\lambda,\lambda;k}^{\text{D0-D2-D6}} = (-1)^k \mathcal{F}_k^{\text{D0-D0}} \times \prod_{i=1}^k \frac{\text{sh}(v - \epsilon_{1,2,3} - \phi_i) \text{sh}(\phi_i - v - \epsilon_{123})^2}{\text{sh}(\phi_i - v - \epsilon_{12,13,23}) \text{sh}(v - \phi_i)^2} \quad (\text{C.41})$$

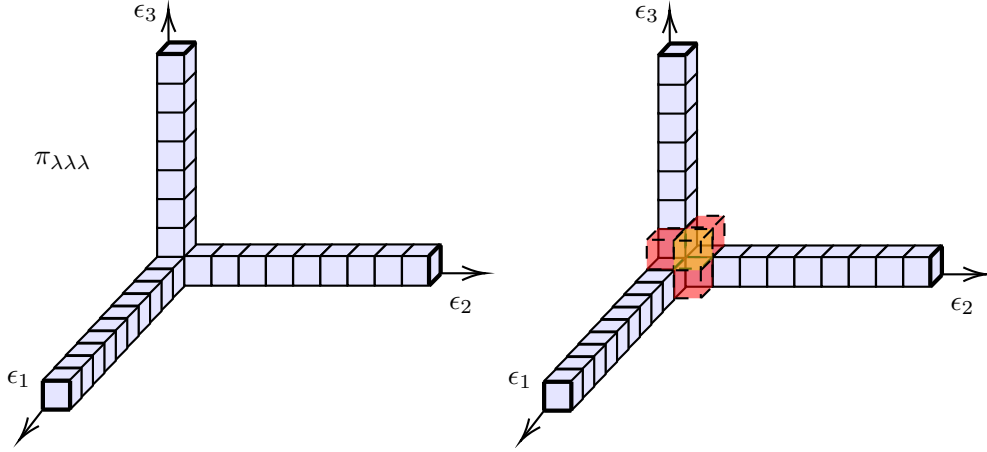


Figure C-5 The left image shows the 3d Young diagram $\pi_{\lambda\lambda\lambda}$ corresponding to the vacuum configuration $(\lambda, \lambda, \lambda)$, where $\lambda = \{(1, 1)\}$ is a 2d Young diagram with a single box. The right image displays the shellboxes with non-trivial charges: red boxes carry charge -1 , and yellow boxes carry charge $+2$.

After applying the JK-residue, there are three poles $\mathcal{X}(1, 2, 2)$, $\mathcal{X}(2, 1, 2)$, $\mathcal{X}(2, 2, 1)$, and their contributions are:

$$\begin{aligned} \mathcal{Z}_{\lambda,\lambda,\lambda}^{\text{D0-D2-D6}}(1, 2, 2) &= - \frac{\text{sh}(\epsilon_1) \text{sh}(\epsilon_{12}) \text{sh}(\epsilon_{13}) \text{sh}(2\epsilon_2 + \epsilon_3) \text{sh}(\epsilon_2 + 2\epsilon_3)}{\text{sh}(\epsilon_{2,3}) \text{sh}(\epsilon_{23}) \text{sh}(\epsilon_1 - \epsilon_2) \text{sh}(\epsilon_1 - \epsilon_3)} \\ \mathcal{Z}_{\lambda,\lambda,\lambda}^{\text{D0-D2-D6}}(2, 1, 2) &= \frac{\text{sh}(\epsilon_2) \text{sh}(\epsilon_{12}) \text{sh}(\epsilon_{23}) \text{sh}(2\epsilon_1 + \epsilon_3) \text{sh}(\epsilon_1 + 2\epsilon_3)}{\text{sh}(\epsilon_{1,3}) \text{sh}(\epsilon_{13}) \text{sh}(\epsilon_1 - \epsilon_2) \text{sh}(\epsilon_2 - \epsilon_3)} \\ \mathcal{Z}_{\lambda,\lambda,\lambda}^{\text{D0-D2-D6}}(2, 2, 1) &= - \frac{\text{sh}(\epsilon_3) \text{sh}(\epsilon_{13}) \text{sh}(\epsilon_{23}) \text{sh}(2\epsilon_1 + \epsilon_2) \text{sh}(\epsilon_1 + 2\epsilon_2)}{\text{sh}(\epsilon_{1,2}) \text{sh}(\epsilon_{12}) \text{sh}(\epsilon_1 - \epsilon_3) \text{sh}(\epsilon_2 - \epsilon_3)} \end{aligned} \quad (\text{C.42})$$

Note that the three contributions are related by cyclic permutation of $(\epsilon_1, \epsilon_2, \epsilon_3)$, as dictated by the S_3 symmetry of the 3-leg vacuum.

2-leg case. The computation proceeds analogously; only the intersection formula simplifies. The $\mathcal{F}$ -factor is:

$$\begin{aligned} \mathcal{F}(x|\pi_{\lambda\mu\emptyset}) &= \frac{\mathcal{F}(x|\pi_{\lambda\emptyset\emptyset}) \mathcal{F}(x|\pi_{\emptyset\mu\emptyset}) \mathcal{F}(x|\pi_{\emptyset\emptyset\emptyset}) \mathcal{F}(x|\pi_{\lambda\emptyset\emptyset} \cap \pi_{\emptyset\mu\emptyset} \cap \pi_{\emptyset\emptyset\emptyset})}{\mathcal{F}(x|\pi_{\lambda\emptyset\emptyset} \cap \pi_{\emptyset\mu\emptyset}) \mathcal{F}(x|\pi_{\lambda\emptyset\emptyset} \cap \pi_{\emptyset\emptyset\emptyset}) \mathcal{F}(x|\pi_{\emptyset\mu\emptyset} \cap \pi_{\emptyset\emptyset\emptyset})} \\ &= \frac{\mathcal{F}(x|\pi_{\lambda\emptyset\emptyset}) \mathcal{F}(x|\pi_{\emptyset\mu\emptyset})}{\mathcal{F}(x|\pi_{\lambda\emptyset\emptyset} \cap \pi_{\emptyset\mu\emptyset})} \end{aligned} \quad (\text{C.43})$$

C.1.5 D0-D8 partition function

We compute the $k = 1$ and $k = 2$ contributions for a single D8-brane ($N = 1$), demonstrating the $\mathcal{F}_{\geq}$ -factor procedure (5.54) and verifying the recursion relation (5.60). Recall that $\mathcal{F}_{\geq}(\mathcal{X}_{\mathcal{B}}(\mathbf{x})|\rho_{\mathcal{A}})$ selects only shell boxes with last coordinate $y_d \leq x_d$. The $k = 2$ calculation additionally clarifies the sign discrepancy between the convention of [11] and ours: the two conventions agree for three of the four $k = 2$ Young diagrams, and differ by a sign for the fourth (ρ_4) due to the replacement $\text{sh}(\phi_i - \phi_j + \epsilon_4) \rightarrow \text{sh}(\phi_i - \phi_j - \epsilon_4)$ in the D0-D0 sector. Throughout, we omit the label $(\mathbf{4}, 1)$ for brevity.

$k = 1$: $\mathcal{F}_{\geq}$ **computation.** The 4d Young diagram $\{(1, 1, 1, 1)\}$ has 15 shell boxes with charges listed in Tab. C-4. For the input $\mathcal{X}_{\underline{4},1}(1, 1, 1, 1)$, the $\mathcal{F}_{\geq}$ definition restricts to

Q = +1	(1, 1, 2, 2)	(1, 2, 1, 2)	(1, 2, 2, 1)	(2, 1, 1, 2)
	(2, 1, 2, 1)	(2, 2, 1, 1)	(2, 2, 2, 2)	
Q = -1	(1, 1, 1, 2)	(1, 1, 2, 1)	(1, 2, 1, 1)	(1, 2, 2, 2)
	(2, 1, 1, 1)	(2, 1, 2, 2)	(2, 2, 1, 2)	(2, 2, 2, 1)

Table C-4 The charges of the shellboxes of the 4d Young diagram $\{(1, 1, 1, 1)\}$.

shell boxes with $x_4 \leq 1$, reducing the 15 boxes to the 7 boxes in Tab. C-5.

Q = +1	(1, 2, 2, 1)	(2, 1, 2, 1)	(2, 2, 1, 1)	
Q = -1	(1, 1, 2, 1)	(1, 2, 1, 1)	(2, 1, 1, 1)	(2, 2, 2, 1)

Table C-5 The charges of the shellboxes of the 4d Young diagram $\{(1, 1, 1, 1)\}$ when the input of $\mathcal{F}_{\geq}$ is $\mathcal{X}_{\underline{4},\alpha}(x_1, x_2, x_3, 1)$.

Thus, we have:

$$\begin{aligned} \mathcal{Z}_{N=1,k=1}^{\text{D0-D8}} &= \mathcal{Z}^{\text{D0-D8}}(\{(1, 1, 1, 1)\}) = \mathcal{F}_{\geq}(\mathcal{X}_{\underline{4},1}(1, 1, 1, 1) | \{(1, 1, 1, 1)\}_{\underline{4},1}) \\ &= - \frac{\text{sh}(\epsilon_{12}) \text{sh}(\epsilon_{13}) \text{sh}(\epsilon_{23})}{\text{sh}(\epsilon_{1,2,3}) \text{sh}(\epsilon_{123})} \end{aligned} \quad (\text{C.44})$$

$k = 2$: **recursion relation (5.60).** The 4d Young diagram $\rho_{\underline{4},1} = \{(1, 1, 1, 1), (2, 1, 1, 1)\}_{\underline{4},1}$ has all boxes with fourth coordinate equal to 1, so $\mathcal{F}_{\geq}$ restricts to shellboxes with $x_4 \leq 1$ as in Tab. C-6.

Q = +1	(1, 2, 2, 1)	(3, 1, 2, 1)	(3, 2, 1, 1)	
Q = -1	(1, 1, 2, 1)	(1, 2, 1, 1)	(3, 1, 1, 1)	(3, 2, 2, 1)

Table C-6 The charges of the shellboxes of the 4d Young diagram $\{(1, 1, 1, 1), (2, 1, 1, 1)\}$ when the input of $\mathcal{F}_{\geq}$ is $\mathcal{X}_{\underline{4},\alpha}(x_1, x_2, x_3, 1)$.

Therefore, we have:

$$\begin{aligned} \mathcal{Z}^{\text{D0-D8}}(\{(1, 1, 1, 1), (2, 1, 1, 1)\}_{\underline{4},1}) &= - \frac{\text{sh}(2\epsilon_1 + \epsilon_2) \text{sh}(2\epsilon_1 + \epsilon_3) \text{sh}(\epsilon_{23})}{\text{sh}(2\epsilon_1) \text{sh}(\epsilon_2) \text{sh}(\epsilon_3) \text{sh}(2\epsilon_1 + \epsilon_{23})} \\ &\quad \times \frac{\text{sh}(\epsilon_{12}) \text{sh}(\epsilon_1 - \epsilon_{23}) \text{sh}(\epsilon_{13})}{\text{sh}(\epsilon_1) \text{sh}(\epsilon_1 - \epsilon_2) \text{sh}(\epsilon_1 - \epsilon_3) \text{sh}(\epsilon_{123})} \end{aligned} \quad (\text{C.45})$$

For the other two Young diagrams $\{(1, 1, 1, 1), (1, 2, 1, 1)\}$ and $\{(1, 1, 1, 1), (1, 1, 2, 1)\}$, the corresponding contributions follow by symmetry in $\epsilon_1, \epsilon_2, \epsilon_3$. However, for $\{(1, 1, 1, 1), (1, 1, 1, 2)\}$ the fourth coordinate of one box differs from the rest, so the full shellbox charges are listed in Tab. C-7.

Fortunately, for both inputs $\mathcal{X}_{\underline{4},\alpha}(1, 1, 1, 1)$ and $\mathcal{X}_{\underline{4},\alpha}(1, 1, 1, 2)$, the last coordinate of both boxes is less than 3. Therefore, the shellboxes actually required are the same as those shown in Tab. C-5. Thus, the contribution of this Young diagram is:

$$\begin{aligned} \mathcal{Z}^{\text{D0-D8}}(\{(1, 1, 1, 1), (1, 1, 1, 2)\}_{\underline{4},1}) &= \frac{\text{sh}(\epsilon_{12}) \text{sh}(\epsilon_{13}) \text{sh}(\epsilon_{23})}{\text{sh}(\epsilon_{1,2,3}) \text{sh}(\epsilon_{123})} \frac{\text{sh}(\epsilon_{12} - \epsilon_4) \text{sh}(\epsilon_{13} - \epsilon_4) \text{sh}(\epsilon_{23} - \epsilon_4)}{\text{sh}(\epsilon_{1,2,3} - \epsilon_4) \text{sh}(\epsilon_{123} - \epsilon_4)} \end{aligned} \quad (\text{C.46})$$

Q = +1	(1, 1, 2, 3)	(1, 2, 1, 3)	(1, 2, 2, 1)	(2, 1, 1, 3)
	(2, 1, 2, 1)	(2, 2, 1, 1)	(2, 2, 2, 3)	
Q = -1	(1, 1, 1, 3)	(1, 1, 2, 1)	(1, 2, 1, 1)	(1, 2, 2, 3)
	(2, 1, 1, 1)	(2, 1, 2, 3)	(2, 2, 1, 3)	(2, 2, 2, 1)

 Table C-7 The charges of the shellboxes of the 4d Young diagram $\{(1, 1, 1, 1), (1, 1, 1, 2)\}$.

Comparing the contributions of $\{(1, 1, 1, 1)\}$ (C.44) and $\{(1, 1, 1, 1), (2, 1, 1, 1)\}$ (C.45) gives:

$$\frac{\mathcal{Z}^{\text{D0-D8}}(\{(1, 1, 1, 1), (2, 1, 1, 1)\}_{\underline{4},1})}{\mathcal{Z}^{\text{D0-D8}}(\{(1, 1, 1, 1)\}_{\underline{4},1})} = \frac{\text{sh}(2\epsilon_1 + \epsilon_2) \text{sh}(\epsilon_1 - \epsilon_2 - \epsilon_3) \text{sh}(2\epsilon_1 + \epsilon_3)}{\text{sh}(2\epsilon_1) \text{sh}(\epsilon_1 - \epsilon_2) \text{sh}(\epsilon_1 - \epsilon_3) \text{sh}(2\epsilon_1 + \epsilon_2 + \epsilon_3)} \quad (\text{C.47})$$

According to (5.60), we obtain $\mathcal{F}_{\geq}(\mathcal{X}_{\underline{4},1}(2, 1, 1, 1) | \{(1, 1, 1, 1), (2, 1, 1, 1)\}_{\underline{4},1})$ from the charges in Tab. C-6, and $\mathcal{F}_{<}(\mathcal{X}_{\underline{4},1}(2, 1, 1, 1) | \{(1, 1, 1, 1)\}_{\underline{4},1})$ from the boxes with last coordinate greater than 1 in Tab. C-4:

$$\begin{aligned} \mathcal{F}_{\geq}(\mathcal{X}_{\underline{4},1}(2, 1, 1, 1) | \{(1, 1, 1, 1), (2, 1, 1, 1)\}_{\underline{4},1}) &= \frac{\text{sh}(\epsilon_{12}) \text{sh}(\epsilon_1 - \epsilon_{23}) \text{sh}(\epsilon_{13})}{\text{sh}(\epsilon_1) \text{sh}(\epsilon_1 - \epsilon_{2,3}) \text{sh}(\epsilon_{123})} \\ \mathcal{F}_{<}(\mathcal{X}_{\underline{4},1}(2, 1, 1, 1) | \{(1, 1, 1, 1)\}_{\underline{4},1}) &= \frac{\text{sh}(\epsilon_1 - \epsilon_{24}) \text{sh}(\epsilon_1 - \epsilon_{34}) \text{sh}(\epsilon_4) \text{sh}(\epsilon_{234})}{\text{sh}(\epsilon_1 - \epsilon_4) \text{sh}(\epsilon_1 - \epsilon_{234}) \text{sh}(\epsilon_{24}) \text{sh}(\epsilon_{34})} \end{aligned} \quad (\text{C.48})$$

Multiplying the two expressions and applying the CY4 condition $\epsilon_{1234} = 0$ verifies the recursion relation (5.60):

$$\begin{aligned} \frac{\mathcal{Z}^{\text{D0-D8}}(\{(1, 1, 1, 1), (2, 1, 1, 1)\}_{\underline{4},1})}{\mathcal{Z}^{\text{D0-D8}}(\{(1, 1, 1, 1)\}_{\underline{4},1})} &= \mathcal{F}_{\geq}(\mathcal{X}_{\underline{4},1}(2, 1, 1, 1) | \{(1, 1, 1, 1), (2, 1, 1, 1)\}_{\underline{4},1}) \\ &\quad \times \mathcal{F}_{<}(\mathcal{X}_{\underline{4},1}(2, 1, 1, 1) | \{(1, 1, 1, 1)\}_{\underline{4},1}) \end{aligned} \quad (\text{C.49})$$

$k = 2$: sign rule. The D0-D0 sector $\tilde{\mathcal{F}}_k^{\text{D0-D0}}$ used in [11] differs from $\mathcal{F}_k^{\text{D0-D0}}$ in (5.52): it replaces $\text{sh}(\phi_i - \phi_j - \epsilon_4)$ in the denominator by $\text{sh}(\phi_i - \phi_j + \epsilon_4)$:

$$\tilde{\mathcal{F}}_k^{\text{D0-D0}} = \frac{1}{k!} \prod_{i \neq j}^k \text{sh}(\phi_i - \phi_j) \prod_{i,j}^k \frac{\text{sh}(\phi_i - \phi_j - \epsilon_{1,2,3} - \epsilon_4)}{\text{sh}(\phi_i - \phi_j - \epsilon_{1,2,3}) \text{sh}(\phi_i - \phi_j + \epsilon_4)} \quad (\text{C.50})$$

The corresponding full partition function is:

$$\tilde{\mathcal{F}}_{N,k}^{\text{D0-D8-D8}} = \tilde{\mathcal{F}}_k^{\text{D0-D0}} \times \prod_{i=1}^k \prod_{\alpha=1}^N \frac{\text{sh}(\phi_i - w_{\underline{4},\alpha})}{\text{sh}(\phi_i - v_{\underline{4},\alpha})} \quad (\text{C.51})$$

For $N = 1$, $k = 2$ there are four poles, corresponding to:

$$\begin{aligned} \rho_1 &= \{(1, 1, 1, 1), (2, 1, 1, 1)\}, & \rho_2 &= \{(1, 1, 1, 1), (1, 2, 1, 1)\}, \\ \rho_3 &= \{(1, 1, 1, 1), (1, 1, 2, 1)\}, & \rho_4 &= \{(1, 1, 1, 1), (1, 1, 1, 2)\} \end{aligned} \quad (\text{C.52})$$

Only ρ_4 requires an additional minus sign relative to our convention, because according to the sign rule:

$$\text{Res}_{\mathcal{X}(\mathbf{x} \in \rho)} \left(\prod_{i=1}^k \frac{d\phi_i}{2\pi i} \right) \tilde{\mathcal{F}}_{N,k}^{\text{D0-D8-}\overline{\text{D8}}} = (-1)^{h(\rho)} \text{Res}_{\mathcal{X}(\mathbf{x} \in \rho)} \left(\prod_{i=1}^k \frac{d\phi_i}{2\pi i} \right) \mathcal{F}_{N,k}^{\text{D0-D8-}\overline{\text{D8}}} \quad (\text{C.53})$$

where:

$$h(\rho) = 1 + |\rho| + \#\{(a, d) \mid (a, a, a, d) \in \rho \text{ and } a \leq d\} \quad (\text{C.54})$$

One finds $h(\rho_1) = h(\rho_2) = h(\rho_3) = 4$ (even, no sign change) and $h(\rho_4) = 5$ (odd, sign change). The sign difference arises because the two D0-D0 sectors differ in a specific factor: $\tilde{\mathcal{F}}$ contains $\text{sh}(\phi_1 - \phi_2 + \epsilon_4)$ in the denominator where $\mathcal{F}$ has $\text{sh}(\phi_1 - \phi_2 - \epsilon_4)$. For the pole $(\phi_1, \phi_2) = (\mathcal{X}(1, 1, 1, 1), \mathcal{X}(1, 1, 1, 2))$ this matters:

$$\begin{aligned} \tilde{\mathcal{F}}_{1,2}^{\text{D0-D8-}\overline{\text{D8}}} \supset \tilde{\mathcal{F}} &= \frac{1}{\text{sh}(\phi_{1,2} - v) \text{sh}(\phi_1 - \phi_2 + \epsilon_4) \text{sh}(\phi_2 - \phi_1 + \epsilon_4)} \\ \mathcal{F}_{1,2}^{\text{D0-D8-}\overline{\text{D8}}} \supset \mathcal{F} &= \frac{1}{\text{sh}(\phi_{1,2} - v) \text{sh}(\phi_1 - \phi_2 - \epsilon_4) \text{sh}(\phi_2 - \phi_1 - \epsilon_4)} \end{aligned} \quad (\text{C.55})$$

At the pole ρ_4 :

$$\begin{aligned} \text{Res}_{\mathcal{X}(\mathbf{x} \in \rho_4)} \tilde{\mathcal{F}} &= \frac{1}{\text{sh}(\epsilon_4) \text{sh}(2\epsilon_4)} \\ \text{Res}_{\mathcal{X}(\mathbf{x} \in \rho_4)} \mathcal{F} &= \frac{1}{\text{sh}(\epsilon_4) \text{sh}(-2\epsilon_4)} = -\text{Res}_{\mathcal{X}(\mathbf{x} \in \rho_4)} \tilde{\mathcal{F}} \end{aligned} \quad (\text{C.56})$$

Since we adopt the convention (5.52), the shell formula is formulated accordingly, and no extra sign factor needs to be tracked in our calculation.

C.1.6 DT4 counting

Three examples illustrate $\mathcal{F}$ -factor computations in DT4 theory: (i) the four-leg case with all legs carrying a single-box 3d Young diagram, demonstrating the cutoff method and showing why charges $Q = +2$ and $Q = -3$ appear when four infinite legs meet at a shared origin; (ii) a mixed one-leg and two-surface case; and (iii) the derivation of the general DT4 integrand (5.84) from the recursion relation (5.60), paralleling the DT3 derivation of Appendix C.1.4.

Four-leg case. Consider the minimal 4d Young diagram $\rho_{\pi_1 \pi_2 \pi_3 \pi_4}$ with all four legs being the single-box 3d Young diagram $\pi = \square = \{(1, 1, 1)\}$:

$$\rho_{\pi_1 \pi_2 \pi_3 \pi_4} = \{(i, 1, 1, 1)\}_{i=1}^{\infty} \cup \{(1, i, 1, 1)\}_{i=1}^{\infty} \cup \{(1, 1, i, 1)\}_{i=1}^{\infty} \cup \{(1, 1, 1, i)\}_{i=1}^{\infty} \quad (\text{C.57})$$

In this case, it is sufficient to take the cutoff at $m > 2$; we may simply set $m = 3$. The resulting truncated minimal 4d Young diagram $\rho_{\pi_1 \pi_2 \pi_3 \pi_4; 3}$ is then given by:

$$\begin{aligned} \rho_{\pi_1 \pi_2 \pi_3 \pi_4; 3} &= \{(1, 1, 1, 1), (2, 1, 1, 1), (3, 1, 1, 1), (1, 2, 1, 1), (1, 3, 1, 1), \\ &\quad (1, 1, 2, 1), (1, 1, 3, 1), (1, 1, 1, 2), (1, 1, 1, 3)\} \end{aligned} \quad (\text{C.58})$$

Q = +1	(1, 1, 2, 4)	(1, 1, 4, 2)	(1, 2, 1, 4)	(1, 2, 4, 1)	(1, 4, 1, 2)	(1, 4, 2, 1)
	(2, 1, 1, 4)	(2, 1, 4, 1)	(2, 2, 2, 4)	(2, 2, 4, 2)	(2, 4, 1, 1)	(2, 4, 2, 2)
	(4, 1, 1, 2)	(4, 1, 2, 1)	(4, 2, 1, 1)	(4, 2, 2, 2)		
Q = -1	(1, 1, 1, 4)	(1, 1, 2, 2)	(1, 1, 4, 1)	(1, 2, 1, 2)	(1, 2, 2, 1)	(1, 2, 2, 4)
	(1, 2, 4, 2)	(1, 4, 1, 1)	(1, 4, 2, 2)	(2, 1, 1, 2)	(2, 1, 2, 1)	(2, 1, 2, 4)
	(2, 1, 4, 2)	(2, 2, 1, 1)	(2, 2, 1, 4)	(2, 2, 4, 1)	(2, 4, 1, 2)	(2, 4, 2, 1)
	(4, 1, 1, 1)	(4, 1, 2, 2)	(4, 2, 1, 2)	(4, 2, 2, 1)		
Q = +2	(1, 2, 2, 2)	(2, 1, 2, 2)	(2, 2, 1, 2)	(2, 2, 2, 1)		
Q = -3	(2, 2, 2, 2)					

 Table C-8 The charges of the shellboxes of the minimal 4d Young diagram $\rho_{\pi_1\pi_2\pi_3\pi_4;3}$.

Then, using the coordinates and charges listed in Tab. C-8, and discarding terms that would yield $\text{sh}(x - v - m\epsilon_a + \dots)$ —namely, the shellboxes whose coordinates contain $m + 1 = 4$ —we obtain the $\mathcal{J}$ -factor. Note that the $Q = +2$ and $Q = -3$ entries in Tab. C-8 arise because the four infinite legs all share the origin box $(1, 1, 1, 1)$, causing multiple contributions to accumulate at a single shell box:

$$\mathcal{J}(x|\rho_{\pi_1\pi_2\pi_3\pi_4}) = \frac{\prod_{A \in \check{4}} \text{sh}(x - v - \epsilon_A)^2}{\text{sh}(x - v - \epsilon_{1234})^3 \prod_{ab \in \underline{6}} \text{sh}(x - v - \epsilon_{ab})} \quad (\text{C.59})$$

Performing the sign reversal $\text{sh}(x) \rightarrow -\text{sh}(-x - \epsilon_{1234})$ on all terms except the addable boxes

$$\mathfrak{U}(\rho_{\pi_1\pi_2\pi_3\pi_4}) = \{(1, 1, 2, 2), (1, 2, 1, 2), (1, 2, 2, 1), (2, 1, 1, 2), (2, 1, 2, 1), (2, 2, 1, 1)\},$$

gives the final $\mathcal{J}_{-\mathfrak{U}}$ -factor:

$$\mathcal{J}_{-\mathfrak{U}}(x|\rho_{\pi_1\pi_2\pi_3\pi_4}) = -\frac{\prod_{a \in \underline{4}} \text{sh}(v - x - \epsilon_a)^2}{\text{sh}(v - x)^3 \prod_{ab \in \underline{6}} \text{sh}(x - v - \epsilon_{ab})} \quad (\text{C.60})$$

The partition function of DT4 counting with minimal 4d Young diagram $\rho_{\pi_1\pi_2\pi_3\pi_4}$ is therefore:

$$\mathcal{J}_{\pi_1\pi_2\pi_3\pi_4;k}^{\text{D0-D2-D8}} = -\mathcal{J}_k^{\text{D0-D0}} \times \prod_{i=1}^k \frac{\prod_{a \in \underline{4}} \text{sh}(v - \phi_i - \epsilon_a)^2}{\text{sh}(v - \phi_i)^3 \prod_{ab \in \underline{6}} \text{sh}(\phi_i - v - \epsilon_{ab})} \quad (\text{C.61})$$

One-leg and two-surface case. We compute the minimal Young diagram $\rho_{\pi_1\lambda_{12}\lambda_{13}}$ with $\pi_1 = \{(1, 1, 1)\}$, $\lambda_{12} = \{(1, 1)\}$, and $\lambda_{13} = \{(1, 1), (1, 2)\}$. This mixed boundary configuration illustrates that the cutoff method works equally well when both leg and surface boundaries are present:

$$\rho_{\pi_1\lambda_{12}\lambda_{13}} = \{(i, 1, 1, 1)\}_{i=1}^\infty \cup \{(i, j, 1, 1)\}_{i,j=1}^\infty \cup \{(i, 1, j, 1), (i, 1, j, 2)\}_{i,j=1}^\infty \quad (\text{C.62})$$

We truncate at $m = 4$ and discard all shellboxes whose coordinates contain $m + 1 = 5$. The remaining shellboxes are listed in Tab. C-9, giving:

$$\mathcal{J}(x|\rho_{\pi_1\lambda_{12}\lambda_{13}}) = \frac{\text{sh}(x - v - \epsilon_2 - 2\epsilon_4) \text{sh}(x - v - \epsilon_{234})}{\text{sh}(x - v - \epsilon_{23}) \text{sh}(x - v - 2\epsilon_4) \text{sh}(x - v - \epsilon_{24})} \quad (\text{C.63})$$

Q = +1	(1, 2, 1, 2)	(1, 1, 1, 3)	(1, 2, 2, 1)
Q = -1	(1, 2, 2, 2)	(1, 2, 1, 3)	

Table C-9 The charges of the shellboxes of the 4d Young diagram $\rho_{\pi_1 \lambda_{12} \lambda_{13}}$ with $\pi_1 = \{(1, 1, 1)\}$, $\lambda_{12} = \{(1, 1)\}$, and $\lambda_{13} = \{(1, 1), (1, 2)\}$.

With the addable boxes $\mathfrak{A}(\rho_{\pi_1 \lambda_{12} \lambda_{13}}) = \{(1, 2, 1, 2), (1, 1, 1, 3), (1, 2, 2, 1)\}$, the $\mathcal{J}_{-\mathfrak{A}}$ -factor is:

$$\mathcal{J}_{-\mathfrak{A}}(x | \rho_{\pi_1 \lambda_{12} \lambda_{13}}) = \frac{\text{sh}(v - x + \epsilon_4 - \epsilon_{13}) \text{sh}(v - x - \epsilon_1)}{\text{sh}(x - v - \epsilon_{23}) \text{sh}(x - v - 2\epsilon_4) \text{sh}(x - v - \epsilon_{24})} \quad (\text{C.64})$$

The partition function of DT4 counting with minimal 4d Young diagram $\rho_{\pi_1 \lambda_{12} \lambda_{13}}$ is therefore:

$$\mathcal{J}_{\pi_1 \lambda_{12} \lambda_{13}; k}^{\text{D0-D2-D4-D8}} = \mathcal{J}_k^{\text{D0-D0}} \times \prod_{i=1}^k \frac{\text{sh}(v - \phi_i + \epsilon_4 - \epsilon_{13}) \text{sh}(v - \phi_i - \epsilon_1)}{\text{sh}(\phi_i - v - \epsilon_{23}) \text{sh}(\phi_i - v - 2\epsilon_4) \text{sh}(\phi_i - v - \epsilon_{24})} \quad (\text{C.65})$$

Derivation of the DT4 integrand (5.84). Analogous to the DT3 derivation of Appendix C.1.4, we consider the DT4 invariant for a lattice set $\tilde{\rho}_{\{\pi_a\}, \{\lambda_{ab}\}} = \{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ of k boxes, arranged so that $\rho_{\{\pi_a\}, \{\lambda_{ab}\}} \cup \{\mathbf{x}_1\}$, $\rho_{\{\pi_a\}, \{\lambda_{ab}\}} \cup \{\mathbf{x}_1\} \cup \{\mathbf{x}_2\}$, $\dots$ are all valid 4d Young diagrams. Iterating the D0-D8 recursion (5.60) and applying the identity (5.61) at each step gives:

$$\begin{aligned} \mathcal{Z}_{\{\pi_a\}, \{\lambda_{ab}\}}(\mathbf{x}_1, \dots, \mathbf{x}_k) &= \mathcal{J}_{\geq}(\mathcal{X}(\mathbf{x}_1) | \rho_{\{\pi_a\}, \{\lambda_{ab}\}} \cup \{\mathbf{x}_1\}) \mathcal{J}_{<}(\mathcal{X}(\mathbf{x}_1) | \rho_{\{\pi_a\}, \{\lambda_{ab}\}}) \\ &\quad \times \mathcal{J}_{\geq}(\mathcal{X}(\mathbf{x}_2) | \rho_{\{\pi_a\}, \{\lambda_{ab}\}} \cup \{\mathbf{x}_1\} \cup \{\mathbf{x}_2\}) \mathcal{J}_{<}(\mathcal{X}(\mathbf{x}_2) | \rho_{\{\pi_a\}, \{\lambda_{ab}\}} \cup \{\mathbf{x}_1\}) \\ &\quad \times \dots \\ &= \left(\frac{\text{sh}(-\epsilon_{14,24,34}) \text{sh}(0)}{\text{sh}(-\epsilon_{1,2,3,4})} \mathcal{J}_{\geq}(\mathcal{X}(\mathbf{x}_1) | \rho_{\{\pi_a\}, \{\lambda_{ab}\}}) \right) \mathcal{J}_{<}(\mathcal{X}(\mathbf{x}_1) | \rho_{\{\pi_a\}, \{\lambda_{ab}\}}) \\ &\quad \times \left(\frac{\text{sh}(-\epsilon_{14,24,34}) \text{sh}(0)}{\text{sh}(-\epsilon_{1,2,3,4})} \mathcal{J}_{\geq}(\mathcal{X}(\mathbf{x}_2) | \rho_{\{\pi_a\}, \{\lambda_{ab}\}} \cup \{\mathbf{x}_1\}) \right) \mathcal{J}_{<}(\mathcal{X}(\mathbf{x}_2) | \rho_{\{\pi_a\}, \{\lambda_{ab}\}} \cup \{\mathbf{x}_1\}) \\ &\quad \times \dots \\ &= \left(\frac{\text{sh}(-\epsilon_{14,24,34}) \text{sh}(0)}{\text{sh}(-\epsilon_{1,2,3,4})} \right)^k \mathcal{J}(\mathcal{X}(\mathbf{x}_1) | \rho_{\{\pi_a\}, \{\lambda_{ab}\}}) \mathcal{J}(\mathcal{X}(\mathbf{x}_2) | \rho_{\{\pi_a\}, \{\lambda_{ab}\}} \cup \{\mathbf{x}_1\}) \times \dots \\ &= \left(\prod_{i,j=1}^k \frac{\text{sh}(\mathcal{X}(\mathbf{x}_i) - \mathcal{X}(\mathbf{x}_j)) \text{sh}(\mathcal{X}(\mathbf{x}_i) - \mathcal{X}(\mathbf{x}_j) - \epsilon_{14,24,34})}{\text{sh}(\mathcal{X}(\mathbf{x}_i) - \mathcal{X}(\mathbf{x}_j) - \epsilon_{1,2,3,4})} \right) \prod_{i=1}^k \mathcal{J}(\mathcal{X}(\mathbf{x}_i) | \rho_{\{\pi_a\}, \{\lambda_{ab}\}}) \end{aligned} \quad (\text{C.66})$$

where in the second equality, we use the relation (5.61). The DT4 partition function thus factorizes into a pairwise interaction kernel and a product of vacuum $\mathcal{J}$ -factors, in precise parallel with the DT3 result of Appendix C.1.4. Substituting ϕ_i for $\mathcal{X}(\mathbf{x}_i)$ and applying the appropriate sign reversals yields the DT4 integrand (5.84).

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